

claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

```
- Source: `claude-wsl-ubuntu`  
- Kind: `claude`  
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d.jsonl`  
- SHA-256: `fafc315f6cb60da8cb771c0a914e3711f3e311cf34bdc536efb0ac390ac8c047`  
- Source modified: `2026-06-30T04:01:54+00:00`  
- Imported at: `2026-07-05T16:48:29+00:00`  
- project: `home-avidullu-projects-khelsutra-guru`  
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`
```

Transcript

1. user (2026-06-29T10:42:58.173Z)

hello. Can you please ensure all the repos in this workspace are sync'd to their respective remote heads?

2. assistant (2026-06-29T10:43:01.673Z)

I'll start by discovering all the git repositories in this workspace.

3. user (2026-06-29T10:43:03.255Z)

```
/home/avidullu/projects/khelsutra-guru/sports-data-collector/.git  
/home/avidullu/projects/khelsutra-guru/rally-annotator/.git  
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.git  
/home/avidullu/projects/khelsutra-guru/sports-obsreport/.git  
/home/avidullu/projects/khelsutra-guru/khelsutra/.git  
/home/avidullu/projects/khelsutra-guru/marketing/.git
```

4. user (2026-06-29T10:43:03.399Z)

```
total 40  
drwxr-xr-x 10 avidullu avidullu 4096 Jun 29 09:48 .  
drwxr-xr-x  5 avidullu avidullu 4096 Jun 29 09:48 ..  
drwxr-xr-x  6 avidullu avidullu 4096 Jun 25 13:06 .venv  
drwxr-xr-x  4 avidullu avidullu 4096 Jun  8 05:58 Annotation Setup  
drwxr-xr-x 22 avidullu avidullu 4096 Jun 27 17:27 badminton-highlight-indexer  
drwxr-xr-x 10 avidullu avidullu 4096 Jun 23 12:41 khelsutra  
drwxr-xr-x  6 avidullu avidullu 4096 Jun 25 11:47 marketing  
drwxr-xr-x  8 avidullu avidullu 4096 Jun 25 11:47 rally-annotator  
drwxr-xr-x  9 avidullu avidullu 4096 Jun 25 11:47 sports-data-collector  
drwxr-xr-x  8 avidullu avidullu 4096 Jun 25 11:47 sports-obsreport
```

5. assistant (2026-06-29T10:43:19.972Z)

Found 6 git repositories. Before touching anything, let me inspect the state of each one ? syncing can be destructive if there are local changes or diverged branches, so I'll gather full status first.

6. user (2026-06-29T10:43:24.749Z)

```
=====  
REPO: sports-data-collector  
--- current branch ---  
master  
--- remotes ---  
origin https://github.com/Khelsutra/sports-data-collector.git (fetch)
```

```

origin https://github.com/Khelsutra/sports-data-collector.git (push)
--- upstream tracking branch ---
origin/master
--- working tree status (porcelain) ---
--- stash list ---
=====
REPO: rally-annotator
--- current branch ---
main
--- remotes ---
origin https://github.com/Khelsutra/rally-annotator.git (fetch)
origin https://github.com/Khelsutra/rally-annotator.git (push)
--- upstream tracking branch ---
origin/main
--- working tree status (porcelain) ---
--- stash list ---
=====
REPO: badminton-highlight-indexer
--- current branch ---
master
--- remotes ---
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (fetch)
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (push)
--- upstream tracking branch ---
origin/master
--- working tree status (porcelain) ---
?? gcs:/
?? gs:/
?? local:/
--- stash list ---
=====
REPO: sports-obsreport
--- current branch ---
main
--- remotes ---
origin https://github.com/Khelsutra/sports-obsreport.git (fetch)
origin https://github.com/Khelsutra/sports-obsreport.git (push)
--- upstream tracking branch ---
origin/main
--- working tree status (porcelain) ---
--- stash list ---
=====
REPO: khelsutra
--- current branch ---
master
--- remotes ---
origin https://github.com/avidullu/khelsutra.git (fetch)
origin https://github.com/avidullu/khelsutra.git (push)
--- upstream tracking branch ---
origin/master
--- working tree status (porcelain) ---
--- stash list ---
=====
REPO: marketing
--- current branch ---
main
--- remotes ---
origin https://github.com/Khelsutra/marketing.git (fetch)
origin https://github.com/Khelsutra/marketing.git (push)
--- upstream tracking branch ---
origin/main
--- working tree status (porcelain) ---
--- stash list ---

```

All repos have a clean tracking setup, working trees are clean except badminton-highlight-indexer which has three untracked directories (`gcs:/`, `gs:/`, `local:/`) ? those won't block a fast-forward. No stashes anywhere. Now let me fetch from all remotes (non-destructive) and compute ahead/behind status.

8. user (2026-06-29T10:43:42.899Z)

```
=====
REPO: sports-data-collector
--- fetch ---
--- ahead/behind vs origin/master (left=ahead local, right=behind) ---
0      1
--- local HEAD / upstream HEAD ---
local:    75bb0fa docs: archive CODE_HEALTH_PLAN ? project DONE (#49)
upstream: a152cf5 fix(cvat): warn on rows silently skipped by load_canonical_csv (#50)
=====
REPO: rally-annotator
--- fetch ---
--- ahead/behind vs origin/main (left=ahead local, right=behind) ---
0      0
--- local HEAD / upstream HEAD ---
local:    3d604bc docs: sync READMEs with current state (multi-agent audit fixes) (#25)
upstream: 3d604bc docs: sync READMEs with current state (multi-agent audit fixes) (#25)
=====
REPO: badminton-highlight-indexer
--- fetch ---
From https://github.com/Khelsutra/badminton-highlight-indexer
- [deleted]      (none)    -> origin/b1-p9/remove-fabricated-motion-data
- [deleted]      (none)    -> origin/b2-p17/surface-provider-failures
- [deleted]      (none)    -> origin/chore/ci-job-timeouts
- [deleted]      (none)    -> origin/chore/fix-gitignore-encoding
- [deleted]      (none)    -> origin/chore/format-gemini-refine-trajectory-hybrid
- [deleted]      (none)    -> origin/chore/nightly-quality-regression
- [deleted]      (none)    -> origin/d13-p12/bytetrack-migration
- [deleted]      (none)    -> origin/docs/decision-snapshot-regression
- [deleted]      (none)    -> origin/docs/nightly-person-cue-lane
- [deleted]      (none)    -> origin/docs/pmf-field-signals
- [deleted]      (none)    -> origin/o1-p11/gemini-remote-file-gc
- [deleted]      (none)    -> origin/proposal/rally-detection-fixes
--- ahead/behind vs origin/master (left=ahead local, right=behind) ---
0      4
--- local HEAD / upstream HEAD ---
local:    ffb1f53 ci: harden the 'System libs' apt step against the runner's Microsoft-repo 403
flake (#391)
upstream: 0c9c693 feat(eval): expose inactivity_temporal_density in the rally_seg_eval A/B CLI
(P2a) (#395)
=====
REPO: sports-obsreport
--- fetch ---
--- ahead/behind vs origin/main (left=ahead local, right=behind) ---
0      0
--- local HEAD / upstream HEAD ---
local:    1f72c1c feat: configurable highlights item_noun (v0.1.3)
upstream: 1f72c1c feat: configurable highlights item_noun (v0.1.3)
=====
REPO: khelsutra
--- fetch ---
--- ahead/behind vs origin/master (left=ahead local, right=behind) ---
0      0
--- local HEAD / upstream HEAD ---
local:    ca8bfcc docs(deploy): consolidate GPU_VM_SERVING.md ? rebase cleanup (#119)
upstream: ca8bfcc docs(deploy): consolidate GPU_VM_SERVING.md ? rebase cleanup (#119)
=====
REPO: marketing
--- fetch ---
```

```
--- ahead/behind vs origin/main (left=ahead local, right=behind) ---
0      0
--- local HEAD / upstream HEAD ---
local:   43d69a8 Campaign-ready: 8 verified artifacts + YouTube channel + $300 budget + GO-
LIVE
upstream: 43d69a8 Campaign-ready: 8 verified artifacts + YouTube channel + $300 budget + GO-
LIVE
```

9. assistant (2026-06-29T10:43:57.728Z)

Clear picture now. Here's the state of all 6 repos:

```
| Repo | Branch | Local commits ahead | Behind remote | Working tree |
|-----|-----|:--:|:--:|-----|
| sports-data-collector | master | 0 | **1** | clean |
| badminton-highlight-indexer | master | 0 | **4** | 3 untracked dirs |
| rally-annotator | main | 0 | 0 | clean ? already synced |
| sports-obsreport | main | 0 | 0 | clean ? already synced |
| khelsutra | master | 0 | 0 | clean ? already synced |
| marketing | main | 0 | 0 | clean ? already synced |
```

This is the safe case: **no repo has diverged** (none have local commits ahead), and no repo has uncommitted work at risk. So a fast-forward is clean and lossless for the two that are behind. I'll use `--ff-only` so it refuses to create merge commits or do anything destructive.

10. user (2026-06-29T10:44:02.428Z)

```
=====
REPO: sports-data-collector ? fast-forwarding to @{upstream}
Updating 75bb0fa..a152cf5
Fast-forward
 src/parsers/cvat/rally_cvat.py | 20 ++++++++
 tests/test_cvat_robustness.py | 32 ++++++++
 2 files changed, 50 insertions(+), 2 deletions(-)
--- new status vs upstream (ahead behind) ---
0      0
--- now at: a152cf5 fix(cvat): warn on rows silently skipped by load_canonical_csv (#50)
=====
REPO: badminton-highlight-indexer ? fast-forwarding to @{upstream}
Updating ffb1f53..0c9c693
Fast-forward
 backend/config/models.py | 7 ++
 backend/eval/rally_seg_eval.py | 18 ++++
 backend/eval/windowing.py | 12 +++
 backend/pipeline/detectors/rally_gate.py | 13 +-
 backend/pipeline/detectors/tracknet_runner.py | 30 +++++-
 backend/pipeline/segmenters/trajectory_hybrid.py | 2 +
 docs/NEXT_STEPS.md | 1 +
 docs/NIGHTLY_ENGINE_REGRESSION.md | 47 ++++++
 docs/RALLY_DETECTION_CODE_REVIEW.md | 59 ++++++
 docs/RALLY_DETECTION_FIXES_PLAN.md | 108 ++++++
 tests/test_rally_gate.py | 17 ++++
 tests/test_rally_seg_eval.py | 48 ++++++
 tests/test_tracknet_runner.py | 34 ++++++
 tests/test_windowing.py | 9 ++
 14 files changed, 396 insertions(+), 9 deletions(-)
 create mode 100644 docs/RALLY_DETECTION_CODE_REVIEW.md
 create mode 100644 docs/RALLY_DETECTION_FIXES_PLAN.md
--- new status vs upstream (ahead behind) ---
0      0
--- now at: 0c9c693 feat(eval): expose inactivity_temporal_density in the rally_seg_eval A/B
CLI (P2a) (#395)
```

11. assistant (2026-06-29T10:44:05.777Z)

Both fast-forwards completed cleanly. Let me do a final verification pass across all 6 repos to confirm everyone is at their remote head.

12. user (2026-06-29T10:44:13.606Z)

REPO	BRANCH	SYNC	HEAD
----	-----	----	----
sports-data-collector	master	in sync	a152cf5
rally-annotator	main	in sync	3d604bc
badminton-highlight-indexer	master	in sync	0c9c693
sports-obsreport	main	in sync	1f72c1c
khelsutra	master	in sync	ca8bfcc
marketing	main	in sync	43d69a8

13. assistant (2026-06-29T10:44:22.345Z)

All 6 repos are now synced to their remote heads. ?

Summary

```
| Repo | Branch | Action taken |
|-----|-----|-----|
| **sports-data-collector** | master | Fast-forwarded 1 commit ? `a152cf5` |
| **badminton-highlight-indexer** | master | Fast-forwarded 4 commits ? `0c9c693` (+ pruned 12
deleted remote branches) |
| **rally-annotator** | main | Already in sync (`3d604bc`) |
| **sports-obsreport** | main | Already in sync (`1f72c1c`) |
| **khelsutra** | master | Already in sync (`ca8bfcc`) |
| **marketing** | main | Already in sync (`43d69a8`) |
```

****How I did it safely:**** I fetched all remotes (non-destructive) and checked ahead/behind status first. Every repo had ****0 local commits ahead**** and a ****clean working tree****, so no local work was at risk. I used ``git merge --ff-only`` for the two repos that were behind ? this only fast-forwards and refuses to create merge commits or rewrite history, so it's lossless.

****One thing worth flagging**** ? badminton-highlight-indexer has three untracked directories with unusual names: ``gcs:/``, ``gs:/``, and ``local:/``. These look like artifacts accidentally created by a tool writing to ``gs://...``-style cloud paths as if they were local filesystem paths. They're not tracked by git and didn't block the sync, but you may want to delete them. Want me to take a look at what's inside them?

14. user (2026-06-29T10:47:09.299Z)

I believe they should be some outputs/tmp space for some GPU or other runs of the past. We also have a gcs bucket with some data in it so make sure to cross check with it as well.

Although the aim is to have you follow the below prompt from the previous session and help me close out an open end. PTAL and proceed

We're closing out the rally-detection quality work ? specifically P2b: the golden-corpus promotion decision for the "finding-B" rally-end density fix ? and then producing a launch-readiness signoff for me to approve.

WHERE TO WORK

- Use the clone at C:\Users\avidu\Projects\badminton-highlight-indexer (NOT any khelsutra-guru\... nested path ? that clone was deleted).
- FIRST: ``git fetch origin && git pull --ff-only`` on master so you're at remote HEAD. master already contains: PR #394 (fixes A+B), #395 (the A/B CLI), and the collector fix. Then branch off origin/master for any change (one PR, no stacking).
- GPU + WSL, the golden labels/videos/trajectory CSVs, and the F:\ data drive are ALL live on this machine now ? so the real-video / golden / GPU test paths that used to be owner-only are runnable this session. Locate the golden manifest + features (likely on F:\ or under .\output); confirm the actual paths before running ? don't assume defaults.

RAMP UP (read before touching code)

- docs/RALLY_DETECTION_FIXES_PLAN.md ? §7 tracker (P2a ? done, P2b pending) and §8 Q2 has the exact A/B runbook + decision rule. This is the source of truth.
- docs/RALLY_DETECTION_CODE_REVIEW.md ? the original 3 findings (A/B/C).
- docs/HOW_RALLY_DETECTION_WORKS.md, docs/CODE_MAP.md ? mental model + navigation.
- Code anchors: backend/pipeline/detectors/tracknet_runner.py::_trim_inactive_tail (the fix, gated on `temporal\_density`); backend/config/models.py (`inactivity\_temporal\_density: bool = False` ? the flag to flip); backend/eval/rally_seg_eval.py::_build_preset (+ the --[vs-]inactivity-temporal-density CLI flags); backend/eval/windowing.py (WindowingPreset carries the flag); backend/eval/promotion.py::promote_if_served_safe (the formal served-safety gate).

WHAT FINDING B IS / WHY THE FLAG

- _trim_inactive_tail's R4 rally-end trim measured "fast-frame density" over the count of active SAMPLES, so under sparse shuttle tracking one fast sample after a tracking gap read as 100% dense and held the rally tail open (inflated end). The fix uses the trailing window's TEMPORAL length instead. It is a NO-OP under dense tracking, so it only moves on clips where the tracker actually drops out. R4 is default-ON in prod, so the fix shipped behind `inactivity\_temporal\_density` (default-OFF) = a zero-regression no-op until measured. P2b is that measurement + the promote/reject decision.

P2b PROCEDURE

1. Ensure golden features exist (regenerate if needed; GPU/WSL):
python -m backend.eval.fusion_golden --manifest <golden_manifest.json> --out-dir <features_dir>
2. Run the A/B at the SERVED operating point (flag OFF vs ON):
python -m backend.eval.rally_seg_eval --manifest <golden_manifest.json> \
--features-dir <features_dir> --preset served --vs served \
--vs-inactivity-temporal-density --iou 0.5
Headline metric = tol_f1 (R2 tolerance-match F1); also inspect |?end| / rally-end precision and the paired sign-test + CI delta. Apply promote_if_served_safe (served_regress_eps=0.0 ? no served-mean drop allowed).
3. DECIDE:
 - If tol_f1 does NOT regress (neutral or better) at SERVED ? PROMOTE: flip IndexingConfig.inactivity_temporal_density default to True (one-line change) + a test update; open one PR.
 - If it regresses ? DO NOT promote: leave the flag OFF (the dormant code stays), and record the numbers as the rejection rationale.Either way: update docs/RALLY_DETECTION_FIXES_PLAN.md §7 (flip P2b to ? DONE or mark REJECTED with the measured deltas) and follow docs/DOC_STATUS.md to honestly update ? archive the plan doc (P1, P2a already done; P2b is the last open row).

FULL VALIDATION (for my signoff)

- Run the complete offline suite: python run_all_tests.py (expect ~1487+ passing).
- Now that GPU/WSL/data are live, ALSO run the real-video / golden / native-WASB / GPU paths that were previously skipped ? enumerate what's runnable and run them; report what passed and anything still not verifiable.
- ruff check . and mypy must be clean (match CI); keep coverage at/above the CI floor.
- Cross-reference the quality table (docs/RALLY_DETECTION_QUALITY_REPORT.md / NEXT_STEPS ship target) so the launch call is evidence-based.

DELIVERABLE = a LAUNCH-READINESS REPORT, then stop for my signoff:

- ? P2b result (the A/B numbers + promote/reject decision + PR link)
- ? full test-suite status (offline + the now-unlocked GPU/golden paths) ? what passed, what couldn't run and why
- ? any open guardrail items (licensing, WASB weight provenance, etc.)
- ? a clear GO / NO-GO recommendation with the reasoning.

Recommend the launch call ? do NOT make it; I sign off.

GUARDRAILS: branch off origin/master; one PR per item; explicit `git add <paths>` (a sibling/owner session may share the checkout); never push to master directly. Honor CLAUDE.md (commercial-clean licensing; paid LLMs = inference only; owner-gated work).

15. assistant (2026-06-29T10:49:28.029Z)

This is a substantial close-out task. Before running anything heavy, I need to (a) resolve an important discrepancy I'm noticing, and (b) confirm the live environment ? the prompt was written for a Windows machine and explicitly says *not* to use the `khelsutra-guru\...\` nested path, yet that's exactly where you've launched me and what I just synced. Let me reconcile that against reality before touching code.

I'll kick off the documentation ramp-up in parallel (background workflow) while I do live-environment + path reconnaissance directly.

16. user (2026-06-29T10:50:03.827Z)

Workflow launched in background. Task ID: wiko3gbys
Summary: Digest rally-detection P2b docs + code anchors into one structured runbook
Transcript dir: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/subagents/workflows/wf_fa7a2077-3a9
Script file: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/workflows/scripts/p2b-rampup-wf_fa7a2077-3a9.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/workflows/scripts/p2b-rampup-wf_fa7a2077-3a9.js"} to iterate without resending the script.)
Run ID: wf_fa7a2077-3a9
To resume after editing the script: Workflow({scriptPath: "/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/workflows/scripts/p2b-rampup-wf_fa7a2077-3a9.js", resumeFromRunId: "wf_fa7a2077-3a9"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

17. user (2026-06-29T10:50:09.879Z)

GPU (nvidia-smi)

Mon Jun 29 10:50:09 2026

+-----+											
NVIDIA-SMI 595.54				Driver Version: 595.79				CUDA Version: 13.2			
+-----+											
GPU		Name		Persistence-M		Bus-Id		Disp.A		Volatile Uncorr. ECC	
Fan		Temp		Perf		Pwr:Usage/Cap		Memory-Usage		GPU-Util Compute M.	
										MIG M.	
+-----+											
0		NVIDIA GeForce RTX 2070 ...		On		00000000:01:00.0		On		N/A	
N/A		49C P8		10W / 80W		1606MiB / 8192MiB				2% Default	
										N/A	
+-----+											

+-----+									
Processes:									
GPU	GI	CI	PID	Type	Process name	GPU Memory			
	ID	ID				Usage			
=====									
No running processes found									
+-----+									

/mnt contents

c
f
wsl
wslg

Windows projects dir (/mnt/c/Users/avidu/Projects)

total 0
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 10:40 .
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 10:38 ..
drwxrwxrwx 1 avidullu avidullu 4096 Jun 16 04:25 Vairagi

```
drwxrwxrwx 1 avidullu avidullu 4096 Jun 22 04:11 _p3_scratch
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 10:40 khelsutra-guru
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 08:32 muneem-ai
```

Windows BHI clone state

fatal: cannot change to '/mnt/c/Users/avidu/Projects/badminton-highlight-indexer': No such file or directory

fatal: cannot change to '/mnt/c/Users/avidu/Projects/badminton-highlight-indexer': No such file or directory

F: data drive (/mnt/f)

18. user (2026-06-29T10:50:17.272Z)

python3 on PATH

```
/home/avidullu/projects/khelsutra-guru/.venv/bin/python
/home/avidullu/projects/khelsutra-guru/.venv/bin/python3
/usr/bin/python3
/bin/python3
```

workspace root venv

```
/home/avidullu/projects/khelsutra-guru/.venv/bin/python
/home/avidullu/projects/khelsutra-guru/.venv/bin/python3
/home/avidullu/projects/khelsutra-guru/.venv/bin/python3.12
```

repo-local venvs

ls: cannot access '/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.venv/bin/python*': No such file or directory

ls: cannot access '/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/venv/bin/python*': No such file or directory

torch/cuda via root venv

torch 2.12.1+cu130 | cuda_available True | NVIDIA GeForce RTX 2070 Super with Max-Q Design

19. user (2026-06-29T10:50:19.223Z)

repo top-level

```
total 192
drwxr-xr-x 22 avidullu avidullu 4096 Jun 27 17:27 .
drwxr-xr-x 10 avidullu avidullu 4096 Jun 29 09:48 ..
-rw-r--r-- 1 avidullu avidullu 598 Jun 25 11:46 .coveragerc
drwxr-xr-x 8 avidullu avidullu 4096 Jun 29 10:44 .git
-rw-r--r-- 1 avidullu avidullu 932 Jun 25 11:46 .gitattributes
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 11:46 .github
-rw-r--r-- 1 avidullu avidullu 1331 Jun 25 16:06 .gitignore
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 13:12 .mypy_cache
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 13:10 .pytest_cache
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 13:12 .ruff_cache
drwxr-xr-x 2 avidullu avidullu 4096 Jun 25 11:46 .vscode
-rw-r--r-- 1 avidullu avidullu 5498 Jun 27 17:27 CLAUDE.md
-rw-r--r-- 1 avidullu avidullu 34880 Jun 25 11:46 README.md
drwxr-xr-x 2 avidullu avidullu 4096 Jun 25 13:10 __pycache__
drwxr-xr-x 20 avidullu avidullu 4096 Jun 27 17:27 backend
-rw-r--r-- 1 avidullu avidullu 2650 Jun 25 11:46 config.json
drwxr-xr-x 5 avidullu avidullu 4096 Jun 25 11:46 deploy
drwxr-xr-x 6 avidullu avidullu 4096 Jun 29 10:44 docs
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 11:46 eval_baselines
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 13:11 gcs:
drwxr-xr-x 4 avidullu avidullu 4096 Jun 25 13:11 gs:
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 13:11 local:
-rw-r--r-- 1 avidullu avidullu 1960 Jun 25 11:46 mypy.ini
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 18:13 output
-rw-r--r-- 1 avidullu avidullu 4748 Jun 27 17:27 pyproject.toml
```



```
-rw-r--r-- 1 avidullu avidullu 240 Jun 25 11:46 pytest.ini
-rw-r--r-- 1 avidullu avidullu 2003 Jun 27 17:27 requirements.txt
-rw-r--r-- 1 avidullu avidullu 789 Jun 25 11:46 ruff.toml
-rw-r--r-- 1 avidullu avidullu 2051 Jun 27 17:27 run_all_tests.py
-rw-r--r-- 1 avidullu avidullu 5572 Jun 27 17:27 run_eval.py
-rw-r--r-- 1 avidullu avidullu 4457 Jun 27 17:27 run_regression.py
drwxr-xr-x 2 avidullu avidullu 4096 Jun 25 11:46 scratch
drwxr-xr-x 3 avidullu avidullu 4096 Jun 27 17:27 scripts
drwxr-xr-x 3 avidullu avidullu 4096 Jun 29 10:44 tests
drwxr-xr-x 3 avidullu avidullu 4096 Jun 27 17:27 tools
drwxr-xr-x 4 avidullu avidullu 4096 Jun 25 13:10 training
```

output dir

```
total 12
drwxr-xr-x 3 avidullu avidullu 4096 Jun 25 18:13 .
drwxr-xr-x 22 avidullu avidullu 4096 Jun 27 17:27 ..
drwxr-xr-x 2 avidullu avidullu 4096 Jun 25 18:13 chunks
```

golden manifests / json manifests (maxdepth 5)

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/eval_baselines/reelcount_manifest.json
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/eval_baselines/fixtures/manifest.json
```

feature dirs

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/features
```

20. user (2026-06-29T10:50:44.287Z)

gcloud / gsutil present?

```
/mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-sdk/bin/gcloud
/mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-sdk/bin/gsutil
```

gcloud auth list

```
Credentialed Accounts
ACTIVE ACCOUNT
*          avi.dullu@gmail.com
```

```
To set the active account, run:
$ gcloud config set account `ACCOUNT`
```

gcloud config

```
[accessibility]
screen_reader = False
[core]
account = avi.dullu@gmail.com
disable_usage_reporting = False
```

```
Your active configuration is: [default]
```

gs:// references in repo config/code

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/CODE_MAP.md:333:| Guard the FULL pipeline + reel-count nightly (GPU) | `@scripts/nightly_reelcount.py::main` ? run the real configured+checkpointed pipeline on gs:// clips, assert reel-count EXACT + quality + report cost, email it; exit 1=regress, 4=stale; `--update-baseline` to re-freeze. Runs on an ephemeral GCP GPU VM (`deploy/gcp/nightly_regression/launch.sh`) that self-deletes |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:4:> a `gs://` clip ? cloud GPU detection ? 22 rallies ? locally-stitched reel, path provably `cloud_run`
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:22:BUCKET=gs://khelsutra-rally-corpus # holds videos/ + models/wasb_badminton_best.pth.tar
```

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:65:mounted read-only for the weights; the job
reads/writes its `gs://` I/O through the storage layer.
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:76: "storage": {"backend": "gcs",
"output_root": "gs://khelsutra-rally-corpus/highlights_run"} }
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:95:VIDEO=gs://khelsutra-rally-
corpus/videos/your_match.mp4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:118:**Batch** over a whole bucket: loop steps
175 over each `gs://?/videos/*.mp4`
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:4:**Last
inventoried:** 2026-06-22 against `gs://khelsutra-rally-corpus`.
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:8:> **?
Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-corpus`, not
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:24:|
Bucket | **`gs://khelsutra-rally-corpus`** (single corpus bucket) |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:30:The
engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gcs://` (or `gs://`),
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:46:|
`models/` | upstream pretrained weights (`wasb_badminton_best.pth.tar`) | **staged to a local
path** (`$WASB_WEIGHTS` / `~/models/?`) for the detector ? not read from `gs://` in place |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:131:all
labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`. Pending the
$7
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/DATA_IN_GCS.md:136:gcloud storage cp "C:\?\Collect\Trajectories\*.csv"
gs://khelsutra-rally-corpus/trajectories/
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:138:#
eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object ($7.3)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:166:have
**no storage-layer fetch**, so a `gs://` manifest cannot be consumed today. Phase-3 must thread
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:184:3.
**Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-corpus/manifests/golden.json`
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:185:
(15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-path
manifest;
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:191:>
authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in gitignored
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:218:
`gs://`) ? it's correct-but-dormant until blocker #2 lands. Add a tiny manifest-validator + a CI
job
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DATA_IN_GCS.md:219:
that authenticates to GCS and runs the litmus against it, or the DoD ("CI reads only from
`gs://`)")
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/DATA_IN_GCS.md:245:reading **only** from `gs://khelsutra-rally-corpus` ? no `C:` /
`F:` / local-Drive path required.
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/NIGHTLY_ENGINE_REGRESSION.md:29:| D7 | **Two buckets:** ephemeral `gs://khelsutra-
rally-nightly` (30-day TTL) for outputs (reports + repo tarball + build staging); golden
clips/labels/weights stay in the permanent no-TTL `gs://khelsutra-rally-corpus` (`nightly-
inputs/`, `weights/`) | owner 2026-06-22 | a bucket-wide TTL on a *dedicated* bucket can never
delete golden data (blast-radius isolation); `create_nightly_bucket.sh` provisions it |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/NIGHTLY_ENGINE_REGRESSION.md:38:- *Reused:*
`backend.pipeline.segmenters.segmenter_registry` + `process_video` (the pipeline) .
`backend.eval.rally_seg_eval.score_intervals` (quality) . `backend.billing` (cost) .
`backend.storage.fetch` (gs:// ? local) . `deploy/gcp/create_gpu_vm.sh` conventions (zone-sweep,
DLVM image, `rally-worker` SA, Ops Agent) . `deploy/gcp/teardown.sh` cost rule . the PR #202
baseline/exit-code/report pattern.
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/NIGHTLY_ENGINE_REGRESSION.md:78:| E5 | **Owner: upload 2-3 short owned clips +
labels + weights to `gs://?/nightly/`** | E1 | ? owner | ? | ? |

```

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/NIGHTLY_ENGINE_REGRESSION.md:98:`gs://?/videos/mahadevpura_smoke_180s.mp4`
(`detector_impl=native`, `skip_ai_handoff=true`):
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/NIGHTLY_ENGINE_REGRESSION.md:103: `gs://` fetch needed a `dst`; the native runner
needs `WASB_REPO_DIR`+`WASB_PYTHON` (not just
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/scripts/nightly_reelcount.py:19: ``eval_baselines/reelcount_manifest.json`` ? the
videos (gs:// / local / gdrive:// URIs) + segmenter
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/scripts/nightly_reelcount.py:336: """Resolve a URI to a LOCAL path. Remote backends
(gdrive:// / gcs:// / s3:// / gdrive://) REQUIRE a
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/scripts/nightly_reelcount.py:455: # fetch INTO the workdir ? gs:// / s3:// /
gdrive:// require a dst (no passthrough). A bad
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CLOUD_GPU_DRYRUN_GUIDE.md:84: gsutil -q cp gs://khelsutra-rally-
corpus/models/wasb_badminton_best.pth.tar ~/models/WASB-SBDT/pretrained_weights/
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CLOUD_GPU_DRYRUN_GUIDE.md:106:gsutil cp "MyMatch.mp4" gs://khelsutra-rally-
corpus/videos/
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CLOUD_GPU_DRYRUN_GUIDE.md:133: gsutil -q cp gs://khelsutra-rally-
corpus/videos/MyMatch.mp4 ~/MyMatch.mp4 # <-- your filename
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CLOUD_GPU_DRYRUN_GUIDE.md:161:gcloud compute ssh rally-gpu --zone="$ZONE"
--command='gsutil cp ~/rally/output/highlights.mp4 gs://khelsutra-rally-
corpus/highlights/MyMatch_highlights.mp4'
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/CLOUD_GPU_DRYRUN_GUIDE.md:164:gsutil cp gs://khelsutra-rally-
corpus/highlights/MyMatch_highlights.mp4 ./MyMatch_highlights.mp4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/gs:/bucket/videos/x.report.md:9:| Video path | gs://bucket/videos/x.mp4 |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/PER_VIDEO_WORKSPACE.md:16:- **Good looks like:** given a `video_id`, a single `ls
<root>/<video_id>/` shows *all* of its artifacts, identically whether `<root>` is a local dir or
`gs://?`. Deleting that one folder fully cleans the video (DB row aside).
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/PER_VIDEO_WORKSPACE.md:26:| D3 | **Two roots:** a *durable* workspace root (may be
`gs://`) for artifacts to keep; a *fast local compute-scratch* mirror (always local) for frame
caches/chunks. Compute writes locally, then **pushes durable artifacts** to the workspace. |
design + back-compat | WSL/native frame extraction never writes to a cloud root (perf +
correctness). |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/PER_VIDEO_WORKSPACE.md:93:2. **Cloud root namespace?** ? *default:*
`gs://<bucket>/<workspace_prefix>/<video_id>/?` with `workspace_prefix="workspaces"` (keeps per-
video folders tidy beside `videos/`, `labels/`, `weights/`; the legacy `outputs/<video_id>/` stays
until P5). ? confirm (affects the deployed bucket contract).
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/SETUP_NEW_MACHINE.md:51:> credit-funded; Paperspace is subscription-gated). The GCS
bucket `gs://khelsutra-rally-corpus` + SA +

```

21. user (2026-06-29T10:50:46.665Z)

```

===== gcs: =====
--- du ---
24K /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gcs:
--- tree (first 50 entries) ---
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gcs:
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gcs:/bucket
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gcs:/bucket/videos
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/gcs:/bucket/videos/x.report.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/gcs:/bucket/videos/x.report.json

```

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gcs:/bucket/videos/x.summary.md
===== gs: =====
--- du ---
52K      /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:
--- tree (first 50 entries) ---
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/b
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/b/clip.report.json
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/b/clip.report.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/b/clip.summary.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos/cli.report.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos/x.report.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos/cli.report.json
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos/x.report.json
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos/x.summary.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/gs:/bucket/videos/cli.summary.md
===== local: =====
--- du ---
232K     /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:
--- tree (first 50 entries) ---
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-3
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-9
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-0
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-10
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-7
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-8
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-13
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-12
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-11
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-6
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-3/test_process_local_scheme_uri_0
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-11/test_process_local_scheme_uri_0
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-6/test_process_local_scheme_uri_0
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-9/test_process_local_scheme_uri_0
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-4/test_process_local_scheme_uri_0
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/local:/tmp/pytest-of-avidullu/pytest-0/test_process_local_scheme_uri_0
```


48:# Local generated artifacts (review reels, eval CSVs, e2e/proof outputs) -- regenerable, never committed

22. assistant (2026-06-29T10:52:02.361Z)

Excellent ? that's a very informative recon pass. Key findings before I continue:

- ****GPU is live****: RTX 2070 Super (8 GB), CUDA 13.2, `torch 2.12.1+cu130`, `cuda\_available True`, idle. ?
- ****The prompt's intended clone doesn't exist here.**** `C:\Users\avidu\Projects\badminton-highlight-indexer` is ****not present**** (`/mnt/c/.../Projects/` has `khelsutra-guru`, `muneem-ai`, `Vairagi`, `\_p3\_scratch` ? no standalone BHI). The only live, synced BHI clone is the one I just fast-forwarded: badminton-highlight-indexer on WSL-ext4 ? which is actually the ***correct*** place for GPU I/O (DrvFs `/mnt/c` is slow).
- ****The `gcs:// gs:// local:` dirs are junk****, exactly as you guessed: stale artifacts where report-writer/storage code resolved `gcs://bucket/...`, `gs://b/...`, `local://tmp/...` URIs as ***literal local paths***. The `local:/tmp/pytest-of-avidullu/...` tree is pure pytest fallout. They use the placeholder name "bucket", not the real corpus ? so they are ****not**** a mirror of real data. Total ~308 KB, regenerable, safe to delete.
- ****The real data lives in `gs://khelsutra-rally-corpus`**** (per `docs/DATA\_IN\_GCS.md`), and gcloud is authenticated as you. `/mnt/f` (F: drive) appears ****empty**** on this machine ? so I can't assume golden videos/CSVs are local; they're in GCS.

Let me verify the GCS bucket inventory and read the data-location source-of-truth, while the ramp-up workflow continues.

23. user (2026-06-29T10:52:03.670Z)

```
1      # Data in GCS ? the corpus source-of-truth map
2
3      **Status:** LIVE / in-progress (migration not complete). **Owner-gated** decisions in
$7.
4      **Last inventoried:** 2026-06-22 against `gs://khelsutra-rally-corpus`.
5      **Goal (owner, 2026-06-22):** run **all** training + inference from the cloud with **all
data in GCS**,
6      so the local Windows box (and the intermittently-mounted external drive) is **no longer
a blocker**.
7
8      > **? Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-
corpus`, not
9      > from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket
yet, it's in the
10     > **gap table ($5)** ? flag it / migrate it, don't hard-code a local path.
11
12     This doc is the **data map**. The *machine* runbooks are
[SETUP_NEW_MACHINE.md](SETUP_NEW_MACHINE.md)
13     (spin up a box) and [deploy/gcp/README.md](../deploy/gcp/README.md) (provision GPU +
train/infer).
14     The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
15     (`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ?
it records
16     *where the bytes are* and *what's still missing*.
17
18     ---
19
20     ## 1. The bucket
21
22     | | |
23     |---|---|
24     | Bucket | **gs://khelsutra-rally-corpus** (single corpus bucket) |
25     | Project | `gen-lang-client-0306026826` (billing *Khelsutra Billing*) |
26     | Region | `asia-south1` (Mumbai), UBLA, co-located with the GPU VMs |
27     | Auth | SA `rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com`
(`roles/storage.objectAdmin`); key at `~/gcp/rally-worker-sa.json` (local only, **never
committed**). Interactive: `gcloud auth login`. |
```

```

28 | Size today | ~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus) |
29
30 The engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gs://` (or
31 `s3://`, `gdrive://`, or a bare local path. Pick the backend with **config + creds, no
code change**.
32
33 ---
34
35 ## 2. Canonical layout (the target contract)
36
37 The cloud worker's input/output contract (`backend/worker/__main__.py`; lineage:
38 `DECOUPLED_COMPUTE/04`):
39
40 | Prefix | Holds | Who reads/writes it |
41 |---|---|---|
42 | `videos/` | source videos / proxies (`<clip>.mp4`) | **input** to `worker infer` &
`worker train` |
43 | `labels/` | golden rally labels (`<clip>.rallies.csv`) | **input** ground truth to
`worker train` + all eval |
44 | `trajectories/` | cached WASB trajectory CSVs | ? **eval/litmus input** (GPU-free),
consumed via `manifests/golden.json`. **NOT** read by `worker train` (it synths or re-extracts)
|
45 | `weights/<version>/` | trained Gen-0 bundles (`weights.json` + a run `manifest.json`)
| **output** of `worker train` |
46 | `models/` | upstream pretrained weights (`wasb_badminton_best.pth.tar`) | **staged to
a local path** (`$WASB_WEIGHTS` / `~/models/?`) for the detector ? not read from `gs://` in
place |
47 | `outputs/<video_id>/` | per-video inference outputs (intervals, etc.) | **output** of
`worker infer` |
48 | `highlights/`, `promo/` | rendered reels + promo artifacts | output / demo |
49
50 > **Note on the manifest.** `worker train` does **not** read a pre-staged corpus
`manifest.json` ? it
51 > *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories
**synthetically
52 > by default** (`--trajectory-source synth`, CPU-mock) or for real via opt-in
`--trajectory-source
53 > detector` (needs `videos/`). A `manifest.json` only appears as an **output** inside
each
54 > `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-
resolvable manifest
55 > (§7.3) ? and `manifests/golden.json` is currently consumed by **no code yet** (eval is
local-FS only;
56 > see Known blockers).
57
58 ---
59
60 ## 3. What's actually in the bucket today (2026-06-22)
61
62 > **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical
`labels/<date>_<name>.rallies.csv`
63 > (= original-6 + `GX010137`/`GX010141` + 7 new; the 7 old `_proxy`/bare dupes were
deleted), and
64 > `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot; the
remaining gap is **videos**
65 > (14 of 15) ? see §6/§7b/§8.
66
67 | Prefix | Present | Drift / notes |
68 |---|---|---|
69 | `labels/` | 7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill
Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy` | ? **inconsistent
naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`,
`GX010141` |
70 | `videos/` | 3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4`

```

```

| ? corpus videos/proxies almost entirely absent |
71 | `trajectories/` | ? | ? **does not exist** (every trajectory is local-only) |
72 | `models/` | `wasb_badminton_best.pth.tar` | ? (? WASB-C3 weight-provenance still an
open legal item before any external share) |
73 | `weights/` | `0.1.0-l4/` | ? first real L4 train bundle |
74 | `outputs/` | one `/`, `parity/` | ? dry-run outputs |
75 | `highlights/`, `promo/` | reels + promo | ? |
76
77 **Drift summary:** the bucket has extra prefixes (`highlights/`, `promo/`, `models/`
alongside
78 `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the
corpus inputs**
79 (`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a
complete corpus.
80
81 ---
82
83 ## 4. How the cloud consumes it
84
85 Provisioning + the verified commands live in
[`deploy/gcp/README.md`](../deploy/gcp/README.md) §3§5.
86 The data-facing entry points (`backend/worker/__main__.py`):
87
88 ```bash
89 # config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
90 # { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826"
} } }
91
92 # INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
93 python -m backend.worker infer \
94     --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
95     --out-uri gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json
96
97 # TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds
manifest on-box,
98 # runs training.gen0.harness, pushes weights/<version>/):
99 RALLY_DETECTOR_IMPL=native python -m backend.worker train \
100     --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
101     --version <v> --trajectory-source detector --segmenter wasb_hybrid --config
config.json
102 ```
103
104 A bare/relative path resolves under a configured `input_root`/`input_backend`
105 (`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-
relative.
106
107 ---
108
109 ## 5. The gap ? what's still local-only (why the box is still a blocker)
110
111 All of this is **tiny except the originals** ? the metadata that blocks cloud eval is <
25 MB total:
112
113 | Data | Where it is now | Size | In bucket? | Needed for |
114 |---|---|---|---|---|
115 | 20 WASB trajectory CSVs | `C:\?\Annotation Setup\Collect\Trajectories\` | **17.5 MB**
| ? none | eval/litmus (cached). *(No `cached` train source exists ? train synths or detector-
extracts.)* |
116 | golden manifest | `output/human_lovo_manifest.json` (C:-absolute paths) | 4.5 KB | ? |
eval corpus definition |
117 | original-6 labels | `F:\[Khelsutra]\Training\Golden Labelled\*.rallies.csv` | ~12 KB
| ? | train + eval |
118 | 7 new-clip labels | repo `output/*.rallies.csv` (gitignored) **** F: archive (backed
up 2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
119 | source videos / proxies | F: archive + Google Drive golden folder | **100s of GB** | ?

```



```

3 only | infer + train (real WASB extraction) |
120
121     > The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually
blocks GPU-free
122     > cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The
**videos** are the
123     > only large migration, and they already exist on **Google Drive** (the golden-corpus
folder), so they
124     > can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path
entirely.
125
126     ---
127
128     ## 6. Migration plan (close the gap)
129
130     **Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker).** Stage all
trajectories,
131     all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`.
Pending the §7
132     naming decision. Sketch:
133
134     ```bash
135     # trajectories (decision-free; absent today)
136     gcloud storage cp "C:\?\Collect\Trajectories\*.csv" gs://khelsutra-rally-
corpus/trajectories/
137     # labels (after the §7.1 naming decision) -> labels/<canonical-name>.rallies.csv
138     # eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object (§7.3)
139     ```
140
141     **Phase 2 ? videos ? `videos/<name>.mp4`** (bucket name = the eval `name`, consistent
with
142     `trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see
`scratch/copy_videos_to_gcs.ps1`):
143
144     | Source | Clips | Size |
145     |---|---|---|
146     | Drive golden folder `?/Golden Label Videos Request - June 15/[Done] Video N\`
(**proxies**) | `mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`,
`Boxhill_Doubles` (7) | ~13.5 GB |
147     | F: archive (**full MP4s** ? proxy-first before upload) | `GX010137`, `GX010141`, +
original-6 (`adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`,
`testlarge_short`, `mahadevpura_singles`) (8) | ~50 GB |
148
149     **Two ways to run it:**
150     - **Decoupled (preferred ? no local box):** `rclone` on a cloud VM with a Google-Drive
remote ?
151     `rclone copy gdrive:"<?/[Done] Video N>/<proxy>.mp4" :gcs:khelsutra-rally-
corpus/videos/<name>.mp4`
152     (one-time owner step: `rclone config` ? `gdrive` remote OAuth). Or `gdown <file-id>`
on the box for
153     the shared folder. This is the path that actually removes the local box.
154     - **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh
scratch/copy_videos_to_gcs.ps1
155     -Execute` (dry-run without `-Execute`; `-Only <name>` for one; skip-existing). Uses
local bandwidth.
156
157     **Priority:** the 7 Drive **proxies** first (already 1080p, sufficient for detection).
The 8 large F:
158     full MP4s should be **proxied to 1080p first** (cheaper storage + faster cloud decode)
before upload.
159     ? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`,
`videos/mahadevpura-1.MP4`)
160     are superseded once `videos/GX030094.mp4` / `videos/mahadevpura_1.mp4` land ? verify-
dupe-then-delete
161     (same as the label cleanup). Track per-clip in §8.

```

```

162
163     **Phase 3 ? make eval/litmus read GCS (real code, NOT a config flip).** The eval drivers
164     (`served_gate_a.evaluate_video`, `calibrate_wasb.load_golden_gts`,
165     `TrackNetRunner.parse_trajectory_csv`)
166     open `spec["trajectory"]`/`spec["labels"]` with plain `open()` and gate on
167     `os.path.isfile()` ? they
168     have **no storage-layer fetch**, so a `gs://` manifest cannot be consumed today. Phase-3
169     must thread
170     `storage.fetch`/`resolve_input_uri` through those paths first. Then: re-probe
171     `fps`/`frame_width` from
172     the bucket videos (don't trust the copied values ? a wrong fps silently mis-aligns every
173     label), and
174     **re-measure the served Gate-A litmus on the GCS 15-clip set as a [[launch-report-
175     convention]] event**
176     (dual-set RAW+golden no-regression, not a silent eval-set swap ? see PR #236).
177
178     ---
179
180     ## 7. Decisions
181
182     1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-
183     DD>_<name>.rallies.csv`, where
184     `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring
185     date** =
186     the *earliest* mtime across all known copies (C: working copy + F: archive ? robust
187     to copy-resets;
188     a restored C: copy shows today, the F: archive keeps the real date). Fixes the
189     `_proxy` inconsistency
190     and disambiguates recycled GoPro names by date. *(Future, when videos are in the
191     bucket: optionally
192     add a `video_id`/MD5 sidecar for the strongest keying ? deferred, needs hashing each
193     video.)*
194
195     2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so
196     cloud/CI eval +
197     the served Gate-A litmus run **GPU-free** (training can still regenerate via
198     `--trajectory-source detector`).
199
200     3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-
201     corpus/manifests/golden.json`
202     (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-
203     path manifest;
204     flipping the litmus to read the GCS manifest is Phase 3.
205
206     4. **Originals migration scope ? OPEN** ? proxies-only vs full 4K, and how much of the
207     15-clip golden
208     set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the
209     local box).
210
211     > Phase-1 was run via a dry-run planner (resolve every clip's trajectory + label, stamp
212     the label's
213     > authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in
214     gitignored
215     > `scratch/`; promote it to `scripts/` if corpus-sync recurs).
216
217     > ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects
218     (`_proxy.rallies.csv`,
219     > bare `GX010128`/`mahadevpura-1`) were confirmed **byte-identical (MD5)** to the new
220     canonical names
221
222     > (content also on C:/F:) and **deleted** ? `labels/` now holds exactly the 15 canonical
223     objects.
224
225     > ? **gcloud gotcha (Phase-2):** `gcloud storage cp` treats `[...]` as a glob, so source
226     paths under
227     > `[Done] Video N/` (Drive) or `[Khelsutra] ?` (F:) fail with "matched no objects." The
228     owner-box
229     > script (`scratch/copy_videos_to_gcs.ps1`) stages such paths through a bracket-free
230     temp file; the
231     > `rclone` cloud path has no such issue.

```

```

201
202 ---
203
204 ## 7b. Known blockers ? must-fix before the DoD (2026-06-22 adversarial review)
205
206 1. **? RESOLVED 2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/319))
207     ? dated labels broke `worker train --trajectory-source detector`.** *History:*
`cmd_train` derived the
208     join key as `fname.replace(".rallies.csv","")` ? `2026-06-22_GX020094`, then looked
up
209     `video_by_stem.get(name)` against bare video stems (`GX020094`), so every clip was
skipped and train
210     aborted with `<2 videos`. **Fix:** the label?clip-name derivation is now the helper
`_label_clip_name()`
211     in [backend/worker/__main__.py](../backend/worker/__main__.py), which strips both the
212     `.rallies.csv`/`.csv` suffix AND a leading `^\d{4}-\d{2}-\d{2}_` prefix, so a dated
label and a legacy
213     bare label both resolve to the bare `videos/<name>` stem (preserves the locked human-
readable naming).
214     Tested: parametrized helper cases + a dated-label?bare-video `cmd_train` join
regression + per-label
215     skip-branch coverage. *(`synth` train + all eval were unaffected ? they don't join
videos.)*
216 2. **Phase-3 eval-on-GCS is net-new plumbing, not a flag** ? see §6 Phase-3. Eval is
local-FS only.
217 3. **`manifests/golden.json` is consumed by zero code today** (train ignores it; eval
can't fetch
218     `gs://`) ? it's correct-but-dormant until blocker #2 lands. Add a tiny manifest-
validator + a CI job
219     that authenticates to GCS and runs the litmus against it, or the DoD ("CI reads only
from `gs://`)
220     stays unverifiable.
221 4. **Gating infra (owner-only, not done):** GPU `GPUS_ALL_REGIONS` quota is still **0**
(no VM ever
222     created ? `deploy/gcp/README` §2); cloud-box secrets (SA JSON + rotated
`GEMINI_API_KEY` + WASB
223     weights staged from `models/`) aren't set up persistently. No real cloud train/infer
runs until these.
224 5. **Reproducibility:** the migration + copy helpers live in gitignored `scratch/`
225     (`migrate_golden_to_gcs.py`, `copy_videos_to_gcs.ps1`) ? promote to `scripts/` so
Phase-2/re-syncs
226     are repeatable from the repo, and add a referenced **proxy-generation** step for the
8 large F: originals.
227
228 ## 8. Progress tracker
229
230 | Item | Status |
231 |---|---|
232 | Bucket + SA + provisioning (deploy/gcp) | ? built |
233 | Cloud worker `infer`/`train` by URI | ? built (`backend/worker`) |
234 | This data-map doc | ? 2026-06-22 |
235 | Phase 1: trajectories ? `trajectories/` (15) | ? 2026-06-22 |
236 | Phase 1: labels reconciled ? `labels/<date>_<name>` (15) | ? 2026-06-22 |
237 | Phase 1: eval manifest ? `manifests/golden.json` | ? 2026-06-22 |
238 | Phase 1: remove 7 stale `_proxy`/bare label objects | ? 2026-06-22 (MD5-verified
dupes, deleted) |
239 | Blocker §7b#1: dated labels join in `worker train --trajectory-source detector` | ?
2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319)) ?
`_label_clip_name` strips the ISO-date prefix |
240 | Phase 2: corpus videos/proxies ? `videos/<name>.mp4` | ? mapping + script done;
**1/15** demo'd (`mahadevpura_1`); 14 remain ? run `rclone` on a cloud box (preferred) or the
owner-box script |
241 | Phase 2: proxy the 8 large F: full-MP4 clips before upload | ? |
242 | Phase 3: eval/litmus default to the GCS manifest | ? |

```

```

243
244     **Definition of done:** a fresh GPU box (or CI) can run inference, the eval suite, and
Gen-0 training
245     reading **only** from `gs://khealsutra-rally-corpus` ? no `C:` / `F:` / local-Drive path
required.
246
247     ---
248
249     ## Related
250     - [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md) ? spin up a box (storage backends in
$7).
251     - [`deploy/gcp/README.md`](../deploy/gcp/README.md) ? GCP GPU provisioning + train/infer
($5).
252     - [`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) .
[`PER_VIDEO_WORKSPACE.md`](PER_VIDEO_WORKSPACE.md) ? locality tiers + `outputs/<video_id>/`.
253     - `docs/archives/past_projects/DECOUPLED_COMPUTE/` ? the cloud-engine design lineage.
254

```

24. user (2026-06-29T10:53:44.434Z)

bucket top-level

```

gs://khealsutra-rally-corpus/archive/
gs://khealsutra-rally-corpus/highlights/
gs://khealsutra-rally-corpus/labels/
gs://khealsutra-rally-corpus/manifests/
gs://khealsutra-rally-corpus/models/
gs://khealsutra-rally-corpus/outputs/
gs://khealsutra-rally-corpus/promo/
gs://khealsutra-rally-corpus/trajectories/
gs://khealsutra-rally-corpus/videos/
gs://khealsutra-rally-corpus/weights/

```

manifests/

```

3840  2026-06-22T10:04:19Z  gs://khealsutra-rally-corpus/manifests/golden.json
TOTAL: 1 objects, 3840 bytes (3.75kiB)

```

videos/ (count + sample)

```

gs://khealsutra-rally-corpus/videos/GX010142_1080p.mp4
gs://khealsutra-rally-corpus/videos/GX010143_1080p.mp4
gs://khealsutra-rally-corpus/videos/GX010147_1080p.mp4
gs://khealsutra-rally-corpus/videos/GX030094_proxy.mp4
gs://khealsutra-rally-corpus/videos/mahadevpura-1.MP4
gs://khealsutra-rally-corpus/videos/mahadevpura_1.mp4
gs://khealsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
gs://khealsutra-rally-corpus/videos/SHANAN/

```

trajectories/ (count + sample)

```

gs://khealsutra-rally-corpus/trajectories/Badminton_BXH_2.csv
gs://khealsutra-rally-corpus/trajectories/Boxhill_Doubles.csv
gs://khealsutra-rally-corpus/trajectories/GX010128.csv
gs://khealsutra-rally-corpus/trajectories/GX010137.csv
gs://khealsutra-rally-corpus/trajectories/GX010141.csv
gs://khealsutra-rally-corpus/trajectories/GX020094.csv
gs://khealsutra-rally-corpus/trajectories/GX030094.csv
gs://khealsutra-rally-corpus/trajectories/adarsh_avi_singles.csv
gs://khealsutra-rally-corpus/trajectories/gbaaddy.csv
gs://khealsutra-rally-corpus/trajectories/kushagra_singles.csv
gs://khealsutra-rally-corpus/trajectories/largetest_doubles.csv
gs://khealsutra-rally-corpus/trajectories/mahadevpura_1.csv
gs://khealsutra-rally-corpus/trajectories/mahadevpura_2.csv
gs://khealsutra-rally-corpus/trajectories/mahadevpura_singles.csv
gs://khealsutra-rally-corpus/trajectories/testlarge_short.csv

```

labels/ (sample)

```
gs://khelsutra-rally-corpus/labels/2026-06-08_adarsh_avi_singles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_gbaaddy.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_kushagra_singles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_targetest_doubles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_testlarge_short.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-09_mahadevpura_singles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-16_GX010128.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-16_mahadevpura_2.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-17_Badminton_BXH_2.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-17_GX030094.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-17_mahadevpura_1.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_Boxhill_Doubles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_GX010137.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_GX010141.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_GX020094.rallies.csv
```

models/

```
6102633 2026-06-20T14:19:50Z gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar
TOTAL: 1 objects, 6102633 bytes (5.82MiB)
```

25. user (2026-06-29T10:54:03.255Z)

golden.json manifest content

```
[
  {
    "name": "adarsh_avi_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/adarsh_avi_singles.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_adarsh_avi_singles.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/kushagra_singles.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_kushagra_singles.rallies.csv"
  },
  {
    "name": "targetest_doubles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/targetest_doubles.csv",
    "fps": 29.97,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_targetest_doubles.rallies.csv"
  },
  {
    "name": "gbaaddy",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/gbaaddy.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_gbaaddy.rallies.csv"
  },
  {
    "name": "testlarge_short",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/testlarge_short.csv",
    "fps": 29.97,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_testlarge_short.rallies.csv"
  },
  {
    "name": "mahadevpura_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/mahadevpura_singles.csv",
    "fps": 29.97,
    "frame_width": 1920,
```

```

    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-09_mahadevpura_singles.rallies.csv"
  },
  {
    "name": "mahadevpura_2",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/mahadevpura_2.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-16_mahadevpura_2.rallies.csv"
  },
  {
    "name": "GX010128",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX010128.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-16_GX010128.rallies.csv"
  },
  {
    "name": "mahadevpura_1",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/mahadevpura_1.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-17_mahadevpura_1.rallies.csv"
  },
  {
    "name": "GX030094",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX030094.csv",
    "fps": 29.97,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-17_GX030094.rallies.csv"
  },
  {
    "name": "Badminton_BXH_2",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/Badminton_BXH_2.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-17_Badminton_BXH_2.rallies.csv"
  },
  {
    "name": "GX010141",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX010141.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_GX010141.rallies.csv"
  },
  {
    "name": "GX010137",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX010137.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_GX010137.rallies.csv"
  },
  {
    "name": "Boxhill_Doubles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/Boxhill_Doubles.csv",
    "fps": 29.97,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_Boxhill_Doubles.rallies.csv"
  },
  {
    "name": "GX020094",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX020094.csv",
    "fps": 29.97,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_GX020094.rallies.csv"
  }
}

```

]

26. user (2026-06-29T10:54:12.020Z)

Windows-side khelsutra-guru BHI clone (possible parallel checkout)

```
total 8
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 10:40 .
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 10:40 ..
drwxrwxrwx 1 avidullu avidullu 4096 Jun 23 04:23 Annotation Setup
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 08:17 badminton-highlight-indexer
drwxrwxrwx 1 avidullu avidullu 4096 Jun 22 18:49 khelsutra
drwxrwxrwx 1 avidullu avidullu 4096 Jun 22 17:56 khelsutra-screencast
drwxrwxrwx 1 avidullu avidullu 4096 Jun 22 09:33 rally-annotator
drwxrwxrwx 1 avidullu avidullu 4096 Jun 29 09:57 sports-data-collector
drwxrwxrwx 1 avidullu avidullu 4096 Jun 7 07:43 sports-obsreport
-rwxrwxrwx 1 avidullu avidullu 5086 Jun 29 10:39 start-here-tmp.txt
43578a7
M .coveragerc
M .gitattributes
M .github/workflows/ci.yml
M .github/workflows/nightly.yml
M .gitignore
M .vscode/settings.json
M CLAUDE.md
M README.md
M backend/annotations/__init__.py
M backend/annotations/automated_provider.py
```

/mnt/f deeper

```
total 8
drwxrwxrwx 2 root root 4096 Jun 12 06:22 .
drwxr-xr-x 6 root root 4096 Jun 12 06:22 ..
f exists?
/mnt/f is not a mountpoint
```

disk space

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sdc	1007G	39G	918G	4%	/
C:\	746G	503G	244G	68%	/mnt/c

27. assistant (2026-06-29T10:55:15.656Z)

I now have what I need to resolve the structure. Let me record the two clarifications and set up a working plan, then read the actual eval code to pin down the exact runnable command (the digest had some ambiguity about feature-gen vs. GPU-free trajectory consumption).

Two things noted:

- **Path confirmed** ? you've cleared the discrepancy; I'm working in badminton-highlight-indexer on WSL-ext4, synced to remote HEAD `0c9c693`. (The Windows-side `/mnt/c/.../khelsutra-guru/badminton-highlight-indexer` is a *separate, dirty* checkout at `43578a7` ? a likely parallel/owner session ? so I'll stay entirely in the WSL clone and won't touch that one.)
- **Authorization noted** ? GPU, Gemini credits, and disk are fair game. Good, because the data lives in `gs://khelsutra-rally-corpus` and I'll need to materialize it locally.

28. user (2026-06-29T10:55:23.161Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

29. user (2026-06-29T10:55:25.674Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-71 of 288 total (41693 tokens, cap 25000). Call Read with offset=72 limit=71 for the next page, or Grep to find a specific section.

Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Digest rally-detection P2b docs + code anchors into one structured
runbook",
3      "agentCount": 7,
4      "logs": [],
5      "result": {
6      "runbook": "This is a synthesis task ? I have all the structured digests I need to
assemble the runbook. No tool calls required; I'll produce the Markdown directly.\n\n# P2b
Runbook ? Promote `inactivity_temporal_density` (R4 finding-B fix)\n\n> **Source of truth:**
`/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_FIXES_PLAN.md` $7 tracker.\n> **Extends:**
`docs/RALLY_DETECTION_QUALITY_REPORT.md`. **Code-review backing:**
`docs/RALLY_DETECTION_CODE_REVIEW.md` (finding B).\n> All paths below are relative to
`/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/` unless
absolute.\n\n--\n\n## 1. CURRENT STATE\n\nFrom the $7 tracker (`"Deliverables & progress
tracker ? source of truth"; legend ? Todo ? ? In progress ? ? Done ? ? Blocked/gated`):\n\n| Row
| Item | Dep | Gated? | Status | PR |\n|----|-----|-----|-----|-----|----|\n| **P1** |
Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-collector`) | ? | No
| ? Done | `sdc#50` |\n| **P2a** | Expose the flag in the `rally_seg_eval` A/B CLI
(`--[vs-]inactivity-temporal-density`) | P0b | No | ? Done | `this PR` |\n| **P2b** | **Promote
P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON if no regression | P2a | ?
owner | ? **Todo** | `?` (none) |\n\n- **P2b is the only open/blocked row.** Its dependency P2a
is done; promotion is **owner-gated** because it needs the golden corpus + GPU/WSL-extracted
trajectory features that do not exist offline, and finding B only shows signal on clips where
the shuttle tracker drops out (not measurable on synthetic/offline fixtures).\n- **Current
default of the flag:**\n\n    backend/config/models.py:199\n
inactivity_temporal_density: bool = False\n\n    Class `IndexingConfig(BaseModel)` (declared
`backend/config/models.py:146`). Default **False** (OFF)**. The runner param mirrors this at
`backend/pipeline/detectors/tracknet_runner.py:329` (`inactivity_temporal_density: bool =
False`).\n\n> **Doc inconsistencies to be aware of (do not `"fix"` silently):** (a) $7 lists
P0a/P0b under PR **394** while the $0/header narrative cites **393**;
```

(b) \$7 marks P1 ? Done (`sdc#50`) but the \$9 Definition-of-Done `rally\_cvat.py` warning-log checkbox is still unchecked.

--\n\n## 2. FINDING B\n\n**Plain explanation.** R4 (`"Inactive Tail Trimming"`, `backend/pipeline/detectors/tracknet\_runner.py::trim\_inactive\_tail`, def line 450) trims a candidate rally window's **end** back to the last frame whose trailing `trail\_frames` window holds enough `"fast"` frames (velocity > `rally\_thresh`). The legacy density check is `fast\_sum / win\_count >= min\_density`, where `win\_count` is the count of **visible/active samples** in the trailing window ? *not* elapsed time. If the detector loses the shuttle for ~1.8 s and then catches a single fast frame at the very end, `win\_count` is tiny (1?2) so `fast\_sum/win\_count` reads 1.0 (100% dense). Sparse trailing points then **defeat the trimmer**, leaving dead-time uncut and artificially inflating the rally **end** boundary. The fix adds the `inactivity\_temporal\_density` flag: when **ON**, the density *denominator* becomes the trailing window's **temporal length** `min(trail\_frames, frames[i] - frames[0] + 1)` instead of the active-sample count, so sparse tracking can no longer inflate density. Resolved as strict temporal density (no separate few-frames fallback). R4 itself is **default-ON in prod** (`window\_inactivity\_end = 1.5`), which is why this real behaviour change ships behind a default-OFF flag.

Why the fix is a no-op under dense tracking. The `group` passed to `trim\_inactive\_tail` contains **only active frames**. Under dense tracking every frame in the trailing span is active, so the temporal denominator `min(trail\_frames, frames[i]-frames[0]+1)` **equals** the sample-count denominator `win\_count`. Identical denominators ? identical output ? flipping the flag changes nothing on well-tracked clips. Behaviour diverges **only** in the sparse-tracking case. Corollary: signal appears only where the shuttle tracker actually drops out, so the A/B cannot be measured on synthetic/offline fixtures ? hence the owner gate on real labelled video.

--\n\n## 3. A/B RUNBOOK\n\n> **Owner box only** (real labelled video + GPU/WSL). The whole point is to compare base = SERVED **flag-OFF** vs candidate = SERVED **flag-ON** at the SERVED operating point.\n> **Placeholders ? replace before running:**\n\n- `` = path to the golden manifest JSON (no concrete path in the plan doc; the served-gate-A regression default is `` = directory of GPU/WSL-extracted golden trajectory features (no concrete path in the plan doc; fusion golden features land as `output/\*\_golden\_features.json`).\n\n**Step 1 ? Generate golden trajectory/fusion features (GPU/WSL).** These are gitignored and must be extracted on the owner box before the A/B can see signal.

--\n\nbash\n# Extract the golden features the A/B reads from (GPU/WSL; produces


```

output/*_golden_features.json).\n# Adjust the manifest/output paths to your corpus
layout.\npython -m backend.eval.golden_fixtures --check          # sanity: confirms fixture
schema is intact\n\n# (feature-gen is the owner-box GPU/WSL extraction step; ensure
<GOLDEN_FEATURES_DIR>\n# is populated with the per-video trajectory feature JSONs before Step
2)\n```\n> The plan doc does not pin an exact feature-gen command; the offline `golden_fixtures
--check` is the committed sanity gate. The actual trajectory-feature extraction is the WSL-
quarantined WASB inference path (`backend/pipeline/detectors/wasb_infer.py`), which is **not
importable on Windows** and runs in WSL2 conda.\n\n**Step 2 ? Run the promotion A/B (the
canonical P2b command, §8 Q2).**\n```\nbash\npython -m backend.eval.rally_seg_eval \\\n
--manifest <GOLDEN_MANIFEST> --features-dir <GOLDEN_FEATURES_DIR> \\\n    --preset served --vs
served --vs-inactivity-temporal-density\n```\n- Both sides resolve to the **served** preset
(`--preset served --vs served`); only the `--vs` (candidate) side flips
`inactivity_temporal_density=True`, isolating the finding-B denominator change at the served
operating point.\n- Wiring: `main()` builds `base = _build_preset(args, "base")` and `cand =
_build_preset(args, "cand")`. For the candidate, `temporal_density =
args.vs_inactivity_temporal_density` or `args.inactivity_temporal_density` then `replace(preset,
inactivity_temporal_density=True)` (`backend/eval/rally_seg_eval.py:602-604,611-612`). The
`store_true` flags default `False`, so absent flags reproduce legacy/served behaviour on both
sides.\n- Flags defined at `backend/eval/rally_seg_eval.py:712-723`.\n\n---\n\n## 4. DECISION
RULE\n\n- **Headline metric:** `tol_f1` (tolF1) ? the R2 tolerance-match F1 at the
**SERVED** operating point. This is the *only* headline metric named in the §8 Q2 / §9 decision
rule.\n- **Promote vs reject (verbatim intent):** base = SERVED flag-OFF, candidate = SERVED
flag-ON; the run "diffs tolF1 at the **SERVED** operating point." ? **Promote** by flipping
`IndexingConfig.inactivity_temporal_density` to `True` (one-line PR) **only if `tol_f1` does not
regress** at SERVED. §9 restates: "flip the flag default-ON if no regression." \n-
**`served_regress_eps`:** **Not named in the plan doc** ? the doc's gate is the qualitative
"tolF1 does not regress." Grounded in code (not the doc), the promotion gate's tolerance is
`served_regress_eps: float = 0.0` at `backend/eval/promotion.py:126` (and `eps=0.0` in
`served_regresses`, `backend/eval/promotion.py:42`), meaning **"0.0 = no served-mean drop
allowed" (docstring line 147). Use `0.0` as the concrete bar unless the owner overrides
it.\n- **Secondary metrics to inspect (harness outputs, NOT promote/reject gates):** The plan
doc elevates none of these; they are diagnostics from the eval harness:\n  - **`|?end|`** ? per-
video rally-end error (`rally_seg_eval` per-video table, `~line 523`); the whole point of
finding B is shrinking inflated ends.\n  - **Paired sign-test + CI / per-tier `delta`** ? module
docstring line 13; sign-test on both tIoU-F1 and R2 `tol_f1` (`~line 358`); aggregate CI via
`_ci(a['tol_f1'])` (`~line 534`).\n  - **`tolP`/`tolR`/`tol_precision`** (tolerance
precision/recall) ? note: a metric literally called "rally-end precision" does **not** exist
in the harness.\n- **`promote_if_served_safe` gate** (`backend/eval/promotion.py:117-224`, the
honest served-safety gate, defaults `min_n=5`, `p_threshold=0.05`, `served_regress_eps=0.0`,
`n_boot=2000`, `seed=0`):\n  1. `base = should_promote_for_user(served_a, served_b, ?)` ? honest
gate on SERVED scores (min-N floor + ?F1 CI excludes 0 + sign test).\n  2. `served =
served_regresses(served_a, served_b, eps=served_regress_eps, ?)`; `served_safe = not
served["regresses"]`. (`served_regresses` ? True when served mean ?F1 < ?eps **or** the
bootstrap CI lies entirely below ?eps.)\n  3. **Non-structural compose:** if `would_promote` and
not `served_safe` ? `promote=False, vetoed=True`; elif `would_promote` ? `promote=True`; else
`promote=False`. **The served guard can only ever remove a promotion, never add one.**\n\n>
Decision boils down to: **promote iff `tol_f1` does not regress at SERVED** (`served_regress_eps
= 0.0`). Inspect `|?end|`, the paired sign-test, and the CI delta to *understand* the result ?
but they are not the gate.\n\n---\n\n## 5. ON PROMOTE (tol_f1 does not regress)\n\n1. **Exact
one-line code change** ? flip the default in `backend/config/models.py:199`:\n  ```\ndiff\n
- inactivity_temporal_density: bool = False\n  + inactivity_temporal_density: bool = True\n
```\n  This is the *only* code change. It propagates automatically: config ?
`trajectory_hybrid` segmenter ? eval `WindowingPreset`
(`backend/eval/windowing.py:137-138,192-193` read it from `IndexingConfig`) ?
`tracknet_runner._trim_inactive_tail`.\n\n2. **Test update.** Tests already cover **both** flag
states (P0b shipped unit-tested ON and OFF). After flipping the default, update any test
asserting the *default-OFF* behaviour, and consider updating the doc-comment block at
`backend/config/models.py:193-198` / the preset doc-comment at `backend/eval/windowing.py:74-76`
to reflect the new default. Re-run the full suite (see §7).\n\n3. **Doc updates:**\n - **§7
tracker:** P2b Status ** ? ? ? ** , PR ` ? ? ?` <promotion PR link>` (§7 is the source of truth).\n
- **§9 Definition-of-Done:** check the box `[ ]` P2 promotion: golden-set A/B at SERVED
windowing, then flip the flag default-ON if no regression.\n - Update the doc header **"Last
updated" date; when all rows land, advance the Status header along **DRAFT ? IN PROGRESS ?
DONE/DELIVERED ? ARCHIVED**.\n - Reindex inbound refs: `docs/README.md` (where §7 is indexed),
and the `RALLY_DETECTION_QUALITY_REPORT.md` cluster.\n - **DOC_STATUS archive protocol** (only

```

when the whole R-series/quality cluster reaches terminal state ? do **not** archive yet per NEXT\_STEPS): run the `docs/DOC\_STATUS.md` §2 due-diligence checklist before any `git mv` to `docs/archives/`: (1) verify claims vs code, tag `[verified]`; (2) flip Status ? DELIVERED/DONE/REJECTED + completion note (what shipped, PRs, paths); (3) update all inbound refs (`docs/README.md`, `CLAUDE.md`, `docs/NEXT\_STEPS.md`, cross-doc links); (4) keep ADR forward/back lineage; (5) `git mv` the whole file to `docs/archives/past\_projects/` (terminal-state only) and note relative links become historical; (6) record the move in DOC\_STATUS §3e + memory current-state.\n\n---\n\n## 6. ON REJECT (tol\_f1 regresses at SERVED)\n\n- **No code change.** Leave `inactivity\_temporal\_density` **default-OFF** (`backend/config/models.py:199` stays `False`). P0b already shipped flag-OFF, so the merged state is a zero-regression no-op in production ? rejecting requires nothing in code.\n- **Record the numbers.** Capture the A/B output (`tol\_f1` base vs candidate, ?tolF1 + CI, paired sign-test, `|?end|`) in the doc so the rejection is auditable.\n- **§7 tracker:** mark P2b as **REJECTED** with a note (or keep ? with a rejection note), PR remains `?`. Per CLAUDE.md, the flag stays gated default-OFF; the fix remains available behind the flag for re-measurement if the corpus grows.\n\n---\n\n## 7. FULL VALIDATION\n\n> Tier-1 = OFFLINE ONLY (no GPU/real-video). Tier-2 (real video + full GPU/WSL/WASB byte-parity + golden corpus) is owner-gated on the self-hosted GPU runner ? exactly the box where P2b is run, so these gates **unlock** during this work.\n\n**Offline suite** (always):  

```
bash\npython run_all_tests.py
```

# full suite, quiet (-q); exit code = pytest's  
`pytest.ini` sets `testpaths = tests` (scratch/ never collected). Fully mocked: no GPU/network/WSL. The runner probes `obsreport` and prints availability (reporting tests self-skip via `pytest.importorskip('obsreport')`).\n- Expected count: documented latest **1492/1492 green** (`docs/CODE\_CLEANUP\_AUDIT\_2026-06-24.md`); a fully-equipped local checkout collects **1513** (parametrize expansion + optional deps present). CI lane reports fewer (obsreport/golden absent).\n\n**Now-unlockable GPU / real-video / golden / WASB gates** (run on the owner box where the golden corpus is present):  

```
tests/test_served_gate_a_regression.py
```

? runs when the manifest at env `SERVED\_GATE\_A\_MANIFEST` (default `not a pytest test; pytest only asserts `wasb\_infer.py` byte-identity (`test\_copy\_infer\_script\_copies\_byte\_identical`).\n- `tests/test\_golden\_fixtures.py` / `tests/test\_golden\_real\_fixtures.py` ? committed de-attributed fixtures; **do** run in CI via `python -m backend.eval.golden\_fixtures --check`. \n- Set the relevant env drivers as needed: `RALLY\_DETECTOR\_IMPL=native|stub`, `WASB\_WEIGHTS/WASB\_DEVICE/WASB\_PYTHON/WASB\_REPO\_DIR/WASB\_SCRATCH`, `SERVED\_GATE\_A\_MANIFEST`.\n\n**Lint / types / coverage** (must stay green):  

```
bash\nruff check --output-format github\nmypy backend training\npython run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70
```

\n- **ruff** (`ruff.toml`): lint-only (no `ruff format`), `target\_version = py38`, rules = defaults + `W,B,I`; `extend-exclude = ['scratch']`; `E402` per-file-ignored for `backend/main.py` and two `scripts/\*`.\n- **mypy** (`mypy.ini`): lenient green baseline; `ignore\_missing\_imports = True`, `explicit\_package\_bases = True`; quarantined-ignore modules: `backend.main`, `backend.worker.\_\_main\_\_`, `backend.pipeline.segmenters.{yolo\_hybrid,gemini,base}`. Ratchet down (remove a quarantine entry) only when you touch that module.\n- **coverage floor:** `--cov-fail-under=70` (calibrated against CI's measured ~72%). **Ratchet up** as coverage grows; never lower without an owner decision. **Omits** in `.coveragerc` (legacy `motion.py`, concrete validators).\n\n---\n\n## 8. GUARDRAILS (CLAUDE.md)\n\n- **Owner-gated work:** No platform/owned-model implementation has started ? do **not** start without explicit owner approval. Hand the heaviest machine-side checks (real video, full GPU/WSL/WASB suite) to the owner who holds the golden corpus + labelled videos; verify pure-logic paths locally. P2b promotion itself is owner-gated.\n- **Commercial-clean licensing** (ratified 2026-06-09): every dependency ? **code, data, AND weights** ? must be **MIT / Apache-2.0 / BSD only**. Enforced by the CI license-scan lane (`pip-licenses`, fails on `GPL|SSPL`). Owned-model base: Qwen3-VL (Apache-2.0) primary, InternVL3 (MIT) fallback; **reject Gemma 3**; Gemma-4 is AMBER/bench-only. (Commercial-ship blocker: C3 WASB checkpoint provenance ? out of P2b scope but noted.)\n- **Paid LLMs** (Gemini/OpenAI/Anthropic) = inference/serving/fallback **ONLY** ? never a training-data source. Exclude minors; per-user training data needs explicit opt-in.\n- **Compute = GCP only** (ADR-013): GPU training + serving on GCP (Cloud Run + GPU per ADR-012). Don't re-evaluate DigitalOcean/Paperspace without a new ADR.\n- **Scope:** v1 = rally + highlight + history + correction loop; gait/biometrics strictly FUTURE (GDPR Art. 9).\n- **Branch / PR discipline:**  
- Branch off **master** (origin/master). **ONE PR per work item; NO PR stacking** ? one small PR per §7 row.\n- **Do**

NOT open a PR unless asked.\*\* Never push to the default branch directly; descriptive commit messages.\n - Tests: `python run\_all\_tests.py` (offline, fully mocked). Python \*\*3.10+\*\* (trackers needs 3.10); ffmpeg required.\n - \*\*Archive policy:\*\* only move terminal-state (DONE/REJECTED) work to `docs/archives/`; honestly update the doc first (verify ? flip status ? update inbound refs ? keep lineage ? `git mv`).\n\n---\n\n## 9. OPEN QUESTIONS / UNKNOWN\n\n\*\*Concrete golden paths not in the plan doc.\*\* `<GOLDEN\_MANIFEST>` and `<GOLDEN\_FEATURES\_DIR>` are placeholders in §8 Q2 with no concrete path. Best available leads (from code/tests, not the plan): manifest default `<repo>/output/human\_lovo\_manifest.json` (env `SERVED\_GATE\_A\_MANIFEST`); features as `output/\*\_golden\_features.json`. \*\*Owner must confirm the actual corpus layout.\*\*\n\n\*\*Exact feature-gen command is unspecified.\*\* The plan doc pins only the `rally\_seg\_eval` A/B command, not the upstream GPU/WSL trajectory-feature extraction command. `python -m backend.eval.golden\_fixtures --check` is the committed \*sanity\* gate, not the extraction step; the real extraction is the WSL-quarantined `backend/pipeline/detectors/wasb\_infer.py` path. \*\*Owner to supply the exact extraction invocation.\*\*\n\n\*\*No numeric regression epsilon/p-value/CI rule in the plan doc.\*\* The gate is the qualitative `tolF1` does not regress.` The only concrete value is `served\_regress\_eps = 0.0` from `backend/eval/promotion.py:126` (code, not doc). Confirm with the owner whether `0.0` (no drop allowed) is the bar or whether a tolerance/p-value applies via `promote\_if\_served\_safe`. \n\n\*\*Secondary metrics are not gates.\*\* `|?end|`, paired sign-test, and CI delta exist in the harness but are \*\*not\*\* elevated by the plan into the promote/reject rule; `rally-end precision` as a named metric does not exist (harness reports `tolP/tolR/tol\_precision`). \n\n\*\*Doc inconsistencies to reconcile during the doc update:\*\* (a) §7 PR \*\*394\*\* vs header \*\*393\*\* for P0a/P0b; (b) §7 P1 ? Done vs §9 unchecked `rally\_cvat.py` warning-log box. Resolve both when touching the tracker. \n\n\*\*Ship-gate target still owner-unset.\*\* The formal ship-gate number (NEXT\_STEPS #7 / #172, suggested shape `LOVO F1@tIoU0.5 ? 0.70` multi-court, p<0.05 vs heuristic) is unrelated to the P2b tolF1-no-regression gate but is the broader blocker; do not conflate. \n\n\*\*Cluster not archive-ready.\*\* Per NEXT\_STEPS/DOC\_STATUS, the R-series/quality doc cluster must \*\*not\*\* be archived until R2 tolF1 is promoted-or-rejected as the gate and the §7.1 single quality number is attributed ? i.e., P2b's outcome is itself a prerequisite for archival.",

```

7 "digests": [
8 {
9 "key": "fixes-plan",
10 "data": {
11 "p2b_current_status": "P2b is NOT done ? it is the only open/blocked row in
the §7 tracker. Verbatim row: `| P2b | **Promote P0b**: golden-set A/B at SERVED windowing ?
flip flag default-ON if no regression | P2a | ? owner | ? | ? |`. Status glyph = ? (Todo),
Gated? = `"? owner`, PR = `"?` (none). It depends on P2a, which is ? done (`this PR`). The
§9 Definition-of-Done item for it is unchecked: `[] **P2 promotion**: golden-set A/B at SERVED
windowing, then flip the flag default-ON if no regression. *(owner-gated, real video ? see §8
Q2)*`. It is owner-gated because it needs the golden corpus + GPU/WSL-extracted trajectory
features that do not exist offline, and finding B is a no-op under dense tracking so the A/B
only shows signal on clips where the shuttle tracker drops out (cannot be measured on
synthetic/offline fixtures).",
12 "ab_runbook_commands": [
13 "python -m backend.eval.rally_seg_eval \"",
14 " --manifest <golden_manifest.json> --features-dir <golden_features_dir>
\"",
15 " --preset served --vs served --vs-inactivity-temporal-density"
16],
17 "decision_rule": "From §8 Q2 / §9, verbatim: base = SERVED with the flag OFF,
candidate = SERVED with it ON; the run `diffs tolF1` at the **SERVED** operating point`.
Decision: `***Promote** by flipping `IndexingConfig.inactivity_temporal_density` to `True` (a
one-line PR) **only if tolF1 does not regress**.` §9 restates: `flip the flag default-ON if no
regression.` The headline gate is tolF1-does-not-regress at the SERVED operating point. The doc
does NOT specify a numeric regression epsilon, p-value, or CI rule for this gate ? those are
absent from the doc (see secondary_metrics).",
18 "headline_metric": "tolF1 (tol_f1) ? the R2 tolerance-match F1 at the SERVED
operating point. This is the only headline metric named in the §8 Q2 / §9 decision rule; the
rule is `promote only if tolF1 does not regress.`",
19 "promote_steps": [
20 "Obtain golden corpus + GPU/WSL-extracted trajectory features (owner-only;
do not exist offline).",
21 "Run the §8 Q2 A/B command: python -m backend.eval.rally_seg_eval --manifest
<golden_manifest.json> --features-dir <golden_features_dir> --preset served --vs served --vs-
```

```

inactivity-temporal-density (base=SERVED flag-OFF, candidate=SERVED flag-ON).",
22 "Check the headline gate: confirm tolF1 does not regress at the SERVED
operating point.",
23 "If no regression: flip IndexingConfig.inactivity_temporal_density from
False to True in backend/config/models.py (line 199) ? a one-line PR. This is the exact one-line
change to promote.",
24 "Update $7 P2b row status ??? with the promotion PR link, and check the $9
P2-promotion box.",
25],
26 "reject_steps": [
27 "The doc gives no explicit reject procedure. Implicit from the decision
rule: if tolF1 regresses at the SERVED operating point, do NOT flip the flag ? leave
inactivity_temporal_density default-OFF (the merged state is already a zero-regression no-op in
production since P0b shipped flag-OFF). No code change is needed to reject; P0b stays gated
default-OFF and P2b remains ? (or is closed as rejected with a note).",
28],
29 "doc_update_steps": [
30 "After promotion, update $7 tracker P2b row: change Status ??? and PR
'?<promotion PR link> (the $7 tracker is the source of truth).",
31 "Check the $9 Definition-of-Done P2-promotion checkbox: '[] **P2
promotion**: golden-set A/B at SERVED windowing, then flip the flag default-ON if no
regression.'",
32 "Update the doc header 'Last updated' date and, when all rows land, flip the
Status header from IN PROGRESS toward DONE per the DRAFT ? IN PROGRESS ? DONE ? archived
lifecycle.",
33 "Note: $1 'Tracking anchors' says $7 is the source of truth and is indexed
in docs/README.md; the plan Extends docs/RALLY_DETECTION_QUALITY_REPORT.md.",
34 "Separately track the still-open sibling-repo item in $9: '[] Warning log
added for malformed rows in rally_cvat.py. (sibling-repo PR, separate)' ? though $7 marks P1 as
? done via sdc#50, creating an inconsistency between $7 (P1 ?) and $9 (rally_cvat warning
unchecked).",
35],
36 "served_regress_eps": "ABSENT from the plan doc. The doc never names
'served_regress_eps'; its decision rule is the qualitative 'tolF1 does not regress'. Grounded in
code (not the doc): backend/eval/promotion.py:126 and backend/eval/serve_contrast.py:112 define
served_regress_eps with default 0.0, meaning '0.0 = no served-mean drop allowed'
(promotion.py:147). That promotion-gate eps default of 0.0 is the closest concrete value, but it
is NOT referenced by RALLY_DETECTION_FIXES_PLAN.md.",
37 "secondary_metrics": [
38 "ABSENT from RALLY_DETECTION_FIXES_PLAN.md ? the requested secondary metrics
[?end], rally-end precision, paired sign-test, and CI delta are NOT mentioned anywhere in the
plan doc. The doc's $8 Q2 / $9 decision rule cites only tolF1.",
39 "These metrics DO exist in the eval harness code (not the doc):
backend/eval/rally_seg_eval.py emits [?end] in its per-video table (line ~523: '| ... | [?start|
| [?end| | split | merge |') and computes a paired sign-test + CI / per-tier delta (module
docstring line 13: 'diff two windowings with a paired sign-test + CI (the per-tier `delta`');
line ~358 'sign-test ... on both tIoU-F1 and the R2 tol_f1'; aggregate CI via _ci(a['tol_f1'])
at line ~534). 'rally-end precision' as a named metric does not appear; the harness reports
tolP/tolR/tol_precision (tolerance precision/recall), not a metric literally called 'rally-end
precision'.",
40 "Conclusion: treat [?end], paired sign-test, and CI-delta as harness outputs
you would inspect when running the A/B, but the PLAN DOC does not elevate any of them into the
promote/reject rule.",
41],
42 "section7_tracker": "$7 'Deliverables & progress tracker ? source of truth'.
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row, branched from
origin/master, no stacking. Rows verbatim: (1) '| P0a | Fix `spread` zero-collapse (behaviour-
neutral, unconditional) | ? | No | ? | #394 |'. (2) '| P0b | `_trim_inactive_tail` temporal
density behind `inactivity_temporal_density` flag (default-OFF) | ? | No | ? | #394 |'. (3) '|
P1 | Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-collector`) | ?
| No | ? | sdc#50 |'. (4) '| P2a | Expose the flag in the `rally_seg_eval` A/B CLI
(`--[vs-]inactivity-temporal-density`) so the promotion run is one command | P0b | No | ? | this
PR |'. (5) '| P2b | **Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON
if no regression | P2a | ? owner | ? | ? |'. SUMMARY: P0a=DONE(#394), P0b=DONE(#394),
P1=DONE(sdc#50), P2a=DONE(this PR), P2b=TODO/owner-gated(no PR). NOTE: P0a and P0b both list PR

```

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#394, though the $0/header narrative says P0 is 'implemented + unit-tested in this same PR
(#393)' ? header references #393 while the $7 table references #394 (an internal inconsistency
in the doc).",
43 "paths_referenced": [
44 "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_FIXES_PLAN.md",
45 "backend/pipeline/detectors/rally_gate.py (P0a _spread fix; $4 + $10 code
anchor)",
46 "backend/pipeline/detectors/tracknet_runner.py (P0b _trim_inactive_tail
temporal_density; $4 + $10 code anchor; flag at line 329/426 confirmed in code)",
47 "src/parsers/cvat/rally_cvat.py (P1 load_canonical_csv warning; sibling
sports-data-collector repo; $4 + $10)",
48 "backend/config/models.py ? IndexingConfig.inactivity_temporal_density (the
EXACT one-line promote flag; verified at line 199, default False)",
49 "backend/eval/rally_seg_eval.py ? the rally_seg_eval A/B CLI run via `python
-m backend.eval.rally_seg_eval` (P2a / $8 Q2); flags --inactivity-temporal-density and --vs-
inactivity-temporal-density (verified lines 587/603/612)",
50 "backend/eval/windowing.py ? WindowingPreset / SERVED preset plumbing
(config ? eval preset; verified)",
51 "backend/pipeline/segmenters/trajectory_hybrid.py ? trajectory_hybrid
plumbing config ? segmenter (verified line 250/305)",
52 "<golden_manifest.json> ? golden manifest, a --manifest placeholder in $8 Q2
(no concrete path given in doc)",
53 "<golden_features_dir> ? golden features dir, a --features-dir placeholder
in $8 Q2 (no concrete path given in doc)",
54 "docs/RALLY_DETECTION_QUALITY_REPORT.md (doc this plan Extends)",
55 "docs/RALLY_DETECTION_CODE_REVIEW.md ($10 internal docs reference)",
56 "docs/README.md (where $7 tracker is indexed)",
57 "backend/eval/promotion.py & backend/eval/serve_contrast.py ? where
served_regress_eps (default 0.0) actually lives; NOT referenced by the plan doc"
58]
59 },
60 },
61 {
62 "key": "code-review-findings",
63 "data": {
64 "finding_a": "{ \"id\": \"A\", \"title\": \"Subtle calculation artifact in _spread
\\\", \"file\": \"backend/pipeline/detectors/rally_gate.py\", \"function\": \"_spread\", \"what_was_wro
ng\": \"Robust spread trims bottom/top 5% via int()-truncated percentile indices: lo = s[max(0,
int(0.05*(len(s)-1))], hi = s[min(len(s)-1, int(0.95*(len(s)-1))]. Because int() truncates
rather than rounds, for short arrays (<20 items) the p95 index collapses to the second-to-last
element, and for len(s)==2 the spread evaluates to strictly 0.0 (zero
deadband).\", \"impact\": \"Not a crash. Currently benign because the play axis is estimated over
the entire video's thousands of trajectory points. Risk is latent: if reused on small window
slices, spread silently collapses to zero.\", \"fix\": \"_spread now rounds the p95 index UP with
math.ceil(), so a 2-element (or any short) input falls back to the full min..max range instead
of collapsing to 0.0. On large arrays ceil-vs-truncate differ by at most one sample, so the
robust p5/p95 trim is unchanged in practice.\", \"behavior_change\": \"None. Behaviour-neutral,
lands unconditionally (not flag-gated).\", \"status\": \"Done\", \"pr\": \"#394 (P0a). Note: review
doc front-matter says PR #393; the fixes-plan progress tracker (source of truth) lists
#394.\", \"definition_of_done\": \"Bug fixed and tested; behaviour-neutral, unconditional.\"}",
65 "finding_b": "{ \"id\": \"B\", \"title\": \"Sensitivity to sparse tracking in
_trim_inactive_tail (rally-end / inactivity temporal-density issue)\", \"file\": \"backend/pipelin
e/detectors/tracknet_runner.py\", \"function\": \"_trim_inactive_tail\", \"heuristic\": \"R4
(Inactive Tail Trimming) ? trims a candidate window's end back to the last point where a
trailing trail_frames window has sufficient density of fast frames (velocity >
rally_thresh).\", \"what_was_wrong\": \"The density check is fast_sum / win_count >= min_density,
but win_count is the number of VISIBLE, ACTIVE frames (samples) in the trailing boundary, NOT
total elapsed time. If the detector loses the shuttle for ~1.8s and picks up only 1 fast frame
at the window's end, win_count is tiny (e.g. 1-2), so fast_sum/win_count can evaluate to 1.0
(100% density). Sparse trailing points then defeat the trimmer, leaving dead-time uncut and
artificially inflating the candidate window's END boundary. Partially mitigated by merge_gap
breaking trajectories, but remains a boundary-logic quirk.\", \"what_the_fix_changes\": \"Adds an
inactivity_temporal_density flag. When ON, the density DENOMINATOR becomes the trailing window's
TEMPORAL length, min(trail_frames, frames[i] - frames[0] + 1), instead of win_count (the count

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of active samples that sparse tracking inflates). Resolved as STRICT temporal density (no separate few-frames fallback ? Q1 resolved).\", \"why\_noop\_under\_dense\_tracking\": \"The 'group' holds only active frames, so under dense tracking the temporal denominator min(trail\_frames, frames[i]-frames[0]+1) equals the sample-count denominator win\_count. The two denominators are identical, so flipping the flag is a no-op on well-tracked clips and changes behaviour ONLY in the sparse-tracking case. Corollary: signal appears only on clips where the shuttle tracker actually drops out ? so it cannot be measured on synthetic/offline fixtures.\", \"why\_flag\_gated\": \"R4 is DEFAULT-ON in production (window\_inactivity\_end = 1.5, promoted at the R1 milestone). An earlier draft wrongly called R4 default-OFF; corrected 2026-06-29. Because R4 is live, the fix is a REAL behaviour change on sparse footage, so it ships behind a default-OFF flag to keep the merge a zero-regression no-op in prod until measured on the golden set.\", \"flag\": {\"name\": \"inactivity\_temporal\_density\", \"default\": \"OFF (False)\", \"plumbing\": \"config (IndexingConfig.inactivity\_temporal\_density) -> trajectory\_hybrid segmenter -> eval WindowingPreset; also exposed in rally\_seg\_eval A/B CLI as --[vs-]inactivity-temporal-density (P2a).\"}, \"status\": \"Implemented + unit-tested in both flag states. Done as P0b. Promotion to default-ON (P2b) is owner-gated and NOT done.\", \"pr\": \"#394 (P0b); CLI toggle P2a marked 'this PR'. (Review doc front-matter cites #393; fixes-plan tracker is source of truth at #394.)\"}],

66           \"finding\_b\_acceptance\": \"{ \"summary\": \"Promotion (P2b) is owner-gated: requires golden corpus + GPU/WSL-extracted trajectory features that do not exist offline.\", \"measurement\_command\": \"python -m backend.eval.rally\_seg\_eval --manifest <golden\_manifest.json> --features-dir <golden\_features\_dir> --preset served --vs served --vs-inactivity-temporal-density\", \"ab\_setup\": \"base = SERVED windowing with flag OFF; candidate = SERVED windowing with flag ON. Measured at the SERVED operating point specifically because R4 is default-ON in prod, so this captures B's real end-boundary effect on sparse footage.\", \"metric\": \"tolF1 (tolerant F1) at the SERVED operating point.\", \"acceptance\_criterion\": \"Promote (flip IndexingConfig.inactivity\_temporal\_density to True, a one-line PR) ONLY IF tolF1 does not regress (no regression at SERVED).\", \"measurability\_caveat\": \"B is a no-op under dense tracking, so the A/B only shows signal on clips where the shuttle tracker actually drops out; cannot be measured on synthetic/offline fixtures ? hence owner-gated on real labelled video.\", \"gate\_status\": \"P2b = Todo, Blocked/gated (owner). Offline unit tests pass in both flag states; golden-corpus regression run is the outstanding owner-gated step.\"},

67           \"finding\_c\": \"{ \"id\": \"C\", \"title\": \"Graceful degradation on incomplete CSV validation\", \"file\": \"src/parsers/cvat/rally\_cvat.py (sibling sports-data-collector repo)\", \"functions\": \"parse\_cvat\_boxes / load\_canonical\_csv\", \"what\_was\_wrong\": \"parse\_cvat\_boxes searches zips for an .xml ending in 'annotations.xml' or else ANY .xml. In load\_canonical\_csv, safe\_int() guards malformed rally\_number fields; a fully malformed row (that doesn't break CSV parsing) is silently 'continue'd. Stable, but a severely malformed CSV could silently drop half its rallies without failing the import gate ? an observability gap.\", \"fix\": \"Add a logger.warning before each 'continue' in load\_canonical\_csv so skipped/malformed rows are surfaced during ingestion. Does not change parsing behaviour, only adds observability.\", \"scope\_note\": \"Lives in the sibling sports-data-collector repo, so it ships as a SEPARATE PR (cannot be in the in-repo #393/#394).\", \"status\": \"Done in sibling repo as P1, PR sdc#50. (Definition-of-done checkbox in fixes plan still shows the warning-log item unchecked, an inconsistency vs the P1 tracker row marked Done.)\"}],

68           }

69           },

70           {

71           \"key\": \"code-anchors\",

30. user (2026-06-29T10:55:28.382Z)

## backend/eval listing

total 496

```
drwxr-xr-x 4 avidullu avidullu 4096 Jun 29 10:44 .
drwxr-xr-x 20 avidullu avidullu 4096 Jun 27 17:27 ..
-rw-r--r-- 1 avidullu avidullu 455 Jun 25 11:46 __init__.py
drwxr-xr-x 2 avidullu avidullu 4096 Jun 29 10:51 __pycache__
-rw-r--r-- 1 avidullu avidullu 24039 Jun 27 17:27 ablation.py
-rw-r--r-- 1 avidullu avidullu 8187 Jun 27 17:27 ablation_pdf.py
-rw-r--r-- 1 avidullu avidullu 2278 Jun 27 17:27 annotations.py
drwxr-xr-x 3 avidullu avidullu 4096 Jun 27 17:27 audio
-rw-r--r-- 1 avidullu avidullu 4790 Jun 27 17:27 batch.py
```

```

-rw-r--r-- 1 avidullu avidullu 6349 Jun 27 17:27 calibrate_local.py
-rw-r--r-- 1 avidullu avidullu 5140 Jun 27 17:27 calibrate_wasb.py
-rw-r--r-- 1 avidullu avidullu 8078 Jun 27 17:27 calibration.py
-rw-r--r-- 1 avidullu avidullu 4639 Jun 27 17:27 classifier.py
-rw-r--r-- 1 avidullu avidullu 8724 Jun 27 17:27 distill_local.py
-rw-r--r-- 1 avidullu avidullu 11745 Jun 27 17:27 eval_partial_labels.py
-rw-r--r-- 1 avidullu avidullu 6735 Jun 27 17:27 eval_stats.py
-rw-r--r-- 1 avidullu avidullu 32996 Jun 27 17:27 experiment.py
-rw-r--r-- 1 avidullu avidullu 11576 Jun 27 17:27 export_wasb_segments.py
-rw-r--r-- 1 avidullu avidullu 6779 Jun 27 17:27 fusion_audio.py
-rw-r--r-- 1 avidullu avidullu 5045 Jun 27 17:27 fusion_compare.py
-rw-r--r-- 1 avidullu avidullu 10637 Jun 27 17:27 fusion_golden.py
-rw-r--r-- 1 avidullu avidullu 13549 Jun 25 16:26 gemini_refine.py
-rw-r--r-- 1 avidullu avidullu 43331 Jun 27 17:27 golden_fixtures.py
-rw-r--r-- 1 avidullu avidullu 19863 Jun 27 17:27 golden_real_fixtures.py
-rw-r--r-- 1 avidullu avidullu 4663 Jun 27 17:27 gt_loader.py
-rw-r--r-- 1 avidullu avidullu 5482 Jun 27 17:27 harness.py
-rw-r--r-- 1 avidullu avidullu 5811 Jun 27 17:27 metrics.py
-rw-r--r-- 1 avidullu avidullu 7945 Jun 27 17:27 per_user_eval.py
-rw-r--r-- 1 avidullu avidullu 9048 Jun 27 17:27 per_user_gate.py
-rw-r--r-- 1 avidullu avidullu 13828 Jun 27 17:27 per_user_regression.py
-rw-r--r-- 1 avidullu avidullu 10135 Jun 27 17:27 promotion.py
-rw-r--r-- 1 avidullu avidullu 35315 Jun 29 10:44 rally_seg_eval.py
-rw-r--r-- 1 avidullu avidullu 7701 Jun 27 17:27 rally_seq_proto.py
-rw-r--r-- 1 avidullu avidullu 9106 Jun 27 17:27 regression.py
-rw-r--r-- 1 avidullu avidullu 6551 Jun 27 17:27 score_calibration.py
-rw-r--r-- 1 avidullu avidullu 10449 Jun 27 17:27 segmentation_metrics.py
-rw-r--r-- 1 avidullu avidullu 8434 Jun 27 17:27 serve_contrast.py
-rw-r--r-- 1 avidullu avidullu 9046 Jun 27 17:27 served_gate_a.py
-rw-r--r-- 1 avidullu avidullu 3658 Jun 27 17:27 splits.py
-rw-r--r-- 1 avidullu avidullu 14920 Jun 27 17:27 tolerance_metrics.py
-rw-r--r-- 1 avidullu avidullu 3694 Jun 27 17:27 training.py
-rw-r--r-- 1 avidullu avidullu 12378 Jun 29 10:44 windowing.py

```

#### does fusion\_golden exist?

```

-rw-r--r-- 1 avidullu avidullu 6779 Jun 27 17:27 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/fusion_audio.py
-rw-r--r-- 1 avidullu avidullu 5045 Jun 27 17:27 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/fusion_compare.py
-rw-r--r-- 1 avidullu avidullu 10637 Jun 27 17:27 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/fusion_golden.py
-rw-r--r-- 1 avidullu avidullu 10637 Jun 27 17:27 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/fusion_golden.py
-rw-r--r-- 1 avidullu avidullu 43331 Jun 27 17:27 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/golden_fixtures.py
-rw-r--r-- 1 avidullu avidullu 19863 Jun 27 17:27 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/golden_real_fixtures.py

```

#### wc -l of key files

```

849 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/rally_seg_eval.py
224 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/promotion.py
245 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/windowing.py
490 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/config/models.py
599 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/pipeline/detectors/tracknet_runner.py
2407 total

```

#### 31. assistant (2026-06-29T10:55:44.439Z)

`fusion\_golden.py` **\*\*does\*\*** exist (the digest agent missed it) ? so the prompt's feature-gen

step is real. Now let me get ground truth on the exact inputs and whether the GPU is even needed (the cached trajectory CSVs suggest the eval may be GPU-free). Reading the actual code paths:

### 32. user (2026-06-29T10:55:44.936Z)

```
1 """Offline golden-set person-feature extractor ? the fusion-P3 measurement's GPU step.
2
3 Mirrors `distill_local`'s CSV-driven flow but ADDS the ~5 person-cue features per
shuttle
4 candidate window, by running `PersonDetector` over the ORIGINAL video. It reuses the
5 EXISTING golden trajectory CSVs (so **no WASB re-run**), needs **no Gemini and no DB**,
and
6 writes one small JSON per video. The LOVO comparison (`fusion_compare.py`) then runs
100% on
7 CPU offline. So the GPU touches exactly one thing ? person detection ? and the lift is
8 measured by the shipped experiment harness; person features enter only as feature
columns the
9 learned scorer weights (no special-casing, default-OFF until the golden ?F1 justifies
them).
10
11 Run (on the GPU box):
12 python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json
--out-dir output
13
14 Each output `output/<name>_golden_features.json` is the replayable feature substrate ?
re-run
15 `fusion_compare` with different variants/thresholds forever without re-detecting on the
GPU.
16 """
17
18 from __future__ import annotations
19
20 import argparse
21 import json
22 import os
23 import time
24 from typing import Any, Dict, List, Optional
25
26 from backend.eval import distill_local
27 from backend.eval.calibrate_wasb import load_golden_gts
28 from backend.eval.training import label_candidates
29 from backend.pipeline.detectors import fusion_features as ff
30 from backend.pipeline.detectors import rally_gate
31 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
32 from backend.utils.run_telemetry import RunTelemetry
33
34 # The 5 fps/resolution-invariant shuttle features distill_local already computes.
35 SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
36 # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
37 PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
38
39
40 def _derive_video_path(spec: Dict[str, Any]) -> str:
41 """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
42 golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming
convention)."""
43 if spec.get("video_path"):
44 return str(spec["video_path"])
45 labels = str(spec["labels"])
46 if labels.endswith(".rallies.csv"):
47 return labels[: -len(".rallies.csv")] + ".MP4"
48 raise ValueError(
49 f"{spec.get('name')}: no video_path and labels {labels!r} isn't *.rallies.csv"
50)
51
```



```

52
53 def extract_video(
54 spec: Dict[str, Any],
55 detector: Optional[Any] = None,
56 frame_skip: int = 3,
57 iou: float = 0.5,
58 log_every_sec: float = 15.0,
59 telemetry: Optional[Any] = None,
60) -> Dict[str, Any]:
61 """One golden video -> a feature record (shuttle + person features per candidate
62 window).
63 Returns ``{name, fps, frame_width,
64 candidates:[{start,end,shuttle{ },person{ },label}], gts}``.
65 `detector` (anything with ``detections_over_windows``) is injectable for GPU-free
66 tests;
67 when None a real `PersonDetector` is built (loads torchvision on first use).
68 `telemetry` (a `RunTelemetry` or None) records per-frame velocity + machine-health
69 snapshots
70 from the detection progress callback so a hard kill leaves a forensic runlog.
71 """
72 fps, fw = float(spec["fps"]), float(spec["frame_width"])
73 traj = str(spec["trajectory"])
74 points = TrackNetRunner.parse_trajectory_csv(traj)
75 intervals, x_shuttle = distill_local.build_candidates(traj, fps, fw)
76 gts = load_golden_gts(str(spec["labels"]))
77 labels = label_candidates(intervals, gts, iou)
78 # Net axis for the person two-sided split ? reuse rally_gate's camera-agnostic
79 estimate
80 # (parity with the shuttle net-crossing gate) rather than hard-coding an
81 orientation.
82 axis, net_px, _span = rally_gate.estimate_play_axis(points)
83 video_path = _derive_video_path(spec)
84
85 if detector is None:
86 from backend.pipeline.detectors.person_detector import PersonDetector
87
88 detector = PersonDetector({"indexing": {"use_gpu": True}})
89
90 # Single-pass person detection over ALL candidate windows (open the video ONCE, read
91 # forward ? no per-window seek). Then compute features per window (pure, no GPU).
92 n = len(intervals)
93 win_frames = [(round(s * fps), round(e * fps)) for (s, e) in intervals]
94 print(
95 f"[fusion_golden] {spec['name']}: {n} windows ? person-detecting in one
96 pass...",
97 flush=True,
98)
99 t0 = time.monotonic()
100 state = {"last": t0}
101
102 def _progress(done: int, total: int) -> None:
103 if telemetry is not None:
104 telemetry.snapshot(done, total, phase=spec["name"])
105 now = time.monotonic()
106 if now - state["last"] >= log_every_sec or done >= total:
107 el = now - t0
108 rate = done / el if el > 0 else 0.0
109 eta = (total - done) / rate if rate > 0 else 0.0
110 pct = 100 * done // total if total else 0
111 print(
112 f"[fusion_golden] {spec['name']}: detect frame {done}/{total} ({pct}%)
113 "
114 f"| {el:.0f}s elapsed | ETA {eta:.0f}s",
115 flush=True,

```

```

109)
110 state["last"] = now
111
112 per_window = detector.detections_over_windows(
113 video_path, win_frames, frame_skip, progress_cb=_progress
114)
115
116 candidates: List[Dict[str, Any]] = []
117 for i, ((s, e), xs) in enumerate(zip(intervals, x_shuttle)):
118 det = per_window[i]
119 det_fw = det.get("frame_width") or fw
120 det_fh = det.get("frame_height") or 0.0
121 # Normalise the (pixel) net coord to 0-1 on the play axis, matching the foot-
point
122 # normalisation fusion_features uses (x/width, y/height).
123 if axis == "x":
124 net_norm = net_px / det_fw if det_fw > 0 else 0.5
125 else:
126 net_norm = net_px / det_fh if det_fh > 0 else 0.5
127 sf, ef = win_frames[i]
128 shuttle_in = [p for p in points if sf <= p.frame <= ef]
129 person = ff.compute_person_features(
130 det["frames"],
131 shuttle_in,
132 det_fw,
133 det_fh,
134 court_polygon=None,
135 net=(axis, net_norm),
136)
137 candidates.append(
138 {
139 "start": float(s),
140 "end": float(e),
141 "shuttle": {k: float(v) for k, v in zip(SHUTTLE_FEATURE_NAMES, xs)},
142 "person": {k: float(v) for k, v in person.items()},
143 "label": int(labels[i]),
144 }
145)
146 return {
147 "name": spec["name"],
148 "fps": fps,
149 "frame_width": fw,
150 "candidates": candidates,
151 "gts": [[float(a), float(b)] for a, b in gts],
152 }
153
154
155 def run(
156 manifest_path: str,
157 out_dir: str,
158 frame_skip: int = 3,
159 iou: float = 0.5,
160 fresh: bool = False,
161 smallest_first: bool = False,
162) -> List[str]:
163 with open(manifest_path, encoding="utf-8") as f:
164 specs = json.load(f)
165 if smallest_first:
166 # Process small videos first for quick feedback (the slow full match lands
last);
167 # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
row/frame).
168 specs = sorted(
169 specs,
170 key=lambda s: (

```

```

171 os.path.getsize(s["trajectory"])
172 if os.path.isfile(s.get("trajectory", ""))
173 else 0
174),
175)
176 os.makedirs(out_dir, exist_ok=True)
177 # Run-telemetry black box: a long GPU run can be OS-killed with no traceback, so
keep a tiny
178 # rotating runlog of velocity + machine health for the post-mortem. Constructing it
must never
179 # break extraction, so guard the build and fall back to no telemetry on any error.
180 telem: Optional[RunTelemetry]
181 try:
182 telem = RunTelemetry(
183 os.path.join(out_dir, "run_telemetry.jsonl"), label="fusion_golden"
184)
185 except Exception: # noqa: BLE001 - telemetry wiring must never break the run it
observes
186 telem = None
187 written: List[str] = []
188 for vi, spec in enumerate(specs, 1):
189 out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
190 # Resumable: skip a video whose JSON already exists (a re-run picks up where a
prior
191 # run left off instead of re-detecting). --fresh forces a redo.
192 if not fresh and os.path.isfile(out):
193 print(
194 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already extracted
"
195 f"-> {out} (use --fresh to redo)",
196 flush=True,
197)
198 written.append(out)
199 continue
200 print(
201 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",
202 flush=True,
203)
204 t0 = time.monotonic()
205 if telem is not None:
206 telem.event("video_start", spec["name"])
207 try:
208 data = extract_video(spec, frame_skip=frame_skip, iou=iou, telemetry=telem)
209 except Exception as e: # noqa: BLE001 - record machine state at death, then re-
raise
210 if telem is not None:
211 telem.event("error", spec["name"], err=str(e))
212 raise
213 n_pos = sum(c["label"] for c in data["candidates"])
214 with open(out, "w", encoding="utf-8") as f:
215 json.dump(data, f, indent=2)
216 if telem is not None:
217 telem.event(
218 "video_done",
219 spec["name"],
220 windows=len(data["candidates"]),
221 positives=n_pos,
222)
223 print(
224 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: DONE in
{time.monotonic() - t0:.0f}s "
225 f"| {len(data['candidates'])} windows ({n_pos} positive) |
{len(data['gts'])} golden rallies "
226 f"-> {out}",
227 flush=True,

```

```

228)
229 written.append(out)
230 return written
231
232
233 def main() -> None:
234 ap = argparse.ArgumentParser(
235 description="Extract golden-set shuttle+person features (fusion P3 GPU step)."
236)
237 ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
238 ap.add_argument("--out-dir", default="output")
239 ap.add_argument("--frame-skip", type=int, default=3)
240 ap.add_argument("--iou", type=float, default=0.5)
241 ap.add_argument(
242 "--fresh", action="store_true", help="re-extract even if a JSON already exists"
243)
244 ap.add_argument(
245 "--smallest-first",
246 action="store_true",
247 help="process smaller videos first for quick feedback (LOVO is order-
independent)",
248)
249 args = ap.parse_args()
250 run(
251 args.manifest,
252 args.out_dir,
253 args.frame_skip,
254 args.iou,
255 fresh=args.fresh,
256 smallest_first=args.smallest_first,
257)
258
259
260 if __name__ == "__main__":
261 main()
262

```

### 33. user (2026-06-29T10:55:45.389Z)

```

1 """Promotion gate with the eval?serve guard: a cue is promotable ONLY if it does not
regress
2 at the SERVED windowing.
3
4 **Why this exists.** ``per_user_gate.should_promote_for_user`` already grades a B?A lift
5 honestly (min-N floor + ?F1 CI excludes 0 + paired sign test). But it grades whatever
scores
6 it is handed ? and the foundational bug was that those scores came from the *recall-
oriented
7 PROPOSAL* windowing (``backend.eval.windowing.PROPOSAL``), which production never
serves. So a
8 cue could clear the bar on a windowing nobody runs (Gate-A: proposal ?F1 **+0.074**)
while
9 regressing on the served one (**?0.008**). Grading harder isn't the fix; **measuring on
the
10 served windowing** is.
11
12 This module composes ? it adds no new statistics. It layers ONE rule on top of
13 ``should_promote_for_user``:
14
15 A cue is promotable only if, at the **SERVED** windowing, it WINS or at least DOES
NOT
16 REGRESS (its served paired ?F1 mean ? ``-served_regress_eps``, and ? when its served
CI is
17 estimable ? that CI's lower bound does not sit clearly in regression territory).
18

```

```

19 A proposal-windowing win is still *reported* (it's useful signal for where to look
next), but it
20 can never carry a promotion on its own. The decision returns BOTH the served and the
proposal
21 verdicts so the eval?serve contrast is explicit and auditable.
22
23 Pure, deterministic, stdlib-only over ``eval_stats``/``per_user_gate``. Additive:
existing eval
24 paths are untouched until a caller routes its A/B through
:func:`promote_if_served_safe`.
25 """
26
27 from __future__ import annotations
28
29 from dataclasses import dataclass, field
30 from typing import Any, Dict, Optional, Sequence
31
32 from backend.eval.eval_stats import paired_delta_ci
33 from backend.eval.per_user_gate import PromotionDecision, should_promote_for_user
34
35 __all__ = ["ServedGateResult", "promote_if_served_safe", "served_regresses"]
36
37
38 def served_regresses(
39 a_scores: Sequence[float],
40 b_scores: Sequence[float],
41 *,
42 eps: float = 0.0,
43 n_boot: int = 2000,
44 seed: int = 0,
45) -> Dict[str, Any]:
46 """Does variant B REGRESS variant A at the served windowing? (the eval?serve guard
core).
47
48 ``a_scores``/``b_scores`` are per-video held-out scores **measured at the SERVED
windowing**
49 (aligned by video). Returns the paired B?A summary (mean ?F1 + bootstrap CI + sign
test) plus
50 a single ``regresses`` boolean:
51
52 - ``regresses=True`` when the served mean ?F1 is below ``-eps`` (a real served
drop), OR
53 the served ?F1 CI lies **entirely** below ``-eps`` (a confidently-negative
band).
54 - ``regresses=False`` when the cue holds the line at the served operating point
(mean ?
55 ``-eps`` and the CI is not a confident net-negative).
56
57 ``eps`` is the regression tolerance: 0.0 = "must not drop at all on the served
mean". A tiny
58 positive ``eps`` lets noise-level wobble through (e.g. 0.005 ? half an F1 point at
n?6).
59 """
60 delta = paired_delta_ci(a_scores, b_scores, n_boot=n_boot, seed=seed)
61 mean_delta = float(delta["mean_delta"])
62 ci_low = float(delta["ci_low"])
63 ci_high = float(delta["ci_high"])
64 # Regression if the served mean dropped beyond tolerance, OR the whole CI is net-
negative.
65 mean_regresses = mean_delta < -eps
66 ci_confidently_negative = ci_high < -eps
67 regresses = bool(mean_regresses or ci_confidently_negative)
68 return {
69 "regresses": regresses,
70 "mean_delta": mean_delta,

```

```

71 "ci_low": ci_low,
72 "ci_high": ci_high,
73 "ci_excludes_zero": bool(delta["ci_excludes_zero"]),
74 "sign_test": delta["sign_test"],
75 "n": int(delta["n"]),
76 "eps": float(eps),
77 }
78
79
80 @dataclass
81 class ServedGateResult:
82 """A promotion decision that BOTH measures the cue on the served windowing AND
reports the
83 proposal-windowing view, so the eval?serve contrast is never hidden.
84
85 - ``promote`` ? final verdict. ``True`` only when the underlying honest gate would
promote
86 *and* the served check did not veto (the cue does not regress at the served
windowing).
87 - ``served_safe`` ? the eval?serve guard's verdict in isolation (cue does not
regress served).
88 - ``vetoed_by_served`` ? ``True`` when the proposal/base gate WOULD have promoted
but the
89 served regression check overruled it (the exact Gate-A failure mode this module
prevents).
90 - ``base_decision`` ? the ``PromotionDecision`` from ``should_promote_for_user`` on
the
91 *promotion-grade* scores (served by default ? see :func:`promote_if_served_safe`).
92 - ``served`` / ``proposal`` ? the paired B?A summary at each windowing (means + CIs
+ sign
93 tests), so a reviewer sees "+0.074 proposal / ?0.008 served" side by side.
94 - ``reason`` ? short human-readable explanation of the final verdict.
95 """
96
97 promote: bool
98 served_safe: bool
99 vetoed_by_served: bool
100 reason: str
101 base_decision: PromotionDecision
102 served: Dict[str, Any]
103 proposal: Optional[Dict[str, Any]] = field(default=None)
104
105 def as_dict(self) -> Dict[str, Any]:
106 return {
107 "promote": self.promote,
108 "served_safe": self.served_safe,
109 "vetoed_by_served": self.vetoed_by_served,
110 "reason": self.reason,
111 "base_decision": self.base_decision.as_dict(),
112 "served": self.served,
113 "proposal": self.proposal,
114 }
115
116
117 def promote_if_served_safe(
118 served_a: Sequence[float],
119 served_b: Sequence[float],
120 *,
121 proposal_a: Optional[Sequence[float]] = None,
122 proposal_b: Optional[Sequence[float]] = None,
123 structural: bool = False,
124 min_n: int = 5,
125 p_threshold: float = 0.05,
126 served_regress_eps: float = 0.0,
127 n_boot: int = 2000,

```

```

128 seed: int = 0,
129) -> ServedGateResult:
130 """Promote a cue ONLY if it does not regress at the SERVED windowing (the
foundational rule).
131
132 The honest bar (`should_promote_for_user`: min-N floor + ?F1 CI excludes 0 + sign
test) is
133 applied to the **served** scores ? the operating point production runs ? so a
proposal-only
134 win can never carry a promotion. The optional `proposal_` scores are scored too
and
135 reported alongside, purely to surface the eval?serve contrast (they NEVER promote on
their
136 own).
137
138 Args:
139 served_a/served_b: per-video held-out scores for variant A (control) and B (the
cue),
140 measured at the **SERVED** windowing. These drive the decision.
141 proposal_a/proposal_b: the same scores at the PROPOSAL windowing (optional).
Reported for
142 contrast; the served regression guard still has the final say.
143 structural: forwarded to `should_promote_for_user` ? a structural guard (e.g.
Gate-A vs
144 held-shuttle FPS) is never *disabled* on "no lift", but it is ALSO not
auto-*promoted* to
145 the served default if it regresses served. So for `structural=True` we report
146 `keep_on` honestly but `promote` reflects the served-safety check.
147 served_regress_eps: served regression tolerance (0.0 = "no served-mean drop
allowed").
148
149 Returns a :class:`ServedGateResult` with the final verdict + both windowings'
evidence.
150 """
151 # 1) The honest gate, applied to the SERVED scores (never the proposal ones).
152 base = should_promote_for_user(
153 served_a,
154 served_b,
155 structural=structural,
156 min_n=min_n,
157 p_threshold=p_threshold,
158 n_boot=n_boot,
159 seed=seed,
160)
161
162 # 2) The eval?serve guard: does B regress A at the served windowing?
163 served = served_regresses(
164 served_a, served_b, eps=served_regress_eps, n_boot=n_boot, seed=seed
165)
166 served_safe = not served["regresses"]
167
168 # 3) Optional proposal-windowing view (reported, never promoting on its own).
169 proposal: Optional[Dict[str, Any]] = None
170 if proposal_a is not None and proposal_b is not None:
171 proposal = served_regresses(
172 proposal_a, proposal_b, eps=served_regress_eps, n_boot=n_boot, seed=seed
173)
174
175 # 4) Compose the verdict. The served guard can only ever REMOVE a promotion, never
add one.
176 if structural:
177 # A structural guard ships ON regardless of measured lift (base.keep_on stays
True), but
178 # it is only auto-promoted to the SERVED default when it is served-safe. If it
regresses

```

```

179 # served, promote=False (don't bake an eval-only win into production) while
keep_on holds.
180 promote = served_safe
181 vetoed = False
182 if served_safe:
183 reason = (
184 "structural guard kept ON; served-safe (does not regress at the served "
185 f"windowing, ?F1 {served['mean_delta']:+.4f}) ? eligible for served
default"
186)
187 else:
188 reason = (
189 "structural guard kept ON, but NOT promoted to the served default: it "
190 f"REGRESSES at the served windowing (?F1 {served['mean_delta']:+.4f}); "
191 "an eval-only win must not be served"
192)
193 else:
194 would_promote = base.promote
195 if would_promote and not served_safe:
196 promote = False
197 vetoed = True
198 reason = (
199 "VETOED by the served-windowing guard: the cue clears the honest bar but
"
200 f"REGRESSES at the served windowing (served ?F1
{served['mean_delta']:+.4f})"
201 f", CI [{served['ci_low']:+.4f}, {served['ci_high']:+.4f})). A win on a
"
202 "windowing production doesn't serve cannot promote."
203)
204 elif would_promote:
205 promote = True
206 vetoed = False
207 reason = (
208 "promoted: clears the honest bar AND is served-safe (served ?F1 "
209 f"{served['mean_delta']:+.4f}). " + base.reason
210)
211 else:
212 promote = False
213 vetoed = False
214 reason = base.reason
215
216 return ServedGateResult(
217 promote=promote,
218 served_safe=served_safe,
219 vetoed_by_served=vetoed,
220 reason=reason,
221 base_decision=base,
222 served=served,
223 proposal=proposal,
224)
225

```

#### 34. user (2026-06-29T10:55:46.340Z)

```

1 """Single-sourced candidate-windowing presets ? the eval?serve foundation fix.
2
3 **The bug this closes.** The offline eval substrate (`distill_local.build_candidates`
?
4 ``fusion_golden`` ? the golden feature JSONs the ablation/compare path re-scores) was
built
5 on a *recall-oriented PROPOSAL* windowing (`vt=0.005, merge_gap=1.0, min_window=0.5`):
6 deliberately permissive so a learned scorer/gate can filter the junk. But the SERVED
pipeline
7 (`TrajectoryHybridSegmenter.process_video`) windows COARSELY (`vt=0.02,

```



```

merge_gap=2.0,
8 min_window=2.0`` ? the ``config.indexing`` defaults). A cue measured on the proposal set
is
9 measured on candidates **production never serves**, so a "win" need not transfer. This
is
10 exactly how Gate-A scored **+0.074** on the proposal set yet **?0.008** on the served
set
11 (``scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md`` vs ``output/gateA_served_verify/``).
12
13 **The fix.** One module owns the two named windowings as :class:`WindowingPreset`s, and
the
14 SERVED preset is **single-sourced from the same Pydantic config defaults production
reads**
15 (:class:`backend.config.models.IndexingConfig`) ? eval and serve can no longer drift.
Helpers
16 build candidates under either preset, and :func:`served_windowing_from_config` lifts the
17 *actual* served windowing out of a live config (overrides included), so an A/B can
target the
18 operating point a given deployment truly serves.
19
20 The companion rule ? *a cue is promotable only if it does NOT regress at the SERVED
21 windowing* ? lives in :mod:`backend.eval.promotion` (which composes ``eval_stats`` +
22 ``per_user_gate``). This module is the **what windowing**; that one is the **promotion
gate**.
23
24 Pure, offline, stdlib + config only. Nothing here changes existing eval behaviour until
a
25 caller opts in (``distill_local`` re-points its module-level presets here, byte-
identically).
26 """
27
28 from __future__ import annotations
29
30 from dataclasses import dataclass
31 from typing import Any, Dict, List, Mapping, Tuple
32
33 from backend.config.models import IndexingConfig
34
35 Interval = Tuple[float, float]
36
37 __all__ = [
38 "WindowingPreset",
39 "SERVED",
40 "PROPOSAL",
41 "PRESETS",
42 "served_windowing_from_config",
43 "build_windows",
44 "windows_to_intervals",
45]
46
47
48 @dataclass(frozen=True)
49 class WindowingPreset:
50 """A named (velocity_thresh, merge_gap, min_window_duration) windowing operating
point.
51
52 ``velocity_thresh`` ? normalised per-second shuttle velocity above which a frame
counts as
53 "in flight"; ``merge_gap`` ? seconds within which active frames coalesce into one
window;
54 ``min_window_duration`` ? windows shorter than this are dropped. These are exactly
the three
55 knobs ``TrackNetRunner.trajectory_to_action_windows`` takes, so a preset *is* the
windowing.
56 """

```

```

57
58 name: str
59 velocity_thresh: float
60 merge_gap: float
61 min_window_duration: float
62 #: 0 = OFF (default). When > 0, windows longer than this are split at their largest
internal
63 #: inactivity gap ? the bounded-windowing over-merge fix (W1,
RALLY_QUALITY_RESEARCH.md §6).
64 max_window_duration: float = 0.0
65 #: When True, ``max_window_duration`` is a HARD cap ? an over-long window with no
internal
66 #: inactivity gap is chopped at its midpoint until ? the cap (continuously-active
fix). OFF by default.
67 hard_cap: bool = False
68 #: R4 rally-END inactivity trim (s, 0 = OFF): trim a window's post-rally slow-drift
tail back to
69 #: the last frame whose trailing window stays dense with FAST motion. The measured
rally-end fix.
70 inactivity_end: float = 0.0
71 #: velocity above which motion counts as "rally" (vs slow drift) for the R4 trim;
min fast-fraction.
72 inactivity_rally_thresh: float = 0.08
73 inactivity_min_density: float = 0.34
74 #: R4 density-denominator fix (finding B), default OFF. When True the R4 fast-
fraction is taken
75 #: over the trailing window's TEMPORAL length, not the count of active samples
(which sparse
76 #: tracking inflates). No-op under dense tracking; gated default-OFF until measured
on SERVED.
77 inactivity_temporal_density: bool = False
78 #: R5 blob-fallback (default OFF). After the cap split + R4 trim, midpoint-chop any
group still
79 #: longer than ``blob_factor`` × ``max_window_duration`` (the gap-less, R4-survived
blob ?
80 #: mahadevpura-1's ~11× residual) and flag it. Runs AFTER R4 (vs ``hard_cap``
before) and the
81 #: factor keeps it surgical (a normal long rally is left alone).
82 blob_fallback: bool = False
83 blob_factor: float = 4.0
84
85 def as_tuple(self) -> Tuple[float, float, float]:
86 """(velocity_thresh, merge_gap, min_window_duration) ? the legacy tuple
shape the
87 ``trajectory_to_action_windows`` / ``distill_local`` call sites already speak.
(The
88 bounded-windowing knob ``max_window_duration`` is passed separately by
``build_windows``.)"""
89 return (self.velocity_thresh, self.merge_gap, self.min_window_duration)
90
91 def to_dict(self) -> Dict[str, Any]:
92 return {
93 "name": self.name,
94 "velocity_thresh": self.velocity_thresh,
95 "merge_gap": self.merge_gap,
96 "min_window_duration": self.min_window_duration,
97 "max_window_duration": self.max_window_duration,
98 "hard_cap": self.hard_cap,
99 "inactivity_end": self.inactivity_end,
100 "inactivity_rally_thresh": self.inactivity_rally_thresh,
101 "inactivity_min_density": self.inactivity_min_density,
102 "inactivity_temporal_density": self.inactivity_temporal_density,
103 "blob_fallback": self.blob_fallback,
104 "blob_factor": self.blob_factor,
105 }

```

```

106
107
108 def _served_preset_from_defaults() -> WindowingPreset:
109 """The SERVED windowing, lifted from the SAME Pydantic config defaults production
reads.
110
111 ``TrajectoryHybridSegmenter.process_video`` reads exactly these three:
112 - velocity: ``idx_cfg[<family>].velocity_threshold`` (default 0.02 ?
wasb/fusion/tracknet)
113 - merge_gap: ``idx_cfg.dead_time_merge_gap`` (default 2.0)
114 - min window: ``idx_cfg.min_action_window_duration`` (default 2.0)
115
116 Sourcing them from :class:`IndexingConfig`'s defaults (not a hand-typed tuple) is
the whole
117 point: change a served default in ``config/models.py`` and the eval SERVED preset
follows,
118 so the two can never silently diverge again.
119 """
120 idx = IndexingConfig() # construct with defaults ? the served operating point
121 return WindowingPreset(
122 name="served",
123 velocity_thresh=float(
124 idx.wasb.velocity_threshold
125), # == fusion/tracknet default 0.02
126 merge_gap=float(idx.dead_time_merge_gap),
127 min_window_duration=float(idx.min_action_window_duration),
128 max_window_duration=float(
129 idx.max_window_duration
130), # W1 bounded-windowing (default 0=OFF)
131 hard_cap=bool(idx.window_hard_cap), # W1 hard-cap fallback (default OFF)
132 inactivity_end=float(
133 idx.window_inactivity_end
134), # R4 rally-end trim (default 0=OFF)
135 inactivity_rally_thresh=float(idx.inactivity_rally_thresh),
136 inactivity_min_density=float(idx.inactivity_min_density),
137 inactivity_temporal_density=bool(
138 idx.inactivity_temporal_density
139), # R4 density fix (finding B, default OFF)
140 blob_fallback=bool(idx.window_blob_fallback), # R5 blob-fallback (default OFF)
141 blob_factor=float(idx.window_blob_factor),
142)
143
144
145 #: The COARSE windowing production actually serves (config defaults; ~0.02/2.0/2.0). A
cue is
146 #: only promotable if it holds up HERE ? the operating point real users get.
147 SERVED: WindowingPreset = _served_preset_from_defaults()
148
149 #: The recall-oriented PROPOSAL windowing the offline substrate is built on
(deliberately
150 #: permissive: propose many, let the classifier/gate filter). NOT what production serves
? a
151 #: win here must be re-confirmed on SERVED before promotion. Mirrors the historical
152 #: ``distill_local.PROPOSAL`` tuple (0.005/1.0/0.5) so the existing golden JSONs stay
valid.
153 PROPOSAL: WindowingPreset = WindowingPreset(
154 name="proposal", velocity_thresh=0.005, merge_gap=1.0, min_window_duration=0.5
155)
156
157 #: Name ? preset, so a CLI / config string ("served" | "proposal") selects one.
158 PRESETS: Dict[str, WindowingPreset] = {SERVED.name: SERVED, PROPOSAL.name: PROPOSAL}
159
160
161 def served_windowing_from_config(config: Any, family: str = "wasb") -> WindowingPreset:
162 """The SERVED windowing a *specific* live config serves (overrides honoured).

```

```

163
164 Reproduces ``TrajectoryHybridSegmenter.process_video``'s knob reads exactly, against
an
165 arbitrary config (the typed :class:`AppConfig` or a plain dict) and segmenter
``family``
166 (``"wasb"`` / ``"fusion"`` / ``"tracknet"``). Use this when a deployment overrode a
served
167 default in ``config.json`` ? the module-level :data:`SERVED` is the *default*
operating point;
168 this is *this config's* operating point. Falls back to the served defaults for any
absent key.
169 """
170 idx = _as_mapping(config, "indexing")
171 family_cfg = _as_mapping(idx, family)
172 return WindowingPreset(
173 name=f"served:{family}",
174 velocity_thresh=float(
175 family_cfg.get("velocity_threshold", SERVED.velocity_thresh)
176),
177 merge_gap=float(idx.get("dead_time_merge_gap", SERVED.merge_gap)),
178 min_window_duration=float(
179 idx.get("min_action_window_duration", SERVED.min_window_duration)
180),
181 max_window_duration=float(
182 idx.get("max_window_duration", SERVED.max_window_duration)
183),
184 hard_cap=bool(idx.get("window_hard_cap", SERVED.hard_cap)),
185 inactivity_end=float(idx.get("window_inactivity_end", SERVED.inactivity_end)),
186 inactivity_rally_thresh=float(
187 idx.get("inactivity_rally_thresh", SERVED.inactivity_rally_thresh)
188),
189 inactivity_min_density=float(
190 idx.get("inactivity_min_density", SERVED.inactivity_min_density)
191),
192 inactivity_temporal_density=bool(
193 idx.get("inactivity_temporal_density", SERVED.inactivity_temporal_density)
194),
195 blob_fallback=bool(idx.get("window_blob_fallback", SERVED.blob_fallback)),
196 blob_factor=float(idx.get("window_blob_factor", SERVED.blob_factor)),
197)
198
199
200 def _as_mapping(obj: Any, key: str) -> Mapping[str, Any]:
201 """``obj[key]`` as a plain mapping, whether ``obj`` is a dict or a typed config
section
202 (``AppConfig.get`` returns nested sections as dicts). Empty mapping if absent/non-
mapping."""
203 if obj is None:
204 return {}
205 if isinstance(obj, Mapping):
206 sub = obj.get(key, {})
207 elif hasattr(obj, "get"):
208 sub = obj.get(key, {})
209 else:
210 sub = getattr(obj, key, {})
211 return sub if isinstance(sub, Mapping) else {}
212
213
214 def build_windows(
215 points: List[Any], fps: float, frame_width: float, preset: WindowingPreset
216) -> List[Dict[str, Any]]:
217 """Trajectory ``points`` ? candidate windows under ``preset`` (the one windowing
call site).
218
219 A thin wrapper over ``TrackNetRunner.trajectory_to_action_windows`` that takes a

```

```

named preset
220 instead of three loose floats, so eval candidate-building and the served path
reference the
221 SAME knobs by name. Imported lazily to keep this module import-light (no detector
deps)."""
222 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
223
224 vt, mg, mwd = preset.as_tuple()
225 return TrackNetRunner.trajectory_to_action_windows(
226 points,
227 fps=fps,
228 frame_width=frame_width,
229 velocity_thresh=vt,
230 merge_gap=mg,
231 min_window_duration=mwd,
232 max_window_duration=preset.max_window_duration,
233 window_hard_cap=preset.hard_cap,
234 window_inactivity_end=preset.inactivity_end,
235 inactivity_rally_thresh=preset.inactivity_rally_thresh,
236 inactivity_min_density=preset.inactivity_min_density,
237 inactivity_temporal_density=preset.inactivity_temporal_density,
238 window_blob_fallback=preset.blob_fallback,
239 window_blob_factor=preset.blob_factor,
240)
241
242
243 def windows_to_intervals(windows: List[Dict[str, Any]]) -> List[Interval]:
244 """`[{start, end, ...}]` ? `[start, end)` ? the interval shape the scorer/gate
speak."""
245 return [(w["start"], w["end"]) for w in windows]
246

```

### 35. user (2026-06-29T10:55:46.898Z)

```

140 # Net line for the person two-sided-motion split (person features only ? the shuttle
141 # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
142 net_axis: Literal["x", "y"] = "y"
143 net_position: float = Field(default=0.5, ge=0.0, le=1.0)
144
145
146 class IndexingConfig(BaseModel):
147 default_segmenter: str = "yolo_hybrid"
148 use_gpu: bool = True
149 # Detector backend for the trajectory hybrids (wasb/tracknet). "auto" = the real
150 # WSL runner (default, production); "stub" = a deterministic CPU mock (no GPU/WSL)
151 # so the whole pipeline is regression-testable on a CPU box / CI; "native" = the
152 # Linux-native WASB runner (no WSL/conda-activate; a GPU box ? M3.5). The
153 # RALLY_DETECTOR_IMPL env var overrides this. See
pipeline/detectors/{stub,native_wasb}_runner.py.
154 detector_impl: str = "auto"
155 motion_threshold: float = 0.08
156 min_segment_duration: float = 3.0
157 dead_time_merge_gap: float = 2.0
158 chunk_duration: float = 90.0
159 chunk_overlap: float = 10.0
160 detector_model: str = "fasterrcnn_resnet50_fpn"
161 detector_score_threshold: float = 0.5
162 yolo_velocity_threshold: float = 0.15
163 yolo_frame_skip: int = 3
164 yolo_tracker: str = "bytetrack.yaml"
165 yolo_prefetch_workers: int = 2
166 min_action_window_duration: float = 2.0
167 # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
= a window
168 # longer than this many seconds is recursively split at its largest internal

```

```

inactivity gap (?
169 # ``window_min_split_gap`` s ? the most likely rally boundary), so one window can't
swallow
170 # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
171 # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13
172 # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
173 # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
174 max_window_duration: float = 6.0
175 window_min_split_gap: float = (
176 0.5 # min internal gap (s) that qualifies as a split point
177)
178 # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
179 # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
180 # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
181 # Only cuts (recall can't drop). OFF by default until measured/promoted.
182 window_hard_cap: bool = False
183 # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
184 # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
185 # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
186 # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
187 # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
188 # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap
189 # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
190 window_inactivity_end: float = 1.5
191 inactivity_rally_thresh: float = 0.20
192 inactivity_min_density: float = 0.30
193 # R4 density DENOMINATOR fix (code-review finding B), default OFF until measured at
the SERVED
194 # operating point. The R4 trim's fast-fraction denominator is the COUNT of active
samples in the
195 # trailing window, which sparse tracking inflates (one fast sample after a gap ?
100% dense ?
196 # tail held open). When True, the denominator is the window's TEMPORAL length
instead. No-op
197 # under dense tracking (sample-count == temporal-span), so this can flip with zero
regression on
198 # well-tracked clips; flag-gated default-OFF since R4 itself is default-ON. See
QUALITY_ITERATIONS.
199 inactivity_temporal_density: bool = False
200 # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
201 # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
202 # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
203 # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
204 # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
205 # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap

```

```

206 # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only
207 # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
208 window_blob_fallback: bool = False
209 # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
210 # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
211 # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
212 # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
213 window_blob_factor: float = 4.0
214 log_yolo_candidates: bool = True

```

### 36. assistant (2026-06-29T10:56:12.050Z)

`fusion\_golden.py` is the **person-fusion** GPU step (runs `PersonDetector` over the *original video* to add person-cue features) ? it needs the actual MP4s, which are mostly absent from GCS. But finding B is a pure *trajectory-windowing* change, so the critical question is whether *rally\_seg\_eval* actually needs those person features or just the trajectory CSVs + labels (all 15 of which are present in GCS, GPU-free). Reading the A/B harness fully to settle this:

### 37. user (2026-06-29T10:56:13.454Z)

```

1 """One-command rally-segmentation eval ? score a windowing against the golden labels
with the
2 over-segmentation-sensitive metric set, honestly.
3
4 This is the harness the rally-quality project (`docs/RALLY_QUALITY_RESEARCH.md` $4.5 /
$7)
5 measures every tier against. It closes the loop the over-merge finding exposed: plain
temporal
6 IoU-F1 is blind to merged mega-windows, so we report it alongside segmental F1@k,
the
7 segment-count ratio, and the explicit `merge_split` breakdown
(`segmentation_metrics`).
8
9 For each golden video it: parses the cached WASB/TrackNet trajectory CSV ? builds
candidate
10 windows under a :class:`~backend.eval.windowing.WindowingPreset` (so it tracks the
SERVED
11 operating point and any future windowing the same way) ? scores the windows-as-rallies
against
12 the ground-truth rally intervals. Aggregates per-video held-out F1 with a bootstrap CI,
and can
13 diff two windowings with a paired sign-test + CI (the per-tier `delta`).
14
15 GROUND TRUTH: the golden manifest (`output/human_lovo_manifest.json`) gives each
video's
16 trajectory path + fps + frame_width; the GT rally intervals come from
17 `output/<name>_golden_features.json` (the `gts` key) so this needs no extra label
files.
18
19 Pure offline (CPU): trajectory parsing + windowing + interval scoring; no GPU/Gemini.
20 """
21
22 from __future__ import annotations
23
24 import argparse
25 import json
26 import os
27 from typing import Any, Dict, List, Optional, Tuple
28
29 from backend.eval import eval_stats, metrics, segmentation_metrics, windowing

```

```

30 from backend.eval.metrics import Interval
31 from backend.eval.windowing import PRESETS, SERVED, WindowingPreset
32
33 DEFAULT_MANIFEST = os.path.join("output", "human_lovo_manifest.json")
34 DEFAULT_FEATURES_DIR = "output"
35 DEFAULT_IOU = 0.5
36
37 #: Long-rally lens threshold (s). A rally is "long" (an eye-catching highlight) when its
duration
38 #: (end ? start) exceeds this. The local detector over-fires, and its false positives
are mostly
39 #: SHORT (motion blips, background-court fragments on multicourt footage); filtering to
long rallies
40 #: strips most of those FPs and reveals the real product quality on the long-rally happy
path. Used
41 #: as the default split point for the stratified report (``--long-tau`` overrides it).
DURATION, not
42 #: shot-count, is the filter: golden ``shots_count`` is unlabeled and the no-AI path
reports it as
43 #: "Unknown", whereas duration is derived from the same start/end labels we already
trust.
44 LONG_RALLY_TAU = 5.0
45
46 #: Multicourt golden clips ? the court-type lookup for the stratified report. The golden
manifest
47 #: (``output/human_lovo_manifest.json``) is gitignored, so this committed set is the
AUDITABLE source
48 #: of the single-vs-multicourt strata, kept in sync with ``docs/GOLDEN_VIDEOS.md`` (the
clips tagged
49 #: "multicourt"). Multicourt footage carries background games on adjacent courts ? the
dominant
50 #: source of short false positives ? so we report it separately. A manifest entry MAY
carry an
51 #: explicit ``court_type`` field, which overrides this set (see :func:`court_type_for`).
52 MULTICOURT_CLIPS = frozenset(
53 {"mahadevpura_2", "GX010128", "Badminton_BXH_2", "Boxhill_Doubles"}
54)
55
56
57 def court_type_for(video: Dict[str, Any]) -> str:
58 """``multicourt`` or ``single`` for a golden video. Honors an explicit manifest
59 ``court_type`` field when present, else falls back to membership in
``MULTICOURT_CLIPS``
60 (keyed by ``name``), else ``single``."""
61 ct = video.get("court_type")
62 if isinstance(ct, str) and ct:
63 return ct
64 return "multicourt" if video.get("name") in MULTICOURT_CLIPS else "single"
65
66
67 def filter_by_min_duration(ivs: List[Interval], min_duration: float) -> List[Interval]:
68 """The long-rally lens: keep only intervals strictly longer than ``min_duration``
seconds.
69
70 ``min_duration <= 0`` is an EXACT passthrough (the default-OFF semantics ? every
real rally has
71 positive duration, so nothing is dropped). It's a FILTER, not a new metric: applied
identically
72 to predictions AND ground truth before any matching, so the tIoU and R2 paths stay
consistent."""
73 if min_duration <= 0.0:
74 return list(ivs)
75 return [iv for iv in ivs if (iv[1] - iv[0]) > min_duration]
76
77

```



```

78 # ----- #
79 # Pure scoring core (no I/O ? unit-testable)
80 # ----- #
81 def score_intervals(
82 preds: List[Interval],
83 gts: List[Interval],
84 iou: float = DEFAULT_IΟΥ,
85 tau_start: Optional[float] = None,
86 tau_end: Optional[float] = None,
87 min_duration: float = 0.0,
88) -> Dict[str, Any]:
89 """The full per-video metric row for predicted vs ground-truth rally intervals.
90
91 Pairs temporal-IoU P/R/F1 + boundary error (`metrics.score`) with the
92 over-segmentation-sensitive set (F1@k, segment-count ratio, merge/split breakdown)
93 so a
94 merged mega-window is *visible* even when IoU-F1 isn't moved.
95
96 Also carries the **R2** task-faithful block (`tolerance_metrics`, the headline
97 going forward;
98 tIoU here is the SECONDARY trend-line per the augment-then-ablation-gate decision):
99 strict-1:1
100 tolerance-match ``tol_f1/tol_precision/tol_recall`` (over-production costs
101 precision) + the
102 start/end boundary-error medians + the over-seg guard counts. See
103 docs/R2_EVAL_METRIC_DESIGN.md.
104
105 ``min_duration`` > 0 applies the **long-rally lens**
106 (:func:`filter_by_min_duration`): BOTH preds
107 and GTs are filtered to rallies longer than that many seconds BEFORE any matching,
108 so the whole
109 row (tIoU set AND R2 block) scores only long rallies. Default 0.0 = OFF (exact
110 passthrough).
111 """
112 from backend.eval import tolerance_metrics as tm
113
114 preds = filter_by_min_duration(preds, min_duration)
115 gts = filter_by_min_duration(gts, min_duration)
116 ts = tm.TAU_START if tau_start is None else tau_start
117 te = tm.TAU_END if tau_end is None else tau_end
118 sc = metrics.score(preds, gts, iou_threshold=iou)
119 f1k = segmentation_metrics.f1_at_overlaps(preds, gts)
120 ms = segmentation_metrics.merge_split_report(preds, gts)
121 tol_matches, tol_p, tol_r, tol_f1 = tm.match_1to1(preds, gts, ts, te)
122 be = tm.boundary_error_report(
123 preds, gts
124) # unconditional best-overlap (the R4-aiming diagnostic)
125 osr = tm.over_seg_report(preds, gts)
126 return {
127 "f1": sc.f1,
128 "precision": sc.precision,
129 "recall": sc.recall,
130 "mean_iou": sc.mean_iou,
131 "mean_start_error": sc.mean_start_error,
132 "mean_end_error": sc.mean_end_error,
133 "f1@0.1": f1k[0.1],
134 "f1@0.25": f1k[0.25],
135 "f1@0.5": f1k[0.5],
136 "segment_count_ratio": segmentation_metrics.segment_count_ratio(preds, gts),
137 "merge_rate": ms["merge_rate"],
138 "split_rate": ms["split_rate"],
139 "merges": ms["merges"],
140 "splits": ms["splits"],
141 "missed": ms["missed"],
142 "spurious": ms["spurious"],

```

```

135 "num_pred": len(preds),
136 "num_gt": len(gts),
137 # --- R2 (tolerance-match) ? the task-faithful headline block ---
138 "tau_start": ts,
139 "tau_end": te,
140 "tol_f1": tol_f1,
141 "tol_precision": tol_p,
142 "tol_recall": tol_r,
143 "tol_matched": float(len(tol_matches)),
144 "dstart_abs_median": be["dstart"]["abs_median"],
145 "dend_abs_median": be["dend"]["abs_median"],
146 "split_count": osr["split_count"],
147 "merge_count": osr["merge_count"],
148 }
149
150
151 _AGG_KEYS = (
152 "f1",
153 "f1@0.25",
154 "f1@0.5",
155 "merge_rate",
156 "split_rate",
157 "segment_count_ratio",
158 "precision",
159 "recall",
160 "tol_f1",
161 "tol_precision",
162 "tol_recall",
163 "dstart_abs_median",
164 "dend_abs_median",
165)
166
167
168 def aggregate(rows: List[Dict[str, Any]]) -> Dict[str, Any]:
169 """Aggregate per-video rows: mean (+ bootstrap 95% CI) of the headline metrics
170 across videos.
171
172 Per-video means (not pooled) keep videos the unit of evidence, matching the
173 harness's
174 leave-one-video-out discipline (windows within a video are correlated)."""
175 out: Dict[str, Any] = {"n": len(rows)}
176 for k in _AGG_KEYS:
177 vals = [r[k] for r in rows]
178 out[k] = eval_stats.mean_with_ci(vals) if vals else None
179 return out
180
181 # ----- #
182 # Golden corpus I/O
183 # ----- #
184
185 def load_golden(
186 manifest_path: str = DEFAULT_MANIFEST, features_dir: str = DEFAULT_FEATURES_DIR
187) -> List[Dict[str, Any]]:
188 """Load each golden video's trajectory path + fps + frame_width (manifest) and GT
189 rally
190 intervals (`<name>_golden_features.json` `gts`). Videos with no GTs are kept but
191 flagged."""
192 with open(manifest_path) as fh:
193 manifest = json.load(fh)
194 out: List[Dict[str, Any]] = []
195 for v in manifest:
196 feat = os.path.join(features_dir, f"{v['name']}_golden_features.json")
197 gts: List[Interval] = []
198 if os.path.exists(feat):
199 with open(feat) as fh:

```

```

196 gts = [
197 (float(s), float(e)) for s, e in (json.load(fh).get("gts") or [])
198]
199 entry: Dict[str, Any] = {
200 "name": v["name"],
201 "trajectory": v["trajectory"],
202 "fps": float(v["fps"]),
203 "frame_width": float(v["frame_width"]),
204 "gts": gts,
205 }
206 # Optional prominent-court gate inputs (multicourt G2; default absent ? the gate
is a no-op):
207 # a normalized court_polygon + optional frame_height + court_height_pad from the
manifest.
208 for k in ("court_polygon", "frame_height", "court_height_pad"):
209 if k in v:
210 entry[k] = v[k]
211 out.append(entry)
212 return out
213
214
215 def windows_for(
216 video: Dict[str, Any], preset: WindowingPreset, min_crossings: int = 0
217) -> List[Interval]:
218 """Parse a video's trajectory, window it under ``preset`` ? predicted rally
intervals.
219
220 ``min_crossings`` > 0 applies the **net-crossings rally-state gate** (Gate-A,
221 ``rally_gate.gate_windows``): drop windows whose shuttle crosses the net fewer than
this many
222 times (a non-rally fragment). This is the cheapest rally-state cue (W2) ? CPU-only,
derived
223 from the trajectory + an auto-estimated net axis; it cleans up the over-split that
bounded
224 windowing (W1) leaves behind.
225
226 NEVER-EMPTY safeguard (mirrors ``FusionSegmenter._select_for_handoff``): if the gate
would
227 drop EVERY window of a video that had ?1, fall back to the ungated windows. The gate
is
228 strictly *pruning* ? it never zeroes out a non-empty video ? so W2-on-W1 is "do no
harm".
229 """
230 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
231
232 points = TrackNetRunner.parse_trajectory_csv(video["trajectory"])
233 wins = windowing.build_windows(points, video["fps"], video["frame_width"], preset)
234 ivs = windowing.windows_to_intervals(wins)
235 # Prominent-court WINDOW gate (multicourt G2, default-OFF): when the video carries a
normalized
236 # court_polygon, drop candidate rallies whose shuttle is MAJORITY outside the
prominent court ? a
237 # background/adjacent-court rally. Window-level (not point-level): a background
rally's central
238 # tail keeps a few in-court points, enough to hold a window, so point-clearing alone
is too leaky
239 # (measured on GX010142 rally #1). No polygon ? no-op. See
docs/PROMINENT_COURT_DETECTION.md.
240 poly = video.get("court_polygon")
241 if poly and ivs:
242 from backend.pipeline.detectors.court_gate import gate_intervals_to_court
243
244 fw = float(video["frame_width"])
245 fh = float(video.get("frame_height") or fw * 9.0 / 16.0)
246 ivs = gate_intervals_to_court(

```

```

247 ivs,
248 points,
249 poly,
250 float(video["fps"]),
251 fw,
252 fh,
253 min_in_court_frac=float(video.get("court_min_in_court_frac", 0.5)),
254 height_pad=float(video.get("court_height_pad", 0.0) or 0.0),
255)
256 if min_crossings > 0 and ivs:
257 from backend.pipeline.detectors.rally_gate import gate_windows
258
259 gated = [
260 (g.start, g.end)
261 for g in gate_windows(
262 points, ivs, video["fps"], min_crossings=min_crossings
263)
264 if g.kept
265]
266 # Gating emptied a non-empty video ? keep the ungated windows.
267 ivs = gated if gated else ivs
268 return ivs
269
270
271 #: One windowed video before scoring: (name, court_type, predicted intervals, GT
intervals).
272 PredictedVideo = Tuple[str, str, List[Interval], List[Interval]]
273
274
275 def predict_all(
276 golden: List[Dict[str, Any]], preset: WindowingPreset, min_crossings: int = 0
277) -> List["PredictedVideo"]:
278 """Window every scorable golden video ONCE (the expensive trajectory-parse +
windowing step),
279 returning ``(name, court_type, preds, gts)`` per video. Lets callers re-score the
same
280 predictions under several lenses (e.g. the all-vs-long stratified report) without
re-windowing."""
281 out: List[PredictedVideo] = []
282 for v in golden:
283 if not v["gts"] or not os.path.exists(v["trajectory"]):
284 continue
285 out.append(
286 (
287 v["name"],
288 court_type_for(v),
289 windows_for(v, preset, min_crossings),
290 v["gts"],
291)
292)
293 return out
294
295
296 def _score_predicted(
297 predicted: List["PredictedVideo"],
298 iou: float,
299 tau_start: Optional[float],
300 tau_end: Optional[float],
301 min_duration: float,
302) -> List[Dict[str, Any]]:
303 """Score pre-windowed videos into per-video rows (carrying ``name`` +
``court_type``)."""
304 rows: List[Dict[str, Any]] = []
305 for name, court, preds, gts in predicted:
306 row = score_intervals(preds, gts, iou, tau_start, tau_end, min_duration)

```

```

307 row["name"] = name
308 row["court_type"] = court
309 rows.append(row)
310 return rows
311
312
313 def evaluate(
314 golden: List[Dict[str, Any]],
315 preset: WindowingPreset,
316 iou: float = DEFAULT_IUO,
317 min_crossings: int = 0,
318 tau_start: Optional[float] = None,
319 tau_end: Optional[float] = None,
320 min_duration: float = 0.0,
321) -> Dict[str, Any]:
322 """Score every golden video under ``preset`` (+ optional net-crossings gate); return
per-video
323 rows + the aggregate. Each row carries both the tIoU set and the R2 tolerance-match
block, plus
324 its ``court_type``. ``min_duration`` > 0 applies the long-rally lens (preds + GTs
filtered to
325 rallies longer than that many seconds) consistently across both metric paths."""
326 rows = _score_predicted(
327 predict_all(golden, preset, min_crossings),
328 iou,
329 tau_start,
330 tau_end,
331 min_duration,
332)
333 return {
334 "preset": preset.to_dict(),
335 "iou": iou,
336 "min_crossings": min_crossings,
337 "min_duration": min_duration,
338 "rows": rows,
339 "aggregate": aggregate(rows),
340 }
341
342
343 def compare(
344 golden: List[Dict[str, Any]],
345 base: WindowingPreset,
346 cand: WindowingPreset,
347 iou: float = DEFAULT_IUO,
348 base_min_crossings: int = 0,
349 cand_min_crossings: int = 0,
350 tau_start: Optional[float] = None,
351 tau_end: Optional[float] = None,
352 guard_weight_merge: Optional[float] = None,
353 guard_weight_split: Optional[float] = None,
354 guard_use_rates: bool = True,
355 min_duration: float = 0.0,
356) -> Dict[str, Any]:
357 """Baseline vs candidate (windowing and/or net-crossings gate): per-video paired ?
(+ CI +
358 sign-test) on both tIoU-F1 and the R2 ``tol_f1`` ? the per-tier 'number'. Also
reports ? on the
359 over-segmentation metrics AND the **R1 net over-seg promotion guard**
(``over_seg_guard``):
360 the honest promote/block verdict for cand-over-base, with the strict per-video rule
reported
361 alongside so the refinement is auditable. ``min_duration`` > 0 scores both sides
through the
362 long-rally lens (same filter applied to base and cand ? it's a scoring lens, not a
windowing

```

```

363 knob, so it is identical on both sides of the A/B)."""
364 from backend.eval import tolerance_metrics as tm
365
366 b = evaluate(
367 golden, base, iou, base_min_crossings, tau_start, tau_end, min_duration
368)
369 c = evaluate(
370 golden, cand, iou, cand_min_crossings, tau_start, tau_end, min_duration
371)
372 by_name_b = {r["name"]: r for r in b["rows"]}
373 paired = [(by_name_b[r["name"]], r) for r in c["rows"] if r["name"] in by_name_b]
374 delta: Dict[str, Any] = {}
375 for k in (
376 "f1",
377 "f1@0.25",
378 "tol_f1",
379 "merge_rate",
380 "split_rate",
381 "segment_count_ratio",
382):
383 delta[k] = eval_stats.paired_delta_ci(
384 [rb[k] for rb, _ in paired], [rc[k] for _, rc in paired]
385)
386 gkw: Dict[str, Any] = {"use_rates": guard_use_rates}
387 if guard_weight_merge is not None:
388 gkw["weight_merge"] = guard_weight_merge
389 if guard_weight_split is not None:
390 gkw["weight_split"] = guard_weight_split
391 guard = tm.over_seg_guard([rb for rb, _ in paired], [rc for _, rc in paired], **gkw)
392 return {
393 "base": b,
394 "cand": c,
395 "n_paired": len(paired),
396 "delta": delta,
397 "over_seg_guard": guard,
398 }
399
400
401 def stratified_report(
402 golden: List[Dict[str, Any]],
403 preset: WindowingPreset,
404 iou: float = DEFAULT_IOU,
405 min_crossings: int = 0,
406 tau_start: Optional[float] = None,
407 tau_end: Optional[float] = None,
408 long_tau: float = LONG_RALLY_TAU,
409) -> Dict[str, Any]:
410 """The long-rally lens payoff: ``{all vs long(>long_tau)} × {single vs multicourt}``
411 aggregates.
412
413 The thesis: local OVER-FIRES, and its false positives are mostly SHORT (motion
414 blips,
415 background-court fragments on multicourt footage). Filtering to long rallies should
416 strip those
417 FPs ? precision (``tol_precision``) should jump, most on multicourt. This table is
418 how we read
419 "do we beat Gemini on the single-match long-rally happy path?".
420
421 Windowing is run ONCE per video (``predict_all``); each cell re-scores the same
422 predictions
423 under its duration lens. A cell aggregates only videos with >=1 golden rally in that
424 stratum
425 (``num_gt`` > 0): a clip with no long rally is not evidence about long-rally quality
426 and its
427 0/0 recall would distort the mean. ``n`` per cell reports the videos that survived

```

```

that gate."""
421 predicted = predict_all(golden, preset, min_crossings)
422 long_label = f"long>{long_tau:g}s"
423 strata = (("all", 0.0), (long_label, long_tau))
424 courts = ("single", "multicourt", "all")
425 cells: Dict[Tuple[str, str], Dict[str, Any]] = {}
426 for dur_label, min_dur in strata:
427 rows = _score_predicted(predicted, iou, tau_start, tau_end, min_dur)
428 for court in courts:
429 sub = [
430 r
431 for r in rows
432 if (court == "all" or r["court_type"] == court) and r["num_gt"] > 0
433]
434 cells[(dur_label, court)] = aggregate(sub)
435 return {
436 "long_tau": long_tau,
437 "labels": [s[0] for s in strata],
438 "courts": list(courts),
439 "cells": cells,
440 "preset": preset.to_dict(),
441 "iou": iou,
442 "min_crossings": min_crossings,
443 }
444
445
446 # ----- #
447 # Reporting
448 # ----- #
449 def _ci(m: Optional[Dict[str, Any]]) -> str:
450 return "?" if not m else f"{m['mean']:.3f} [{m['ci_low']:.3f},{m['ci_high']:.3f}]"
451
452
453 def _format_guard(g: Dict[str, Any]) -> str:
454 """Render the R1 net over-seg promotion-guard verdict (with the strict rule for
455 contrast)."""
456 basis = "rate" if g["use_rates"] else "count"
457 lines = [
458 f"\n## R1 over-seg promotion guard ({basis}-based, "
459 f"w_merge={g['weight_merge']}, w_split={g['weight_split']})",
460 f" NET verdict: {'PASS' if g['passes'] else 'BLOCK'} "
461 f"(net cost {g['base_net']:.3f} ? {g['cand_net']:.3f}, ? {g['net_delta']:+.3f};",
462 f"no_merge_rate_regress={g['no_merge_rate_regress']})",
463 f" strict per-video rule (R2 §2.3.3, for contrast): "
464 f"{'pass' if g['strict_passes'] else 'BLOCK'} "
465 f"(count increases ? merges:{len(g['new_merges'])})",
466 f"splits:{len(g['new_splits'])})",
467]
468 if g["merge_rate_regress"]:
469 names = ", ".join(
470 f"{r['name']}({r['base']:.2f}?{r['cand']:.2f})"
471 for r in g["merge_rate_regress"]
472)
473 lines.append(f" ? merge_rate REGRESSED on: {names}")
474 return "\n".join(lines)
475
476 def format_report(result: Dict[str, Any]) -> str:
477 p = result["preset"]
478 bounded = []
479 if p.get("max_window_duration"):
480 bounded.append(
481 f"cap={p['max_window_duration']} + ((hard)" if p.get("hard_cap") else ""

```

```

482 if p.get("inactivity_end"):
483 bounded.append(
484 f"inact_end={p['inactivity_end']}/rt={p['inactivity_rally_thresh']}"
485 f"/md={p['inactivity_min_density']}"
486)
487 if p.get("blob_fallback"):
488 bounded.append("blob_fallback")
489 extra = (" " + " ".join(bounded)) if bounded else ""
490 md = result.get("min_duration", 0.0) or 0.0
491 md_tag = f", LONG-RALLY LENS min_dur>{md:g}s" if md > 0 else ""
492 lines = [
493 f"# Rally-segmentation eval ? windowing '{p['name']}' "
494 f"(vt={p['velocity_thresh']}, merge_gap={p['merge_gap']}, "
495 f"min_win={p['min_window_duration']}{extra}), IoU={result['iou']}{md_tag}",
496 "",
497]
498 lines.append(
499 "| video | F1 | F1@0.25 | F1@0.5 | merges | merge_rate | seg_ratio | pred/gt |"
500)
501 lines.append("---|---|---|---|---|---|---|---|")
502 for r in result["rows"]:
503 lines.append(
504 f"| {r['name']} | {r['f1']:.3f} | {r['f1@0.25']:.3f} | {r['f1@0.5']:.3f} | "
505 f"{int(r['merges'])} | {r['merge_rate']:.2f} | "
506 f"{r['segment_count_ratio']:.2f} | "
507 f"{r['num_pred']}/{r['num_gt']} |"
508)
509 a = result["aggregate"]
510 lines += [
511 "",
512 f"***Aggregate (n={a['n']}, mean [95% CI]):** "
513 f"F1 {_ci(a['f1'])} · F1@0.25 {_ci(a['f1@0.25'])} · F1@0.5 {_ci(a['f1@0.5'])} · "
514 f"merge_rate {_ci(a['merge_rate'])} · seg_ratio "
515 f"{_ci(a['segment_count_ratio'])}",
516]
517 # --- R2 (tolerance-match) block ? the task-faithful headline (tIoU above is
518 secondary) ---
519 rows = result["rows"]
520 if rows and "tol_f1" in rows[0]:
521 ts, te = rows[0]["tau_start"], rows[0]["tau_end"]
522 lines += [
523 "",
524 f"## R2 ? tolerance-match (?_start={ts}s, ?_end={te}s, strict 1:1)",
525 "",
526 "| video | tolF1 | tolP | tolR | |?start| | |?end| | split | merge |",
527 "|---|---|---|---|---|---|---|---|",
528]
529 for r in rows:
530 lines.append(
531 f"| {r['name']} | {r['tol_f1']:.3f} | {r['tol_precision']:.2f} | "
532 f"{r['tol_recall']:.2f} | {r['dstart_abs_median']:.2f} | "
533 f"{r['dend_abs_median']:.2f} | {int(r['split_count'])} | "
534 f"{int(r['merge_count'])} |"
535)
536 lines += [
537 "",
538 f"***R2 aggregate (n={a['n']}):** tolF1 {_ci(a['tol_f1'])} · "
539 f"tolP {_ci(a['tol_precision'])} · tolR {_ci(a['tol_recall'])} · "
540 f"|?start| {_ci(a['dstart_abs_median'])} · |?end| "
541 f"{_ci(a['dend_abs_median'])} "
542 f"(start?solved / end=the gap)",
543]
544]
545 return "\n".join(lines)

```



```

541
542 def format_stratified(strat: Dict[str, Any]) -> str:
543 """Render the long-rally stratified table: {all vs long} × {single, multicourt, all-
544 courts}.
545 ``tolP`` is the column that carries the story ? if filtering to long rallies strips
546 local's
547 short false positives, precision rises (most on multicourt). ``n`` is the videos
548 with >=1 golden
549 rally in that stratum."""
550 long_tau = strat["long_tau"]
551 cells = strat["cells"]
552 lines = [
553 f"\n## Long-rally lens ? stratified (long = duration > {long_tau:g}s; "
554 "n = videos with >=1 golden rally in the stratum)",
555 "",
556 "| stratum | court | n | tolF1 | tolP | tolR | tIoU F1@0.5 | seg_ratio |",
557 "|---|---|---|---|---|---|---|---|",
558]
559 for dur_label in strat["labels"]:
560 for court in strat["courts"]:
561 a = cells[(dur_label, court)]
562 lines.append(
563 f"| {dur_label} | {court} | {a['n']} | {_ci(a['tol_f1'])} | "
564 f"{_ci(a['tol_precision'])} | {_ci(a['tol_recall'])} | "
565 f"{_ci(a['f1@0.5'])} | {_ci(a['segment_count_ratio'])} | "
566)
567 return "\n".join(lines)
568
569 def _resolve_preset(name: str) -> WindowingPreset:
570 if name in PRESETS:
571 return PRESETS[name]
572 raise SystemExit(f"unknown preset '{name}'; choices: {sorted(PRESETS)}")
573
574 def _build_preset(args: argparse.Namespace, side: str) -> WindowingPreset:
575 """Resolve the ``base`` (``--preset``) or ``cand`` (``--vs``) windowing from
576 ``args`` by
577 applying the ``dataclasses.replace`` overrides for cap / hard-cap / blob-fallback /
578 inactivity. ``cand`` falls back to the base override when its ``--vs-`` variant is
579 unset
580 (so a baseline-vs-candidate promotion check runs in one shot)."""
581 from dataclasses import replace
582
583 if side == "base":
584 preset = _resolve_preset(args.preset)
585 cap = args.max_window_duration
586 hard_cap = args.hard_cap
587 blob_fallback = args.blob_fallback
588 inactivity_end = args.inactivity_end
589 temporal_density = args.inactivity_temporal_density
590 else:
591 preset = _resolve_preset(args.vs)
592 cap = (
593 args.vs_max_window_duration
594 if args.vs_max_window_duration is not None
595 else args.max_window_duration
596)
597 hard_cap = args.vs_hard_cap or args.hard_cap
598 blob_fallback = args.vs_blob_fallback or args.blob_fallback
599 inactivity_end = (
600 args.vs_inactivity_end
601 if args.vs_inactivity_end is not None
602 else args.inactivity_end

```

```

601)
602 temporal_density = (
603 args.vs_inactivity_temporal_density or args.inactivity_temporal_density
604)
605 if cap is not None:
606 preset = replace(preset, max_window_duration=cap)
607 if hard_cap:
608 preset = replace(preset, hard_cap=True)
609 if blob_fallback:
610 preset = replace(preset, blob_fallback=True)
611 if temporal_density:
612 preset = replace(preset, inactivity_temporal_density=True)
613 if args.blob_factor is not None:
614 preset = replace(preset, blob_factor=args.blob_factor)
615 if inactivity_end is not None:
616 preset = replace(preset, inactivity_end=inactivity_end)
617 if args.inactivity_rally_thresh is not None:
618 preset = replace(preset, inactivity_rally_thresh=args.inactivity_rally_thresh)
619 if args.inactivity_min_density is not None:
620 preset = replace(preset, inactivity_min_density=args.inactivity_min_density)
621 return preset
622
623
624 def main() -> None:
625 ap = argparse.ArgumentParser(
626 description="Rally-segmentation eval vs golden labels (offline)."
627)
628 ap.add_argument("--manifest", default=DEFAULT_MANIFEST)
629 ap.add_argument("--features-dir", default=DEFAULT_FEATURES_DIR)
630 ap.add_argument(
631 "--preset", default=SERVED.name, help="windowing preset (default: served)"
632)
633 ap.add_argument("--vs", default=None, help="second preset to diff against --preset")
634 ap.add_argument("--iou", type=float, default=DEFAULT_IOU)
635 ap.add_argument(
636 "--min-crossings",
637 type=int,
638 default=0,
639 help="net-crossings rally-state gate on --preset (0=off; W2 Gate-A)",
640)
641 ap.add_argument(
642 "--vs-min-crossings",
643 type=int,
644 default=0,
645 help="net-crossings gate on --vs (default: same as --min-crossings)",
646)
647 ap.add_argument(
648 "--max-window-duration",
649 type=float,
650 default=None,
651 help="override bounded-windowing cap (s) on BOTH presets (W1; e.g. 12)",
652)
653 ap.add_argument(
654 "--vs-max-window-duration",
655 type=float,
656 default=None,
657 help="cap (s) on --vs ONLY (default: same as --max-window-duration); lets a "
658 "BASELINE (no cap) vs CANDIDATE (capped) promotion check run in one shot",
659)
660 ap.add_argument(
661 "--hard-cap",
662 action="store_true",
663 help="enforce the cap as a HARD cap on --preset (chop continuously-active "
664 "windows at the cap when no inactivity gap exists; W1 hard-cap fallback)",
665)

```

```

666 ap.add_argument(
667 "--vs-hard-cap",
668 action="store_true",
669 help="hard-cap fallback on --vs (default: same as --hard-cap)",
670)
671 ap.add_argument(
672 "--blob-fallback",
673 action="store_true",
674 help="R5 blob-fallback on --preset (after R4, force-chop any group still over
the "
675 "cap at its midpoint + flag/log it; the continuously-active blob last resort)",
676)
677 ap.add_argument(
678 "--vs-blob-fallback",
679 action="store_true",
680 help="R5 blob-fallback on --vs (default: same as --blob-fallback)",
681)
682 ap.add_argument(
683 "--blob-factor",
684 type=float,
685 default=None,
686 help="R5 magnitude gate (both presets): chop only groups > this × cap (default
4.0)",
687)
688 ap.add_argument(
689 "--inactivity-end",
690 type=float,
691 default=None,
692 help="R4 rally-end inactivity trim (s) on --preset (0=off; trims post-rally
drift tail)",
693)
694 ap.add_argument(
695 "--vs-inactivity-end",
696 type=float,
697 default=None,
698 help="R4 inactivity trim on --vs (default: same as --inactivity-end)",
699)
700 ap.add_argument(
701 "--inactivity-rally-thresh",
702 type=float,
703 default=None,
704 help="R4 velocity above which motion counts as 'rally' (both presets; default
0.08)",
705)
706 ap.add_argument(
707 "--inactivity-min-density",
708 type=float,
709 default=None,
710 help="R4 min trailing fast-fraction to stay 'in rally' (both presets; default
0.34)",
711)
712 ap.add_argument(
713 "--inactivity-temporal-density",
714 action="store_true",
715 help="R4 density over the trailing window's TEMPORAL length, not the active-
sample "
716 "count, on --preset (finding-B fix; no-op under dense tracking, trims sparse-
track tails)",
717)
718 ap.add_argument(
719 "--vs-inactivity-temporal-density",
720 action="store_true",
721 help="R4 temporal-density on --vs (default: same as --inactivity-temporal-
density); use "
722 "'--preset served --vs served --vs-inactivity-temporal-density' for the P2

```

```

promotion A/B",
723)
724 ap.add_argument(
725 "--tau-start",
726 type=float,
727 default=None,
728 help="R2 tolerance-match start window (s); default 2.0 (forgives the service
window)",
729)
730 ap.add_argument(
731 "--tau-end",
732 type=float,
733 default=None,
734 help="R2 tolerance-match end window (s); default 1.5 (keeps the rally-end
honest)",
735)
736 ap.add_argument(
737 "--guard-weight-merge",
738 type=float,
739 default=None,
740 help="R1 over-seg guard: merge cost weight (default 2.0; merge hides a rally)",
741)
742 ap.add_argument(
743 "--guard-weight-split",
744 type=float,
745 default=None,
746 help="R1 over-seg guard: split cost weight (default 1.0; split only fragments)",
747)
748 ap.add_argument(
749 "--guard-counts",
750 action="store_true",
751 help="R1 over-seg guard scores raw counts instead of normalized rates "
752 "(rates are the default ? comparable across a structural mega?bounded change)",
753)
754 ap.add_argument(
755 "--min-duration",
756 type=float,
757 default=0.0,
758 help="LONG-RALLY LENS (0=OFF): score ONLY rallies longer than this many seconds,
"
759 "filtering BOTH predictions and golden GTs before matching. Strips local's "
760 "short false-positive blips to reveal long-rally quality. Applies across the "
761 "tIoU + R2 paths and to --vs (same filter on both sides).",
762)
763 ap.add_argument(
764 "--stratify",
765 action="store_true",
766 help="also print the long-rally stratified report: {all vs long(>--long-tau)} ×
"
767 "{single vs multicourt} aggregates (court type from GOLDEN_VIDEOS.md / the "
768 "manifest court_type field). The actual long-rally payoff table.",
769)
770 ap.add_argument(
771 "--long-tau",
772 type=float,
773 default=LONG_RALLY_TAU,
774 help=f"the 'long' rally duration threshold (s) used by --stratify "
775 f"(default {LONG_RALLY_TAU:g}; owner's North-star happy-path is >5s).",
776)
777 ap.add_argument(
778 "--json-out", default=None, help="optional path to write the full result JSON"
779)
780 args = ap.parse_args()
781
782 golden = load_golden(args.manifest, args.features_dir)

```

```

783 base = _build_preset(args, "base")
784 if args.vs:
785 vs_mc = args.vs_min_crossings or args.min_crossings
786 cand = _build_preset(args, "cand")
787 result = compare(
788 golden,
789 base,
790 cand,
791 args.iou,
792 base_min_crossings=args.min_crossings,
793 cand_min_crossings=vs_mc,
794 tau_start=args.tau_start,
795 tau_end=args.tau_end,
796 guard_weight_merge=args.guard_weight_merge,
797 guard_weight_split=args.guard_weight_split,
798 guard_use_rates=not args.guard_counts,
799 min_duration=args.min_duration,
800)
801 print(format_report(result["base"]))
802 print("\n" + format_report(result["cand"]))
803 d = result["delta"]
804 print(
805 f"\n## ? ({args.vs} mc={vs_mc} ? {args.preset} mc={args.min_crossings}), "
806 f"n_paired={result['n_paired']}"
807)
808 for k, dd in d.items():
809 st = dd["sign_test"]
810 print(
811 f" ?{k}: {dd['mean_delta']:+.3f} "
812 f"CI?0={dd['ci_excludes_zero']} sign(W/L)={st['wins']}/{st['losses']} "
813 f"p={st['p_value']:.3f}"
814)
815 print(_format_guard(result["over_seg_guard"]))
816 else:
817 result = evaluate(
818 golden,
819 base,
820 args.iou,
821 args.min_crossings,
822 tau_start=args.tau_start,
823 tau_end=args.tau_end,
824 min_duration=args.min_duration,
825)
826 print(format_report(result))
827 if args.stratify:
828 # The long-rally payoff table always reports the BASE preset (the served/current
829 # split by duration × court type. Independent of --min-duration (it does its own
830 # split).
831 print(
832 format_stratified(
833 stratified_report(
834 golden,
835 base,
836 args.iou,
837 args.min_crossings,
838 tau_start=args.tau_start,
839 tau_end=args.tau_end,
840 long_tau=args.long_tau,
841)
842)
843)
844 if args.json_out:
845 with open(args.json_out, "w", encoding="utf-8") as fh:

```

```

844 json.dump(result, fh, indent=2)
845 print(f"\n[written {args.json_out}]\n")
846
847
848 if __name__ == "__main__":
849 main()
850

```

### 38. assistant (2026-06-29T10:56:58.392Z)

Now the data flow is crystal-clear, and it changes the whole plan for the better:

**\*\*`rally\_seg\_eval` reads only two things\*\*** (`load\_golden`, lines 183-212):

1. `--manifest` ? each video's **`trajectory`** path + `fps` + `frame_width``
2. `--features-dir` ? <name>_golden_features.json`, but **only the `gts` key** (ground-truth rally intervals) ? the person/shuttle features are never touched by this harness.`

The trajectory CSV is parsed directly (line 232). **\*\*So the A/B needs neither the GPU nor the video files\*\*** ? it needs the cached trajectory CSVs (all 15 in GCS) and the golden GTs (derivable from the 15 label CSVs in GCS via `load_golden_gts``). The expensive GPU WASB extraction was already done and cached as those `trajectories/*.csv``. `fusion_golden`` (GPU+video) is only needed for the *person-fusion* track, which is irrelevant to finding B.

This means I can run the real, faithful P2b A/B entirely offline. Let me confirm the two parsing helpers and the actual CSV formats before building:

### 39. user (2026-06-29T10:56:59.811Z)

```

1 """Per-video calibration driver: WASB trajectory + golden rallies -> tuned thresholds +
"+X%".
2
3 Wires the already-tested pieces together:
4 WASB trajectory CSV (Frame,Visibility,X,Y)
5 -> TrackNetRunner.trajectory_to_action_windows(cfg) [predict_fn]
6 -> backend.eval.calibration (improvement_report / cross_validate)
7
8 Run (after a full-video WASB pass exists):
9 python -m backend.eval.calibrate_wasb --trajectory full_traj.csv \
10 --labels "golden_labels_badminton.csv" --fps 59.94 --frame-width 1920
11
12 Honesty note: the grid is *selected* on the full GT, so `improvement_report` on that
13 same GT is an **optimistic** upper bound. The trustworthy number is the
14 cross-validated held-out **recall** (and, once a 2nd labelled video exists,
15 `leave_one_video_out` for precision/F1 generalization).
16 """
17
18 from __future__ import annotations
19
20 import argparse
21 from typing import Callable, List, Tuple
22
23 from backend.eval.calibration import (
24 cross_validate,
25 evaluate,
26 improvement_report,
27 select_best,
28)
29 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
30
31 Interval = Tuple[float, float]
32 Config = Tuple[float, float, float] # (velocity_thresh, merge_gap, min_window_duration)
33
34 BASELINE: Config = (
35 0.02,
36 2.0,

```

```

37 2.0,
38) # the WasbConfig / trajectory_to_action_windows defaults
39
40
41 def load_golden_gts(csv_path: str) -> List[Interval]:
42 """Rally [start,end] seconds from a golden CSV ? delegates to the canonical loader.
43
44 Accepts BOTH golden conventions now (start_time/end_time AND the collector export's
45 start,end ? the historical fork is closed; see gt_loader.py / ADR-008). Keeps this
46 driver's legacy lenient semantics (skip bad rows), but skipped rows are now logged.
47 """
48 from backend.eval.gt_loader import load_gt_intervals
49
50 return load_gt_intervals(csv_path, strict=False)
51
52
53 def make_predict_fn(
54 trajectory_csv: str, fps: float, frame_width: float
55) -> Callable[[Config], List[Interval]]:
56 """predict_fn(cfg) -> rally windows, from a cached WASB trajectory (parsed once)."""
57 points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
58
59 def predict_fn(cfg: Config) -> List[Interval]:
60 vt, mg, mwd = cfg
61 wins = TrackNetRunner.trajectory_to_action_windows(
62 points,
63 fps=fps,
64 frame_width=frame_width,
65 velocity_thresh=vt,
66 merge_gap=mg,
67 min_window_duration=mwd,
68)
69 return [(w["start"], w["end"]) for w in wins]
70
71 return predict_fn
72
73
74 def default_grid() -> List[Config]:
75 return [
76 (vt, mg, mwd)
77 for vt in (0.005, 0.01, 0.02, 0.03, 0.05)
78 for mg in (1.0, 2.0, 3.0)
79 for mwd in (1.0, 2.0, 3.0)
80]
81
82
83 def run(
84 trajectory_csv: str,
85 labels_csv: str,
86 fps: float,
87 frame_width: float,
88 iou: float = 0.5,
89) -> dict:
90 gts = load_golden_gts(labels_csv)
91 if not gts:
92 raise SystemExit(f"no usable golden rallies in {labels_csv}")
93 predict_fn = make_predict_fn(trajectory_csv, fps, frame_width)
94 grid = default_grid()
95
96 base = evaluate(predict_fn(BASELINE), gts, iou)
97 best_cfg, best_f1 = select_best(
98 predict_fn, grid, gts, metric="f1", iou_threshold=iou
99)
100 fit = improvement_report(
101 predict_fn, BASELINE, best_cfg, gts, iou

```

```

102) # OPTIMISTIC (fit on full GT)
103 cv = cross_validate(predict_fn, grid, gts, k=5) # HONEST held-out recall
104
105 print("=== WASB rally-segmentation calibration ===")
106 print(f"golden rallies: {len(gts)} | IoU thresh: {iou}")
107 print(
108 f"baseline cfg {BASELINE}: precision={base.precision:.3f}
recall={base.recall:.3f} "
109 f"f1={base.f1:.3f} meanIoU={base.mean_iou:.3f}"
110)
111 print(f"tuned cfg {best_cfg}: f1={best_f1:.3f}")
112 print(
113 f" precision {fit['baseline']['precision']:.3f} ->
{fit['tuned']['precision']:.3f} "
114 f"({fit['precision_delta_pct']:+.1f}% , fit-on-full = optimistic)"
115)
116 print(
117 f" f1 {fit['baseline']['f1']:.3f} -> {fit['tuned']['f1']:.3f} "
118 f"({fit['f1_delta_pct']:+.1f}%)"
119)
120 print(
121 f"HONEST (cross-val, {cv.get('folds', '?')} folds): held-out recall "
122 f"{cv.get('mean_test', float('nan')):.3f} ± {cv.get('std_test', 0):.3f}, "
123 f"generalization_gap {cv.get('generalization_gap', float('nan')):+.3f}"
124)
125 print(
126 "(For precision/F1 generalization, add a 2nd labelled video ->
leave_one_video_out.)"
127)
128 return {"baseline": base.as_dict(), "best_cfg": best_cfg, "fit": fit, "cv": cv}
129
130
131 def main():
132 ap = argparse.ArgumentParser()
133 ap.add_argument(
134 "--trajectory", required=True, help="WASB trajectory CSV (Frame,Visibility,X,Y)"
135)
136 ap.add_argument(
137 "--labels", required=True, help="golden labels CSV (start_time/end_time cols)"
138)
139 ap.add_argument("--fps", type=float, default=59.94)
140 ap.add_argument("--frame-width", type=float, default=1920.0)
141 ap.add_argument("--iou", type=float, default=0.5)
142 args = ap.parse_args()
143 run(args.trajectory, args.labels, args.fps, args.frame_width, args.iou)
144
145
146 if __name__ == "__main__":
147 main()
148

```

40. user (2026-06-29T10:57:06.351Z)

#### parse\_trajectory\_csv location + body

```

264: def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
--- _trim_inactive_tail + flag locations ---
314: def trajectory_to_action_windows(
329: inactivity_temporal_density: bool = False,
426: temporal_density=inactivity_temporal_density,
450: def _trim_inactive_tail(

```

41. user (2026-06-29T10:57:58.365Z)

#### sample trajectory CSV header (adarsh\_avi\_singles)



```
Frame,Visibility,X,Y
1,1,1069.8391342163086,489.07470703125
2,1,1069.3811416625977,502.46131896972656
3,1,1071.0103225708008,511.7584419250488
4,1,1073.2082748413086,512.3460388183594
```

#### line count of that trajectory

```
84481
```

#### sample label CSV (2026-06-08\_adarsh\_avi\_singles.rallies.csv)

```
rally_number,start_time,end_time,ending_reason,sport
1,37.434,72.194,winner,badminton
2,79.936,112.954,winner,badminton
3,130.193,134.947,winner,badminton
4,139.697,144.944,unforced_error,badminton
5,152.196,157.198,unforced_error,badminton
6,161.694,165.950,unforced_error,badminton
7,170.817,176.327,forced_error,badminton
8,182.829,186.083,unforced_error,badminton
9,193.576,199.831,forced_error,badminton
```

#### 42. assistant (2026-06-29T10:58:37.949Z)

All confirmed ? trajectory CSVs are `Frame,Visibility,X,Y` (the WASB format `parse\_trajectory\_csv` expects), labels are `rally\_number,start\_time,end\_time,...` (what `load\_gt\_intervals` reads). The plan is faithful and fully offline. Let me materialize the corpus and smoke-test that the venv can import the harness.

#### 43. user (2026-06-29T10:58:45.731Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

#### 44. user (2026-06-29T10:59:47.527Z)

#### downloading trajectories (15 CSVs, ~17MB)

```
.....
```

```
Average throughput: 3.9MiB/s
```

#### downloading labels

```
..
```

```
Average throughput: 104.8kiB/s
```

#### downloading manifest

```
.
```

#### inventory

```
trajectories: 15 files
Badminton_BXH_2.csv
Boxhill_Doubles.csv
GX010128.csv
GX010137.csv
GX010141.csv
GX020094.csv
GX030094.csv
adarsh_avi_singles.csv
gbaaddy.csv
kushagra_singles.csv
largetest_doubles.csv
mahadevpura_1.csv
mahadevpura_2.csv
mahadevpura_singles.csv
testlarge_short.csv
```

```
labels: 15 files
15M /home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/trajectories
68K /home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/labels
```

#### 45. user (2026-06-29T10:59:49.838Z)

##### smoke-test: can the venv import the eval harness from the repo?

```
imports OK; harness at /home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/rally_seg_eval.py
SERVED preset: WindowingPreset(name='served', velocity_thresh=0.02, merge_gap=2.0,
min_window_duration=2.0, max_window_duration=6.0, hard_cap=False, inactivity_end=1.5,
inactivity_rally_thresh=0.2, inactivity_min_density=0.3, inactivity_temporal_density=False,
blob_fallback=False, blob_factor=4.0)
flag default inactivity_temporal_density = False
```

#### 46. assistant (2026-06-29T11:00:20.046Z)

Corpus materialized (15 trajectories + 15 labels), and the venv imports the harness cleanly. SERVED preset confirmed (R4 trim ON at `inactivity\_end=1.5`, `temporal\_density=False`), flag default `False`. Now I'll build a local-path manifest + the gts feature JSONs using the repo's \*own\* `load\_gt\_intervals`, so the ground truth is byte-identical to what `fusion\_golden` would have written.

#### 47. assistant (2026-06-29T11:00:31.280Z)

```
#!/usr/bin/env python
"""Build a LOCAL-path golden manifest + gts feature JSONs for the P2b rally_seg_eval A/B.
```

Why this exists: rally\_seg\_eval reads (a) each clip's trajectory CSV from --manifest and (b) ONLY the "gts" key of <name>\_golden\_features.json from --features-dir. It needs neither the GPU nor the source videos. The cached WASB trajectories + golden labels both live in gs://khelsutra-rally-corpus and are materialized locally here; the GTs are derived with the repo's OWN load\_gt\_intervals so they are byte-identical to fusion\_golden's output.

```
Inputs (downloaded from gs://khelsutra-rally-corpus, alongside this script):
golden_gcs.json ? 15-clip manifest (gs:// trajectory+label paths)
trajectories/<name>.csv ? cached WASB trajectories (Frame,Visibility,X,Y)
labels/<date><name>.rallies.csv ? golden rally labels
Outputs (consumed by python -m backend.eval.rally_seg_eval):
manifest_local.json ? trajectory/labels rewritten to local abs paths
features/<name>_golden_features.json ? {"gts": [[start,end],...]}
"""
```

```
import json
import os
```

```
from backend.eval.gt_loader import load_gt_intervals
```

```
DEST = os.path.dirname(os.path.abspath(__file__))
TRAJ = os.path.join(DEST, "trajectories")
LBL = os.path.join(DEST, "labels")
FEAT = os.path.join(DEST, "features")
os.makedirs(FEAT, exist_ok=True)
```

```
with open(os.path.join(DEST, "golden_gcs.json")) as f:
 specs = json.load(f)
```

```
local_manifest = []
summary = []
for s in specs:
 name = s["name"]
 traj_local = os.path.join(TRAJ, os.path.basename(s["trajectory"]))
 lbl_local = os.path.join(LBL, os.path.basename(s["labels"]))
```

```

assert os.path.isfile(traj_local), f"missing trajectory {traj_local}"
assert os.path.isfile(lbl_local), f"missing labels {lbl_local}"
gts = load_gt_intervals(lbl_local, strict=False)
feat = {
 "name": name,
 "fps": float(s["fps"]),
 "frame_width": float(s["frame_width"]),
 "gts": [[float(a), float(b)] for a, b in gts],
}
with open(os.path.join(FEAT, f"{name}_golden_features.json"), "w") as f:
 json.dump(feat, f, indent=2)
local_manifest.append(
 {
 "name": name,
 "trajectory": traj_local,
 "fps": float(s["fps"]),
 "frame_width": float(s["frame_width"]),
 "labels": lbl_local,
 }
)
summary.append((name, len(gts)))

with open(os.path.join(DEST, "manifest_local.json"), "w") as f:
 json.dump(local_manifest, f, indent=2)

print(f"built manifest_local.json ({len(local_manifest)} clips) + {len(summary)} feature JSONs")
for name, n in sorted(summary):
 print(f" {name:24s} {n:3d} golden rallies")
print("TOTAL golden rallies:", sum(n for _, n in summary))

```

#### 48. user (2026-06-29T11:00:32.327Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b\_golden/build\_local\_manifest.py (file state is current in your context ? no need to Read it back)

#### 49. user (2026-06-29T11:00:37.466Z)

built manifest\_local.json (15 clips) + 15 feature JSONs

Badminton_BXH_2	44 golden rallies
Boxhill_Doubles	43 golden rallies
GX010128	52 golden rallies
GX010137	34 golden rallies
GX010141	29 golden rallies
GX020094	40 golden rallies
GX030094	41 golden rallies
adarsh_avi_singles	96 golden rallies
gbaaddy	48 golden rallies
kushagra_singles	32 golden rallies
largetest_doubles	49 golden rallies
mahadevpura_1	10 golden rallies
mahadevpura_2	40 golden rallies
mahadevpura_singles	31 golden rallies
testlarge_short	6 golden rallies
TOTAL golden rallies: 595	

#### 50. assistant (2026-06-29T11:00:46.477Z)

15 clips, 595 golden rallies, all non-empty. Now the actual P2b A/B ? base = SERVED (flag OFF) vs candidate = SERVED (flag ON), at IoU 0.5, writing full JSON for the promotion-gate computation:

#### 51. user (2026-06-29T11:00:54.882Z)

## Rally-segmentation eval ? windowing 'served' (vt=0.02, merge\_gap=2.0, min\_win=2.0,

video	F1	F1@0.25	F1@0.5	merges	merge_rate	seg_ratio	pred/gt
adarsh_avi_singles	0.679	0.787	0.679	0	0.00	1.30	125/96
kushagra_singles	0.416	0.675	0.416	3	0.19	1.41	45/32
largetest_doubles	0.667	0.800	0.667	0	0.00	1.45	71/49
gbaaddy	0.417	0.639	0.417	1	0.04	2.00	96/48
testlarge_short	0.300	0.400	0.300	0	0.00	2.33	14/6
mahadevpura_singles	0.602	0.747	0.602	0	0.00	1.68	52/31
mahadevpura_2	0.244	0.488	0.244	9	0.53	1.05	42/40
GX010128	0.602	0.722	0.602	0	0.00	1.56	81/52
mahadevpura_1	0.000	0.000	0.000	2	0.80	0.20	2/10
GX030094	0.468	0.645	0.468	0	0.00	2.02	83/41
Badminton_BXH_2	0.443	0.672	0.443	2	0.09	1.77	78/44
GX010141	0.510	0.745	0.510	6	0.45	0.76	22/29
GX010137	0.822	0.904	0.822	0	0.00	1.15	39/34
Boxhill_Doubles	0.431	0.554	0.431	0	0.00	2.02	87/43
GX020094	0.424	0.667	0.424	0	0.00	1.48	59/40

**\*\*Aggregate (n=15, mean [95% CI]):\*\*** F1 0.468 [0.367,0.560] · F1@0.25 0.630 [0.516,0.721] · F1@0.5 0.468 [0.367,0.560] · merge\_rate 0.140 [0.033,0.274] · seg\_ratio 1.478 [1.188,1.715]

## R2 ? tolerance-match (?\_start=2.0s, ?\_end=1.5s, strict 1:1)

video	tolF1	tolP	tolR	?start	?end	split	merge
adarsh_avi_singles	0.552	0.49	0.64	0.94	0.54	13	0
kushagra_singles	0.078	0.07	0.09	2.09	2.52	6	3
largetest_doubles	0.583	0.49	0.71	0.70	0.65	5	0
gbaaddy	0.278	0.21	0.42	0.93	1.10	8	1
testlarge_short	0.200	0.14	0.33	2.71	2.58	1	0
mahadevpura_singles	0.530	0.42	0.71	0.79	0.52	7	0
mahadevpura_2	0.122	0.12	0.12	3.06	4.60	4	9
GX010128	0.466	0.38	0.60	0.71	0.73	2	0
mahadevpura_1	0.000	0.00	0.00	32.58	27.00	0	2
GX030094	0.419	0.31	0.63	0.79	0.96	6	0
Badminton_BXH_2	0.328	0.26	0.45	1.15	0.71	4	2
GX010141	0.431	0.50	0.38	0.78	0.68	0	6
GX010137	0.685	0.64	0.74	0.39	0.44	1	0
Boxhill_Doubles	0.262	0.20	0.40	0.79	0.97	9	0
GX020094	0.364	0.31	0.45	0.74	0.58	7	0

**\*\*R2 aggregate (n=15):\*\*** tolF1 0.353 [0.252,0.449] · tolP 0.302 [0.211,0.391] · tolR 0.445 [0.325,0.557] · ?start 3.277 [0.900,7.645] · ?end 2.972 [0.826,6.618] (start?solved / end=the gap)

## Rally-segmentation eval ? windowing 'served' (vt=0.02, merge\_gap=2.0, min\_win=2.0,

video	F1	F1@0.25	F1@0.5	merges	merge_rate	seg_ratio	pred/gt
adarsh_avi_singles	0.694	0.796	0.694	0	0.00	1.25	120/96
kushagra_singles	0.453	0.693	0.453	3	0.19	1.34	43/32
largetest_doubles	0.631	0.793	0.631	0	0.00	1.27	62/49
gbaaddy	0.448	0.657	0.448	1	0.04	1.79	86/48
testlarge_short	0.333	0.444	0.333	0	0.00	2.00	12/6
mahadevpura_singles	0.525	0.725	0.525	0	0.00	1.58	49/31
mahadevpura_2	0.250	0.525	0.250	9	0.53	1.00	40/40
GX010128	0.582	0.716	0.582	0	0.00	1.58	82/52
mahadevpura_1	0.000	0.000	0.000	2	0.80	0.20	2/10
GX030094	0.466	0.621	0.466	0	0.00	1.83	75/41
Badminton_BXH_2	0.459	0.656	0.459	2	0.09	1.77	78/44
GX010141	0.549	0.784	0.549	5	0.38	0.76	22/29

GX010137	0.800	0.857	0.800	0	0.00	1.06	36/34
Boxhill_Doubles	0.422	0.562	0.422	0	0.00	1.98	85/43
GX020094	0.472	0.697	0.472	0	0.00	1.23	49/40

**\*\*Aggregate (n=15, mean [95% CI]):\*\*** F1 0.472 [0.376,0.558] · F1@0.25 0.635 [0.527,0.721] · F1@0.5 0.472 [0.376,0.558] · merge\_rate 0.135 [0.031,0.268] · seg\_ratio 1.375 [1.115,1.590]

## R2 ? tolerance-match (?\_start=2.0s, ?\_end=1.5s, strict 1:1)

video	tolF1	tolP	tolR	?start	?end	split	merge
adarsh_avi_singles	0.574	0.52	0.65	0.94	0.48	12	0
kushagra_singles	0.107	0.09	0.12	2.13	1.79	6	3
largetest_doubles	0.541	0.48	0.61	0.69	0.74	4	0
gbaaddy	0.284	0.22	0.40	0.93	0.78	7	1
testlarge_short	0.333	0.25	0.50	1.32	0.40	1	0
mahadevpura_singles	0.525	0.43	0.68	0.81	0.51	6	0
mahadevpura_2	0.150	0.15	0.15	3.06	4.45	3	9
GX010128	0.478	0.39	0.62	0.67	0.62	2	0
mahadevpura_1	0.000	0.00	0.00	32.58	26.95	0	2
GX030094	0.362	0.28	0.51	0.80	0.95	6	0
Badminton_BXH_2	0.344	0.27	0.48	1.22	0.51	4	2
GX010141	0.471	0.55	0.41	0.78	0.51	0	5
GX010137	0.600	0.58	0.62	0.42	0.40	1	0
Boxhill_Doubles	0.281	0.21	0.42	0.79	1.13	8	0
GX020094	0.404	0.37	0.45	0.74	0.57	6	0

**\*\*R2 aggregate (n=15):\*\*** tolF1 0.364 [0.271,0.450] · tolP 0.319 [0.231,0.402] · tolR 0.441 [0.332,0.537] · |?start| 3.192 [0.836,7.536] · |?end| 2.719 [0.624,6.366] (start?solved / end=the gap)

## ? (served mc=0 ? served mc=0), n\_paired=15

?f1: +0.004 [-0.014,+0.020] CI?0=False sign(W/L)=8.0/6.0 p=0.791  
 ?f1@0.25: +0.005 [-0.008,+0.018] CI?0=False sign(W/L)=8.0/6.0 p=0.791  
 ?tol\_f1: +0.010 [-0.014,+0.034] CI?0=False sign(W/L)=10.0/4.0 p=0.180  
 ?merge\_rate: -0.005 [-0.014,+0.000] CI?0=False sign(W/L)=0.0/1.0 p=1.000  
 ?split\_rate: -0.010 [-0.017,-0.005] CI?0=True sign(W/L)=0.0/7.0 p=0.016  
 ?segment\_count\_ratio: -0.103 [-0.159,-0.055] CI?0=True sign(W/L)=1.0/11.0 p=0.006

## R1 over-seg promotion guard (rate-based, w\_merge=2.0, w\_split=1.0)

NET verdict: PASS (net cost 0.397 ? 0.378, ? -0.020; no\_merge\_rate\_regress=True)  
 strict per-video rule (R2 §2.3.3, for contrast): pass (count increases ? merges:0 splits:0)

[written /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b\_golden/ab\_served\_temporal\_density.json]

### 52. assistant (2026-06-29T11:01:49.534Z)

The A/B ran cleanly on all 15 clips. Headline result:

- **\*\*tolF1 (SERVED):** base 0.353 ? candidate 0.364, ?tolF1 = +0.010\*\* [-0.014, +0.034], sign 10W/4L, p=0.180 ? **\*\*neutral-to-positive, does not regress\*\***.
- **\*\*|?end| improved 2.97 ? 2.72s\*\*** (the rally-end error finding B targets).
- **\*\*Over-segmentation significantly reduced\*\***: ?segment\_count\_ratio ?0.103 (p=0.006), ?split\_rate ?0.010 (p=0.016); merge\_rate flat-to-better.
- **\*\*R1 over-seg guard: PASS\*\*** (net cost 0.397 ? 0.378).

Now let me compute the formal verdict with the repo's own `promote\_if\_served\_safe` gate rather than eyeballing it ? pairing per-video tolF1 by name:

### 53. assistant (2026-06-29T11:02:00.255Z)

#!/usr/bin/env python

```

"""Apply the formal P2b promotion gate to the served A/B result, using the repo's own
backend.eval.promotion code (no hand-math). Reads ab_served_temporal_density.json, pairs
per-video tolF1 (base=flag-OFF vs cand=flag-ON) by name, and prints:
- served_regresses(eps=0.0): the plan's literal gate ("promote iff tolF1 does not regress")
- promote_if_served_safe(structural=True): finding B is a do-no-harm correctness fix ? the
 served-safe reading the plan states ("neutral or better ? promote")
- promote_if_served_safe(structural=False): the stricter honest-LIFT bar, for contrast
"""

import json
import os

from backend.eval.eval_stats import paired_delta_ci
from backend.eval.promotion import promote_if_served_safe, served_regresses

DEST = os.path.dirname(os.path.abspath(__file__))
with open(os.path.join(DEST, "ab_served_temporal_density.json")) as f:
 res = json.load(f)

base_by = {r["name"]: r["tol_f1"] for r in res["base"]["rows"]}
cand_by = {r["name"]: r["tol_f1"] for r in res["cand"]["rows"]}
names = [n for n in base_by if n in cand_by]
a = [base_by[n] for n in names] # served flag-OFF
b = [cand_by[n] for n in names] # served flag-ON

print(f"n_paired = {len(names)}")
print(f"mean tolF1 base(OFF) = {sum(a)/len(a):.4f} cand(ON) = {sum(b)/len(b):.4f}")
d = paired_delta_ci(a, b, n_boot=2000, seed=0)
print(
 f"?tolF1 = {d['mean_delta']:+.4f} CI[{d['ci_low']:+.4f},{d['ci_high']:+.4f}] "
 f"CI?0={d['ci_excludes_zero']} sign W/L={d['sign_test']['wins']}/{d['sign_test']['losses']}
"
 f"p={d['sign_test']['p_value']:.3f}"
)
print()
reg = served_regresses(a, b, eps=0.0)
print("== plan's literal gate: served_regresses(eps=0.0) ==")
print(f" regresses = {reg['regresses']} (mean_delta {reg['mean_delta']:+.4f}, "
 f" CI[{reg['ci_low']:+.4f},{reg['ci_high']:+.4f}])")
print(f" -> SERVED-SAFE (does not regress) = {not reg['regresses']}")
print()
for structural in (True, False):
 g = promote_if_served_safe(a, b, structural=structural, served_regress_eps=0.0)
 print(f"== promote_if_served_safe(structural={structural}) ==")
 print(f" promote = {g.promote}")
 print(f" served_safe = {g.served_safe}")
 print(f" vetoed = {g.vetoed_by_served}")
 print(f" reason = {g.reason}")
 print()

```

#### 54. user (2026-06-29T11:02:01.327Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b\_golden/decide.py (file state is current in your context ? no need to Read it back)

#### 55. user (2026-06-29T11:02:07.739Z)

```

n_paired = 15
mean tolF1 base(OFF) = 0.3532 cand(ON) = 0.3636
?tolF1 = +0.0104 CI[-0.0139,+0.0341] CI?0=False sign W/L=10.0/4.0 p=0.180

== plan's literal gate: served_regresses(eps=0.0) ==
 regresses = False (mean_delta +0.0104, CI[-0.0139,+0.0341])
 -> SERVED-SAFE (does not regress) = True

```

```

== promote_if_served_safe(structural=True) ==
 promote = True
 served_safe = True
 vetoed = False
 reason = structural guard kept ON; served-safe (does not regress at the served
windowing, ?F1 +0.0104) ? eligible for served default

== promote_if_served_safe(structural=False) ==
 promote = False
 served_safe = True
 vetoed = False
 reason = insufficient evidence (?F1 95% CI [-0.0139, 0.0341] spans 0; sign test does not
agree (10-4, p=0.1796)); falling back to shipped baseline

```

## 56. assistant (2026-06-29T11:02:50.283Z)

**\*\*Formal verdict: PROMOTE.\*\*** The repo's own gate confirms it:

```

- **Plan's literal gate** ? `served_regresses(eps=0.0)` ? **does not regress** (?tolF1 +0.0104,
CI upper +0.034); **served-safe = True**.
- **`promote_if_served_safe(structural=True)` ? `promote = True`** ? "served-safe ? eligible for
served default." This is the correct reading: finding B is a *do-no-harm correctness fix* to an
existing default-ON heuristic (R4), gated OFF only to keep the merge a zero-regression no-op
until measured. The plan's rule is explicit: "neutral or better ? PROMOTE."
- The stricter `structural=False` bar returns "insufficient evidence of *lift*" (CI spans 0) ?
but **not vetoed, not a regression**. That bar is for a brand-new cue that must prove its worth;
it's the honest caveat I'll surface to you, not a blocker.

```

Net: tolF1 neutral-to-positive (no regression), over-segmentation **\*\*significantly\*\*** reduced, rally-end error improved, over-seg guard PASS. That satisfies the promotion rule. Proceeding to promote per the prompt's explicit instruction (the launch GO/NO-GO is the only thing I'll leave for your signoff).

Let me locate the test assertions and doc sections I need to update, and confirm the git baseline.

## 57. user (2026-06-29T11:02:57.656Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

## 58. user (2026-06-29T11:03:03.726Z)

### tests referencing the flag

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_tracknet_runner.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_windowing.py

```

### test assertions on the DEFAULT (False) of the flag

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py:59: inactivity_temporal_density=False,
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py:60: vs_inactivity_temporal_density=False,
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py:68: assert SERVED.inactivity_temporal_density is False
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py:70: assert _build_preset(args,
"base").inactivity_temporal_density is False
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py:71: assert _build_preset(args,
"can").inactivity_temporal_density is False
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_rally_seg_eval.py:77: assert _build_preset(args,
"base").inactivity_temporal_density is False

```

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_windowing.py:115:
assert W.SERVED.inactivity_temporal_density is False # default OFF
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_windowing.py:133:
"inactivity_temporal_density": False, # finding-B density fix, default OFF
```

### docs referencing the flag

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY_DETECTION_CODE_REVIEW.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/PACKAGING_PLAN.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT_STEPS.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY_DETECTION_FIXES_PLAN.md
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/README.md
```

### git baseline

```
0c9c693 feat(eval): expose inactivity_temporal_density in the rally_seg_eval A/B CLI (P2a)
(#395)
--- untracked (should be only junk + p2b_golden) ---
?? gcs:/
?? gs:/
?? local:/
```

### 59. assistant (2026-06-29T11:03:29.343Z)

`output/p2b\_golden/` is correctly gitignored (only the junk dirs are untracked), and I'm cleanly at origin/master `0c9c693`. Now let me read the exact test assertions and doc sections that encode the old default, so I update them faithfully. (Note: `\_build\_preset` only ever flips the flag \*on\*, never off ? so flipping the default has test implications I need to handle carefully.)

### 60. user (2026-06-29T11:03:30.310Z)

```
1 """Tests for backend.eval.rally_seg_eval ? the pure scoring/aggregation core (no golden
files,
2 no trajectory I/O). The windowing + trajectory parsing it composes are tested in their
own modules.
3 """
4
5 import argparse
6
7 import pytest
8
9 from backend.eval.rally_seg_eval import (
10 _build_preset,
11 aggregate,
12 compare,
13 court_type_for,
14 evaluate,
15 filter_by_min_duration,
16 score_intervals,
17 stratified_report,
18 windows_for,
19)
20 from backend.eval.windowing import SERVED, WindowingPreset
21
22 GTS = [(0.0, 10.0), (20.0, 30.0), (40.0, 50.0)]
23 # A permissive preset so a short synthetic trajectory still forms a window.
24 _PRESET = WindowingPreset(
25 name="t", velocity_thresh=0.02, merge_gap=2.0, min_window_duration=0.5
26)
27
28
29 def _write_oscillating_traj(path, fps=30, secs=4, width=1280):
30 """A fast shuttle oscillating smoothly across the net (x=width/2) via a triangle
```



```

wave ?
31 continuous high velocity (active) + many net crossings ? one gated-survivable
window. """
32 lo, hi, period = 200, width - 200, 20 # frames per full oscillation
33 rows = ["Frame,Visibility,X,Y"]
34 for i in range(int(fps * secs)):
35 phase = (i % period) / period # 0..1
36 tri = 2 * phase if phase < 0.5 else 2 * (1 - phase) # triangle 0?1?0
37 rows.append(f"{i},1,{lo + (hi - lo) * tri:.1f},300")
38 path.write_text("\n".join(rows))
39 return str(path)
40
41
42 def _eval_args(**overrides):
43 """An argparse.Namespace with the windowing-knob defaults `_build_preset` reads
44 (matching the CLI defaults: None for floats, False for store_true flags)."""
45 base = dict(
46 preset="served",
47 vs="served",
48 max_window_duration=None,
49 vs_max_window_duration=None,
50 hard_cap=False,
51 vs_hard_cap=False,
52 blob_fallback=False,
53 vs_blob_fallback=False,
54 blob_factor=None,
55 inactivity_end=None,
56 vs_inactivity_end=None,
57 inactivity_rally_thresh=None,
58 inactivity_min_density=None,
59 inactivity_temporal_density=False,
60 vs_inactivity_temporal_density=False,
61)
62 base.update(overrides)
63 return argparse.Namespace(**base)
64
65
66 def test_build_preset_temporal_density_default_off():
67 # The SERVED preset (and a default A/B) carries the finding-B flag OFF.
68 assert SERVED.inactivity_temporal_density is False
69 args = _eval_args()
70 assert _build_preset(args, "base").inactivity_temporal_density is False
71 assert _build_preset(args, "cand").inactivity_temporal_density is False
72
73
74 def test_build_preset_temporal_density_promotion_ab():
75 # The canonical P2 promotion A/B: base = served (flag OFF), cand = served + flag ON.
76 args = _eval_args(vs_inactivity_temporal_density=True)
77 assert _build_preset(args, "base").inactivity_temporal_density is False
78 assert _build_preset(args, "cand").inactivity_temporal_density is True
79 # --inactivity-temporal-density sets BOTH sides (cand inherits the base override).
80 both = _eval_args(inactivity_temporal_density=True)
81 assert _build_preset(both, "base").inactivity_temporal_density is True
82 assert _build_preset(both, "cand").inactivity_temporal_density is True
83
84
85 def test_score_intervals_perfect():
86 row = score_intervals(GTS, GTS)
87 assert row["f1"] == pytest.approx(1.0)
88 assert row["f1@0.25"] == pytest.approx(1.0)
89 assert row["merge_rate"] == 0.0 and row["split_rate"] == 0.0
90 assert row["merges"] == 0 and row["num_pred"] == 3 and row["num_gt"] == 3
91
92
93 def test_score_intervals_surfaces_over_merge():

```

```

94 # One mega-window swallowing two rallies: tIoU-F1 is hurt AND merge_rate makes it
explicit.
95 preds = [(0.0, 30.0), (40.0, 50.0)]

```

## 61. user (2026-06-29T11:03:31.667Z)

```

100 W.served_windowing_from_config(None, "wasb").as_tuple() == W.SERVED.as_tuple()
101)
102
103
104 def test_served_windowing_from_config_honors_r5_blob_fallback():
105 # The R5 blob-fallback + its magnitude gate lift out of a live config (so the
eventual default
106 # flip and any per-deployment override take effect on the served path).
107 d = {"indexing": {"window_blob_fallback": True, "window_blob_factor": 5.0}}
108 p = W.served_windowing_from_config(d, "wasb")
109 assert p.blob_fallback is True and p.blob_factor == 5.0
110
111
112 def test_served_windowing_from_config_honors_temporal_density_flag():
113 # The finding-B density fix lifts out of a live config so the owner can A/B it at
the SERVED
114 # operating point and flip the default once it's measured on the golden set.
115 assert W.SERVED.inactivity_temporal_density is False # default OFF
116 d = {"indexing": {"inactivity_temporal_density": True}}
117 assert W.served_windowing_from_config(d, "wasb").inactivity_temporal_density is True
118
119
120 def test_preset_roundtrip_dict():
121 # R1 milestone FLIPPED ON (2026-06-18): the SERVED defaults now carry W1 cap=6 + R4.
122 d = W.SERVED.to_dict()
123 assert d == {
124 "name": "served",
125 "velocity_thresh": 0.02,
126 "merge_gap": 2.0,
127 "min_window_duration": 2.0,
128 "max_window_duration": 6.0,
129 "hard_cap": False,
130 "inactivity_end": 1.5,
131 "inactivity_rally_thresh": 0.20,
132 "inactivity_min_density": 0.30,
133 "inactivity_temporal_density": False, # finding-B density fix, default OFF
134 "blob_fallback": False,
135 "blob_factor": 4.0,
136 }
137
138
139 def test_served_preset_is_r1_milestone_default():
140 # R1 milestone FLIPPED ON (2026-06-18, PR #204): W1 bounded windowing (cap=6) + the
R4
141 # rally-END inactivity trim are now the shipped SERVED defaults (the smallest safe
quality
142 # flip, n=13 baseline?milestone ?tolF1 +0.222, sign 12/0, p=0.0005). The W1.5 hard-
cap, the
143 # R5 blob-fallback, and W2 net-crossings all stay OFF.
144 assert W.SERVED.max_window_duration == 6.0

```

## 62. user (2026-06-29T11:03:36.388Z)

### test\_tracknet\_runner.py temporal\_density context

```

280: def test_trim_inactive_tail_temporal_density_flag_trims_sparse_late_sample():
298: temporal_density=True,
303: def test_trim_inactive_tail_temporal_density_is_noop_on_dense_tracking():
309: group, 30, 0.3, 0.2, temporal_density=True

```

## RALLY\_DETECTION\_FIXES\_PLAN.md ? section anchors / P2b / status / dates

```
5:# Refactor Rally Detection Edge Cases & Parser Stability
7:> **Status:** `IN PROGRESS` . **Owner:** `Agent` . **Created:** `2026-06-19` . **Last
updated:** `2026-06-29`
23:## 0. TL;DR
26:## 1. Problem & goal
34:## 4. Design / architecture
55:## 6. Honest limits ? what this does NOT do
58:## 7. Deliverables & progress tracker ? **source of truth**
63:| P0a | Fix `_spread` zero-collapse (behaviour-neutral, unconditional) | ? | No | ? | #394 |
64:| P0b | `_trim_inactive_tail` temporal density behind `inactivity_temporal_density` flag
(default-OFF) | ? | No | ? | #394 |
65:| P1 | Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-
collector`) | ? | No | ? | sdc#50 |
66:| P2a | Expose the flag in the `rally_seg_eval` A/B CLI (`--[vs-]inactivity-temporal-
density`) so the promotion run is one command | P0b | No | ? | this PR |
67:| P2b | **Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON if no
regression | P2a | ? owner | ? | ? |
69:## 8. Open questions ? owner / external
75:2. **Owner promotion gate (P2b) ? pending.** P0b ships flag-OFF, so the merge is a
76: **zero-regression no-op in production**. The A/B is now a single command (P2a wired the
toggle
93:## 9. Definition of done
102:## 10. References
```

### plan doc total lines

```
108 /home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_FIXES_PLAN.md
```

### 63. assistant (2026-06-29T11:03:40.621Z)

Let me read the full plan doc and the tracknet test region so my edits are precise.

### 64. user (2026-06-29T11:03:41.148Z)

```
1 <!--
2 PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3 -->
4
5 # Refactor Rally Detection Edge Cases & Parser Stability
6
7 > **Status:** `IN PROGRESS` . **Owner:** `Agent` . **Created:** `2026-06-19` . **Last
updated:** `2026-06-29`
8 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`
9 > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`.
10 > **Relation to existing docs:** Extends `RALLY_DETECTION_QUALITY_REPORT.md`.
11 >
12 > **2026-06-29 update:** the three findings were re-verified against current `master`
(post the
13 > 38-commit advance that touched both `rally_gate.py` and `tracknet_runner.py`) ? **all
three were
14 > still present and unfixed.** P0 (the two in-repo edge cases) is **implemented + unit-
tested in this
15 > same PR (#393)**: the `_spread` fix (A) is behaviour-neutral and lands
unconditionally; the
16 > `_trim_inactive_tail` density fix (B) is a behaviour change on sparse footage (R4 is
default-ON
17 > in prod) so it lands behind a **default-OFF** `inactivity_temporal_density` flag,
promoted only
18 > after a golden-set measurement ($8 Q2). P1 lives in the sibling `sports-data-
collector` repo, so it
19 > ships as a **separate PR there** (cannot be in #393).
20
```

```

21 ---
22
23 ## 0. TL;DR
24 The rally detection pipeline is largely robust, but a deep code review highlighted three
minor edge cases/bugs: a potential divide-by-zero or zero-spread in `_spread` for tiny
sequences, a density calculation vulnerability for sparse tracks in `_trim_inactive_tail`, and
silent skipping of malformed rows in the CVAT parsing script. This plan outlines the fixes for
these issues to increase stability.
25
26 ## 1. Problem & goal
27 The current implementation of the candidate windowing handles normal inputs extremely
well (including handling blobs), but could fail or behave erratically under specific conditions:
28 - Zero-spread calculation: `_spread` in `rally_gate.py` evaluates to `0.0` for
arrays of length 2, which could cause a zero deadband if used on short clips.
29 - Sparse trailing frames: `_trim_inactive_tail` calculates frame density based on
`visible` frames. A long gap followed by a single active frame gives 100% density, artificially
inflating the rally end.
30 - Silent failure on malformed CSV: `load_canonical_csv` in `rally_cvat.py` silently
ignores rows without a valid `rally_number`.
31
32 Goal: Fix these three issues to harden the rally detection and ingestion pipelines
without changing the existing heuristic thresholds.
33
34 ## 4. Design / architecture
35 The changes are entirely localized to the identified functions:
36 - `rally_gate.py` ? done (#393): `_spread` now rounds the p95 index UP with
`math.ceil()`, so a
37 2-element (or any short) input falls back to the full min..max range instead of
collapsing to
38 `0.0`. On large arrays ceil-vs-truncate differ by at most one sample, so the robust
p5/p95 trim is
39 unchanged in practice.
40 - `tracknet_runner.py` ? done (#393), flag-gated default-OFF: `_trim_inactive_tail`
gains an
41 `temporal_density` flag. When ON, the density denominator is the trailing window's
temporal
42 length** (`min(trail_frames, frames[i] - frames[0] + 1)`) instead of `win_count` (the
count of
43 active *samples*, which sparse tracking inflates). Because `group` holds only active
frames, with
44 dense tracking the two denominators are identical ? so flipping the flag is a no-op
on
45 well-tracked clips** and changes only the sparse-tracking case. Correction
(2026-06-29): R4 is
46 default-ON in production (`window_inactivity_end = 1.5`, promoted at the R1
milestone ? an
47 earlier draft of this doc wrongly called it default-OFF), so the fix is a behaviour
change* on
48 sparse footage and ships behind the new `inactivity_temporal_density` flag (default-
OFF**, plumbed
49 config ? `trajectory_hybrid` ? eval `WindowingPreset`), to be promoted only after a
golden-set
50 measurement at the SERVED operating point ($8 Q2).
51 - `rally_cvat.py` ? separate PR (sibling `sports-data-collector` repo, not this
one): introduce a
52 `logger.warning` before each `continue` in `load_canonical_csv` so skipped/malformed
rows are
53 surfaced during ingestion.
54
55 ## 6. Honest limits ? what this does NOT do
56 This will NOT change the core AI validation (Phase 3) or the existing `velocity_thresh`
values. It solely tightens up boundary calculation edge cases. It will not suddenly "fix"
rallies that the tracking model missed entirely due to lighting or occlusion.
57
58 ## 7. Deliverables & progress tracker ? source of truth

```

```

59 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
60
61 | ID | Deliverable | Depends on | Gated? | Status | PR |
62 |----|-----|-----|-----|-----|----|
63 | P0a | Fix `_spread` zero-collapse (behaviour-neutral, unconditional) | ? | No | ? |
#394 |
64 | P0b | `_trim_inactive_tail` temporal density behind `inactivity_temporal_density` flag
(default-OFF) | ? | No | ? | #394 |
65 | P1 | Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-
collector`) | ? | No | ? | sdc#50 |
66 | P2a | Expose the flag in the `rally_seg_eval` A/B CLI (`--[vs-]inactivity-temporal-
density`) so the promotion run is one command | P0b | No | ? | this PR |
67 | P2b | **Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON if no
regression | P2a | ? owner | ? | ? |
68
69 ## 8. Open questions ? owner / external
70 1. ~~Do we want `_trim_inactive_tail` to fall back to the existing behavior if there are
very few
71 frames, or strictly enforce a minimum temporal density?~~ **Resolved (#393):** strict
temporal
72 density. Because `group` carries only active frames, the temporal denominator equals
the
73 sample-count denominator under dense tracking, so this is behaviour-preserving for
well-tracked
74 clips and only corrects the sparse case ? no separate few-frames fallback needed.
75 2. **Owner promotion gate (P2b) ? pending.** P0b ships flag-OFF, so the merge is a
76 **zero-regression no-op in production**. The A/B is now a single command (P2a wired
the toggle
77 into `rally_seg_eval`):
78
79 ```
80 python -m backend.eval.rally_seg_eval \
81 --manifest <golden_manifest.json> --features-dir <golden_features_dir> \
82 --preset served --vs served --vs-inactivity-temporal-density
83 ```
84
85 (base = SERVED with the flag OFF, candidate = SERVED with it ON; diffs tolF1 at the
SERVED
86 operating point ? R4 is default-ON in prod, so this measures B's real end-boundary
effect on
87 sparse footage.) **Promote** by flipping `IndexingConfig.inactivity_temporal_density`
to `True`
88 (a one-line PR) **only if tolF1 does not regress**. This step is owner-only: it needs
the golden
89 corpus + GPU/WSL-extracted trajectory features, which don't exist offline. Note
finding B is a
90 **no-op under dense tracking**, so the A/B only shows signal on clips where the
shuttle tracker
91 actually drops out ? which is exactly why it can't be measured on synthetic/offline
fixtures.
92
93 ## 9. Definition of done
94 - [x] `_spread` bug fixed and tested (behaviour-neutral, unconditional).
95 - [x] `_trim_inactive_tail` density fix implemented behind `inactivity_temporal_density`
(default-OFF),
96 plumbed config ? segmenter ? eval preset, tested in both flag states.
97 - [] **P2 promotion**: golden-set A/B at SERVED windowing, then flip the flag default-
ON if no
98 regression. *(owner-gated, real video ? see §8 Q2)*
99 - [] Warning log added for malformed rows in `rally_cvat.py`. *(sibling-repo PR,
separate)*
100 - [x] Tests updated and passing *(offline suite; golden-corpus regression is owner-gated
? see Q2)*.
101

```

```

102 ## 10. References
103 **Internal code anchors `[verified]`:**
104 - `backend/pipeline/detectors/rally_gate.py`
105 - `backend/pipeline/detectors/tracknet_runner.py`
106 - `src/parsers/cvat/rally_cvat.py`
107 **Internal docs:**
108 - `docs/RALLY_DETECTION_CODE_REVIEW.md`
109

```

#### 65. user (2026-06-29T11:03:42.015Z)

```

274) # fully-fast: untouched
275 assert (
276 TrackNetRunner._trim_inactive_tail(allfast, 0, 0.3, 0.2) == allfast
277) # trail=0: OFF/no-op
278
279
280 def test_trim_inactive_tail_temporal_density_flag_trims_sparse_late_sample():
281 # Code-review finding B, flag-gated default-OFF. Dense rally 0..40, a long detection
282 gap,
283 # then one stray fast sample at frame 400 (sparse tail).
284 group = [(i, 0.5) for i in range(0, 41)] + [(400, 0.5)]
285 # Default (legacy denominator = active-sample count): the lone late sample reads as
286 100%
287 # dense and HOLDS the tail open ? the pre-fix behaviour, preserved so the merge is a
288 no-op.
289 legacy = TrackNetRunner._trim_inactive_tail(
290 group, trail_frames=30, min_density=0.34, rally_thresh=0.2
291)
292 assert legacy[-1][0] == 400 # stray sample kept (bug behaviour, default)
293
294 # Flag ON (temporal denominator): the stray sample is 1/trail_frames dense ? trimmed
295 off.
296 fixed = TrackNetRunner._trim_inactive_tail(
297 group,
298 trail_frames=30,
299 min_density=0.34,
300 rally_thresh=0.2,
301 temporal_density=True,
302)
303 assert fixed[-1][0] == 40 # stray sparse sample at frame 400 trimmed off
304
305
306 def test_trim_inactive_tail_temporal_density_is_noop_on_dense_tracking():
307 # The fix must be behaviour-NEUTRAL when tracking is dense (contiguous frames): the
308 # active-sample count equals the temporal span, so both denominators give the same
309 trim.
310 group = [(i, 0.5) for i in range(0, 61)] + [(i, 0.03) for i in range(61, 201)]
311 legacy = TrackNetRunner._trim_inactive_tail(group, 30, 0.3, 0.2)
312 temporal = TrackNetRunner._trim_inactive_tail(
313 group, 30, 0.3, 0.2, temporal_density=True
314)
315 assert legacy == temporal
316
317
318 def test_window_inactivity_end_trims_through_api():
319 # one window of fast motion then slow drift (both visible & > velocity_thresh ? one
320 group).

```

#### 66. assistant (2026-06-29T11:05:28.007Z)

I now have the full picture for the edits. Let me first confirm I can actually push/open a PR (the deliverable needs a PR link), and create the promotion branch off master.

## 67. user (2026-06-29T11:05:35.408Z)

### gh availability + auth

```
/usr/bin/gh
github.com
? Logged in to github.com account avidullu (/home/avidullu/.config/gh/hosts.yml)
- Active account: true
- Git operations protocol: https
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo', 'workflow'
```

### git remote + credential helper

```
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (fetch)
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (push)
```

### current branch / cleanliness (tracked only)

```
master
```

### create promotion branch off origin/master

```
Switched to a new branch 'p2b/promote-inactivity-temporal-density'
p2b/promote-inactivity-temporal-density
```

## 68. assistant (2026-06-29T11:05:41.562Z)

`gh` is authed (`avidullu`, `repo`+`workflow` scopes) and the branch is created. Now the edits. First the one-line code change (flag default + an honest, numbers-backed comment):

## 69. user (2026-06-29T11:05:50.510Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/config/models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

## 70. assistant (2026-06-29T11:05:52.915Z)

Now the test updates ? reflecting the promoted default-ON honestly:

## 71. user (2026-06-29T11:06:02.945Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_rally\_seg\_eval.py has been updated successfully. (file state is current in your context ? no need to Read it back)

## 72. user (2026-06-29T11:06:11.043Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

## 73. user (2026-06-29T11:06:15.854Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

## 74. assistant (2026-06-29T11:07:12.935Z)

Now the doc updates to `RALLY\_DETECTION\_FIXES\_PLAN.md` ? honest per the DOC\_STATUS protocol (flip status, record the measured numbers, check the DoD boxes):

## 75. user (2026-06-29T11:07:16.666Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-

indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

76. user (2026-06-29T11:07:26.280Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

77. user (2026-06-29T11:07:31.970Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

78. assistant (2026-06-29T11:07:33.999Z)

Now the \$8 Q2 and \$9 Definition-of-Done updates:

79. user (2026-06-29T11:07:38.652Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

80. user (2026-06-29T11:07:48.385Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

81. user (2026-06-29T11:07:54.921Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

82. assistant (2026-06-29T11:08:05.863Z)

All edits in place. Now validation ? starting with the three changed test files for fast feedback, then I'll run the full suite:

83. user (2026-06-29T11:08:10.887Z)

ERROR: file or directory not found: tests/test\_models.py

no tests ran in 0.00s

84. user (2026-06-29T11:08:23.912Z)

..... [100%]  
68 passed in 2.26s  
any config-default tests referencing the flag elsewhere?  
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_windowing.py

85. assistant (2026-06-29T11:08:46.213Z)

68 passed. Now the full offline suite plus ruff and mypy (in parallel):

86. user (2026-06-29T11:10:00.035Z)

..... [ 38%]



```

..... [42%]
..... [47%]
..... [52%]
..... [57%]
..... [61%]
..... [66%]
..... [71%]
.....SSSS..... [76%]
..... [80%]
..... [85%]
..... [90%]
..... [95%]
..... [99%]
. [100%]
===== warnings summary =====
../.venv/lib/python3.12/site-packages/fastapi/testclient.py:1
/home/avidullu/projects/khelsutra-guru/.venv/lib/python3.12/site-
packages/fastapi/testclient.py:1: StarletteDeprecationWarning: Using `httpx` with
`starlette.testclient` is deprecated; install `httpx2` instead.
 from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
1503 passed, 10 skipped, 1 warning in 66.61s (0:01:06)
=====
RESULT: PASS - all tests green
=====

```

**87. user (2026-06-29T11:10:06.046Z)**

**ruff check .**

All checks passed!

**mypy backend training**

```

backend/annotations/base.py:88: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend/pipeline/validators/base.py:44: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend/sports/__init__.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend/providers/__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 138 source files

```

**88. assistant (2026-06-29T11:10:46.117Z)**

Outstanding ? clean across the board:

```

- **Full offline suite: 1503 passed, 10 skipped** (66s), RESULT: PASS.
- **ruff: All checks passed.**
- **mypy: no issues in 138 source files.**

```

The flag flip broke nothing. Let me verify exactly what I'm about to commit and capture the 10 skip reasons (to know which now-unlockable paths remain), then commit the promotion with an explicit file list.

**89. user (2026-06-29T11:10:53.092Z)**

**tracked changes (should be exactly 4 files)**

```

backend/config/models.py | 17 ++++++-----
docs/RALLY_DETECTION_FIXES_PLAN.md | 37 ++++++-----
tests/test_rally_seg_eval.py | 19 ++++++-----
tests/test_windowing.py | 14 ++++++-----
4 files changed, 57 insertions(+), 30 deletions(-)

```

**the one-line code change**

```
- inactivity_temporal_density: bool = False
+ inactivity_temporal_density: bool = True
```

#### untracked (must NOT be staged)

```
?? gcs:/
?? gs:/
?? local:/
```

#### 90. user (2026-06-29T11:12:06.656Z)

```
SKIPPED [1] tests/test_ablation.py:400: could not import 'reportlab': No module named
'reportlab'
SKIPPED [1] tests/test_ablation.py:525: golden output/*_golden_features.json absent (run
fusion_golden first)
SKIPPED [1] tests/test_ablation.py:555: golden output/*_golden_features.json absent
SKIPPED [1] tests/test_ablation.py:629: could not import 'reportlab': No module named
'reportlab'
SKIPPED [1] tests/test_app_home.py:193: Windows drive-letter semantics only meaningful on nt
SKIPPED [1] tests/test_served_gate_a_regression.py:91: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:106: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:123: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:135: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:155: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/human_lovo_manifest.json'); runs on the owner box
===== 1503 passed, 10 skipped, 1 warning in 64.77s (0:01:04) =====
```

#### 91. assistant (2026-06-29T11:12:32.426Z)

The diff is exactly the 4 intended files, the one-line flip is correct, and the junk dirs are untracked (won't be staged). The 10 skips break down as: 5 `served\_gate\_a\_regression` (golden absent at the \*default\* path ? \*\*unlockable\*\*, I have the trajectories locally), 2 `ablation` (need full fusion features = videos+GPU), 2 `ablation` PDF (need `reportlab`), 1 Windows-only.

Let me commit the promotion (explicit file list) and read the served-gate test in parallel to plan the unlock:

#### 92. user (2026-06-29T11:12:44.432Z)

```
f188070 feat(config): promote inactivity_temporal_density to default-ON (P2b)
backend/config/models.py | 17 ++++++-----
docs/RALLY_DETECTION_FIXES_PLAN.md | 37 ++++++-----
tests/test_rally_seg_eval.py | 19 ++++++-----
tests/test_windowing.py | 14 ++++++-----
4 files changed, 57 insertions(+), 30 deletions(-)
```

#### 93. user (2026-06-29T11:12:44.461Z)

```
1 """SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real
footage.
2
3 The flagged gap (#146): the served default flip (`default_segmenter=fusion_hybrid` +
4 ``min_crossings=2``) is **mock-tested only** on real video (`test_fusion_hybrid.py`
drives a
5 MagicMock runner). The offline ablation litmus
(`test_ablation.test_litmus_reproduces_gate_a`)
```

```

6 validates the Gate-A *signal* ? but on the **permissive proposal windowing**
7 (`distill_local.PROPOSAL`), NOT the production operating point. This module closes the
gap: it
8 runs the real ``FusionSegmenter`` decision seams (`_enrich_candidate_stats`` net-
crossings +
9 `_select_for_handoff`` Gate A) over windows from the **production**
10 ``trajectory_to_action_windows`` config, on the cached golden WASB trajectories, and
asserts
11 the served path's measured behaviour.
12
13 GPU-free / Gemini-free (person cue OFF) ? it stops at the AI-handoff boundary. Skips
cleanly in
14 CI where the golden trajectories are absent; runs on the owner box where they are
present, over
15 **whatever golden clips are currently on disk** (`_golden_specs`` requires >= 6).
16
17 CORPUS NOTE ? re-grounded 2026-06-22 (the golden set is mutable and grows over time;
this is the
18 SECOND re-grounding ? #236 pinned the 2026-06-18 9-clip snapshot, which the corpus has
since
19 outgrown). The original 6 *single-court* golden clips this litmus was first calibrated
on
20 (2026-06-14) have been retired from the box ? their labels are gone ? so the present
golden set is
21 the **15 multicourt clips** now on disk (mahadevpura_1/2/singles, GX0101xx, GX0x0094,
22 Badminton_BXH_2, Boxhill_Doubles, adarsh_avi_singles, kushagra_singles,
largetest_doubles, gbaaddy,
23 testlarge_short). On that corpus the served path is **recall-starved**: WASB tracks a
single shuttle
24 while several courts rally at once, so ``trajectory_to_action_windows`` emits FEWER
windows than
25 there are rallies (served over-seg ratio < 1, i.e. under-segmentation). Because Gate A
only ever
26 DROPS windows, on this under-segmented corpus it is mildly **F1-negative** (mean served
?F1 ?
27 -0.023), not the ~neutral -0.008 the original 6 showed. Under the permissive PROPOSAL
windowing the
28 same gate is ~F1-neutral (?F1 ? +0.03) while still cutting over-segmentation
(1.36?0.79).
29
30 The canonical **+0.074 / 6-of-6 / p=0.031** Gate-A win was a property of the
original-6 corpus
31 and is NOT reproduced here (those clips are no longer on the box); it is preserved in
the archive
32 (`docs/archives/quality-iterations/2026-06-first-ab-experiment/`). These tests now pin
the
33 **present-corpus** behaviour so a future windowing/corpus change that silently shifts it
trips the
34 litmus. The pinned magnitudes are a date-stamped snapshot of the present golden set and
will need
35 re-measurement as the corpus grows.
36 ""
37
38 from __future__ import annotations
39
40 import json
41 import os
42
43 import pytest
44
45 # cv2 is imported transitively by trajectory_hybrid (the served engine). On a box
without it
46 # (CI), skip the whole module rather than erroring at import.
47 pytest.importorskip("cv2")
48

```

```

49 from backend.eval import segmentation_metrics as _segm # noqa: E402
50 from backend.eval import served_gate_a as sga # noqa: E402
51 from backend.eval.calibrate_wasb import load_golden_gts # noqa: E402
52 from backend.eval.metrics import score # noqa: E402
53 from backend.pipeline.detectors import rally_gate # noqa: E402
54 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner # noqa: E402
55
56 # The golden manifest lives in the MAIN checkout's output/ (gitignored; absent in a
fresh
57 # worktree / CI). Resolve it relative to this repo root first, with an env override for
an
58 # out-of-tree checkout.
59 _REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
60 MANIFEST = os.environ.get(
61 "SERVED_GATE_A_MANIFEST",
62 os.path.join(_REPO_ROOT, "output", "human_lovo_manifest.json"),
63)
64
65
66 def _golden_specs():
67 """Return the manifest specs whose trajectory + label files are all present, else
[]."""
68 if not os.path.isfile(MANIFEST):
69 return []
70 with open(MANIFEST, encoding="utf-8") as f:
71 specs = json.load(f)
72 ok = [
73 s
74 for s in specs
75 if os.path.isfile(str(s.get("trajectory", "")))
76 and os.path.isfile(str(s.get("labels", "")))
77]
78 return ok if len(ok) >= 6 else []
79
80
81 _SPECS = _golden_specs()
82 _skip = pytest.mark.skipif(
83 not _SPECS,
84 reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
85)
86
87
88 # ----- served-path
litmus
89
90
91 @_skip
92 def test_served_path_runs_on_all_golden_videos():
93 """The production FusionSegmenter decision path runs end-to-end (no GPU/Gemini) on
EVERY
94 present golden trajectory and yields a handoff window list for each, for both gate
settings.
95 Asserts against the present corpus count (>= 6), not a stale fixed 6 ? the golden
set grows."""
96 results = sga.evaluate_manifest(_SPECS)
97 assert len(results) == len(_SPECS)
98 assert len(results) >= 6
99 for r in results:
100 assert r.num_gt > 0
101 assert (
102 r.n_handoff_off >= r.n_handoff_on
103) # Gate A only ever drops windows, never adds
104
105
106 @_skip

```

```

107 def test_served_gate_a_does_not_materially_regress_f1():
108 """SERVED no-material-regression guard: on the production windowing Gate A must not
MATERIALLY
109 hurt F1. On the present recall-starved multicourt corpus the served path is under-
segmented
110 (over-seg ratio < 1), so Gate A ? which only DROPS windows ? trims a few borderline-
true
111 windows and is mildly negative, NOT the ~neutral it was on the original single-court
6. We pin
112 a band that admits this measured mild loss but trips on a material regression."""
113 results = sga.evaluate_manifest(_SPECS)
114 agg = sga.aggregate(results)
115 # Honest measured value (2026-06-22, present 15-clip multicourt golden set): mean
served
116 # ?F1 ? -0.023 (was ? -0.034 on the 9-clip / ? -0.008 on the retired original-6).
Band admits
117 # the mild recall-starved loss while still tripping if Gate A materially destroys
served F1.
118 assert -0.06 <= agg["mean_delta_f1"] <= 0.03, agg
119 # And Gate A must never collapse served F1 to ~half the OFF baseline (measured ratio
? 0.90).
120 assert agg["mean_f1_on"] >= 0.5 * agg["mean_f1_off"] - 1e-9, agg
121
122
123 @skip
124 def test_served_gate_a_does_not_increase_over_segmentation():
125 """Gate A only ever DROPS windows, so the served over-seg ratio must not rise when
it's on
126 (the direction-of-effect invariant ? even where the absolute lift is ~neutral)."""
127 results = sga.evaluate_manifest(_SPECS)
128 agg = sga.aggregate(results)
129 assert agg["mean_seg_ratio_on"] <= agg["mean_seg_ratio_off"] + 1e-9, agg
130 # Per-video too: the gate is monotone (kept windows ? all windows).
131 for r in results:
132 assert r.seg_ratio_on <= r.seg_ratio_off + 1e-9, r
133
134
135 @skip
136 def test_served_uses_production_windowing_operating_point():
137 """Pin the served operating point to config.json's indexing defaults ? if someone
retunes
138 production windowing, this litmus's premise (and the divergence it documents) is
revisited
139 deliberately, not silently."""
140 assert sga.PROD_VELOCITY_THRESH == 0.02
141 assert sga.PROD_MERGE_GAP == 2.0
142 assert sga.PROD_MIN_WINDOW == 2.0
143
144
145 # ----- Gate A under the permissive PROPOSAL
windowing
146
147 # distill_local.PROPOSAL ? the permissive, recall-oriented candidate windowing the
OFFLINE
148 # ablation/fusion_compare harness builds on (low velocity thresh, short merge/min-dur ?
many
149 # small windows). On the original-6 single-court corpus the Gate-A lift here was +0.074
(6/6,
150 # p=0.031; archived). On the present multicourt corpus that lift is GONE ? the gate is
~F1-neutral
151 # but still cuts over-segmentation. We pin the present-corpus behaviour, not the retired
+0.074.
152 _PROPOSAL = (0.005, 1.0, 0.5)
153
154

```

```

155 @_skip
156 def test_proposal_windowing_gate_a_is_f1_neutral_and_cuts_over_seg():
157 """CONTRAST pin: under the OFFLINE harness's permissive proposal windowing, the SAME
158 ``rally_gate`` net-crossing gate the served seam uses behaves DIFFERENTLY than under
the
159 production windowing. The durable invariant (any corpus): the gate reduces over-
segmentation
160 (seg_on < seg_off), since it only drops windows. On the present multicourt corpus it
is

```

#### 94. assistant (2026-06-29T11:13:17.724Z)

Commit is in (`f188070`). The served-gate test reads `SERVED\_GATE\_A\_MANIFEST` and needs 26 specs with trajectory+labels present ? my local manifest satisfies that, so I can unlock those 5 real-data tests. Let me push + open the PR, and run that unlock in parallel:

#### 95. user (2026-06-29T11:13:34.690Z)

##### push branch

```

remote: Create a pull request for 'p2b/promote-inactivity-temporal-density' on GitHub by
visiting:
remote: https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/p2b/promote-
inactivity-temporal-density
remote:
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch] p2b/promote-inactivity-temporal-density -> p2b/promote-inactivity-temporal-
density
branch 'p2b/promote-inactivity-temporal-density' set up to track 'origin/p2b/promote-inactivity-
temporal-density'.

```

##### create PR

```

Warning: 3 uncommitted changes
https://github.com/Khelsutra/badminton-highlight-indexer/pull/396

```

#### 96. user (2026-06-29T11:13:42.294Z)

##### now-unlocked served Gate-A regression on the real 15-clip golden set

```

 cw, points, fps, frame_width, video_path, idx_cfg
)
 # Gate-A signal ? net-crossing count for this window. Pure, GPU-free, always persisted
 # (single service feed crosses ~once; a real rally crosses >=2x). axis/net auto-
estimated
 # over the whole trajectory by rally_gate (camera-agnostic); reused across windows.
 gated = rally_gate.gate_windows(
 points,
 [(cw["start"], cw["end"])],
 fps,
 min_crossings=0,
 estimate=self._axis_estimate(points),
)
 stats["net_crossings"] = float(gated[0].crossings) if gated else 0.0

> fcfg = idx_cfg.fusion
 ^^^^^^^^^^^^^^^^^^
E AttributeError: 'dict' object has no attribute 'fusion'

backend/pipeline/segmenters/fusion_hybrid.py:135: AttributeError
3 failed, 2 passed in 5.53s

```

#### 97. assistant (2026-06-29T11:14:14.670Z)

**\*\*PR #396 is open\*\***: <https://github.com/Khelsutra/badminton-highlight-indexer/pull/396>

The served Gate-A unlock surfaced something important: **\*\*3 of 5 failed ? but not from my**

change.\*\* The error is `AttributeError: 'dict' object has no attribute 'fusion'` at  
`fusion\_hybrid.py:135` (`idx\_cfg.fusion`), a config-shape mismatch entirely unrelated to the  
`inactivity\_temporal\_density` bool. This is a pre-existing issue exposed only because these  
tests normally skip (golden absent). Let me confirm it's pre-existing and understand whether  
it's a \*production\* bug or a \*test-harness\* bug ? that distinction matters for the launch call.  
Also verifying the PR diff is clean:

#### 98. user (2026-06-29T11:14:15.138Z)

```
1 """SERVED Gate-A verification: run the *production* ``FusionSegmenter`` decision seams
2 on a cached WASB trajectory and score the resulting AI-handoff windows vs golden labels.
3
4 The offline litmus (``backend.eval.ablation`` / ``fusion_compare``) measures Gate A on
5 the
6 **permissive proposal windowing** (``distill_local.PROPOSAL`` = recall-oriented) re-
7 scored
8 through the experiment harness's ``_gate_mask``. That validates the *signal*. It does
9 NOT
10 exercise the **served path**: in production the windows come from
11 ``trajectory_to_action_windows`` at the production operating point
12 (``velocity_threshold=0.02, dead_time_merge_gap=2.0, min_action_window_duration=2.0``),
13 the
14 net-crossing count is computed by ``FusionSegmenter._enrich_candidate_stats``, and the
15 gate is
16 ``FusionSegmenter._select_for_handoff``. THIS module drives exactly those production
17 methods ?
18 no reimplementations of the windowing or the gate ? on a cached trajectory CSV, so the
19 verification reflects what ``process_video`` actually hands to Gemini.
20
21 GPU-free and Gemini-free: it stops at the handoff-selection boundary (the windows that
22 WOULD
23 be sent to the AI). With the person cue OFF (the default) nothing here loads torch or
24 touches
25 the GPU ? it just re-scores cached WASB trajectories.
26
27 Run (owner box, golden trajectories present):
28 python -m backend.eval.served_gate_a --manifest output/human_lovo_manifest.json
29
30 """
31
32 from __future__ import annotations
33
34 import argparse
35 import json
36 from dataclasses import dataclass
37 from typing import Any, Dict, List, Optional, Tuple
38
39 from backend.eval import segmentation_metrics as _segm
40 from backend.eval.calibrate_wasb import load_golden_gts
41 from backend.eval.metrics import Interval, Score, score
42 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
43 from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
44
45 # Production windowing operating point (config.json indexing defaults ? the values
46 # TrajectoryHybridSegmenter.process_video reads for the served fusion run).
47 PROD_VELOCITY_THRESH = 0.02
48 PROD_MERGE_GAP = 2.0
49 PROD_MIN_WINDOW = 2.0
50
51 def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:
52 """The ``indexing`` config dict the served FusionSegmenter seams read ? mirrors
53 config.json's ``indexing.fusion`` block, with Gate A at ``min_crossings`` and the
54 person cue OFF (so no GPU / torch is touched)."""
55 return {
56 "fusion": {
```

```

49 "substrate": "wasb",
50 "min_crossings": min_crossings,
51 "min_players": 0,
52 "cue_flags": {},
53 },
54 "wasb": {"velocity_threshold": PROD_VELOCITY_THRESH},
55 "dead_time_merge_gap": PROD_MERGE_GAP,
56 "min_action_window_duration": PROD_MIN_WINDOW,
57 }
58
59
60 def served_handoff_windows(
61 trajectory_csv: str, fps: float, frame_width: float, min_crossings: int
62) -> List[Interval]:
63 """Run the PRODUCTION FusionSegmenter decision path on a cached trajectory and
return the
64 windows that would reach the AI handoff (i.e. the served rally candidates).
65
66 Uses the real ``FusionSegmenter`` seam methods ? ``_enrich_candidate_stats``
(computes
67 ``net_crossings``) and ``_select_for_handoff`` (Gate A) ? over windows from the
production
68 ``trajectory_to_action_windows`` operating point. No GPU, no Gemini: the person cue
is OFF.
69
70 """
71 points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
72 windows = TrackNetRunner.trajectory_to_action_windows(
73 points,
74 fps=fps,
75 frame_width=frame_width,
76 velocity_thresh=PROD_VELOCITY_THRESH,
77 merge_gap=PROD_MERGE_GAP,
78 min_window_duration=PROD_MIN_WINDOW,
79)
80 idx_cfg = _served_idx_cfg(min_crossings)
81 # Construct the real served segmenter (db=None, config carries the indexing block).
We only
82 # call the pure decision seams ? no process_video, no DB, no provider ? exactly as
the
83 # registry constructs it for `name`/`description`.
84 seg = FusionSegmenter(None, {"indexing": idx_cfg, "sport": "badminton"})
85 # The served per-window stats (incl. net_crossings) via the production enrichment
hook.
86 idx_cfg_typed: Any = idx_cfg # type: ignore[annotation-type-mismatch]
87 window_stats = [
88 seg._enrich_candidate_stats(
89 cw, points, fps, frame_width, video_path="", idx_cfg=idx_cfg_typed
90)
91 for cw in windows
92]
93 keep = seg._select_for_handoff(windows, window_stats, idx_cfg_typed)
94 return [(windows[i]["start"], windows[i]["end"]) for i in keep]
95
96 @dataclass
97 class ServedVideoResult:
98 name: str
99 num_gt: int
100 on: Score # Gate-A ON (min_crossings=2)
101 off: Score # Gate-A OFF (min_crossings=0, control)
102 seg_ratio_on: float # |preds| / |gts| with Gate-A ON (over-segmentation)
103 seg_ratio_off: float
104 n_handoff_on: int
105 n_handoff_off: int
106

```



```

107 @property
108 def delta_f1(self) -> float:
109 return self.on.f1 - self.off.f1
110
111
112 def evaluate_video(
113 spec: Dict[str, Any], iou: float = 0.5, on_crossings: int = 2
114) -> ServedVideoResult:
115 """Score the served Gate-A ON (min_crossings=on_crossings) vs OFF (0) handoff
116 windows for
117 one golden video against its golden labels."""
118 fps, fw = float(spec["fps"]), float(spec["frame_width"])
119 traj, labels = str(spec["trajectory"]), str(spec["labels"])
120 gts = load_golden_gts(labels)
121 preds_on = served_handoff_windows(traj, fps, fw, min_crossings=on_crossings)
122 preds_off = served_handoff_windows(traj, fps, fw, min_crossings=0)
123 return ServedVideoResult(
124 name=str(spec["name"]),
125 num_gt=len(gts),
126 on=score(preds_on, gts, iou),
127 off=score(preds_off, gts, iou),
128 seg_ratio_on=_segm.segment_count_ratio(preds_on, gts),
129 seg_ratio_off=_segm.segment_count_ratio(preds_off, gts),
130 n_handoff_on=len(preds_on),
131 n_handoff_off=len(preds_off),
132)
133
134 def evaluate_manifest(
135 specs: List[Dict[str, Any]], iou: float = 0.5, on_crossings: int = 2
136) -> List[ServedVideoResult]:
137 return [evaluate_video(s, iou, on_crossings) for s in specs]
138
139
140 def aggregate(results: List[ServedVideoResult]) -> Dict[str, Any]:
141 """Mean ON/OFF F1 + over-seg ratio, the paired ?F1, and the per-video win count."""
142 n = len(results)
143
144 def _mean(getter) -> float:
145 return sum(getter(r) for r in results) / n if n else 0.0
146
147 deltas = [r.delta_f1 for r in results]
148 wins = sum(1 for d in deltas if d > 1e-9)
149 losses = sum(1 for d in deltas if d < -1e-9)
150 return {
151 "num_videos": n,
152 "mean_f1_on": _mean(lambda r: r.on.f1),
153 "mean_f1_off": _mean(lambda r: r.off.f1),
154 "mean_delta_f1": _mean(lambda r: r.delta_f1),
155 "mean_precision_on": _mean(lambda r: r.on.precision),
156 "mean_precision_off": _mean(lambda r: r.off.precision),
157 "mean_recall_on": _mean(lambda r: r.on.recall),
158 "mean_recall_off": _mean(lambda r: r.off.recall),
159 "mean_seg_ratio_on": _mean(lambda r: r.seg_ratio_on),
160 "mean_seg_ratio_off": _mean(lambda r: r.seg_ratio_off),
161 "wins": wins,
162 "losses": losses,
163 }
164
165
166 def format_table(
167 results: List[ServedVideoResult], agg: Dict[str, Any], iou: float
168) -> str:
169 lines = [
170 f"=== SERVED Gate-A verification (production FusionSegmenter path, IoU>={iou})"

```

```

===",
171 f" {'video':<24s} {'gt':>3s} {'F1_off':>7s} {'F1_on':>7s} {'?F1':>7s} "
172 f"{'segR_off':>9s} {'segR_on':>8s} {'hand_off':>9s} {'hand_on':>8s}",
173]
174 for r in results:
175 lines.append(
176 f" {'r.name':<24s} {r.num_gt:>3d} {r.off.f1:>7.3f} {r.on.f1:>7.3f} "
177 f"{'r.delta_f1':>7.3f} {r.seg_ratio_off:>9.2f} {r.seg_ratio_on:>8.2f} "
178 f"{'r.n_handoff_off':>9d} {r.n_handoff_on:>8d}"
179)
180 lines += [
181 f" {'-' * 92}",
182 f" MEAN F1 {agg['mean_f1_off']:.3f} (off) -> {agg['mean_f1_on']:.3f} (on) "
183 f"?F1={agg['mean_delta_f1']:+.3f} "
184 f"P {agg['mean_precision_off']:.3f}->{agg['mean_precision_on']:.3f} "
185 f"R {agg['mean_recall_off']:.3f}->{agg['mean_recall_on']:.3f}",
186 f" over-seg ratio {agg['mean_seg_ratio_off']:.2f} (off) -> "
187 f"{agg['mean_seg_ratio_on']:.2f} (on) | wins {agg['wins']}/{agg['num_videos']}"
188 f"(losses {agg['losses']})",
189]
190 return "\n".join(lines)
191
192
193 def run(
194 manifest_path: str, iou: float = 0.5
195) -> Tuple[List[ServedVideoResult], Dict[str, Any]]:
196 with open(manifest_path, encoding="utf-8") as f:
197 specs = json.load(f)
198 results = evaluate_manifest(specs, iou=iou)
199 return results, aggregate(results)
200
201
202 def main(argv: Optional[List[str]] = None) -> None:
203 ap = argparse.ArgumentParser(
204 description="Verify served Gate-A on real footage via the production
205 FusionSegmenter path."
206)
207 ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
208 ap.add_argument("--iou", type=float, default=0.5)
209 args = ap.parse_args(argv)
210 results, agg = run(args.manifest, iou=args.iou)
211 print(format_table(results, agg, args.iou))
212
213 if __name__ == "__main__":
214 main()
215

```

## 99. user (2026-06-29T11:14:16.225Z)

```

118 idx_cfg: IndexingConfig,
119) -> Dict[str, Any]:
120 stats = super()._enrich_candidate_stats(
121 cw, points, fps, frame_width, video_path, idx_cfg
122)
123 # Gate-A signal ? net-crossing count for this window. Pure, GPU-free, always
124 persisted
125 # (single service feed crosses ~once; a real rally crosses >=2x). axis/net auto-
126 estimated
127 # over the whole trajectory by rally_gate (camera-agnostic); reused across
128 windows.
129 gated = rally_gate.gate_windows(
130 points,
131 [(cw["start"], cw["end"])],

```

```

129 fps,
130 min_crossings=0,
131 estimate=self._axis_estimate(points),
132)
133 stats["net_crossings"] = float(gated[0].crossings) if gated else 0.0
134
135 fcfg = idx_cfg.fusion
136 if fcfg.cue_flags.get("person", False):
137 stats.update(
138 self._person_features(cw, points, fps, frame_width, video_path, fcfg)
139)
140 return stats
141
142 # P3 (learned fusion) first-step landing note (per MULTI_SIGNAL_FUSION_PLAN +
NEXT_STEPS 2026-06-14):
143 # Person features (from _person_features when cue_flags.person) are now persisted
for the harness
144 # `compare` LOVO litmus on the 6 golden videos. Eager-person-compute optimisation
(compute only)
145 # for shuttle-seeded windows + padding, not whole video) is already in place via the
lazy
146 # self._person + windowed detections_per_frame. Next: actual harness run + lift
measurement
147 # (owner/hardware) before any served promotion. Flag-gated (default person OFF). See
148 # docs/NEXT_STEPS.md "Rally-detection quality track" and EXPERIMENT_HARNESS.md for
discipline.
149
150 def _axis_estimate(self, points: List[Any]) -> tuple:
151 """Cache rally_gate's (axis, net, span) play-axis estimate per trajectory so
it's
152 computed once per video, not once per candidate window (it depends only on
points)."""

```

#### 100. assistant (2026-06-29T11:15:37.903Z)

Now it's clear: this is a **test-harness bug**, not a production bug.

``served_gate_a._served_idx_cfg`` builds the indexing config as a **dict** (and casts it to ``Any`` with a ``type: ignore`` to silence the mismatch), but ``FusionSegmenter._enrich_candidate_stats`` accesses ``idx_cfg.fusion`` as a **typed attribute** (line 135). Production ``process_video`` passes a typed ``IndexingConfig``, so production is fine ? but this means the **only litmus** that validates the served `FusionSegmenter` path on real footage is itself broken and has never actually run green on real data (it skips in CI, crashes when run). Pre-existing, unrelated to P2b. A real verification gap worth tracking.

Let me verify the PR diff is clean, then update the doc with the real PR number:

#### 101. user (2026-06-29T11:15:49.923Z)

##### files in PR #396 (should be exactly 4)

```

backend/config/models.py
docs/RALLY_DETECTION_FIXES_PLAN.md
tests/test_rally_seg_eval.py
tests/test_windowing.py

```

##### confirm 2 served\_gate\_a tests that DID pass (the non-evaluate\_manifest ones)

3 failed, 2 passed in 5.62s

#### 102. user (2026-06-29T11:15:50.988Z)

The file `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY_DETECTION_FIXES_PLAN.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 103. assistant (2026-06-29T11:15:53.975Z)

PR diff is clean (exactly 4 files). Let me finish the doc PR-number references and commit the update:

**104. user (2026-06-29T11:15:58.018Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

**105. user (2026-06-29T11:16:00.480Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

**106. user (2026-06-29T11:16:07.688Z)**

pushed; PR commits:  
ebd019c docs: link P2b promotion to PR #396  
f188070 feat(config): promote inactivity\_temporal\_density to default-ON (P2b)

**107. assistant (2026-06-29T11:16:35.407Z)**

PR #396 now carries both commits. Now let me file the two GitHub issues ? the stray-dirs one you asked for, plus the served-gate verification gap I just found (both tracked for later, not fixed in this session):

**108. user (2026-06-29T11:16:51.220Z)**

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/397>  
----  
<https://github.com/Khelsutra/badminton-highlight-indexer/issues/398>

**109. assistant (2026-06-29T11:17:44.890Z)**

Both issues filed: **##397** (stray dirs) and **##398** (served-gate harness crash).

I've run everything runnable with the data present. The one remaining "now-unlockable" path is the video-backed GPU fusion ablation ? let me check whether the golden videos are actually available to demonstrate the GPU person-detection path, or whether it's blocked by the known corpus-video migration gap:

**110. user (2026-06-29T11:17:51.312Z)**

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

**111. user (2026-06-29T11:18:32.006Z)**

**golden-clip video availability in GCS (15 golden names vs videos/ present)**

gs://khelsutra-rally-corpus/videos/GX010142\_1080p.mp4  
gs://khelsutra-rally-corpus/videos/GX010143\_1080p.mp4  
gs://khelsutra-rally-corpus/videos/GX010147\_1080p.mp4  
gs://khelsutra-rally-corpus/videos/GX030094\_proxy.mp4  
gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4  
gs://khelsutra-rally-corpus/videos/mahadevpura\_1.mp4  
gs://khelsutra-rally-corpus/videos/mahadevpura\_smoke\_180s.mp4  
gs://khelsutra-rally-corpus/videos/SHANAN/

**match check (does a video plausibly exist per golden name?)**

adarsh\_avi\_singles: (no video)  
kushagra\_singles: (no video)  
largetest\_doubles: (no video)

```
gbaaddy: (no video)
testlarge_short: (no video)
mahadevpura_singles: (no video)
mahadevpura_2: (no video)
GX010128: (no video)
mahadevpura_1: VIDEO PRESENT
GX030094: VIDEO PRESENT
Badminton_BXH_2: (no video)
GX010141: (no video)
GX010137: (no video)
Boxhill_Doubles: (no video)
GX020094: (no video)
```

#### sizes of any present golden videos

```
1429944539 2026-06-20T14:27:35Z gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4
548827049 2026-06-22T10:21:57Z gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
90194321 2026-06-20T14:20:32Z gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
```

#### 112. assistant (2026-06-29T11:19:01.627Z)

Confirmed: only \*\*2 of 15\*\* golden videos are in GCS (`mahadevpura\_1` 548MB, `GX030094` 1.4GB) ? so the video-backed GPU \*\*fusion ablation cannot run\*\* here, blocked by the known corpus-video migration gap (DATA\_IN\_GCS \$5/\$7b: 14/15 videos not yet migrated), \*\*not\*\* by hardware. I can still demonstrate the GPU person-detection path is operational on a single clip. Let me first verify the GPU deps are importable before committing to a 548MB download:

#### 113. user (2026-06-29T11:19:09.888Z)

```
torch: 2.12.1+cu130
torchvision: 0.27.1+cu130
cv2: 4.13.0
cuda: True NVIDIA GeForce RTX 2070 Super with Max-Q Design
PersonDetector import: OK
READY
```

#### 114. assistant (2026-06-29T11:19:21.384Z)

All GPU deps present (torch 2.12 + torchvision 0.27 + cv2 4.13, CUDA available, PersonDetector imports). Let me run a real single-clip GPU smoke on `mahadevpura\_1` to prove the production person-detection feature path is operational. First the video download:

#### 115. user (2026-06-29T11:27:08.105Z)

Command running in background with ID: blah9bwi0. Output is being written to: /tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/blah9bwi0.output. You will be notified when it completes. To check interim output, use Read on that file path.

#### 116. assistant (2026-06-29T11:27:27.986Z)

The 548 MB download is running in the background. While it transfers, let me prepare the single-clip manifest (with explicit `video\_path`) so the GPU run is ready to fire the moment the video lands:

#### 117. assistant (2026-06-29T11:27:30.346Z)

```
[
 {
 "name": "mahadevpura_1",
 "trajectory": "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b_golden/trajectories/mahadevpura_1.csv",
 "labels": "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b_golden/labels/2026-06-17_mahadevpura_1.rallies.csv",
 "fps": 59.94,
```

```

 "frame_width": 1920.0,
 "video_path": "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/videos/mahadevpura_1.mp4"
 }
]

```

#### 118. user (2026-06-29T11:27:31.438Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b\_golden/manifest\_gpu\_smoke.json (file state is current in your context ? no need to Read it back)

#### 119. assistant (2026-06-29T11:27:45.143Z)

Manifest is staged. The video download is running in the background (~2-3 min at observed throughput); I'll be notified when it completes, then run the GPU person-detection smoke on `mahadevpura\_1` and write up the full launch-readiness report.

**\*\*Quick status while that transfers\*\*** ? the core deliverable is done and on GitHub:

- ? **\*\*P2b decided: PROMOTE.\*\*** Golden A/B at SERVED (15 clips, 595 rallies): tolF1 0.353?0.364 (no regression), over-seg significantly reduced, guard PASS. Verified with the repo's own `promote\_if\_served\_safe`.

- ? **\*\*PR [#396](https://github.com/Khelsutra/badminton-highlight-indexer/pull/396)\*\*** opened (flag flip + tests + docs; suite 1503?/10 skip, ruff+mypy clean).

- ? **\*\*Issues [#397](https://github.com/Khelsutra/badminton-highlight-indexer/issues/397)\*\*** (stray dirs) and **\*\*[#398](https://github.com/Khelsutra/badminton-highlight-indexer/issues/398)\*\*** (a real pre-existing served-path verification gap I found) filed.

Standing by for the download to finish.

#### 120. user (2026-06-29T11:35:09.950Z)

there is an F:\GoPro Recorded Videos on a mounted disk which holds *\*all\** the data we ever used for training, eval etc. can you do a deep pass on it and index it so that any repo can refer it and it also removes the dedup. Can fork off a separate chip to isolate it separately from this wokrr

#### 121. assistant (2026-06-29T11:35:44.125Z)

Good ? that's a substantial, separable task. Before I "fork off a separate chip" for it, I need to confirm the drive is actually mounted (on my earlier recon `/mnt/f` existed but was **\*\*empty / not a mountpoint\*\*** ? it may have been mounted since). Let me verify the path and scope the data, and check on the running download:

#### 122. user (2026-06-29T11:35:50.929Z)

##### mounted disks

Filesystem	Size	Used	Avail	Use%	Mounted on
none	7.8G	4.0K	7.8G	1%	/mnt/wsl
none	7.8G	76K	7.8G	1%	/mnt/wslg
none	7.8G	76K	7.8G	1%	/mnt/wslg/versions.txt
none	7.8G	76K	7.8G	1%	/mnt/wslg/doc
C:\	746G	504G	243G	68%	/mnt/c

##### /mnt contents

```

c
f
wsl
wslg

```

##### /mnt/f now?

/mnt/f is not a mountpoint

```
total 8
drwxrwxrwx 2 root root 4096 Jun 12 06:22 .
drwxr-xr-x 6 root root 4096 Jun 12 06:22 ..
```

#### look for 'GoPro Recorded Videos' across mounts

##### download status (mahadevpura\_1.mp4)

```
total 528680
drwxr-xr-x 2 avidullu avidullu 4096 Jun 29 11:19 .
drwxr-xr-x 6 avidullu avidullu 4096 Jun 29 11:27 ..
-rw-r--r-- 1 avidullu avidullu 548827049 Jun 29 11:21 mahadevpura_1.mp4_.gstmp
```

##### downloading mahadevpura\_1.mp4 (548MB)

#### 123. assistant (2026-06-29T11:36:16.759Z)

The **\*\*F:** drive is not currently mounted into this WSL instance**\*\*** ? ``/mnt/f`` exists but is empty and ``df`` shows only ``C:`` mounted. The GoPro archive must be visible to Windows but not yet bridged into WSL. Let me confirm Windows sees F: and try to mount it (non-interactively so it can't hang):

#### 124. user (2026-06-29T11:36:22.622Z)

##### does Windows see an F: drive?

Name	GB_Used	GB_Free
C	503.6	242.2
E	0	0
F	1148.5	714.5
G	515.7	230.1

##### attempt non-interactive drvfs mount of F: -> /mnt/f

```
sudo: a password is required
exit: 0
```

##### recheck /mnt/f

#### 125. assistant (2026-06-29T11:37:44.495Z)

Windows sees the drive clearly: **\*\*F: = ~1.9 TB (1148 GB used, 714 GB free)\*\*** ? that's the GoPro master archive. But it's **\*\*not bridged into WSL\*\***, and I can't mount it myself: ``sudo mount -t drvfs F: /mnt/f`` requires your password (non-interactive sudo failed). So I need one decision from you on access, and one on how aggressive to be with dedup on what is your master copy of **\*all\*** training/eval data.

Two things to note about my plan:

- **\*\*"Fork off a separate chip"\*** ? I'll launch an isolated background agent dedicated to the F: index+dedup, so it runs independently of the P2b/launch-readiness work (which stays intact ? PR #396 is already up).
- **\*\*Safety\*\***: on a master archive holding "all the data we ever used," I will **\*\*index by content hash (MD5 = the repo's `video\_id`)** first and treat deletion as gated**\*\*** ? I won't destroy your only copy of anything without your explicit go-ahead.

#### 126. user (2026-06-29T11:37:47.700Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

#### 127. user (2026-06-29T11:47:43.534Z)

Your questions have been answered: "F: (the 1.9 TB GoPro archive) isn't mounted into WSL and I can't sudo-mount it. How should the forked agent reach it?"="I'll mount it (recommended)", "How aggressive should the dedup be on this master archive (it holds your only copy of the

data)?="Report + plan only (recommended)". You can now continue with these answers in mind.

#### 128. assistant (2026-06-29T11:48:34.476Z)

Got it ? **\*\*mount it yourself, and report-only dedup\*\*** (non-destructive). That's the safe choice on a master archive.

Please run this in this WSL session (sudo needs your password, which I can't supply):

```
```bash
sudo mount -t drvfs F: /mnt/f
```
```

Once it's mounted, just say "mounted" (or anything) and I'll **\*\*launch the isolated background agent immediately\*\***. Its scope will be strictly non-destructive:

- **\*\*Catalog\*\*** every video under `F:\GoPro Recorded Videos` (path, size, mtime, codec/duration where cheap) keyed by **\*\*MD5 = the repo's `video\_id`\*\*** ? staged hashing (size ? partial-hash ? full-MD5 on collisions) so it doesn't blindly read all 1.1 TB.
- **\*\*Dedup report\*\***: exact-duplicate groups (confirmed by MD5), reclaimable space, and a proposed canonical-to-keep per group ? **\*\*deletes nothing\*\***.
- **\*\*Cross-reference\*\*** against `gs://khelsutra-rally-corpus` + the 15 golden clips: which files are already migrated vs. the 14/15 still-missing corpus videos (closing the DATA\_IN\_GCS \$5 gap).
- **\*\*Output\*\*** a single `master\_video\_index.json` + a human-readable report any repo can point at (I'll propose publishing the index to GCS for cross-repo reference after you review it).

Meanwhile, let me check on the GPU-smoke download and finish the launch-readiness report in parallel:

#### 129. user (2026-06-29T11:48:41.698Z)

##### download dir state

```
total 528680
drwxr-xr-x 2 avidullu avidullu 4096 Jun 29 11:19 .
drwxr-xr-x 6 avidullu avidullu 4096 Jun 29 11:27 ..
-rw-r--r-- 1 avidullu avidullu 548827049 Jun 29 11:21 mahadevpura_1.mp4_.gstmp
```

##### task output tail

##### downloading mahadevpura\_1.mp4 (548MB)

##### final mp4 present + playable header?

still finalizing (.gstmp present: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b\_golden/videos/mahadevpura\_1.mp4\_.gstmp)

#### 130. assistant (2026-06-29T11:49:44.652Z)

The download's bytes are fully present (size matches GCS exactly) but gcloud is still finalizing the rename ? I'll let it complete on its own rather than interfere. The GPU is already proven live (deps import, CUDA available), so the smoke is optional polish, not load-bearing.

Before I write the launch call, I need the **\*\*happy-path quality numbers\*\*** and the **\*\*ship-gate target\*\*** to make it evidence-based (my A/B gave overall served quality but not the long-rally x court breakdown). Running the stratified eval (now reflecting the *\*promoted\** config) and pulling the ship target:

#### 131. user (2026-06-29T11:49:53.638Z)

stratified served eval (POST-promotion config: temporal\_density now ON) ? long-rally x court

**Long-rally lens ? stratified (long = duration > 5s; n = videos with >=1 golden rally in the stratum)**

```
stratum	court	n	tolF1	tolP	tolR	tIoU F1@0.5	seg_ratio
all	single	11	0.382 [0.270,0.483]	0.343 [0.238,0.445]	0.450 [0.321,0.563]	0.488	
```



[0.368,0.593] | 1.300 [0.992,1.580] |  
| all | multicourt | 4 | 0.313 [0.199,0.429] | 0.255 [0.180,0.346] | 0.415 [0.232,0.566] | 0.428  
[0.302,0.542] | 1.582 [1.193,1.877] |  
| all | all | 15 | 0.364 [0.271,0.450] | 0.319 [0.231,0.402] | 0.441 [0.332,0.537] | 0.472  
[0.376,0.558] | 1.375 [1.115,1.590] |  
| long>5s | single | 11 | 0.266 [0.171,0.368] | 0.330 [0.191,0.494] | 0.244 [0.159,0.329] |  
0.412 [0.278,0.532] | 0.840 [0.583,1.117] |  
| long>5s | multicourt | 4 | 0.258 [0.134,0.354] | 0.217 [0.110,0.307] | 0.323 [0.166,0.429] |  
0.439 [0.266,0.575] | 1.520 [1.322,1.708] |  
| long>5s | all | 15 | 0.264 [0.180,0.345] | 0.299 [0.187,0.425] | 0.265 [0.189,0.338] | 0.419  
[0.311,0.516] | 1.022 [0.783,1.273] |

### 132. user (2026-06-29T11:49:55.316Z)

#### ship-gate target references (NEXT\_STEPS + QUALITY\_REPORT)

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:11:0b. \*\*?  
[packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app  
? `docs/PACKAGING\_PLAN.md`.\*\* The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer  
--app` serves the \*\*bundled SPA\*\* single-origin, \*\*torch-free\*\*, with app-home state / ffmpeg  
resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,  
P2-launcher/P2a/P2d/P2e). \*\*? Next (fresh focused pushes):\*\* \*(1)\*\* P3 \*\*GPU validation +  
screenrecast\*\* (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB  
0-rally tuned-config blocker), \*(2)\*\* P2b-install (uv/script + truly-local default config),  
\*(3)\*\* P2c first-run wizard (cross-repo khelsutra SPA from `api/capabilities`), then P3-cloud  
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-  
session-packaging-app]]`.

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:18:5.  
\*\*[Claude] Per-video workspace P1?P6\*\* (P1 v6 DB migration ? P2 worker ? P3 per-video routing of  
`detector/+`tmp/` \*`[the `output/chunks\_` de-hardcode base = ? done via #282]\* ? P4 ? P5  
Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger  
migration ? author the version bump once.

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:20:7.

\*\*[owner] Set the ship-gate target (#172)\*\* ? needs corpus growth.

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:53:\*\*M0  
in-container quick wins are COMPLETE:\*\* ship-gate provisional blocker (\*\*#170\*\*), train/eval  
split guard (\*\*#171\*\*), the \*\*doc-consistency sweep\*\* (\*\*#174\*\* ? corrected 16 drifted docs vs  
the code), and the \*\*P3 person-feature fusion litmus machinery is already present + suite-  
green\*\* (`backend/eval/fusion\_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI /  
paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red  
repo-wide due to a \*\*GitHub Actions minutes/billing outage\*\* (account-level, not code) ? work is  
local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes /  
spending limit).

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:61:- Ship-  
gate target calibration ? \*\*#172\*\* (owner decision; needs corpus growth).

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:85:wrapper  
to 0.66 ? that is the owned model's ship target\*\* (its FPs are background-game pickups + dead-  
time merges,

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:119:

Priority order for new videos: \*(a) multi-court badminton\*\* (the target distribution AND where  
the ship target

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:125:2.

\*\*[owner decision] Set the ship-gate target\*\* ? now informed: floor = heuristic 0.480  
(necessary), \*\*multi-court

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md:127:

(not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-  
court folds with

---- QUALITY\_REPORT headline ----

/home/avidullu/projects/khelsutra-guru/badminton-highlight-

indexer/docs/RALLY\_DETECTION\_QUALITY\_REPORT.md:19:learns to weight ? never a hand-coded branch ?  
and ships \*\*default-OFF\*\*, promoted only on a measured

/home/avidullu/projects/khelsutra-guru/badminton-highlight-

indexer/docs/RALLY\_DETECTION\_QUALITY\_REPORT.md:24:Since the last checkpoint (2026-06-14) the  
headline result moved from "one promoted-then-reverted cue"

/home/avidullu/projects/khelsutra-guru/badminton-highlight-

indexer/docs/RALLY\_DETECTION\_QUALITY\_REPORT.md:41:## Headline results at a glance

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:49:| **Gemini cloud reference** | clean **0.93**
/ multi-court **0.66** (the owned-model ship target is the 0.66) |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:77:fusion gates target, and exactly why we had to
build an over-segmentation-aware metric before we could
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:107:ships first) and a **feature-level path**
(the learned route), where cues emit small,
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:108:scale/translation-invariant per-window
features (counts, ratios, zone-membership, two-sidedness ? never
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:155:We found plain temporal-IoU-F1@0.5 is the
wrong ruler in both directions: for a ~6 s rally it
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:163:construction: it *lowers* our headline number
(tIoU-F1 0.440 ? tolerance-F1 0.323 on the milestone) by
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:192:ship target**, not the clean ceiling.
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:199:| Gen-0 owned learned head (LOVO, n=6) |
0.744 | 0.561 |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:204:0.66 target, not an architecture-sized one.
(The aggregate **heuristic LOVO floor** across n=6 is
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:206:ship=False**.)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:208:### 4.2 The over-segmentation milestone (this
cycle's headline)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:224:| Comparison | ?tolF1 | 95 % CI | sign-test |
significant? |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:261:| mahadevpura-2 (40) | F1@0.5 / F1@0.25 |
0.557 / 0.835 | 0.296 / 0.722 |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:262:| GX010128 (52) | F1@0.5 / F1@0.25 | 0.813 /
0.857 | 0.457 / 0.756 |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:265:(F1@0.5)**. On GX010128 local even **out-
recalls Gemini** ? it overlaps all 52 golden rallies; Gemini
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:273:| GX010128 | 0.78 | 0.98 | **2.12** | 0.76 |
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md:278:mahadevpura-2 hit 1.03x of Gemini's rally
count yet F1@0.5 only 0.150 with W1+W2 ? right count, wrong

```

### 133. assistant (2026-06-29T11:50:55.378Z)

\*\*\*"mounted"\*\*\* ? let me verify F: is accessible and scope the GoPro archive, then immediately launch the isolated background agent for the index+dedup.

### 134. user (2026-06-29T11:50:59.128Z)

/mnt/f now mounted?

/mnt/f is a mountpoint

F: 1.9T 1.2T 715G 62% /mnt/f

top-level of F:

total 17412

drwxrwxrwx 1 root root 512 Aug 22 2021 \$RECYCLE.BIN

drwxrwxrwx 1 root root 512 Dec 31 1979 .

```
drwxr-xr-x 6 root root 4096 Jun 12 06:22 ..
drwxrwxrwx 1 root root 512 Oct 31 2024 .Spotlight-V100
drwxrwxrwx 1 root root 512 Oct 31 2024 .fseventsd
drwxrwxrwx 1 root root 512 Oct 31 2024 Data
-rwxrwxrwx 1 root root 7966967 Jun 27 2020 Install Security for Mac.zip
-rwxrwxrwx 1 root root 8933456 Jun 27 2020 Install Security for Windows.exe
drwxrwxrwx 1 root root 512 Aug 22 2021 System Volume Information
drwxrwxrwx 1 root root 512 Mar 6 08:29 [Khelsutra] GoPro Hero Black 12
```

### locate 'GoPro Recorded Videos'

```
/mnt/f/[Khelsutra] GoPro Hero Black 12
/mnt/f/Data/Initial GoPro with Sports Recordings
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos
```

### 135. assistant (2026-06-29T11:51:44.896Z)

F: is mounted (1.9 TB, 1.2 TB used) and the archive is at `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos` (plus a second data folder `/mnt/f/Data/Initial GoPro with Sports Recordings`). Launching the isolated background agent now ? strictly read-only, report-only:

### 136. user (2026-06-29T11:52:19.038Z)

Async agent launched successfully.

agentId: a5598ac50dde6368 (internal ID - do not mention to user. Use SendMessage with to: 'a5598ac50dde6368', summary: '<5-10 word recap>' to continue this agent.)

The agent is working in the background. You will be notified automatically when it completes.

Do not duplicate this agent's work ? avoid working with the same files or topics it is using.

output\_file: /tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/a5598ac50dde6368.output

Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and reading it will overflow your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.

### 137. assistant (2026-06-29T11:52:54.037Z)

The F: index+dedup agent is now running in the background, fully isolated (read-only, report-only). I'll be notified when it finishes. The P2b/launch work is untouched by it.

Now the deliverable you asked for. The GPU-smoke video download is still finalizing (optional polish ? GPU liveness is already proven), so here is the report; I'll append the smoke result if/when it lands.

### 138. user (2026-06-29T11:52:57.129Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

### 139. assistant (2026-06-29T11:53:41.732Z)

## ? Launch-Readiness Report ? Rally-Detection Quality Track (P2b close-out)

\*Prepared 2026-06-29. Worked in the WSL clone at [badminton-highlight-indexer](badminton-highlight-indexer/), synced to remote HEAD `0c9c693`. \*\*Recommendation only ? the launch call is yours.\*\*

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### 1. P2b result ? the finding-B promotion decision

\*\*Decision: PROMOTE\*\* ? `IndexingConfig.inactivity\_temporal\_density` flipped `False ? True`.

\*\*PR [#396](https://github.com/Khelsutra/badminton-highlight-indexer/pull/396)\*\*.

I ran the canonical A/B at the \*\*SERVED\*\* operating point on the full 15-clip golden corpus (595

rallies). I did **not** need the GPU or the source videos: the harness consumes the **\*cached WASB trajectory CSVs\* + golden GTs**, all of which I pulled from ``gs://khelsutra-rally-corpus`` (GTs built via the repo's own ``load_gt_intervals``, so byte-identical to ``fusion_golden``).

```
metric (SERVED)	base (OFF)	candidate (ON)	?	verdict
tolF1 (headline)	0.353	0.364	**+0.010** · CI [?0.014,+0.034] · 10W/4L · p=0.18	
neutral-to-positive, **no regression**				
`\|?end\|` (s)	2.97	2.72	?0.25	rally-end error improved
segment_count_ratio	?	?	**?0.103**	p=0.006 ? over-seg ?
split_rate	?	?	**?0.010**	p=0.016 ?
R1 over-seg guard	?	?	net 0.397?0.378	**PASS**
```

The repo's own gate confirms it: ``served_regresses(eps=0.0) = False`` ? ``promote_if_served_safe(structural=True) = promote``. Per the §8 Q2 rule (**\*promote iff tolF1 does not regress\***) ? **\*\*PROMOTE\*\***.

**\*\*Honest caveat:\*\*** the stricter honest-**\*lift\*** bar (``structural=False``) returns "no significant lift" (CI spans 0). This is a **\*\*do-no-harm correctness fix\*\*** ? it removes a real bug (sparse-track tails inflating rally ends) with a measured reduction in over-segmentation and zero regression ? not a measured quality **\*win\***. That is exactly the bar the plan set for it.

---

## 2. Test-suite status

**\*\*Offline suite (authoritative):** ``1503 passed, 10 skipped`` · ``ruff` clean` · ``mypy` clean` (138 files).**\*\*** The flag flip broke nothing; tests updated to the new default (both flag states still covered).

**\*\*Now-unlocked paths (GPU/data live this session):\*\***

```
Path	Result
P2b A/B on real 15-clip golden trajectories	? **ran** (GPU-free; the expensive WASB
extraction was already cached in GCS)	
GPU hardware + stack	? **live & verified** ? RTX 2070 Super, CUDA 13.2, `torch 2.12+cu130`,
`torchvision`, `cv2` all import	
`served_gate_a_regression` (production served path on real footage)	?? **2/5 pass, 3 ERROR**
? pre-existing harness bug, see §3	
Video-backed **fusion ablation** (`test_ablation` LITMUS)	? **cannot run** ? needs ?6 video-
backed clips; only **2 of 15** corpus videos are in GCS (the migration gap), not a hardware	
limit	
Native-WASB **GPU byte-parity**	? owner-box manual diff (the pytest byte-identity portion is
green in the suite)	
2x `ablation` PDF tests	skipped ? `reportlab` not installed (trivial; low value)
1x `test_app_home`	skipped ? Windows-only
```

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## 3. Open guardrail items

- \*\*? WASB C3 checkpoint provenance (licensing).\*\*** ``wasb_badminton_best.pth.tar`` provenance is an unresolved legal item before any external share (CLAUDE.md commercial-clean rule; flagged in QUALITY\_REPORT). **\*\*Commercial-launch blocker.\*\***
- \*\*? Ship-gate target is owner-unset (#172).\*\*** There is no ratified bar to declare GO against. Suggested shape: `*LOVO F1@tIoU0.5 ? 0.70 multi-court, p<0.05 vs heuristic*`; heuristic floor = 0.480.
- \*\*? Served-path verification gap (new ? issue [#398](https://github.com/Khelsutra/badminton-highlight-indexer/issues/398)).\*\*** The **\*only\*** litmus that exercises the production ``FusionSegmenter`` served path on real footage **\*\*crashes\*\*** (``dict`` object has no attribute ``fusion``, ``fusion_hybrid.py:135``). Root cause is a test-harness config-shape bug (``served_gate_a`` builds a dict; the override reads ``fusion``); **\*\*production passes a typed config so is unaffected\*\*** ? but it means that path is currently **\*\*unvalidated on real footage\*\***

outside a MagicMock. Pre-existing, unrelated to P2b.

4. **Corpus-video migration gap.** **14 of 15** golden videos are not yet in GCS (DATA\_IN\_GCS §5/§7b). Blocks GPU-free cloud eval/reproducibility and the fusion ablation. **(Your F: archive index ? now running ? directly addresses this.)**

5. **Cloud GPU quota = 0** (DATA\_IN\_GCS §7b#4) ? no cloud GPU VM ever provisioned; no real cloud train/infer run yet.

6. **CI red repo-wide** ? a GitHub Actions **billing/minutes outage** (account-level, not code; NEXT\_STEPS M0).

7. **Repo hygiene** ? stray `gcs:/ gs:/ local:/` dirs in repo root (issue [#397](https://github.com/Khelsutra/badminton-highlight-indexer/issues/397)).

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## 4. Evidence-based quality picture

The number I measured is the **local heuristic windowing floor**, which is **not** the served-quality tier:

```
Tier	F1@tIoU0.5 (multi-court)	Status
Local heuristic (this A/B, post-promotion)	all-courts 0.47, multicourt 0.43, long-rally>5s 0.42	the free FLOOR ? below any plausible ship bar
Gemini cloud (serves the quality tier today)	0.66 multi-court (0.93 clean)	the owned-model ship target
Gen-0 owned learned head (LOVO, n=6)	0.744	clears 0.70 ? but owner-gated, n=6 only, not in production
```

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## 5. GO / NO-GO recommendation

- **GO** ? merge PR #396 (the P2b slice). **Clean, validated, do-no-harm; suite green; the last open row of the fixes plan is closed and the plan honestly marked DONE.**

- **NO-GO** ? external/commercial rally-detection product launch, **for now**. **Reasons, in priority order:** **(1)** WASB weight provenance (licensing) is unresolved [#1]; **(2)** the ship-gate target is unratified [#2], and the only tier that would meet it (the owned Gen-0 head, 0.744) is owner-gated, n=6, and not in production ? the local floor (~0.47) is well under any bar; **(3)** the production served path is unverified on real footage [#398]; **(4)** cloud GPU isn't provisioned [#5].

- A **Gemini-served** launch is quality-supported (0.66?0.93) but still gated by **#1** (WASB provenance) and **#398**.

**Highest-leverage unblocks (yours to make):** set the ship-gate target (#172), and resolve the WASB weight-provenance licensing question. Those two convert most of the above from "unknown" to a decidable GO.

**I'm stopping here for your signoff on the launch call.**

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### Parallel work in flight

- **F: archive index+dedup** ? background agent running (read-only, report-only); I'll report the catalog, dedup groups + reclaimable space, and the GCS migration-gap list when it finishes.

- **GPU fusion smoke** (`mahadevpura\_1`) ? download finalizing; optional, since GPU liveness is already proven. I'll append it or drop it.

A couple of things worth your eye when you have a moment: should I **install `reportlab`** to close the 2 PDF-test skips, and do you want the served-gate fix (#398) prioritized as its own PR? Neither blocks P2b.

**140. user (2026-06-29T12:01:35.375Z)**

<task-notification>

<task-id>a5598ac50dde6368</task-id>

```
<tool-use-id>toolu_01LzABpPipZRcfuD3jbYihrv</tool-use-id>
<output-file>/tmp/claude-1000/-home-avidullu-projects-khelsutra-
guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/a5598ac50ddee6368.output</output-file>
<status>completed</status>
<summary>Agent "Index + dedup F: GoPro archive" finished</summary>
<note>A task-notification fires each time this agent stops with no live background children of
its own. The user can send it another message and resume it, so the same task-id may notify more
than once.</note>
<result>The waiter is still running. I'll await its completion notification rather than continue
manual polling. The confirm full-MD5 pass needs ~30 more minutes to read the remaining large
files. I'll pause and wait for the system to notify me when the background job `bhon4nb6e`
completes.</result>
<usage><subagent_tokens>78417</subagent_tokens><tool_uses>44</tool_uses><duration_ms>554765</dur
ation_ms></usage>
</task-notification>
```

#### 141. assistant (2026-06-29T12:01:57.158Z)

The F: agent paused itself mid-run ? it kicked off a background full-MD5 confirmation pass (`bhon4nb6e`, ~30 min) and stopped to wait, which returned control to me. Let me check what it has already written and whether that hashing job is still alive, without touching its transcript:

#### 142. user (2026-06-29T12:02:03.174Z)

##### artifacts so far

total 764

drwxr-xr-x	2	avidullu	avidullu	4096	Jun 29 12:00	.
drwxr-xr-x	11	avidullu	avidullu	4096	Jun 29 11:52	..
-rw-r--r--	1	avidullu	avidullu	13730	Jun 29 11:59	build_outputs.py
-rw-r--r--	1	avidullu	avidullu	197350	Jun 29 11:53	catalog_all_files.nul
-rw-r--r--	1	avidullu	avidullu	245895	Jun 29 11:53	catalog_raw.tsv
-rw-r--r--	1	avidullu	avidullu	0	Jun 29 11:53	classify.err
-rw-r--r--	1	avidullu	avidullu	3054	Jun 29 11:53	classify.py
-rw-r--r--	1	avidullu	avidullu	2985	Jun 29 11:54	confirm_groups.json
-rw-r--r--	1	avidullu	avidullu	30414	Jun 29 12:00	dedup_report.md
-rw-r--r--	1	avidullu	avidullu	2168	Jun 29 11:53	dedup_stage1.py
-rw-r--r--	1	avidullu	avidullu	2872	Jun 29 11:54	dedup_stage2.py
-rw-r--r--	1	avidullu	avidullu	26326	Jun 29 11:55	ffprobe_meta.json
-rw-r--r--	1	avidullu	avidullu	2205	Jun 29 11:55	ffprobe_meta.py
-rw-r--r--	1	avidullu	avidullu	1598	Jun 29 12:01	full_md5.json
-rw-r--r--	1	avidullu	avidullu	3260	Jun 29 11:54	fullmd5.py
-rw-r--r--	1	avidullu	avidullu	2002	Jun 29 12:01	fullmd5_confirm.out
-rw-r--r--	1	avidullu	avidullu	7747	Jun 29 11:56	gcs_full_ls.txt
-rw-r--r--	1	avidullu	avidullu	6288	Jun 29 12:00	gcs_migration_gap.md
-rw-r--r--	1	avidullu	avidullu	744	Jun 29 11:56	gcs_videos_ls.txt
-rw-r--r--	1	avidullu	avidullu	3840	Jun 29 11:57	golden_manifest.json
-rw-r--r--	1	avidullu	avidullu	5827	Jun 29 11:58	golden_map.json
-rw-r--r--	1	avidullu	avidullu	113632	Jun 29 12:00	master_video_index.json
-rw-r--r--	1	avidullu	avidullu	3823	Jun 29 11:54	partial_sigs.json
-rw-r--r--	1	avidullu	avidullu	6971	Jun 29 12:01	progress.log
-rw-r--r--	1	avidullu	avidullu	1632	Jun 29 11:53	size_groups.json
-rw-r--r--	1	avidullu	avidullu	0	Jun 29 11:54	stage2.err
-rw-r--r--	1	avidullu	avidullu	49111	Jun 29 11:53	videos.json

##### progress.log tail

```
[2026-06-29T11:56:37] hashed [26/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/ELT Pickleball/GX010044.MP4 -> 18a64359554e075d2436c464a65fef1f
[2026-06-29T11:56:42] hashed [27/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/ELT Pickleball/Pickleball chestie footage.MP4 ->
54c447a1b1c3750c413157e5a3a9a82a
[2026-06-29T11:56:43] hashed [28/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/ELT Pickleball/Pickleball recording ELT.MP4 ->
8027cdfcc97a046b860b63ddfd2680dd
```

```
[2026-06-29T11:56:47] hashed [29/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/GX010051.MP4 -> abd67ef19530514d1b6be95309c9886d
[2026-06-29T11:56:48] hashed [30/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/GX010053.MP4 -> 915473794e0909c9f95a01e4511a5e98
[2026-06-29T11:57:00+00:00] golden manifest fetched
[2026-06-29T11:57:01] hashed [31/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/Mom Jan 26 Singing.MP4 -> dd604b1931695473b1222511fdb7fc30
[2026-06-29T11:57:34] hashed [32/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton - Boxhill - 17 June/GX010137.MP4 -> 3d19ef835a69be93acdd75ef636f345a
[2026-06-29T11:58:01] hashed [33/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton - Boxhill - 17 June/GX010141.MP4 -> 219a84a4e1a985edf5fb039142c18b8d
[2026-06-29T11:58:58] hashed [34/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 1 June 2026/100GOPRO/GX010108.MP4 -> 751b6266be89dce2e9a39946caa9325c
[2026-06-29T11:59:55] hashed [35/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/GX010128.MP4 -> 8a5a404019e41c0473e7f10d17d86863
[2026-06-29T12:00:04+00:00] preliminary outputs built (partial md5)
[2026-06-29T12:00:08] hashed [36/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/mahadevpura-1.MP4 -> 3b06b9eb623ec65d64d2fbda99a24539
[2026-06-29T12:00:56] hashed [37/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/mahadevpura-2.MP4 -> 929dd84c15e602320ac27a2455c35422
[2026-06-29T12:01:53] hashed [38/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June/100GOPRO/Badminton/100GOPRO/GX010130.MP4 ->
3fd4daffee1863c7178eb6ffca684955
```

is the MD5 confirm pass still running?

any md5/hash background scripts alive?

```
41626 403 /bin/bash -c source /home/avidullu/.claude/shell-snapshots/snapshot-
bash-1782729783114-9kwgea.sh 2>/dev/null || true && shopt -u extglob 2>/dev/null || true && eval
'cd /home/avidullu/projects/khelsutra-guru/f_drive_index/ echo "[$(date -Iseconds)] launching
fullmd5 confirm mode in background" >> progress.log python3 fullmd5.py confirm >
fullmd5_confirm.out 2>&1 echo "EXIT=$?" tail -5 fullmd5_confirm.out' < /dev/null && pwd -P >|
/tmp/claude-c57b-cwd
```

**143. assistant (2026-06-29T12:03:04.403Z)**

The agent did serious work and checkpointed everything ? and its full-MD5 confirm pass (PID 41626) is **\*\*still alive and progressing\*\*** (at file 38/94, writing to `progress.log`), running independently. It already produced preliminary artifacts. Let me review what it found and learn its finalize interface, then set up a tracked wait so I'm notified when the hashing completes:

**144. user (2026-06-29T12:03:06.446Z)**

```
1 # GCS Migration Gap ? golden/corpus videos on F: vs gs://khelsutra-rally-corpus/videos/
2
3 _Generated: 2026-06-29T12:00:04.301088_
4
5 ## Finding
6 All 15 golden clips have **labels + trajectories** in the bucket, but the bucket
`videos/` folder contains only a handful of derived/proxy/smoke objects ? the **full-res golden
source videos are essentially NOT migrated** (closes DATA_IN_GCS $5).
7
8 ## gs://khelsutra-rally-corpus/videos/ inventory
9 - `gs://khelsutra-rally-corpus/videos/GX010142_1080p.mp4` ? 767,766,440 B ? kind:
100p-derived
10 - `gs://khelsutra-rally-corpus/videos/GX010143_1080p.mp4` ? 632,948,235 B ? kind:
100p-derived
11 - `gs://khelsutra-rally-corpus/videos/GX010147_1080p.mp4` ? 659,514,891 B ? kind:
100p-derived
12 - `gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4` ? 1,429,944,539 B ? kind:
proxy
13 - `gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4` ? 548,827,049 B ? kind: proxy
14 - `gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4` ? 548,827,049 B ? kind:
proxy(dup-name)
```

```

15 - `gs://khehsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` ? 90,194,321 B ? kind:
smoke-clip
16 - `gs://khehsutra-rally-corpus/videos/SHANAN/VID_1.mp4` ? 691,158,901 B ? kind: shanan
17 - `gs://khehsutra-rally-corpus/videos/SHANAN/VID_2.mp4` ? 2,794,800,788 B ? kind: shanan
18 - `gs://khehsutra-rally-corpus/videos/SHANAN/VID_3.mp4` ? 2,627,544,458 B ? kind: shanan
19
20 ## 15 golden clips: present on F: and migration status
21
22 | golden_name | on F: (full-res source) | in GCS videos/ as full-res? | needs upload |
23 |---|---|---|---|
24 | adarsh_avi_singles | YES | NO (only proxy/derived if any) | YES |
25 | kushagra_singles | YES | NO (only proxy/derived if any) | YES |
26 | targetest_doubles | YES | NO (only proxy/derived if any) | YES |
27 | gbaaddy | YES | NO (only proxy/derived if any) | YES |
28 | testlarge_short | YES | NO (only proxy/derived if any) | YES |
29 | mahadevpura_singles | YES | NO (only proxy/derived if any) | YES |
30 | mahadevpura_2 | YES | NO (only proxy/derived if any) | YES |
31 | GX010128 | YES | NO (only proxy/derived if any) | YES |
32 | mahadevpura_1 | YES | NO (only proxy/derived if any) | YES |
33 | GX030094 | YES | NO (only proxy/derived if any) | YES |
34 | Badminton_BXH_2 | YES | NO (only proxy/derived if any) | YES |
35 | GX010141 | YES | NO (only proxy/derived if any) | YES |
36 | GX010137 | YES | NO (only proxy/derived if any) | YES |
37 | Boxhill_Doubles | YES | NO (only proxy/derived if any) | YES |
38 | GX020094 | YES | NO (only proxy/derived if any) | YES |
39
40 **15/15 golden clips have a full-res source on F:. 0/15 full-res golden sources are in
gs://.../videos/. => 15 still need GCS upload.**
41
42 ## Per-golden F: source paths (for upload staging)
43 - **adarsh_avi_singles** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Training/Golden Labelled/Adarsh and Avi.MP4` (size 11,445,052,772)
44 - **kushagra_singles** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Training/Golden Labelled/KushagraYashAviFirstVideo.MP4` (size 8,058,214,679)
45 - **targetest_doubles** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Training/Golden Labelled/LargeTestVideo.MP4` (size 11,531,882,761)
46 - **gbaaddy** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Training/Golden Labelled/gBaaddy20May2026.MP4` (size 7,075,098,663)
47 - **testlarge_short** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Training/Golden Labelled/TestLargeVideo.MP4` (size 1,510,427,943)
48 - **mahadevpura_singles** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Training/Golden Labelled/MahadevpuraSingles.MP4` (size 196,182,380)
49 - **mahadevpura_2** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/mahadevpura-2.MP4` (size 9,227,260,503)
50 - **GX010128** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/GX010128.MP4`
51 - **mahadevpura_1** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/mahadevpura-1.MP4` (size 2,611,863,875) NOTE: full-res 2.6GB;
GCS videos/mahadevpura_1.mp4 (548MB) is the PROXY, equals F:
KhelSutra_work/kushagra_june12_proxies/mahadevpura-1.MP4
52 - **GX030094** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 2 May 2026/100GOPRO/GX030094.MP4` (size 10,611,615,335) NOTE: GCS
videos/GX030094_proxy.mp4 (1.4GB) is the PROXY, not this 10.6GB full-res
53 - **Badminton_BXH_2** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/GoPro 20 May 2026/100GOPRO/Badminton BXH 2.MP4` (size 10,925,859,511)
54 - **GX010141** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton - Boxhill - 17 June/GX010141.MP4`
55 - **GX010137** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton - Boxhill - 17 June/GX010137.MP4`
56 - **Boxhill_Doubles** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 4 May 2026/100GOPRO/GX010097.MP4` (size 11,532,555,534)
57 - **GX020094** -> `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 2 May 2026/100GOPRO/GX020094.MP4` (size 11,533,390,543)
58
59 ## CRITICAL upload caveat (bracket glob)

```



```

60 `gcloud storage cp` treats `[...]` in a path as a character-class GLOB, so the bracketed
source root `/mnt/f/[Khelsutra] GoPro Hero Black 12/...` will NOT upload directly ? it expands
to nothing / errors. Stage through a bracket-free path.
61
62 ### Suggested bracket-free staging recipe (DO NOT RUN here ? owner-gated)
63 ```bash
64 # 1. Create a bracket-free staging dir OUTSIDE /mnt/f (read-only on F:):
65 STAGE=$HOME/gcs_upload_stage; mkdir -p "$STAGE"
66 # 2. Hard-link or copy (copy across filesystems) each golden full-res into STAGE
67 # with a clean canonical name. Example (copy, since F: is a different FS):
68 cp "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/Adarsh and Avi.MP4" \
69 "$STAGE/adarsh_avi_singles.mp4"
70 # 3. Upload from the bracket-free staging path:
71 gcloud storage cp "$STAGE/adarsh_avi_singles.mp4" gs://khelsutra-rally-
corpus/videos/adarsh_avi_singles.mp4
72 # 4. Verify size/crc32c, then clean STAGE.
73 # Alternative: gcloud storage cp supports no-glob via copying the file after rename;
74 # there is no reliable --no-glob flag, so always stage bracket-free.
75 ```

```

#### 145. user (2026-06-29T12:03:06.654Z)

```

1 # F: Drive Dedup Report (READ-ONLY ? report/plan only, NOTHING executed)
2
3 _Generated: 2026-06-29T12:00:04.300608_
4 _Canonical id = full-file MD5 (backend/utils/hashing.py::compute_video_id)._
5
6 ## Totals scanned
7 - Total media files: **1795**
8 - Total bytes: **1,231,768,920,373** (1.12 TB)
9 - VIDEO: 220 files, 1.05 TB
10 - PROXY: 163 files, 30.96 GB
11 - THUMB: 163 files, 1.29 MB
12 - OTHER: 1249 files, 40.44 GB
13
14 ## Video dedup summary
15 - VIDEO files: **220**
16 - Unique videos (canonical): **180**
17 - Duplicate (redundant) copies: **40**
18 - Duplicate groups: **37** (14 confirmed by full-MD5, 23 provisional by size+partial-
sig)
19 - **Total reclaimable bytes (if redundant copies removed): 254,056,638,359 (236.61 GB)**
20
21 ## Method (staged, efficient on 1.2 TB)
22 1. Group VIDEO by exact byte size ? distinct size = unique (no hash).
23 2. Same-size groups -> partial signature md5(first 16MB ++ last 16MB) via seek reads.
24 3. Surviving (size,partial-sig) collisions -> FULL md5sum to CONFIRM exact dup =
video_id.
25
26 ## Duplicate groups (PROPOSED canonical-to-keep = earliest mtime, tie-break shortest
path)
27
28 ### dg001 (CONFIRMED full-MD5)
29 - video_id (full MD5): `18a64359554e075d2436c464a65fef1f`
30 - size each: 417,255,008 bytes (397.93 MB)
31 - KEEP (canonical): `/mnt/f/Data/Initial GoPro with Sports Recordings/GX010044.MP4`
(mtime 2025-12-30T14:29:02)
32 - redundant copies (1), reclaimable 397.93 MB:
33 - DELETE-CANDIDATE: `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/ELT Pickleball/GX010044.MP4` (mtime 2026-03-06T08:32:48)
34
35 ### dg002 (CONFIRMED full-MD5)
36 - video_id (full MD5): `1a6a25481c0e1025420fa129ecfeab2b`
37 - size each: 300,902,350 bytes (286.96 MB)

```

```

38 - KEEP (canonical): `/mnt/f/Data/Initial GoPro with Sports Recordings/GX010043.MP4`
(mtime 2025-12-30T14:11:20)
39 - redundant copies (1), reclaimable 286.96 MB:
40 - DELETE-CANDIDATE: `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/ELT Pickleball/GX010043.MP4` (mtime 2026-03-06T08:32:40)
41
42 ### dg003 (CONFIRMED full-MD5)
43 - video_id (full MD5): `28a81f9c2b26ea16634a7d39cf46aaa8`
44 - size each: 28,958,174 bytes (27.62 MB)
45 - KEEP (canonical): `/mnt/f/Data/Initial GoPro with Sports Recordings/GX010049.MP4`
(mtime 2026-01-15T04:44:38)
46 - redundant copies (1), reclaimable 27.62 MB:
47 - DELETE-CANDIDATE: `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/Cricket/Cricket Half pitch - 2.MP4` (mtime 2026-03-06T08:33:26)
48
49 ### dg004 (CONFIRMED full-MD5)
50 - video_id (full MD5): `2a14be8cdfd12424726526d064aba47d`
51 - size each: 226,442,602 bytes (215.95 MB)
52 - KEEP (canonical): `/mnt/f/Data/Initial GoPro with Sports Recordings/GX010040.MP4`
(mtime 2025-12-30T13:59:36)
53 - redundant copies (1), reclaimable 215.95 MB:
54 - DELETE-CANDIDATE: `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/ELT Pickleball/GX010040.MP4` (mtime 2026-03-06T08:32:36)
55
56 ### dg005 (CONFIRMED full-MD5)
57 - video_id (full MD5): `2c0ce3a7cfba85f2edda66e525e77d7e`
58 - size each: 662,437,897 bytes (631.75 MB)
59 - KEEP (canonical): `/mnt/f/Data/Initial GoPro with Sports Recordings/GX010029.MP4`
(mtime 2025-12-02T00:57:52)
60 - redundant copies (1), reclaimable 631.75 MB:
61 - DELETE-CANDIDATE: `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/Bike/Bike going to Mayura.MP4` (mtime 2026-03-06T08:32:18)
62
63 ### dg006 (CONFIRMED full-MD5)
64 - video_id (full MD5): `43792f9b2ef854faf6caf9ca8d085542`
65 - size each: 1,360,746,043 bytes (1.27 GB)
66 - KEEP (canonical): `/mnt/f/Data/Initial GoPro with Sports Recordings/GX010050.MP4`
(mtime 2026-01-15T05:19:28)
67 - redundant copies (1), reclaimable 1.27 GB:
68 - DELETE-CANDIDATE: `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/2026 March 6/100GOPRO/Cricket/Cricket Half pitch.MP4` (mtime 2026-03-06T08:33:48)
69
70 ### dg007 (CONFIRMED full-MD5)

```

#### 146. assistant (2026-06-29T12:03:39.380Z)

The agent's output is high quality. Preliminary findings:

- **\*\*1795 media files / 1.12 TB\*\*** cataloged. **\*\*220 videos ? 180 unique, 40 redundant; ~237 GB reclaimable\*\*** (14 dup groups already MD5-confirmed, 23 pending the running hash pass).
- **\*\*Migration gap nailed\*\***: all **\*\*15 golden clips have full-res sources on F:\*\*** (exact paths captured), and **\*\*0/15 full-res are in GCS\*\*** ? the bucket only has proxies/derived. It even caught that `mahadevpura\_1`/`GX030094` in GCS are **\*proxies\*** of much larger full-res originals. Bracket-glob upload caveat + staging recipe documented.

Let me confirm the hash job is still alive (so my wait isn't pointless) and check its progress:

#### 147. user (2026-06-29T12:03:44.274Z)

is PID 41626 (md5 confirm) still alive?

```

PID ELAPSED STAT COMMAND
41626 501 Ss /bin/bash -c source /home/avidullu/.claude/shell-snapshots/snapshot-
bash-1782729783114-9kwgea.sh 2>/dev/null || true && shopt -u extglob 2>/dev/null || true && eval
'cd /home/avidullu/projects/khelsutra-guru/f_drive_index/ echo "[$(date -Iseconds)] launching

```

```
fullmd5 confirm mode in background" >> progress.log python3 fullmd5.py confirm >
fullmd5_confirm.out 2>&1 echo "EXIT=$?" tail -5 fullmd5_confirm.out' < /dev/null && pwd -P >|
/tmp/claude-c57b-cwd
```

**progress: hashed N/94**

39

**last 3 progress lines + freshness**

```
[2026-06-29T12:00:56] hashed [37/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June 2026/mahadevpura-2.MP4 -> 929dd84c15e602320ac27a2455c35422
[2026-06-29T12:01:53] hashed [38/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June/100G0PRO/Badminton/100G0PRO/GX010130.MP4 ->
3fd4daffee1863c7178eb6ffca684955
[2026-06-29T12:02:49] hashed [39/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June/100G0PRO/Badminton/100G0PRO/GX010131.MP4 ->
5c2a0678a5319c52cabeb9d27136c2b3
now: 2026-06-29T12:03:44+00:00
```

**child python hashing proc?**

```
41626 501 /bin/bash -c source /home/avidullu/.claude/shell-snapshots/snapshot-
bash-1782729783114-9kwgea.sh 2>/dev/null || true && shopt -u extglob 2>/dev/null || true && eval
'cd /home/avidullu/projects/khelsutra-guru/f_drive_index/ echo "[$(date -Iseconds)] launching
fullmd5 confirm mode in background" >> progress.log python3 fullmd5.py confirm >
fullmd5_confirm.out 2>&1 echo "EXIT=$?" tail -5 fullmd5_confirm.out' < /dev/null && pwd -P >|
/tmp/claude-c57b-cwd
41647 501 python3 fullmd5.py confirm
```

**148. assistant (2026-06-29T12:04:06.248Z)**

Confirmed alive and progressing (PIDs 41626/41647, at 39/94). The remaining files include the huge 10 GB+ golden originals, so ~25?35 min left. Let me check the agent's build script so I can auto-finalize when hashing completes, then set up a tracked wait:

**149. user (2026-06-29T12:04:06.739Z)**

```
1 #!/usr/bin/env python3
2 """Build master_video_index.json, dedup_report.md, gcs_migration_gap.md from
3 the checkpointed intermediate artifacts. Re-runnable; reflects however much
4 full-MD5 has completed so far (provisional ids for not-yet-hashed uniques).
5 READ-ONLY w.r.t. /mnt/f.
6 """
7 import json, os, datetime
8 from collections import defaultdict
9
10 BASE = '/home/avidullu/projects/khelsutra-guru/f_drive_index'
11 videos = json.load(open(f'{BASE}/videos.json'))
12 size_groups = json.load(open(f'{BASE}/size_groups.json')) # {size: [idx]} >=2
13 partial = json.load(open(f'{BASE}/partial_sigs.json')) # {idx: psig}
14 confirm_groups = json.load(open(f'{BASE}/confirm_groups.json')) # {"size|psig": [idx]}
15 >=2
16 full_md5 = json.load(open(f'{BASE}/full_md5.json')) if
os.path.exists(f'{BASE}/full_md5.json') else {}
17 ffmpegmeta = json.load(open(f'{BASE}/ffmpegprobe_meta.json')) if
os.path.exists(f'{BASE}/ffmpegprobe_meta.json') else {}
18 golden_map = json.load(open(f'{BASE}/golden_map.json'))
19
20 # ---- GCS objects (parsed from full ls) ----
21 gcs_videos = golden_map['gcs_videos_objects'] # list of {obj,size,kind}
22 gcs_sizes = defaultdict(list)
23 gcs_basenames = {}
24 for o in gcs_videos:
25 gcs_sizes[o['size']].append(o['obj'])
26 gcs_basenames[os.path.basename(o['obj']).lower()] = o['obj']
27
28 # ---- golden name -> set of F primary/also paths ----
```

```

28 golden_paths = {} # path -> golden_name
29 for g in golden_map['golden']:
30 for p in [g.get('f_primary')] + g.get('also', []):
31 if p:
32 cp = p.split(' ')[0] # strip annotation
33 golden_paths[cp] = g['name']
34
35 # ---- Build video_id per file (full md5 if available, else provisional) ----
36 def file_video_id(i):
37 fm = full_md5.get(str(i))
38 if fm and not str(fm).startswith('ERROR'):
39 return fm, False
40 # provisional: size:<bytes>:p:<partialsig>
41 v = videos[i]
42 ps = partial.get(str(i))
43 if ps:
44 return f"size:{v['size_bytes']}:p:{ps}", True
45 return f"size:{v['size_bytes']}", True
46
47 # ---- Cluster duplicates by full md5 where available ----
48 # Within each confirm_group, if all members have full md5, cluster by md5.
49 # If md5 missing, fall back to (size,psig) as the dup cluster key (provisional dup).
50 dup_clusters = defaultdict(list) # cluster_key -> [idx]
51 member_of = {} # idx -> cluster_key
52 for key, idxs in confirm_groups.items():
53 # try full md5 clustering
54 md5s = {}
55 for i in idxs:

```

#### 150. assistant (2026-06-29T12:04:35.522Z)

`build\_outputs.py` is explicitly **re-runnable** and upgrades provisional to confirmed as `full\_md5.json` fills ? so I can auto-finalize without resuming the subagent. Let me launch a tracked background waiter that finalizes the moment hashing completes, and check on the stalled GPU-smoke download:

#### 151. user (2026-06-29T12:04:41.437Z)

Command running in background with ID: by1lanqjr. Output is being written to: /tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/by1lanqjr.output. You will be notified when it completes. To check interim output, use Read on that file path.

#### 152. user (2026-06-29T12:04:44.581Z)

##### GPU-smoke download final state

```

total 528680
drwxr-xr-x 2 avidullu avidullu 4096 Jun 29 11:19 .
drwxr-xr-x 6 avidullu avidullu 4096 Jun 29 11:27 ..
-rw-r--r-- 1 avidullu avidullu 548827049 Jun 29 11:21 mahadevpura_1.mp4_.gstmp

```

##### is the gcloud download still running?

```

32454 2640 /bin/bash -c source /home/avidullu/.claude/shell-snapshots/snapshot-
bash-1782729783114-9kwgea.sh 2>/dev/null || true && shopt -u extglob 2>/dev/null || true && eval
'DEST=/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b_golden mkdir
-p "$DEST/videos" echo "### downloading mahadevpura_1.mp4 (548MB)" time timeout 420 gcloud
storage cp gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4 "$DEST/videos/mahadevpura_1.mp4"
2>&1 | tail -3 ls -la "$DEST/videos/" 2>&1' < /dev/null && pwd -P >| /tmp/claude-72bd-cwd
32553 2623 python3.12 /mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-
sdk/lib/gcloud.py storage cp gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/videos/mahadevpura_1.mp4
32557 2623 python3.12 /mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-
sdk/lib/gcloud.py storage cp gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4

```

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/videos/mahadevpura_1.mp4
32559 2623 python3.12 /mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-
sdk/lib/gcloud.py storage cp gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/videos/mahadevpura_1.mp4
32562 2623 python3.12 /mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-
sdk/lib/gcloud.py storage cp gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/videos/mahadevpura_1.mp4
32571 2623 python3.12 /mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-
sdk/lib/gcloud.py storage cp gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/output/p2b_golden/videos/mahadevpura_1.mp4
```

**download task tail**

**downloading mahadevpura\_1.mp4 (548MB)**

**153. assistant (2026-06-29T12:05:10.078Z)**

The GPU-smoke download is **\*\*hung\*\*** ? gcloud has been retrying for 44 minutes (the `timeout` didn't reap its parallel child processes) and never finalized. It's optional polish and GPU liveness is already proven (CUDA available, all deps import), so I'll kill it cleanly rather than let it burn resources alongside the F: hashing job:

**154. user (2026-06-29T12:05:15.904Z)**

Exit code 144

**killing the hung mahadevpura\_1 download (optional GPU smoke ? abandoning)**

**155. assistant (2026-06-29T12:05:33.456Z)**

The kill succeeded (the download task reported `failed exit 144` ? killed), though my verification steps got cut off. Let me re-verify cleanup and confirm the F: hashing + finalize-waiter are healthy and unaffected:

**156. user (2026-06-29T12:05:39.174Z)**

**any mahadevpura download procs left?**

**clean partial**

total 8

drwxr-xr-x 2 avidullu avidullu 4096 Jun 29 12:05 .

drwxr-xr-x 6 avidullu avidullu 4096 Jun 29 11:27 ..

**F: hash job 41626 alive + progress N/94**

41626 613

43

**last hash line**

[2026-06-29T12:05:27] hashed [43/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June/100G0PRO/Badminton/100G0PRO/GX020130.MP4 -> eddd8e4d90dc65fedd81724a7277ba23

**finalize-waiter by1lanqjr alive?**

**157. assistant (2026-06-29T12:05:57.095Z)**

Cleanup confirmed ? download killed, `.gstmp` removed, and the F: hash job is healthy (43/94, still progressing). Let me make sure my kill didn't take out the finalize-waiter as collateral:

**158. user (2026-06-29T12:06:04.060Z)**

**look for the finalize-waiter (waits on 41626 then runs build\_outputs.py)**

also check the harness task list for by1lanqjr via its output file

```
-rw-r--r-- 1 avidullu avidullu 0 Jun 29 12:04 /tmp/claude-1000/-home-avidullu-projects-
khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/by1lanqjr.output
```

#### 159. assistant (2026-06-29T12:06:30.645Z)

The finalize-waiter died (collateral from the kill signal ? its sibling background session got caught), but the \*\*F: hash job (41626) is alive and progressing\*\*. Let me relaunch a fresh, fully-detached finalize-waiter:

#### 160. user (2026-06-29T12:06:38.282Z)

```
finalize-waiter relaunched (detached). pid_group started.
[1]+ Done setsid bash -c '
IDX=/home/avidullu/projects/khelsutra-guru/f_drive_index
while kill -0 41626 2>/dev/null; do sleep 60; done
{
 echo "[wait] md5 confirm pass exited at $(date -Iseconds)"
 tail -4 "$IDX/fullmd5_confirm.out" 2>&1
 echo "### rebuilding final outputs"
 python3 "$IDX/build_outputs.py" 2>&1 | tail -6
 echo "=== FINAL DEDUP SUMMARY ==="
 sed -n "6,20p" "$IDX/dedup_report.md" 2>&1
 echo "=== DONE FINALIZE ==="
} >> "$IDX/finalize.out" 2>&1
' < /dev/null > /dev/null 2>&1
```

confirm a waiter/sleep is now present

```
46752 52 sleep 60
47216 1 bash -c IDX=/home/avidullu/projects/khelsutra-guru/f_drive_index while kill -0
41626 2>/dev/null; do sleep 60; done { echo "[wait] md5 confirm pass exited at $(date
-Iseconds)" tail -4 "$IDX/fullmd5_confirm.out" 2>&1 echo "### rebuilding final outputs"
python3 "$IDX/build_outputs.py" 2>&1 | tail -6 echo "=== FINAL DEDUP SUMMARY ===" sed -n
"6,20p" "$IDX/dedup_report.md" 2>&1 echo "=== DONE FINALIZE ===" } >> "$IDX/finalize.out" 2>&1
47217 1 sleep 60
```

#### 161. assistant (2026-06-29T12:07:02.988Z)

The detached finalize-waiter is running robustly (PID 47216). It's `setsid`-detached so it survives signals, but that means it won't auto-notify me ? so let me add a lightweight harness-tracked poller that watches for its completion marker and pings me when the final numbers are ready:

#### 162. user (2026-06-29T12:07:04.825Z)

Command running in background with ID: bgjgsibwq. Output is being written to:  
/tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/bgjgsibwq.output. You will be notified when it completes. To check interim output, use Read on that file path.

#### 163. user (2026-06-29T12:07:07.897Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

#### 164. assistant (2026-06-29T12:07:36.818Z)

Everything's now running unattended with notifications wired up. Here's where things stand.

### ? Primary work (P2b) ? done, awaiting your signoff

- \*\*PR [#396](https://github.com/Khelsutra/badminton-highlight-indexer/pull/396)\*\* ? finding-B promoted to default-ON; suite 15037/10-skip, ruff+mypy clean.
- Launch-readiness report delivered above: \*\*GO to merge P2b; NO-GO on a product launch\*\* until the ship-gate target (#172) and WASB weight provenance are resolved. Issues

```

[#397](https://github.com/Khelsutra/badminton-highlight-indexer/issues/397) ,
[#398](https://github.com/Khelsutra/badminton-highlight-indexer/issues/398) filed.
- **GPU smoke abandoned** ? the `mahadevpura_1` download hung for 44 min (gcloud retry loop the
`timeout` couldn't reap); I killed it and cleaned up. GPU liveness is already proven (CUDA
available, torch/torchvision/cv2 import), so the smoke was only polish ? not worth re-fighting.

```

## ? F: archive index + dedup ? isolated agent, interim results in

Running read-only, report-only. \*\*Nothing deleted or moved.\*\* Preliminary pass already complete; a full-MD5 confirmation pass is finishing (43/94 hashed, ~25 min left), after which I'll auto-finalize and report exact numbers.

```

What it found so far (artifacts in `/home/avidullu/projects/khelsutra-guru/f_drive_index/`):
- **1,795 media files / 1.12 TB** cataloged (220 videos · 163 proxies · 163 thumbs · 1,249
other).
- **Dedup: 220 videos ? 180 unique, 40 redundant copies, ~237 GB reclaimable** across 37 groups
(14 MD5-confirmed, 23 confirming now). Most dups are the same GoPro clips living in both
`Data/Initial GoPro?` and `GoPro Recorded Videos/?`.
- **Migration gap nailed** : all **15/15 golden clips have full-res sources on F:** (exact paths
captured) but **0/15 full-res are in GCS** ? the bucket only holds proxies/derived. It even
caught that the GCS `mahadevpura_1` (548 MB) and `GX030094_proxy` (1.4 GB) are *proxies* of 2.6
GB / 10.6 GB originals. Bracket-glob (`[Khelsutra]`) upload caveat + a safe staging recipe are
documented.
- Deliverables: `master_video_index.json` (keyed by `video_id`=MD5, the cross-repo reference),
`dedup_report.md`, `gcs_migration_gap.md`, `catalog_raw.tsv`.

```

**\*\*I'll ping you with the finalized numbers\*\*** (confirmed dup groups + exact reclaimable bytes) when the hash pass lands. At that point, with your go-ahead, I can act on the report ? e.g. **\*\*migrate the 15 full-res golden originals to GCS\*\*** (closes the DATA\_IN\_GCS gap and unblocks the cloud/video-backed eval that's currently impossible), and/or execute the dedup as **\*\*quarantine\*\*** (reversible) once you've reviewed the plan.

Two open quick-decisions for you whenever (neither blocks anything): publish `master\_video\_index.json` to `gs://khelsutra-rally-corpus/manifests/` so any repo can reference it, and whether to prioritize the #398 served-gate fix as its own PR.

### 165. user (2026-06-29T12:09:57.470Z)

```

<task-notification>
<task-id>by1lanqjr</task-id>
<tool-use-id>toolu_01LMF4tKa1GXXM8BwTVwqQWv</tool-use-id>
<output-file>/tmp/claude-1000/-home-avidullu-projects-khelsutra-
guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/by1lanqjr.output</output-file>
<status>killed</status>
<summary>Background command "Wait for hashing to finish, then rebuild final F: index artifacts"
was stopped</summary>
</task-notification>

```

### 166. assistant (2026-06-29T12:10:13.853Z)

No response requested.

### 167. user (2026-06-29T12:10:13.989Z)

```

<task-notification>
<task-id>bgjgsibwq</task-id>
<tool-use-id>toolu_01TDvjvqR66h4UvXWqdMYBxX</tool-use-id>
<status>stopped</status>
<summary>No completion record was found for this background shell command from the previous
session. It may have been stopped (via the UI, Monitor timeout, or agent teardown ? these leave
no transcript marker), or it may have been running when the previous Claude Code process exited.
Check the output file for partial results before assuming it completed.</summary>
</task-notification>

```

### 168. assistant (2026-06-29T12:10:31.318Z)

Both my waiters were lost to a session boundary (the previous Claude Code process exited). The key question is whether the `setsid`-detached finalizer and the hash job survived. Let me check the actual state on disk and in the process table:

#### 169. user (2026-06-29T12:10:37.111Z)

now: 2026-06-29T12:10:37+00:00

hash job 41626 + finalizer 47216 still alive?

```
PID ELAPSED COMMAND
47216 232 bash -c IDX=/home/avidullu/projects/khelsutra-guru/f_drive_index while kill -0
41626 2>/dev/null; do sleep 60; done { echo "[wait] md5 confirm pass exited at $(date
-Iseconds)" tail -4 "$IDX/fullmd5_confirm.out" 2>&1 echo "### rebuilding final outputs"
python3 "$IDX/build_outputs.py" 2>&1 | tail -6 echo "=== FINAL DEDUP SUMMARY ===" sed -n
"6,20p" "$IDX/dedup_report.md" 2>&1 echo "=== DONE FINALIZE ===" } >> "$IDX/finalize.out" 2>&1
```

any fullmd5/build\_outputs procs?

```
47216 232 bash -c IDX=/home/avidullu/projects/khelsutra-guru/f_drive_index while kill -0
41626 2>/dev/null; do sleep 60; done { echo "[wait] md5 confirm pass exited at $(date
-Iseconds)" tail -4 "$IDX/fullmd5_confirm.out" 2>&1 echo "### rebuilding final outputs"
python3 "$IDX/build_outputs.py" 2>&1 | tail -6 echo "=== FINAL DEDUP SUMMARY ===" sed -n
"6,20p" "$IDX/dedup_report.md" 2>&1 echo "=== DONE FINALIZE ===" } >> "$IDX/finalize.out" 2>&1
```

hashing progress (N/94)

49

```
[2026-06-29T12:08:24] hashed [48/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June/100GOPRO/Badminton/Adarsh v Vasant.MP4 ->
6618d504a9e9ee4a26eb7b43ef396982
[2026-06-29T12:09:21] hashed [49/94] /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded
Videos/Badminton 12 June/100GOPRO/Badminton/All 1 v 1.MP4 -> 65a6d2bfd909f4734139f71baaea33a6
```

finalize.out present? (means finalizer ran)

```
ls: cannot access '/home/avidullu/projects/khelsutra-guru/f_drive_index/finalize.out': No such
file or directory
--- finalize.out ---
cat: /home/avidullu/projects/khelsutra-guru/f_drive_index/finalize.out: No such file or
directory
```

#### 170. assistant (2026-06-29T12:12:25.883Z)

Both detached processes' status is mixed: my `setsid` finalizer (47216) **\*\*survived\*\*** the session boundary, but the subagent's hash job (41626) **\*\*died\*\*** at 49/94. Good news ? the hashing prioritized dup-candidates first, so dedup is likely fully resolved; the unhashed remainder is large *\*unique\** golden originals (cosmetic provisional IDs). Let me stop the dangling finalizer, rebuild the report from current state, and assess:

#### 171. user (2026-06-29T12:12:34.538Z)

is hash job 41626 truly dead?

41626 GONE

stop the dangling detached finalizer group 47216

47216 GONE

full\_md5.json entries

49 entries

confirm\_groups (dup candidates needing full md5)

groups: 41 candidate files: 86

rebuild final report from current hashes

dup clusters: 34 (confirmed 18 / prov 16 )

redundant copies: 35

reclaimable bytes: 200082934940 186.34 GB

full\_md5 hashed: 49

=== DEDUP SUMMARY ===



## Totals scanned

- Total media files: \*\*1795\*\*
- Total bytes: \*\*1,231,768,920,373\*\* (1.12 TB)
  - VIDEO: 220 files, 1.05 TB
  - PROXY: 163 files, 30.96 GB
  - THUMB: 163 files, 1.29 MB
  - OTHER: 1249 files, 40.44 GB

## Video dedup summary

- VIDEO files: \*\*220\*\*
- Unique videos (canonical): \*\*185\*\*
- Duplicate (redundant) copies: \*\*35\*\*
- Duplicate groups: \*\*34\*\* (18 confirmed by full-MD5, 16 provisional by size+partial-sig)
- \*\*Total reclaimable bytes (if redundant copies removed): 200,082,934,940 (186.34 GB)\*\*

=== provisional dup groups remaining ===

1

### 172. assistant (2026-06-29T12:13:15.505Z)

Resuming ? thanks, that explains the blip (Windows extension-update reload killed the subagent's hash job at 49/94, though my `setuid`-detached finalizer survived). State is intact and recoverable.

Current rebuilt numbers: \*\*220 videos ? 185 unique, 35 redundant, 186 GB reclaimable\*\*, but \*\*16 of 34 dup groups are still provisional\*\* (size+partial-sig, not yet full-MD5-confirmed) because the hash pass died partway. Let me confirm the hash script is resumable, then relaunch it robustly (`setuid`-detached, which just proved it survives these blips) to finish confirmation + the golden-original video\_ids:

### 173. user (2026-06-29T12:13:16.366Z)

```
1 #!/usr/bin/env python3
2 """Compute full-file MD5 (= video_id, per backend/utils/hashing.py::compute_video_id)
3 for a given set of video indices. Checkpoints to full_md5.json after EACH file so
4 the run is resumable and partial results survive. READ-ONLY.
5
6 Usage: fullmd5.py <mode>
7 mode=confirm -> only the files in confirm_groups (dup candidates) + golden-name
matches
8 mode=all -> all VIDEO files not yet hashed (progressive, for remaining uniques)
9 """
10 import json, os, hashlib, sys, time
11
12 BASE = '/home/avidullu/projects/khelsutra-guru/f_drive_index'
13 videos = json.load(open(f'{BASE}/videos.json'))
14 mode = sys.argv[1] if len(sys.argv) > 1 else 'confirm'
15
16 GOLDEN = {'adarsh_avi_singles', 'kushagra_singles', 'largetest_doubles', 'gbaaddy',
17 'testlarge_short', 'mahadevpura_singles', 'mahadevpura_2', 'gx010128', 'kushagra_singles',
18 'mahadevpura_1', 'gx030094', 'badminton_bxh_2', 'gx010141', 'gx010137', 'boxhill_doubles', 'gx020094'}
19
20 md5_file = f'{BASE}/full_md5.json'
21 if os.path.exists(md5_file):
22 md5map = json.load(open(md5_file))
23 else:
24 md5map = {}
25
26 # Determine target indices
27 targets = set()
28 if mode == 'confirm':
29 cg = json.load(open(f'{BASE}/confirm_groups.json'))
30 for idxs in cg.values():
```

```

31 targets.update(idxs)
32 # plus golden-name basename matches (loose normalization)
33 for v in videos:
34 b = os.path.splitext(os.path.basename(v['path']))[0].lower()
35 norm = b.replace(' ', '').replace('-', '').replace('_', '')
36 for g in GOLDEN:
37 gn = g.replace('_', '').replace('-', '')
38 if gn in norm or norm in gn:
39 targets.add(v['idx'])
40 break
41 elif mode == 'all':
42 targets = set(range(len(videos)))
43 else:
44 print('unknown mode'); sys.exit(1)
45
46 todo = [i for i in sorted(targets) if str(i) not in md5map]
47 print('mode=%s targets=%d already_hashed=%d todo=%d' % (
48 mode, len(targets), len(targets)-len(todo), len(todo)), flush=True)
49
50 def md5_full(path):
51 h = hashlib.md5()
52 with open(path, 'rb') as f:
53 for block in iter(lambda: f.read(8*1024*1024), b''):
54 h.update(block)
55 return h.hexdigest()
56
57 errlog = open(f'{BASE}/progress.log', 'a')
58 done = 0
59 t0 = time.time()
60 for i in todo:
61 v = videos[i]
62 st = time.time()
63 try:
64 digest = md5_full(v['path'])
65 md5map[str(i)] = digest
66 except Exception as e:
67 md5map[str(i)] = 'ERROR:' + str(e)[:100]
68 errlog.write('[%s] FULLMD5 ERROR %s : %s\n' %
69 (time.strftime('%Y-%m-%dT%H:%M:%S'), v['path'], e))
70 errlog.flush()
71 # checkpoint after EACH file
72 json.dump(md5map, open(md5_file, 'w'), indent=0)
73 done += 1
74 dt = time.time() - st
75 gb = v['size_bytes']/1e9
76 print(' [%d/%d] %.2fGB in %.1fs (%.0fMB/s) %s' % (
77 done, len(todo), gb, dt, gb*1000/max(dt, 0.001), os.path.basename(v['path'])),
78 flush=True)
79 errlog.write('[%s] hashed [%d/%d] %s -> %s\n' % (
80 time.strftime('%Y-%m-%dT%H:%M:%S'), done, len(todo), v['path'],
81 md5map[str(i)][0:32]))
82 errlog.flush()
83
84 print('DONE mode=%s hashed_this_run=%d total_elapsed=%.0fs' % (mode, done,
85 time.time()-t0), flush=True)
86 errlog.write('[%s] fullmd5 mode=%s DONE hashed=%d elapsed=%.0fs\n' % (
87 time.strftime('%Y-%m-%dT%H:%M:%S'), mode, done, time.time()-t0))
88 errlog.close()
89

```

#### 174. assistant (2026-06-29T12:13:55.667Z)

`fullmd5.py` is fully resumable (checkpoints after each file, skips the 49 already done).  
 Relaunching it `setsid`-detached ? the pattern that just proved it survives Windows reloads ?  
 chained to auto-finalize, plus a harness-tracked poller for notification:

#### 175. user (2026-06-29T12:14:03.611Z)

```
[1]+ Done setsid bash -c '
IDX=/home/avidullu/projects/khelsutra-guru/f_drive_index
cd "$IDX"
echo "[resume] fullmd5 confirm restart $(date -Iseconds)" >> finalize.out
python3 fullmd5.py confirm >> fullmd5_confirm.out 2>&1
echo "[resume] hashing complete, building final outputs $(date -Iseconds)" >> finalize.out
python3 build_outputs.py >> finalize.out 2>&1
{
 echo "=== FINAL DEDUP SUMMARY ==="
 sed -n "6,20p" "$IDX/dedup_report.md"
 echo "=== DONE FINALIZE ==="
} >> finalize.out 2>&1
' < /dev/null > /dev/null 2>&1
```

**relaunched. fullmd5 todo count:**

mode=confirm targets=94 already\_hashed=49 todo=45

**confirm hashing proc alive:**

```
50283 2 bash -c IDX=/home/avidullu/projects/khelsutra-guru/f_drive_index cd "$IDX" echo
"[resume] fullmd5 confirm restart $(date -Iseconds)" >> finalize.out python3 fullmd5.py confirm
>> fullmd5_confirm.out 2>&1 echo "[resume] hashing complete, building final outputs $(date
-Iseconds)" >> finalize.out python3 build_outputs.py >> finalize.out 2>&1 { echo "=== FINAL
DEDUP SUMMARY ===" sed -n "6,20p" "$IDX/dedup_report.md" echo "=== DONE FINALIZE ===" } >>
finalize.out 2>&1
```

#### 176. user (2026-06-29T12:14:03.623Z)

Command running in background with ID: bpudjuz2j. Output is being written to:  
/tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/bpudjuz2j.output. You will be notified when it completes. To check interim output, use Read on that file path.

#### 177. assistant (2026-06-29T12:14:19.978Z)

Resumed cleanly ? the blip was just the Windows extension-update reload; nothing was lost. All on-disk state survived, and the P2b work (PR #396) was already on GitHub so it's untouched.

Here's where we are:

#### 178. user (2026-06-29T12:14:21.921Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

#### 179. assistant (2026-06-29T12:14:31.796Z)

**\*\*Resumed.\*\* Recovery status:**

- **\*\*P2b / PR [#396](https://github.com/Khelsutra/badminton-highlight-indexer/pull/396)\*\*** ? unaffected (pushed to GitHub before the blip). Issues #397/#398 stand. Launch-readiness report stands as delivered, awaiting your signoff.  
- **\*\*F: index+dedup\*\*** ? the Windows reload killed the subagent's hash job at 49/94, but every artifact survived on disk and the script checkpoints per-file, so I relaunched it cleanly: `targets=94, already\_hashed=49, todo=45`. It's now running **\*\*setsid-detached\*\*** (the pattern that demonstrably survived this exact reload) and will **\*\*auto-finalize\*\*** (rebuild the report with full MD5s) when done. A harness poller will notify me ? **\*\*~15?20 min\*\*** (the remainder includes the big 10 GB golden originals).

**\*\*Interim dedup numbers\*\*** (rebuilt from the 49 hashed so far; 16 groups still pending full-MD5 confirmation):

- **\*\*220 videos ? 185 unique, 35 redundant copies, ~186 GB reclaimable\*\*** (18 groups MD5-confirmed, 16 provisional-by-strong-signature pending).  
- Migration gap is already final and solid: **\*\*15/15 golden full-res sources located on F:, 0/15 full-res in GCS.\*\***

Nothing is deleted or moved ? still strictly report-only. I'll come back with the **final confirmed numbers + the dedup plan + the golden-upload plan** when the hash pass lands. No action needed from you meanwhile; the two open quick-decisions (publish the index to GCS; prioritize the #398 fix) can wait until then.

180. user (2026-06-29T12:21:49.263Z)

I think we definitely will need to publish to GCS but I have a subtask as well for you. I think if you can propose a "recorded video storage map" such that I can replicate the same map + the associated videos across my Google Drive (G:\) and local storage (F:\) or any other new medium as long as it can be mounted on windows or linux. So we will have GCS and G: as the cloud source and backup while F: is where I actually get the videos from my GoPro. This will give us 3 copies and then we can afford to lose one copy (likely GCS because after training the videos there are just incurring storage cost and I have a 2-5TB Google Drive storage plan already so F: and G: will be the N + 1 and we can load onto GCS as and when we need for training. Proxy was uploaded for the same reason to reduce the unnecessary cost. Feel free to propose adding a new isolated github repo for this (similar to data collector and raly annotator) which is exclusively for this storage management and possible cost savings similar to reducing dedup (which you found) and unstaging from GCS once the training passes have been rattified.

181. assistant (2026-06-29T12:23:37.463Z)

Great framing ? this is really a **3-copy, content-addressed corpus** with GCS as the disposable training tier, and the index I'm already building is the natural seed for it. Let me write up a concrete proposal you can react to, then ask you the couple of decisions that are genuinely yours (the repo, mainly). The F: hashing keeps running in the background while I do this.

Let me draft the spec as a reviewable doc:

182. assistant (2026-06-29T12:24:27.801Z)

## Recorded-Video Storage Map ? proposal (v0)

A single, medium-agnostic layout + manifest so the **same map and the same videos** replicate identically across **F:\ (local GoPro source)**, **G:\ (Google Drive, durable backup)**, **GCS (disposable training tier)**, and any future mountable medium. Seeded by the F: index already built  
(`master\_video\_index.json`).

---

### 1. Goals & invariants

- **3 copies, N+1:** every *ratified* video lives on **2 durable** tiers (F: + G:). GCS is a 3rd, **disposable** copy ? we can lose it without data loss.
- **GCS is a cache, not a vault.** Hold cheap **proxies** there persistently; stage **full-res** only while a training pass needs it; **evict** after the pass is ratified to stop paying for it.
- **Cost down:** dedup (?186 GB found on F:), proxy-only cloud by default, GCS lifecycle eviction.
- **Safety is enforced in code:** never delete any copy unless the tool has *verified* (by hash) that **2 other durable copies exist**. "Afford to lose one" is an invariant, not a hope.
- **Licensing-clean** (MIT/Apache/BSD only), cross-platform (Windows + Linux/WSL), tool-driven.

### 2. Tiers & roles

Tier	Medium	Role	Holds	Access
F:	local disk (GoPro offload)	ingest source + durable copy #1	full-res masters (+ raw `_ingest/`)	filesystem / rsync
G:	Google Drive 2?5 TB	durable copy #2 (the "always have it")	full-res masters +	

```
proxies | Drive client (Win) · `rclone` (Linux) |
| **GCS** | `gs://khelsutra-rally-corpus` | **disposable** training cache | **proxies** always;
full-res **only while training** | `gcloud storage` |
| **new** | any mountable drive/medium | drop-in extra copy | whatever you sync to it |
filesystem / rclone |
```

A new medium becomes a tier with one `index` + one `sync` ? because the layout is identical everywhere.

### 3. Canonical layout (identical on every tier, rooted at `

```
...
<root>/
 _ingest/ # F: ONLY ? raw GoPro card dumps, as-is, transient (never
mirrored)
 videos/
 full/<canonical_name>.mp4 # full-res master (F:, G:, GCS-only-while-training)
 proxy/<canonical_name>.mp4 # 1080p proxy (GCS always; G: optional)
 derived/
 trajectories/<canonical_name>.csv
 labels/<canonical_name>.rallies.csv
 weights/<version>/... # trained bundles (unchanged from DATA_IN_GCS)
 manifests/
 corpus_index.json # THE MAP (source of truth; see §5)
 STORAGE_MAP.md # this doc, travels with the data
...
```

Flat + human-readable on purpose: object stores are flat, and replication is trivial when the layout matches byte-for-byte. The messy GoPro nesting (`100GOPRO/?`, chapters, `.LRV`/`.THM`) stays only in `\_ingest/`; promotion to `videos/full/` normalizes it.

### 4. Naming & identity (the bridge between human-navigable and content-addressed)

- **\*\*canonical\_name\*\*** = the human key already used by the corpus (`adarsh\_avi\_singles`, `GX010128`, ?).
- It is the **\*\*path\*\***. A naming authority lives in the manifest (no guessing from GoPro's recycled `GX0101NN`).
- **\*\*video\_id\*\*** = full-file MD5 (the repo's `compute\_video\_id`) ? the **\*\*integrity + dedup\*\*** key, recorded in the manifest. Plus `crc32c` (what GCS returns) for cheap cross-tier verify without re-reading full files locally.
- A video is the **\*same\*** video on every tier iff its `video\_id` matches ? that's how `verify`/`dedup`/`evict` reason safely.

### 5. The manifest = the map (evolves `master\_video\_index.json`)

One JSON any repo reads to resolve a name ? wherever the bytes are cheapest/available.

```
```jsonc
{
  "schema_version": 1,
  "videos": [{
    "canonical_name": "adarsh_avi_singles",
    "video_id": "<md5 of full-res>",
    "golden": true,
    "full": { "rel_path": "videos/full/adarsh_avi_singles.mp4", "size": 11445052772, "md5":
"...", "crc32c": "...", "duration_s": 1490, "fps": 59.94, "res": "3840x2160" },
    "proxy": { "rel_path": "videos/proxy/adarsh_avi_singles.mp4", "size": 548000000, "md5":
"...", "crc32c": "...", "source_video_id": "<md5 of full-res>" },
    "tiers": { "F": "full", "G": "full", "GCS": "proxy" },          // verified presence per
```

```

tier
  "verified": { "F": "2026-06-29", "G": null, "GCS": "2026-06-22" },
  "derived": { "labels": "derived/labels/2026-06-08_adarsh_avi_singles.rallies.csv",
"trajectory": "derived/trajectories/adarsh_avi_singles.csv" },
  "source_capture": { "device": "GoPro Hero Black 12", "capture_date": "2026-06-08",
"gowpro_original": "GX010108.MP4", "f_ingest_path": "/mnt/f/[Khelsutra] ../GX010108.MP4" },
  "lifecycle": { "gcs_full_state": "evicted", "trained_ratified": true }
}
}
...

`tiers` + `verified` make replication declarative: "make tier X match the manifest." Drift =
`verify` finds an entry whose hash doesn't match or is missing.

```

6. Proxy model

Each master may have a 1080p proxy (`ffmpeg` transcode, deterministic settings recorded in the manifest). Proxies are sufficient for WASB detection/eval, ~1/20th the bytes, and are the **default** cloud object. Full-res is the archival truth on F:/G:. (This formalizes why `mahadevpura\_1` / `GX030094` are proxies in GCS today.)

7. Lifecycle & the cost saver

Per-video `lifecycle.gcs\_full\_state`: `never ? staged ? evicted`.

- **stage**: push full-res (or proxy) to GCS for a training/eval run.
- **ratify**: mark `trained\_ratified=true` when a pass using it is accepted.
- **evict**: delete the GCS **full-res** object once ratified ? **guarded** by: full-res verified present

on **both** F: and G: first. Proxy + labels + trajectories stay in GCS (cheap, keeps cloud eval runnable). GCS bucket **lifecycle rule** auto-expires `videos/full/\*` after N days as a backstop.

8. `storagectl` ? the one tool (verbs, tier-agnostic)

```

| Verb | What | |
|---|---|---|
| `index <root>` | scan a mounted tier, hash (staged: size?partial?full), update the manifest |
| `verify <tier>` | every manifest entry present on tier with matching hash; report missing/corrupt/extra |
| `sync <src> <dst>` | copy missing/changed entries src?dst (rsync · `rclone` Drive · `gcloud storage rsync`); idempotent |
| `dedup <root> [--quarantine|--delete]` | the dedup pass (already built); report-only by default |
| `proxy <names?>` | generate 1080p proxies + record transcode settings |
| `stage <names?> --to gcs [--full|--proxy]` | push for a training run |
| `evict <names?> --from gcs --require-ratified` | cost saver; refuses unless ?2 durable copies verify |
| `status` | per-video tier matrix + per-tier bytes + reclaimable + N+1 violations |

```

Backends abstracted: a tier is a filesystem path, an `rclone` remote, or a `gs://` prefix ? same verbs.

Bracket-glob caveat ([Khelsutra]) handled in the FS backend (stage through bracket-free paths).

9. Proposed new repo (isolated, like `sports-data-collector` / `rally-annotator`)

Purpose: corpus custody + replication + cost control ? nothing else. Name candidates: `rally-corpus-vault` · `sports-media-vault` · `corpus-custodian` · `media-storage-map`.

```

...
<repo>/

```

```

storagectl/          # the CLI (pure-Python stdlib + shells to rclone/gcloud/ffmpeg)
schema/corpus_index.schema.json
STORAGE_MAP.md       # this spec (the canonical doc)
manifests/           # the committed corpus_index.json (the map; big binary blobs never
committed)
tests/               # verify/dedup/evict-guard unit tests (mocked tiers)
.github/workflows/   # schema-validate + verify dry-run + license scan
...

```

```

- **Consumed by other repos** by reading `corpus_index.json` (resolve
`canonical_name`/`video_id` ?
  best available tier). `badminton-highlight-indexer` keeps its `gs://` eval manifest but can
*derive*
  it from this map.
- **Seeded today** by the F: index I just built (`master_video_index.json`, `dedup_report.md`,
  `gcs_migration_gap.md`) ? they drop straight in.
- Licensing-clean: stdlib + `rclone` (MIT) + `gcloud` + `ffmpeg` invoked as tools (no GPL
  linkage).

```

10. Bootstrap plan (once approved)

1. Scaffold the repo + move this doc + the F: index artifacts in.
2. Finish the F: full-MD5 pass (running) ? authoritative `corpus\_index.json` with F: tier filled.
3. ****Dedup F:**** (?186 GB) ? quarantine first, then delete after you review.
4. ****Mirror F?:G:**** the canonical `videos/full/` (rclone) ? durable N+1 reached.
5. ****GCS:**** upload the 15 golden full-res (closes the migration gap) for the next training pass; set the `videos/full/` lifecycle-evict rule; keep proxies + labels + trajectories resident.
6. Wire `verify` into CI so drift/corruption surfaces automatically.

11. Open decisions (yours)

- New repo: create now? which name?
- G: full-res for ****all**** captures, or only the ****golden/ratified**** subset (rest stay F:-only)?
- Proxy spec (resolution/bitrate) ? 1080p to match existing proxies?
- Canonical-name authority for non-golden GoPro clips (date+original-stem, e.g. `2026-06-08\_GX010108`)?

183. user (2026-06-29T12:24:28.885Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/f_drive_index/STORAGE_MAP_PROPOSAL.md (file state is current in your context ? no need to Read it back)

184. assistant (2026-06-29T12:25:00.356Z)

Proposal written to [STORAGE_MAP_PROPOSAL.md](f_drive_index/STORAGE_MAP_PROPOSAL.md). Here's the shape of it.

****The core idea:**** one ****medium-agnostic canonical layout**** + one ****manifest (the map)**** that replicates byte-identically across F:/G:/GCS/any-new-drive. Identity is dual-keyed ? human `canonical\_name` for the **path** (navigable), `video\_id`=MD5 + `crc32c` for **integrity/dedup/verify**. The index I'm building right now is literally the seed manifest.

****Your tiering, made into invariants:****

- ****F:**** ingest source + durable copy #1 . ****G:**** durable copy #2 (the "always have it") .
- **GCS:**** **disposable** training cache.
- ****GCS** holds proxies persistently; full-res only while a pass is training, then `evict`-after-ratify ? that's your cost saver, formalized (and exactly why the existing GCS objects are proxies).
- ****N+1 enforced in code:**** the `evict`/`delete` verbs **refuse** to drop any copy unless they've hash-verified ?2 other durable copies exist. "Afford to lose one" becomes a guarantee, not a hope.

****One tool, `storagectl`, with tier-agnostic verbs (`index` . `verify` . `sync` . `dedup` . `proxy` .**

stage · evict · status`) backing onto filesystem / `rclone` (Drive) / `gcloud` (GCS) ? all MIT/Apache/BSD-clean, Windows+Linux. It absorbs the dedup pass (~186 GB) and the bracket-glob handling I already hit.

****New repo:**** yes ? a clean sibling to `sports-data-collector`/`rally-annotator`, scoped ***only*** to corpus custody + replication + cost control, seeded by today's F: index artifacts. Other repos just read `corpus\_index.json` to resolve a name ? cheapest available tier.

I'll bring the concrete bootstrap/execution plan (dedup ? G: mirror ? GCS golden upload ? lifecycle rule) once the F: hash pass finishes. Two decisions are genuinely yours and shape what I scaffold:

185. user (2026-06-29T13:00:42.319Z)

Your questions have been answered: "Create the isolated storage-management repo now, and under which name? (sibling to sports-data-collector / rally-annotator)"="rally-corpus-vault", "What should the durable G:\ (Drive) backup hold ? i.e. what gets the full 3-copy treatment?"="Golden + ratified only". You can now continue with these answers in mind.

186. assistant (2026-06-29T13:01:18.421Z)

The F: finalization completed (the poller fired). Let me grab the ****final**** dedup numbers, then scaffold ****`rally-corpus-vault`**** (Golden+ratified G: scope) per your choices:

187. user (2026-06-29T13:01:18.723Z)

```
1      [resume] fullmd5 confirm restart 2026-06-29T12:14:01+00:00
2      [resume] hashing complete, building final outputs 2026-06-29T12:43:02+00:00
3      Outputs written.
4      master_video_index.json entries (unique videos): 175
5      VIDEO files total: 220
6      dup clusters: 41 (confirmed 41 / prov 0 )
7      redundant copies: 45
8      reclaimable bytes: 270365122649 251.80 GB
9      full_md5 hashed: 94
10     === FINAL DEDUP SUMMARY ===
11     ## Totals scanned
12     - Total media files: **1795**
13     - Total bytes: **1,231,768,920,373** (1.12 TB)
14       - VIDEO: 220 files, 1.05 TB
15       - PROXY: 163 files, 30.96 GB
16       - THUMB: 163 files, 1.29 MB
17       - OTHER: 1249 files, 40.44 GB
18
19     ## Video dedup summary
20     - VIDEO files: **220**
21     - Unique videos (canonical): **175**
22     - Duplicate (redundant) copies: **45**
23     - Duplicate groups: **41** (41 confirmed by full-MD5, 0 provisional by size+partial-sig)
24     - **Total reclaimable bytes (if redundant copies removed): 270,365,122,649 (251.80 GB)**
25
26     === DONE FINALIZE ===
27
```

188. assistant (2026-06-29T13:02:16.237Z)

****F: index/dedup is now fully confirmed**** (all 94 hash-candidates done, 0 provisional): ****220** videos ? 175 unique, 45 redundant, 41 MD5-confirmed dup groups, 251.8 GB reclaimable.**

Now scaffolding ****`rally-corpus-vault`**** (G: scope = golden+ratified). Setting up the repo structure and seeding it from today's index:

189. user (2026-06-29T13:02:21.898Z)


```
scaffold dirs:
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.git
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.git/branches
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.git/hooks
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.git/info
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.git/objects
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.git/refs
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.github
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.github/workflows
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/schema
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/seed
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl
/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/tests
seed files:
total 404
drwxr-xr-x 2 avidullu avidullu 4096 Jun 29 13:02 .
drwxr-xr-x 8 avidullu avidullu 4096 Jun 29 13:02 ..
-rw-r--r-- 1 avidullu avidullu 245895 Jun 29 13:02 f_drive_catalog_raw.tsv
-rw-r--r-- 1 avidullu avidullu 32466 Jun 29 13:02 f_drive_dedup_report.md
-rw-r--r-- 1 avidullu avidullu 110685 Jun 29 13:02 f_drive_video_index.json
-rw-r--r-- 1 avidullu avidullu 6288 Jun 29 13:02 gcs_migration_gap.md
```

190. assistant (2026-06-29T13:02:51.921Z)

Repo skeleton created and seeded. Now writing the foundational files ? license, spec, schema, and metadata:

191. assistant (2026-06-29T13:02:56.034Z)

MIT License

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192. user (2026-06-29T13:02:57.105Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/LICENSE (file state is current in your context ? no need to Read it back)

193. assistant (2026-06-29T13:02:59.718Z)

Python

```
__pycache__/
*.pyc
.venv/
```

```
.pytest_cache/  
.ruff_cache/  
.mypy_cache/  
*.egg-info/
```

Never commit media or huge regenerable scans

```
*.mp4  
*.MP4  
*.mov  
*.MOV  
*.lrv  
*.LRV  
*.thm  
*.THM  
*.insv  
/scratch/
```

Local tier config may carry machine-specific mount paths / remotes

```
vault.local.toml
```

Large regenerable catalogs (the committed manifest is the source of truth)

```
seed/f_drive_catalog_raw.tsv
```

194. user (2026-06-29T13:03:00.787Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.gitignore (file state is current in your context ? no need to Read it back)

195. user (2026-06-29T13:03:16.678Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

196. assistant (2026-06-29T13:03:26.443Z)

```
[build-system]  
requires = ["setuptools>=61.0"]  
build-backend = "setuptools.build_meta"
```

```
[project]  
name = "rally-corpus-vault"  
version = "0.1.0"  
description = "Medium-agnostic storage map + custody tooling for the recorded-video corpus (F: /  
Google Drive / GCS)."  
readme = "README.md"  
requires-python = ">=3.10"  
license = { text = "MIT" }  
authors = [{ name = "Khelsutra" }]
```

Pure stdlib at runtime; external media moves shell out to rclone / gcloud / rsync / ffmpeg (invoked as tools, never linked) to keep the dependency surface MIT/Apache/BSD-clear

```
dependencies = []
```

```
[project.optional-dependencies]  
dev = ["pytest>=7", "ruff", "jsonschema"]
```

```
[project.scripts]  
storagectl = "storagectl.storagectl:main"
```

```
[tool.setuptools]  
packages = ["storagectl"]
```

```
[tool.ruff]
target-version = "py310"
```

197. user (2026-06-29T13:03:27.478Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/pyproject.toml (file state is current in your context ? no need to Read it back)

198. assistant (2026-06-29T13:03:49.136Z)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://github.com/Khelsutra/rally-corpus-vault/schema/corpus_index.schema.json",
  "title": "Corpus Index (the storage map)",
  "type": "object",
  "required": ["schema_version", "videos"],
  "properties": {
    "schema_version": { "type": "integer", "const": 1 },
    "generated": { "type": "string", "description": "ISO-8601 timestamp" },
    "canonical_layout": {
      "type": "object",
      "description": "Relative paths under each tier root; identical on every tier.",
      "properties": {
        "full": { "type": "string", "default": "videos/full/{canonical_name}.mp4" },
        "proxy": { "type": "string", "default": "videos/proxy/{canonical_name}.mp4" },
        "labels": { "type": "string", "default": "derived/labels/{labels_name}" },
        "trajectory": { "type": "string", "default": "derived/trajectories/{canonical_name}.csv" }
      }
    },
    "tiers": {
      "type": "array",
      "description": "Declared storage tiers (a tier is an FS path, an rclone remote, or a gs:// prefix).",
      "items": {
        "type": "object",
        "required": ["name", "kind", "durable"],
        "properties": {
          "name": { "type": "string", "examples": ["F", "G", "GCS"] },
          "kind": { "enum": ["fs", "rclone", "gcs"] },
          "root": { "type": "string", "description": "FS path / rclone remote:path / gs:// prefix" },
          "durable": { "type": "boolean", "description": "true = a copy that counts toward the N+1 guarantee (F, G); false = disposable cache (GCS)" }
        }
      }
    },
    "videos": {
      "type": "array",
      "items": { "$ref": "#/$defs/video" }
    }
  },
  "$defs": {
    "filecopy": {
      "type": "object",
      "required": ["rel_path", "size", "md5"],
      "properties": {
        "rel_path": { "type": "string" },
        "size": { "type": "integer", "minimum": 0 },
        "md5": { "type": "string", "pattern": "^[a-f0-9]{32}$", "description": "video_id when this is the full-res master" },
        "crc32c": { "type": "string", "description": "base64 crc32c (matches gcloud) for cheap cross-tier verify" },
        "duration_s": { "type": "number" },
        "fps": { "type": "number" },

```

```

    "res": { "type": "string", "examples": ["3840x2160", "1920x1080"] },
    "source_video_id": { "type": "string", "description": "for proxies: the md5 of the full-
res they were transcoded from" }
  },
  "video": {
    "type": "object",
    "required": ["canonical_name", "video_id"],
    "properties": {
      "canonical_name": { "type": "string", "description": "human key = the path stem on every
tier" },
      "video_id": { "type": "string", "pattern": "^[a-f0-9]{32}$", "description": "full-res
MD5 (== backend/utils/hashing.py::compute_video_id)" },
      "golden": { "type": "boolean", "default": false },
      "full": { "$ref": "#/$defs/filecopy" },
      "proxy": { "$ref": "#/$defs/filecopy" },
      "tiers": {
        "type": "object",
        "description": "verified presence per tier name -> what is held there",
        "additionalProperties": { "enum": ["full", "proxy", "absent"] }
      },
      "verified": {
        "type": "object",
        "description": "per tier name -> ISO date of last hash-verify (null = present but
unverified)",
        "additionalProperties": { "type": ["string", "null"] }
      },
      "derived": {
        "type": "object",
        "properties": {
          "labels": { "type": "string" },
          "trajectory": { "type": "string" }
        }
      }
    },
    "source_capture": {
      "type": "object",
      "properties": {
        "device": { "type": "string" },
        "capture_date": { "type": "string" },
        "gopro_original": { "type": "string" },
        "f_ingest_path": { "type": "string" }
      }
    },
    "lifecycle": {
      "type": "object",
      "properties": {
        "gcs_full_state": { "enum": ["never", "staged", "evicted"], "default": "never" },
        "trained_ratified": { "type": "boolean", "default": false }
      }
    }
  }
}
}
}
}

```

199. user (2026-06-29T13:03:50.171Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/schema/corpus_index.schema.json (file state is current in your context ? no need to Read it back)

200. assistant (2026-06-29T13:04:15.062Z)

rally-corpus-vault

Custody, replication, and cost control for the recorded-video corpus ? one **medium-agnostic storage map** so the **same layout and the same videos** replicate identically across **F:** (local GoPro source), **Google Drive (G:\, durable backup)**, **GCS (disposable training cache)**, and any future mountable medium.

This is an isolated sibling to ``sports-data-collector`` and ``rally-annotator``: it does corpus storage management and nothing else. Other repos consume it by reading ``manifests/corpus_index.json`` to resolve a video name ? wherever the bytes are cheapest/available.

Why

- **3 copies, N+1.** Every ratified video lives on ?2 **durable** tiers (F: + G:). GCS is a 3rd, **disposable** copy we can lose without data loss.
- **GCS is a cache, not a vault.** Cheap **proxies** stay resident; full-res is **staged only** while a training pass needs it, then **evicted** after the pass is ratified to stop paying storage.
- **Safety in code.** ``evict`/`delete`` refuse to drop any copy unless they have **hash-verified** that ?2 other durable copies exist. "Afford to lose one" is an enforced invariant.
- **Cost down.** Dedup (the F: scan found **251.8 GB** reclaimable across 41 confirmed duplicate groups), proxy-only cloud by default, GCS lifecycle eviction.

The full design is in **STORAGE_MAP.md** ? read that first.

Layout (identical on every tier)

```
...
<root>/
  videos/full/<canonical_name>.mp4      # full-res master   (F:, G:; GCS only while training)
  videos/proxy/<canonical_name>.mp4     # 1080p proxy       (GCS always; G: optional)
  derived/{labels,trajectories}/...
  weights/<version>/...
  manifests/corpus_index.json           # THE MAP (source of truth)
...
```

Quickstart

```
```bash
pip install -e . # console script: storagectl
cp vault.example.toml vault.local.toml # point tiers F:/G:/GCS at your mounts/remotes

storagectl index F # scan a tier, hash, update the manifest
storagectl status # per-video tier matrix + reclaimable + N+1 violations
storagectl dedup F # report duplicate groups (add --quarantine / --delete)
storagectl verify G # check a tier matches the manifest by hash
storagectl stage GX010128 --to GCS --full # push full-res for a training run
storagectl evict GX010128 --from GCS --require-ratified # cost saver (guarded by N+1)
```
```

Seed data

``seed/`` carries the 2026-06-29 first scan of F: that bootstrapped this repo:

- ``f_drive_video_index.json`` ? 175 unique videos keyed by MD5 video_id
- ``f_drive_dedup_report.md`` ? 41 confirmed dup groups, 251.8 GB reclaimable
- ``gcs_migration_gap.md`` ? 15/15 golden full-res on F:, 0/15 full-res in GCS (the upload backlog)

``storagectl index`` regenerates the canonical ``manifests/corpus_index.json`` from live tier scans.

Licensing

MIT. Runtime is pure stdlib; media moves shell out to ``rclone`` (MIT), ``gcloud``, ``rsync``, and ``ffmpeg`` as external tools (never linked) ? keeping the dependency surface MIT/Apache/BSD-clean per the program-wide licensing guardrail.

201. user (2026-06-29T13:04:16.138Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/README.md (file state is current in your context ? no need to Read it back)

202. assistant (2026-06-29T13:04:52.814Z)

Recorded-Video Storage Map (canonical spec)

> **Status:** v1 · decisions locked 2026-06-29. The map any mountable medium replicates.

1. Invariants

- 3 copies, N+1.** Every **ratified** video has **2 durable** copies (F: + G:). GCS is a 3rd, **disposable** copy.
- GCS is a cache.** Proxies resident; full-res staged only during a training pass; **evicted** after **ratification**. Proxies + labels + trajectories stay (keeps GPU-free cloud eval runnable).
- No copy is deleted** unless the tool has **hash-verified** **2** other durable copies remain.
- Licensing-clean**, cross-platform (Windows + Linux/WSL), tool-driven.

2. Tiers (locked)

| Tier | Medium | Durable? | Role | Holds |
|--------------|-------------------------------|----------|-------------------------|---|
| F: | local disk (GoPro offload) | ? | ingest source + copy #1 | full-res masters (+ raw `_ingest/`) |
| G: | Google Drive 2?5 TB | ? | copy #2 | full-res of golden + ratified only (locked scope) |
| GCS | `gs://khelsutra-rally-corpus` | ? | disposable | training cache proxies always; full-res only while training |
| (new) | any mountable drive | ? | if you choose | drop-in extra whatever you `sync` to it |

G: scope decision (2026-06-29): golden + ratified only. Raw/unused captures stay F:-only; they get mirrored to G: the moment they become training-relevant (labelled or used in a ratified pass). A new medium becomes a tier with one `index` + one `sync` ? the layout is identical everywhere.

3. Canonical layout (byte-identical on every tier)

...

```
<root>/
  _ingest/                                # F: ONLY ? raw GoPro card dumps as-is, transient, never
  mirrored
    videos/
      full/<canonical_name>.mp4          # full-res master
      proxy/<canonical_name>.mp4         # 1080p proxy
    derived/
      trajectories/<canonical_name>.csv
      labels/<labels_name>.rallies.csv
    weights/<version>/...
    manifests/corpus_index.json          # THE MAP
    STORAGE_MAP.md
  ...
```

Flat + human-readable: object stores are flat and replication is trivial when the layout matches. The messy GoPro nesting (`100GOPRO/?`, chapters, `.LRV`/`.THM`) lives only in `\_ingest/`; promotion to `videos/full/` normalizes it.

4. Identity & naming (locked)

- `canonical\_name`** = the path stem on every tier (navigable).
- golden clips:** the corpus name (`adarsh\_avi\_singles`, `GX010128`, ?).

- ****non-golden GoPro clips:**** ``<capture_date>_<gopro_stem>`` (e.g. ``2026-06-08_GX010108``) ?
disambiguates

GoPro's recycled ``GX0101NN`` by date. (Authority is the manifest, never guessed at read time.)

- ****`video\_id`** = full-file MD5** (== ``compute_video_id``) ? the integrity + dedup key.

- ****`crc32c`**** (what ``gcloud`` returns) recorded too, for cheap cross-tier verify without re-reading
full files locally.

- Two files are the *same* video iff ``video_id`` matches ? how ``verify`/`dedup`/`evict`` reason safely.

5. The manifest (``manifests/corpus_index.json``)

Schema: ``schema/corpus_index.schema.json``. One entry per unique video; ``tiers`/`verified`` record per-tier presence (hash-checked); ``lifecycle`` tracks ``gcs_full_state`` (``never` ? staged ? evicted``) and ``trained_ratified``. This file is the source of truth; replication is declarative ? ****make tier X match the manifest****; drift = a ``verify`` mismatch.

6. Proxies (locked: 1080p)

Each master may carry a 1080p proxy (``ffmpeg``, deterministic settings recorded in the manifest), sufficient for WASB detection/eval at ~1/20th the bytes, and the ****default cloud object****. Full-res is

the archival truth on F:/G:. (Formalizes why ``mahadevpura_1`/`GX030094`` are proxies in GCS today.)

7. Lifecycle & the cost saver

``stage` ? (train) ? ratify ? evict``:

- ****stage****: push full-res (or proxy) to GCS for a run.

- ****ratify****: set ``trained_ratified=true`` once a pass using it is accepted.

- ****evict****: delete the GCS full-res ? ****guarded****: refuses unless full-res is verified present on

****both**** F: and G:. GCS bucket lifecycle rule auto-expires ``videos/full/*`` after N days as a backstop.

8. ``storagectl`` verbs

| Verb | What | Mutates? |

|---|---|---|

| ``index <tier>`` | scan + staged-hash + update manifest | manifest only |

| ``verify <tier>`` | entries present on tier w/ matching hash; report missing/corrupt/extra | no |

| ``status`` | per-video tier matrix + bytes + reclaimable + ****N+1 violations**** | no |

| ``dedup <fs-tier>`` | exact-MD5 dup groups; ``--quarantine`` / ``--delete`` (guarded) | optional |

| ``proxy <names?>`` | 1080p transcode (ffmpeg) | adds proxy |

| ``sync <src> <dst>`` | copy missing/changed entries (rsync · rclone · gcloud rsync); idempotent

| ``dst tier`` |

| ``stage <names?> --to GCS [--full|--proxy]`` | push for a run | GCS |

| ``evict <names?> --from GCS --require-ratified`` | cost saver; ****refuses unless ?2 durable copies verify**** | GCS |

Backends: a tier is an FS path, an ``rclone`` remote, or a ``gs://`` prefix ? same verbs. The FS backend

handles the ****bracket-glob caveat**** (``gcloud``/shell treat ``[Khelsutra]`` as a glob ? always stage through bracket-free paths).

9. Bootstrap (this repo, 2026-06-29)

1. ? First F: scan ? ``seed/`` (175 unique videos, 41 dup groups / ****251.8 GB**** reclaimable, 15/15 golden full-res located on F:, 0/15 full-res in GCS).

2. Build ``manifests/corpus_index.json`` from the seed (F: tier filled).

3. ****Dedup F:**** quarantine first ? delete after review (~251.8 GB).

4. ****Mirror F:G:**** golden+ratified ``videos/full/`` via ``rclone`` ? N+1 reached.

5. ****GCS:**** upload the 15 golden full-res (closes the gap) for the next pass; set the ``videos/full/*`` lifecycle-evict rule; keep proxies + labels + trajectories resident.

6. Wire `verify` into CI so drift/corruption surfaces automatically.

10. How other repos consume it

Read `manifests/corpus\_index.json`; resolve `canonical\_name`/`video\_id` ? the cheapest available tier.

`badminton-highlight-indexer` keeps its `gs://` eval manifest but can *derive* it from this map, so the

eval corpus and the storage truth never drift.

203. user (2026-06-29T13:04:53.892Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/STORAGE_MAP.md (file state is current in your context ? no need to Read it back)

204. assistant (2026-06-29T13:05:18.292Z)

Now the CLI itself ? the safety-critical logic (the N+1 evict guard, dedup) implemented as pure, testable functions, with media moves shelling out to rclone/gcloud/rsync/ffmpeg:

205. assistant (2026-06-29T13:06:13.371Z)

```
#!/usr/bin/env python3
"""storagectl ? custody/replication/cost-control for the recorded-video corpus.
```

A tier is a filesystem path, an rclone remote, or a gs:// prefix; the canonical layout is identical on every tier (see STORAGE_MAP.md), so replication is declarative: "make tier X match the manifest".

Runtime is pure stdlib. Media moves shell out to rsync / rclone / gcloud / ffmpeg (tools, not linked).

The safety-critical logic (the N+1 evict guard, the dedup canonical pick) is pure and unit-tested.

```
"""
```

```
from __future__ import annotations
```

```
import argparse
import hashlib
import json
import os
import subprocess
import sys
from dataclasses import dataclass, field
from typing import Any, Dict, List, Optional, Tuple
```

```
PARTIAL = 16 * 1024 * 1024 # bytes head+tail for the cheap signature
```

```
BLOCK = 8 * 1024 * 1024
```

----- config / manifest

```
@dataclass
```

```
class Tier:
```

```
    name: str
    kind: str # "fs" | "rclone" | "gcs"
    root: str
    durable: bool
```

```
def load_config(path: str) -> Dict[str, Tier]:
```

```
    with open(path, encoding="utf-8") as f:
```

```
        cfg = json.load(f)
```

```
    return {t["name"]: Tier(t["name"], t["kind"], t["root"], bool(t["durable"])) for t in
cfg["tiers"]}
```



```
def load_manifest(path: str) -> Dict[str, Any]:
    if not os.path.exists(path):
        return {"schema_version": 1, "videos": []}
    with open(path, encoding="utf-8") as f:
        return json.load(f)

def save_manifest(path: str, manifest: Dict[str, Any]) -> None:
    os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
    with open(path, "w", encoding="utf-8") as f:
        json.dump(manifest, f, indent=2)
```

----- hashing

```
def md5_full(path: str) -> str:
    h = hashlib.md5()
    with open(path, "rb") as f:
        for block in iter(lambda: f.read(BLOCK), b""):
            h.update(block)
    return h.hexdigest()

def partial_sig(path: str, size: int) -> str:
    """md5(first 16MB ++ last 16MB) ? cheap pre-filter so we only full-hash genuine
    collisions."""
    h = hashlib.md5()
    with open(path, "rb") as f:
        h.update(f.read(PARTIAL))
        if size > PARTIAL:
            f.seek(max(0, size - PARTIAL))
            h.update(f.read(PARTIAL))
    return h.hexdigest()
```

----- pure: dedup

```
@dataclass
class FileRec:
    path: str
    size: int
    mtime: float
    md5: Optional[str] = None

def dedup_plan(files: List[FileRec]) -> List[Dict[str, Any]]:
    """Exact-content duplicate groups (keyed by full MD5). Canonical-to-keep = earliest mtime,
    tie-break shortest path. Pure + deterministic. `files` must already carry full md5."""
    by_md5: Dict[str, List[FileRec]] = {}
    for fr in files:
        if fr.md5:
            by_md5.setdefault(fr.md5, []).append(fr)
    groups: List[Dict[str, Any]] = []
    for md5, recs in sorted(by_md5.items()):
        if len(recs) < 2:
            continue
        ordered = sorted(recs, key=lambda r: (r.mtime, len(r.path)))
        keep, redundant = ordered[0], ordered[1:]
        groups.append(
            {
                "video_id": md5,
                "keep": keep.path,
                "redundant": [r.path for r in redundant],
                "reclaimable_bytes": sum(r.size for r in redundant),
            }
        )
```

```
)
return groups
```

----- pure: the N+1 guard

```
def durable_full_copies(video: Dict[str, Any], tiers: Dict[str, Tier]) -> List[str]:
    """Durable tiers that hold a VERIFIED full-res copy of this video."""
    out = []
    tier_state = video.get("tiers", {})
    verified = video.get("verified", {})
    for name, t in tiers.items():
        if t.durable and tier_state.get(name) == "full" and verified.get(name):
            out.append(name)
    return out

def can_evict(
    video: Dict[str, Any],
    from_tier: str,
    tiers: Dict[str, Tier],
    require_ratified: bool = True,
) -> Tuple[bool, str]:
    """May we drop ``video``'s full-res from ``from_tier`` without breaking N+1?

    Rule: refuse unless ?2 durable tiers hold a *verified* full-res copy. Evicting a disposable
    tier (GCS) never reduces the durable count, but we still require the durable copies to exist
    and be hash-verified first ? that is the whole point of the guarantee.
    """
    if require_ratified and not video.get("lifecycle", {}).get("trained_ratified"):
        return False, "not ratified (lifecycle.trained_ratified != true)"
    if tiers.get(from_tier) and tiers[from_tier].durable:
        # Evicting a DURABLE tier: require ?2 OTHER durable verified copies to remain.
        remaining = [t for t in durable_full_copies(video, tiers) if t != from_tier]
        if len(remaining) >= 2:
            return True, f"ok ? {len(remaining)} other durable verified copies: {remaining}"
        return False, f"refuse ? only {len(remaining)} other durable verified copies: {remaining}"
    # Evicting a disposable tier (GCS): require ?2 durable verified copies to exist.
    durable = durable_full_copies(video, tiers)
    if len(durable) >= 2:
        return True, f"ok ? {len(durable)} durable verified copies: {durable}"
    return False, f"refuse ? only {len(durable)} durable verified copies: {durable} (need ?2)"

def n_plus_one_violations(manifest: Dict[str, Any], tiers: Dict[str, Tier]) -> List[str]:
    """Golden/ratified videos that do NOT yet have ?2 durable verified full-res copies."""
    bad = []
    for v in manifest.get("videos", []):
        needs = v.get("golden") or v.get("lifecycle", {}).get("trained_ratified")
        if needs and len(durable_full_copies(v, tiers)) < 2:
            bad.append(v["canonical_name"])
    return bad
```

----- backends (shell out)

```
def _run(cmd: List[str], dry: bool) -> int:
    print(("DRY " if dry else "RUN ") + " ".join(repr(c) if " " in c else c for c in cmd))
    if dry:
        return 0
    return subprocess.call(cmd)

def fs_scan(root: str) -> List[FileRec]:
```

```

"""Walk a filesystem tier's videos/full + _ingest, returning FileRecs (no hashing yet)."""
recs: List[FileRec] = []
for base, _dirs, files in os.walk(root):
    for fn in files:
        if os.path.splitext(fn)[1].lower() not in (".mp4", ".mov", ".avi", ".mkv", ".m4v"):
            continue
        p = os.path.join(base, fn)
        try:
            st = os.stat(p)
        except OSError:
            continue
        recs.append(FileRec(p, st.st_size, st.st_mtime))
return recs

def hash_for_dedup(recs: List[FileRec]) -> None:
    """Staged hashing in place: size groups ? partial sig ? full md5 only on real collisions."""
    by_size: Dict[int, List[FileRec]] = {}
    for r in recs:
        by_size.setdefault(r.size, []).append(r)
    for size, group in by_size.items():
        if len(group) < 2:
            continue
        by_sig: Dict[str, List[FileRec]] = {}
        for r in group:
            try:
                by_sig.setdefault(partial_sig(r.path, size), []).append(r)
            except OSError:
                continue
        for sig_group in by_sig.values():
            if len(sig_group) < 2:
                continue
            for r in sig_group:
                try:
                    r.md5 = md5_full(r.path)
                except OSError:
                    r.md5 = None

```

----- commands

```

def cmd_dedup(args: argparse.Namespace, tiers: Dict[str, Tier]) -> int:
    tier = tiers[args.tier]
    if tier.kind != "fs":
        print(f"dedup only supports fs tiers (got {tier.kind})", file=sys.stderr)
        return 2
    recs = fs_scan(tier.root)
    hash_for_dedup(recs)
    groups = dedup_plan(recs)
    total = sum(g["reclaimable_bytes"] for g in groups)
    print(f"# dedup {args.tier}: {len(recs)} videos, {len(groups)} duplicate groups, "
          f"{total/1e9:.2f} GB reclaimable")
    for g in groups:
        print(f"\n{g['video_id']} keep: {g['keep']}")
        for r in g["redundant"]:
            print(f" redundant: {r}")
    if args.quarantine or args.delete:
        qdir = os.path.join(tier.root, "_dedup_quarantine")
        for g in groups:
            for r in g["redundant"]:
                if args.quarantine:
                    dst = os.path.join(qdir, os.path.relpath(r, tier.root))
                    os.makedirs(os.path.dirname(dst), exist_ok=True) if not args.dry else None
                    _run(["mv", r, dst], args.dry)
                elif args.delete:

```

```

        _run(["rm", "-f", r], args.dry)

    return 0

def cmd_status(args: argparse.Namespace, tiers: Dict[str, Tier]) -> int:
    m = load_manifest(args.manifest)
    vids = m.get("videos", [])
    print(f"# corpus: {len(vids)} unique videos across tiers {list(tiers)}")
    per_tier_bytes: Dict[str, int] = {t: 0 for t in tiers}
    for v in vids:
        for t in tiers:
            if v.get("tiers", {}).get(t) == "full" and v.get("full"):
                per_tier_bytes[t] += v["full"].get("size", 0)
    for t, b in per_tier_bytes.items():
        print(f" {t}: {b/1e9:.1f} GB full-res")
    viol = n_plus_one_violations(m, tiers)
    print(f"\nN+1 violations (golden/ratified with <2 durable verified copies): {len(viol)}")
    for n in viol[:50]:
        print(f" ! {n}")
    return 0

def cmd_verify(args: argparse.Namespace, tiers: Dict[str, Tier]) -> int:
    tier = tiers[args.tier]
    m = load_manifest(args.manifest)
    missing, mismatch, ok = [], [], 0
    for v in m.get("videos", []):
        if v.get("tiers", {}).get(tier.name) != "full" or not v.get("full"):
            continue
        if tier.kind == "fs":
            p = os.path.join(tier.root, v["full"]["rel_path"])
            if not os.path.isfile(p):
                missing.append(v["canonical_name"])
            elif md5_full(p) != v["video_id"]:
                mismatch.append(v["canonical_name"])
            else:
                ok += 1
        else:
            print(f" (verify for {tier.kind} uses crc32c via the backend ? TODO wire {tier.kind})")
    print(f"# verify {tier.name}: {ok} ok, {len(missing)} missing, {len(mismatch)} hash-mismatch")
    for n in missing:
        print(f" MISSING {n}")
    for n in mismatch:
        print(f" CORRUPT {n}")
    return 1 if (missing or mismatch) else 0

def cmd_evict(args: argparse.Namespace, tiers: Dict[str, Tier]) -> int:
    m = load_manifest(args.manifest)
    by_name = {v["canonical_name"]: v for v in m.get("videos", [])}
    rc = 0
    for name in args.names:
        v = by_name.get(name)
        if not v:
            print(f" ? {name}: not in manifest")
            rc = 1
            continue
        ok, reason = can_evict(v, args.from_tier, tiers, require_ratified=args.require_ratified)
        if not ok:
            print(f" REFUSE {name}: {reason}")
            rc = 1
            continue
        tier = tiers[args.from_tier]

```

```

    rel = v["full"]["rel_path"]
    if tier.kind == "gcs":
        _run(["gcloud", "storage", "rm", f"{tier.root.rstrip('/')}/{rel}"], args.dry)
    elif tier.kind == "fs":
        _run(["rm", "-f", os.path.join(tier.root, rel)], args.dry)
    elif tier.kind == "rclone":
        _run(["rclone", "delete", f"{tier.root.rstrip('/')}/{rel}"], args.dry)
    if not args.dry:
        v.setdefault("lifecycle", {})[gcs_full_state] = "evicted" if tier.kind == "gcs"
else v["lifecycle"].get(gcs_full_state, "never")
    v.setdefault("tiers", {})[args.from_tier] = "proxy" if tier.kind == "gcs" else
"absent"
    print(f"  EVICT {name}: {reason}")
    if not args.dry:
        save_manifest(args.manifest, m)
    return rc

def cmd_stub(verb: str) -> int:
    print(f"`{verb}` backend is specified in STORAGE_MAP.md §8 and not yet wired in v0.1 "
          f"(shells to rsync/rclone/gcloud/ffmpeg). Contributions welcome.")
    return 0

```

----- CLI

```

def main(argv: Optional[List[str]] = None) -> int:
    ap = argparse.ArgumentParser(prog="storagectl", description=__doc__.splitlines()[0])
    ap.add_argument("--config", default="vault.local.json", help="tier config (default
vault.local.json)")
    ap.add_argument("--manifest", default="manifests/corpus_index.json")
    ap.add_argument("--dry", action="store_true", help="print actions, change nothing")
    sub = ap.add_subparsers(dest="cmd", required=True)

    p = sub.add_parser("dedup", help="report (and optionally quarantine/delete) duplicate videos
on an fs tier")
    p.add_argument("tier")
    p.add_argument("--quarantine", action="store_true")
    p.add_argument("--delete", action="store_true")

    sub.add_parser("status", help="per-tier bytes + N+1 violations").add_argument # noqa: B018
    sub.add_parser("status", help=argparse.SUPPRESS) if False else None

    pv = sub.add_parser("verify", help="check a tier matches the manifest by hash")
    pv.add_argument("tier")

    pe = sub.add_parser("evict", help="drop full-res from a tier (N+1 guarded)")
    pe.add_argument("names", nargs="+")
    pe.add_argument("--from", dest="from_tier", required=True)
    pe.add_argument("--require-ratified", action="store_true", default=True)
    pe.add_argument("--force-unratified", dest="require_ratified", action="store_false")

    for verb in ("index", "sync", "stage", "proxy"):
        sub.add_parser(verb, help=f"{verb} (see STORAGE_MAP.md §8)")

    args = ap.parse_args(argv)
    tiers = load_config(args.config) if os.path.exists(args.config) else {}

    if args.cmd == "dedup":
        return cmd_dedup(args, tiers)
    if args.cmd == "status":
        return cmd_status(args, tiers)
    if args.cmd == "verify":
        return cmd_verify(args, tiers)
    if args.cmd == "evict":

```

```
    return cmd_evict(args, tiers)
return cmd_stub(args.cmd)
```

```
if __name__ == "__main__":
    raise SystemExit(main())
```

206. user (2026-06-29T13:06:14.451Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py (file state is current in your context ? no need to Read it back)

207. assistant (2026-06-29T13:06:27.654Z)

Let me fix a botched line in the argparse setup (I duplicated the `status` subparser):

208. user (2026-06-29T13:06:31.618Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

209. assistant (2026-06-29T13:06:50.263Z)

Now the package init, example tier config, the safety tests, and CI:

210. assistant (2026-06-29T13:06:51.750Z)

```
"""storagectl ? recorded-video corpus custody / replication / cost control."""
```

```
from .storagectl import ( # noqa: F401
    Tier,
    can_evict,
    dedup_plan,
    durable_full_copies,
    n_plus_one_violations,
)
```

```
__version__ = "0.1.0"
```

211. user (2026-06-29T13:06:52.770Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/__init__.py (file state is current in your context ? no need to Read it back)

212. assistant (2026-06-29T13:07:04.149Z)

```
{
  "_comment": "Copy to vault.local.json and point each tier at YOUR mount/remote. A tier 'root'
is the CANONICAL root that directly contains videos/full, videos/proxy, derived/, manifests/.
For an as-yet-unnormalized F: GoPro dump you can point dedup at the raw folder; verify/sync
expect the canonical root.",
  "tiers": [
    { "name": "F",    "kind": "fs",    "root": "/mnt/f/[Khelsutra] GoPro Hero Black 12/corpus",
"durable": true },
    { "name": "G",    "kind": "rclone", "root": "gdrive:rally-corpus",
"durable": true },
    { "name": "GCS",  "kind": "gcs",   "root": "gs://khelsutra-rally-corpus",
"durable": false }
  ]
}
```

213. user (2026-06-29T13:07:05.216Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/vault.example.json (file state is current in your context ? no need to Read it back)

214. assistant (2026-06-29T13:07:25.847Z)

"""Tests for the safety-critical pure logic: the N+1 evict guard and the dedup canonical pick.

These are the parts where a bug loses data, so they are unit-tested independently of any tier I/O.

"""

```
from storagectl.storagectl import (
    FileRec,
    Tier,
    can_evict,
    dedup_plan,
    durable_full_copies,
    n_plus_one_violations,
)
```

```
TIERS = {
    "F": Tier("F", "fs", "/mnt/f/corpus", durable=True),
    "G": Tier("G", "rclone", "gdrive:rally-corpus", durable=True),
    "GCS": Tier("GCS", "gcs", "gs://khelsutra-rally-corpus", durable=False),
}
```

```
def _vid(**over):
    v = {
        "canonical_name": "clip",
        "video_id": "a" * 32,
        "full": {"rel_path": "videos/full/clip.mp4", "size": 100, "md5": "a" * 32},
        "tiers": {"F": "full", "G": "full", "GCS": "full"},
        "verified": {"F": "2026-06-29", "G": "2026-06-29", "GCS": "2026-06-22"},
        "lifecycle": {"trained_ratified": True},
    }
    v.update(over)
    return v
```

----- N+1 evict guard

```
def test_evict_gcs_ok_when_two_durable_verified():
    ok, _ = can_evict(_vid(), "GCS", TIERS)
    assert ok is True
```

```
def test_evict_refused_when_not_ratified():
    ok, why = can_evict(_vid(lifecycle={"trained_ratified": False}), "GCS", TIERS)
    assert ok is False and "ratified" in why
```

```
def test_evict_refused_when_only_one_durable_verified():
    # G: present but UNVERIFIED ? only F: counts ? refuse.
    ok, why = can_evict(_vid(verified={"F": "2026-06-29", "G": None, "GCS": "x"}), "GCS", TIERS)
    assert ok is False and "need ?2" in why
```

```
def test_evict_refused_when_durable_copy_absent():
    ok, _ = can_evict(_vid(tiers={"F": "full", "G": "absent", "GCS": "full"}), "GCS", TIERS)
    assert ok is False
```

```
def test_evict_durable_tier_needs_two_OTHER_durable():
    # Evicting F: itself ? only G: remains durable+verified (1) ? refuse (need ?2 others).
    ok, _ = can_evict(_vid(), "F", TIERS)
```

```

assert ok is False

def test_durable_full_copies_excludes_disposable_and_unverified():
    v = _vid(verified={"F": "d", "G": None, "GCS": "d"})
    assert durable_full_copies(v, TIERS) == ["F"] # G unverified, GCS not durable

def test_n_plus_one_violation_flags_golden_with_one_copy():
    m = {"videos": [_vid(golden=True, tiers={"F": "full", "G": "absent", "GCS": "full"})]}
    assert n_plus_one_violations(m, TIERS) == ["clip"]

def test_n_plus_one_clean_when_two_durable():
    m = {"videos": [_vid(golden=True)]}
    assert n_plus_one_violations(m, TIERS) == []

```

----- dedup canonical pick

```

def test_dedup_picks_earliest_mtime_canonical():
    files = [
        FileRec("/a/copy_b.mp4", 10, mtime=200.0, md5="x" * 32),
        FileRec("/a/copy_a.mp4", 10, mtime=100.0, md5="x" * 32), # earliest ? keep
    ]
    g = dedup_plan(files)
    assert len(g) == 1
    assert g[0]["keep"] == "/a/copy_a.mp4"
    assert g[0]["redundant"] == ["/a/copy_b.mp4"]
    assert g[0]["reclaimable_bytes"] == 10

def test_dedup_tiebreak_shortest_path_on_equal_mtime():
    files = [
        FileRec("/deep/nested/long_name.mp4", 5, mtime=100.0, md5="y" * 32),
        FileRec("/x.mp4", 5, mtime=100.0, md5="y" * 32), # shorter path ? keep
    ]
    g = dedup_plan(files)
    assert g[0]["keep"] == "/x.mp4"

def test_dedup_ignores_singletons():
    files = [FileRec("/only.mp4", 1, mtime=1.0, md5="z" * 32)]
    assert dedup_plan(files) == []

```

215. user (2026-06-29T13:07:26.921Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/tests/test_safety.py (file state is current in your context ? no need to Read it back)

216. assistant (2026-06-29T13:07:34.138Z)

```

name: ci
on:
  push:
    branches: [master]
  pull_request:

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
      with:

```



```

    python-version: "3.11"
- name: Install
  run: pip install -e ".[dev]"
- name: Lint
  run: ruff check .
- name: Validate manifest schema (if present)
  run: |
    python - <<'PY'
    import json, glob, sys
    try:
        import jsonschema
    except ImportError:
        print("jsonschema not installed; skipping"); sys.exit(0)
    schema = json.load(open("schema/corpus_index.schema.json"))
    for m in glob.glob("manifests/*.json"):
        jsonschema.validate(json.load(open(m)), schema)
        print("valid:", m)
    PY
- name: Tests
  run: pytest -q

```

217. user (2026-06-29T13:07:36.165Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/.github/workflows/ci.yml (file state is current in your context ? no need to Read it back)

218. assistant (2026-06-29T13:08:15.432Z)

Let me fix the README config reference and run the safety tests to confirm the core works:

219. user (2026-06-29T13:08:18.218Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

220. user (2026-06-29T13:08:22.590Z)

import + CLI help smoke

import ok 0.1.0

CLI parses OK

safety tests

..... [100%]
11 passed in 0.02s

221. assistant (2026-06-29T13:08:46.815Z)

Import works, CLI parses, **11/11 safety tests pass**. Committing the scaffold:

222. user (2026-06-29T13:08:56.035Z)

Exit code 128

staged files (catalog tsv should be ignored):

```

A .github/workflows/ci.yml
A .gitignore
A LICENSE
A README.md
A STORAGE_MAP.md
A pyproject.toml
A schema/corpus_index.schema.json
A seed/f_drive_dedup_report.md
A seed/f_drive_video_index.json
A seed/gcs_migration_gap.md
A storagectl/__init__.py

```

```
A storagectl/storagectl.py
A tests/test_safety.py
A vault.example.json
Author identity unknown
```

*** Please tell me who you are.

Run

```
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
```

to set your account's default identity.

Omit --global to set the identity only in this repository.

fatal: empty ident name (for <avidullu@AVIS-MSI.>) not allowed

tree (committed):

```
.github/workflows/ci.yml
.gitignore
LICENSE
README.md
STORAGE_MAP.md
pyproject.toml
schema/corpus_index.schema.json
seed/f_drive_dedup_report.md
seed/f_drive_video_index.json
seed/gcs_migration_gap.md
storagectl/__init__.py
storagectl/storagectl.py
tests/test_safety.py
vault.example.json
```

log:

fatal: your current branch 'master' does not have any commits yet

223. assistant (2026-06-29T13:09:13.338Z)

Just need a git identity for this fresh repo (staging worked; catalog correctly ignored). Let me match whatever the sibling repos use and commit:

224. user (2026-06-29T13:09:22.165Z)

sibling repo identity

name=Avi Dullu email=avi.dullu@gmail.com

ceed158 feat: scaffold rally-corpus-vault ? storage map + storagectl + safety core

committed files: 14

225. assistant (2026-06-29T13:09:57.637Z)

Committed (`ceed158`, 14 files). Before I wrap up, let me make sure CI would actually be green ? running ruff on the new repo (CI runs it), since I advertised it:

226. user (2026-06-29T13:10:01.147Z)

```
F401 [*] `dataclasses.field` imported but unused
--> storagectl/storagectl.py:18:36
|
16 | import subprocess
17 | import sys
18 | from dataclasses import dataclass, field
|                                     ^^^^^
19 | from typing import Any, Dict, List, Optional, Tuple
|
help: Remove unused import: `dataclasses.field`
```

Found 1 error.
[*] 1 fixable with the `--fix` option.

227. assistant (2026-06-29T13:10:11.386Z)

One unused import ? fixing and amending the commit:

228. user (2026-06-29T13:10:13.511Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

229. user (2026-06-29T13:10:26.931Z)

ruff recheck

All checks passed!

tests recheck

..... [100%]
11 passed in 0.02s

final commit

c41545a feat: scaffold rally-corpus-vault ? storage map + storagectl + safety core

230. assistant (2026-06-29T13:10:51.710Z)

Clean ? ruff passes, 11 tests green, amended to `c41545a`. The scaffold is CI-ready.

231. user (2026-06-29T13:10:55.533Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

232. assistant (2026-06-29T13:11:11.964Z)

****`rally-corpus-vault` is built and committed locally**** (`c41545a`, CI-clean) ? your storage-map proposal made real:

- ****[STORAGE_MAP.md](rally-corpus-vault/STORAGE_MAP.md)**** ? the canonical spec with your decisions locked (name, G: = golden+ratified, 1080p proxies, date+stem naming).
- ****`storagectl`**** ? `dedup` / `status` / `verify` / `evict` working;
`index`/`sync`/`stage`/`proxy` honestly stubbed for the next pass.
- ****The safety core is real and tested****: `can\_evict` enforces "never drop a copy unless ?2 durable copies hash-verify" ? ****11/11 tests green****, ruff clean.
- ****schema****, MIT license, CI (ruff + schema-validate + pytest), and ****`seed`**** carrying today's F: scan as the bootstrap data.

Consolidated session status

| Thread | State |
|---|---|
| ****P2b promotion**** | PR [#396](https://github.com/Khelsutra/badminton-highlight-indexer/pull/396) up ? ****awaiting your launch signoff**** |
| Issues filed | [#397](https://github.com/Khelsutra/badminton-highlight-indexer/issues/397)
stray dirs · [#398](https://github.com/Khelsutra/badminton-highlight-indexer/issues/398) served-gate harness bug |
| ****F: index/dedup**** | Final: ****175 unique videos, 41 dup groups, 251.8 GB reclaimable****; 15/15 golden full-res on F:, 0/15 in GCS. Artifacts in `f\_drive\_index/` |
| ****rally-corpus-vault**** | scaffolded + committed locally; needs a remote |

Ready when you say go (all gated on your confirmation ? nothing outward/destructive done yet)

1. ****Push the repo**** to a new GitHub remote (the question below).
2. ****Publish the index to GCS**** ? I'd build the canonical `corpus\_index.json` (implement `index`), then push to `gs://khelsutra-rally-corpus/manifests/`.
3. ****Bootstrap the 3-copy corpus****: dedup F: ? ****quarantine**** the 252 GB (reversible, you review

before any delete) ? mirror golden full-res F:?G: (rclone) ? upload the 15 golden full-res to GCS (closes the gap, unblocks cloud/video eval) ? set the GCS `videos/full/\*` lifecycle-evict rule.

One outward decision to proceed:

233. user (2026-06-29T13:42:52.421Z)

Your questions have been answered: "rally-corpus-vault is committed locally and CI-clean. Create the GitHub remote and push it?"="Private under Khelsutra". You can now continue with these answers in mind.

234. assistant (2026-06-29T13:43:08.176Z)

Creating the private remote under Khelsutra and pushing:

235. user (2026-06-29T13:43:27.353Z)

```
https://github.com/Khelsutra/rally-corpus-vault
To https://github.com/Khelsutra/rally-corpus-vault.git
 * [new branch]      HEAD -> master
branch 'master' set up to track 'origin/master'.
remote + tracking
origin https://github.com/Khelsutra/rally-corpus-vault.git (fetch)
origin https://github.com/Khelsutra/rally-corpus-vault.git (push)
master
c41545a feat: scaffold rally-corpus-vault ? storage map + storagectl + safety core
```

236. assistant (2026-06-29T13:44:02.693Z)

? **github.com/Khelsutra/rally-corpus-vault** is live (private, under Khelsutra) ? pushed and tracking.

237. user (2026-06-29T13:44:06.470Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

238. assistant (2026-06-29T13:44:23.573Z)

Everything you asked for this session is landed. Where it all stands:

Done

- **P2b promotion** ? PR [#396](https://github.com/Khelsutra/badminton-highlight-indexer/pull/396) (finding-B promoted to default-ON; A/B showed no regression + reduced over-segmentation; suite 1503?, ruff/mypy clean). **Awaiting your launch signoff** ? my recommendation: GO to merge P2b, NO-GO on a product launch until the ship-gate (#172) and WASB weight provenance are resolved.
- **Issues** [#397](https://github.com/Khelsutra/badminton-highlight-indexer/issues/397) (stray dirs) and [#398](https://github.com/Khelsutra/badminton-highlight-indexer/issues/398) (the real served-gate verification bug I found).
- **F: archive** fully indexed: **175 unique videos, 41 confirmed dup groups, 251.8 GB reclaimable**, and the full migration gap mapped (15/15 golden full-res on F:, 0/15 in GCS). Artifacts in `f\_drive\_index`.
- **[rally-corpus-vault](https://github.com/Khelsutra/rally-corpus-vault)** ? the storage map made real: the spec, `storagectl` with a tested N+1 safety core, schema, CI, MIT license, seeded from the F: scan.

Your move (all gated ? nothing outward/destructive run without your go)

1. **Publish the index to GCS** ? I'll implement `storagectl index`, build the canonical `corpus\_index.json`, push it to `gs://?/manifests/`.
2. **Bootstrap the 3-copy corpus** ? dedup F: ? **quarantine** the 252 GB (you review before any delete) ? mirror golden full-res F:?G: ? upload the 15 golden full-res to GCS ? set the

`videos/full/\*` lifecycle-evict rule.

Nothing on F: has been moved or deleted; it's all still report-only. Just say which of those to start, or point me at the next thing.

Two small loose ends from earlier whenever you want them: I left the `mahadevpura\_1` GPU smoke abandoned (GPU is proven live regardless), and `reportlab` is still uninstalled (2 PDF tests skip) ? neither blocks anything.

239. user (2026-06-29T14:00:42.377Z)

Can you take a look at the comments on the PR and then merge.

240. assistant (2026-06-29T14:00:59.017Z)

Let me pull the PR's comments, reviews, inline review threads, and check status before merging.

241. user (2026-06-29T14:01:06.525Z)

===== PR #396 overview =====

```
{"isDraft":false,
"mergeStateStatus":"CLEAN",
"mergeable":"MERGEABLE",
"number":396,
"reviewDecision":"",
"state":"OPEN",
"title":"feat(config): promote inactivity_temporal_density to default-ON (P2b)",
"url":"https://github.com/Khelsutra/badminton-highlight-indexer/pull/396"}
```

===== Reviews (approve/request-changes) =====

[COMMENTED] avidullu: LGTM. Two comments mostly around process but approval to merge. Feel free to file a bug for it for now and merge this PR.

===== Issue comments (general) =====

242. assistant (2026-06-29T14:01:17.921Z)

The user reviewed it: `"LGTM. Two comments mostly around process but approval to merge. Feel free to file a bug for it for now and merge this PR."` PR is `MERGEABLE` / `CLEAN`. Let me read those two inline comments so I can file the bug they want before merging:

243. user (2026-06-29T14:01:20.343Z)

===== Inline review comments on the diff =====

--- avidullu on backend/config/models.py:201

If already not there can we please add this as a "launch bit flip PR must also include moving the documentation of the project to docs/archives/ appropriately including any quality iterations, project documents, ADRs etc."?

Additionally, I think we can add a CI to the repo which blocks PR submission on identifying any 404 links (i.e. the launch PR must also update hrefs across repo)

--- avidullu on docs/RALLY_DETECTION_FIXES_PLAN.md:110

where is this rule codified? Can you point me to the link?

244. assistant (2026-06-29T14:01:48.462Z)

Two process comments, both actionable. The user said file a bug and merge ? but one is a direct question I should answer first. Let me see exactly what line 110 references and whether that "rule" is actually codified anywhere:

245. user (2026-06-29T14:01:49.868Z)

100 corpus + GPU/WSL-extracted trajectory features, which don't exist offline. Note finding B is a

```

101      **no-op under dense tracking**, so the A/B only shows signal on clips where the
shuttle tracker
102      actually drops out ? which is exactly why it can't be measured on synthetic/offline
fixtures.
103
104      **Measured 2026-06-29 (PR #396).** Ran exactly the command above on the 15-clip
golden set
105      (trajectories + labels from `gs://khealsutra-rally-corpus`; GTs via the repo's own
106      `load_gt_intervals`). Result at SERVED: **tolF1 0.353 ? 0.364** (?tolF1 **+0.010**,
95% CI
107      [?0.014, +0.034], sign **10W/4L**, p=0.18 ? neutral-to-positive, **does not
regress**); |?end|
108      **2.97 ? 2.72 s**; seg_count_ratio **?0.103** (p=0.006), split_rate **?0.010**
(p=0.016); **R1
109      over-seg guard PASS** (net 0.397?0.378). Formal gate:
`served_regresses(eps=0.0)=False`,
110      `promote_if_served_safe(structural=True) ? promote`. Per the rule (promote iff tolF1
does not
111      regress) ? **PROMOTED** to default-ON. The stricter honest-LIFT bar reports "no
significant lift"
112      (CI spans 0) but **no regression** ? i.e. a do-no-harm correctness fix, not a
measured win.
113
114      ## 9. Definition of done
115      - [x] `_spread` bug fixed and tested (behaviour-neutral, unconditional).
116      - [x] `_trim_inactive_tail` density fix implemented behind `inactivity_temporal_density`
(default-OFF),
117      plumbed config ? segmenter ? eval preset, tested in both flag states.
118      - [x] **P2 promotion**: golden-set A/B at SERVED windowing, flag flipped default-ON ? no
regression
119      (?tolF1 +0.010, 10W/4L; seg_ratio ?0.103 p=0.006; over-seg guard PASS).
*(2026-06-29, PR #396 ? see §8 Q2)*
120      - [x] Warning log added for malformed rows in `rally_cvat.py`. *(sibling-repo PR
`sdc#50`)*
121      - [x] Tests updated and passing *(offline suite; golden-corpus regression is owner-gated
? see Q2)*.
122
123      ## 10. References
124      **Internal code anchors `[verified]`:**

```

246. user (2026-06-29T14:01:54.625Z)

where is the 'archive only terminal-state / honestly update first' rule codified?

```

/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/CLAUDE.md:23:
**`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only move
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/CLAUDE.md:24:  **terminal-
state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and **before
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/CLAUDE.md:25:  archiving,
HONESTLY update the doc first** (verify claims vs code ? flip status + a completion
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/CLAUDE.md:26:  note ? update
inbound refs ? keep the lineage ? then `git mv`), per the checklist + living
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/docs/DOC_STATUS.md:3:>
**Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**.
**Last full review:** 2026-06-21.
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/docs/DOC_STATUS.md:17:## 2.
Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/docs/DOC_STATUS.md:23:5.
**`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs
stay at `docs/` top level). Relative links inside become historical (note this in the doc's
header).
/home/avidullu/projects/khealsutra-guru/badminton-highlight-indexer/docs/DOC_STATUS.md:26:>
**Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered
on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`),
reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to
`docs/archives/past_projects/DECOUPLED_COMPUTE/`.

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DOC_STATUS.md:82:## 4.
Open due-diligence findings / actions (2026-06-21 review)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DOC_STATUS.md:93:Update
a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-
slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive
policy and [[working-agreement-tracked-project-docs]].

DOC_STATUS.md exists + headline

```
-rw-r--r-- 1 avidullu avidullu 11972 Jun 25 11:46 /home/avidullu/projects/khelsutra-  
guru/badminton-highlight-indexer/docs/DOC_STATUS.md  
1:# Documentation Status & Archival Register  
4:> **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone /  
delivered / archived), and re-review at each clean-slate.  
5:> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as  
work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly  
updating it first** (owner directive, 2026-06-21).  
9:## 0. How to use this  
11:- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of  
truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival  
checklist.  
12:- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-  
readiness and points to each tracked doc's §7.  
14:## 1. Lifecycle legend  
17:## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*  
18:Before a foundational/tracked doc moves to `docs/archives/`:  
19:1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts  
`[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it  
first.)  
23:5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY;  
live docs stay at `docs/` top level). Relative links inside become historical (note this in the  
doc's header).  
26:> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5  
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward  
WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014  
lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.  
28:## 3. The register  
30:### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the  
completeness source-of-truth)  
35:| `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation  
pending) | Promote-or-reject toF1 as the gate (ablation) ? then archive with the R-cluster |
```

247. assistant (2026-06-29T14:03:09.064Z)

Both comments are clear, and the user pre-authorized "file a bug and merge." Comment 1 is two
process asks (auto-archive docs on a launch PR + a broken-link CI). Comment 2 is a question ?
"where is this rule codified?" ? and the answer is real: the rule *is* codified in
`backend/eval/promotion.py` (`promote\_if\_served\_safe` / `served\_regresses(eps=0.0)`), the plan's
§8 Q2 is just the prose. The doc should link to it.

Let me file the tracking bug, grab the comment IDs to reply in-thread, and check the repo's
merge methods ? in parallel:

248. user (2026-06-29T14:03:13.504Z)

inline comment IDs

```
3492196978      backend/config/models.py:201  
3492228166      docs/RALLY_DETECTION_FIXES_PLAN.md:110
```

allowed merge methods

```
{ "delete_branch_on_merge": true, "merge_commit": true, "rebase": true, "squash": true }
```

249. user (2026-06-29T14:03:22.319Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/399>

250. assistant (2026-06-29T14:04:10.930Z)

Issue ****#399**** filed; comment IDs and merge methods confirmed (squash matches the repo's ``(#NN)`` history). Now: reply to both threads (answer the codification question, link the issue), then squash-merge and clean up locally.

251. user (2026-06-29T14:04:53.243Z)

reply to comment 2 (where is the rule codified)

replied

reply to comment 1 (archival convention + 404 CI)

replied

squash-merge PR #396 (delete remote branch)

verify merged

```
{"mergeCommit":"39dff1d67255bf9a903c1e57b6ec0b9e1bc469ce","mergedAt":"2026-06-29T14:04:50Z","state":"MERGED"}
```

252. assistant (2026-06-29T14:05:02.320Z)

Merged ? squash commit ``39dff1d``, remote branch deleted, both comment threads answered. Now cleaning up the local checkout (sync master, drop the merged feature branch):

253. user (2026-06-29T14:05:12.169Z)

Switched to branch 'master'

Your branch is up to date with 'origin/master'.

```
docs/RALLY_DETECTION_FIXES_PLAN.md | 37 ++++++-----
tests/test_rally_seg_eval.py        | 19 +++++-----
tests/test_windowing.py             | 14 +++++-----
4 files changed, 57 insertions(+), 30 deletions(-)
Deleted branch p2b/promote-inactivity-temporal-density (was ebd019c).
```

local master now at:

```
39dff1d feat(config): promote inactivity_temporal_density to default-ON (P2b) (#396)
0c9c693 feat(eval): expose inactivity_temporal_density in the rally_seg_eval A/B CLI (P2a) (#395)
```

confirm the promoted default landed on master:

```
202: inactivity_temporal_density: bool = True
```

254. assistant (2026-06-29T14:05:32.061Z)

Done ? ****PR #396 is merged.**** ?

- ****Reviewed both comments.**** Comment 2 ("where is this rule codified?") ? answered in-thread: the rule is codified in `[backend/eval/promotion.py]` (`badminton-highlight-indexer/backend/eval/promotion.py`) (``promote_if_served_safe` + `served_regresses(eps=0.0)` = "no served-mean drop allowed"`); §8 Q2 is just the prose.
- ****Filed the bug**** you asked for ? [#399](https://github.com/Khelsutra/badminton-highlight-indexer/issues/399) ? capturing both process items from comment 1 (launch-PR must auto-archive project docs per DOC_STATUS §2; CI link-checker that blocks broken/404 hrefs) plus the rule?code cross-link point. Replied on both threads.
- ****Squash-merged**** ? master ``39dff1d`` ****feat(config): promote inactivity_temporal_density to default-ON (P2b) (#396)****; deleted the remote + local branch; synced local master and confirmed ``inactivity_temporal_density: bool = True`` landed.

P2b is fully closed out and shipped. The only thing still open on your side is the broader ****GO/NO-GO launch call**** (my rec stands: the rally-detection **quality** gate isn't met yet ? WASB provenance + ship-gate target #172), and the ****corpus bootstrap**** (dedup quarantine / G: mirror / golden?GCS upload) whenever you want me to start it.

255. user (2026-06-29T14:09:33.868Z)

I cannot check your context size, how much more is left?

256. assistant (2026-06-29T14:09:56.396Z)

Honestly, I can't measure my own remaining context ? there's no tool that reports my token usage or window size to me from inside the conversation, so any number I gave you would be made up. What I can tell you:

- I'm running on a large (1M-token) window, and while this session has been long (lots of file reads, the F: scan, the repo scaffold), nothing has felt truncated yet ? I still have the full thread in view.
- The harness ****auto-summarizes/compacts**** older context when it grows long, so I'll keep working seamlessly even if it fills up. You won't hit a hard wall mid-task.
- The practical signal to watch: if I start misremembering earlier specifics (wrong file paths, numbers, decisions), that's the cue context got compacted ? call it out and I'll re-check from disk/GitHub.

The reassuring part: ****nothing important lives only in my context.**** It's all durably persisted ?

- PR #396 merged to master; issues #397/#398/#399 on GitHub;
- `rally-corpus-vault` pushed to the private remote;
- the F: index/dedup artifacts in `f\_drive\_index/`, and the proposal/spec docs on disk.

So even a fresh session loses nothing.

If you'd like insurance for continuity, I can write a short ****handoff note to memory**** (or a `HANDOFF.md`) capturing exactly where each thread stands ? P2b done, launch GO/NO-GO pending, the corpus bootstrap steps queued. Want me to do that?

257. user (2026-06-29T16:19:28.223Z)

Then it must be fine to continue here. what were the pending items apart from the video indexing and data moves? Before we do the video directory migration, can you tell me what the repo supports right now? Also please use "G:\My Drive\KheISutraGuru\Sports Videos Corpus" which is the Google Drive folder root for all video storage that we will use. It would be best if the metadata etc. is also written there. What I am interested in is what are the steps to follow for me to add the next vide reliably using this new mechanism?

258. assistant (2026-06-29T16:21:10.740Z)

Good ? let's keep going here. Let me check whether G: is reachable from WSL and whether that corpus folder exists, since it changes how I wire the tooling:

259. user (2026-06-29T16:21:17.770Z)

is G: mounted into WSL?

```
mountpoint: /mnt/g: No such file or directory
c
f
wsl
wslg
```

does the corpus root exist? (G:\My Drive\KheISutraGuru\Sports Videos Corpus)

```
ls: cannot access '/mnt/g/My Drive/KheISutraGuru': No such file or directory
ls: cannot access '/mnt/g/My Drive/KheISutraGuru/Sports Videos Corpus': No such file or directory
```

F: still mounted?

```
/mnt/f is a mountpoint
```

Windows sees G: as (from earlier): G 515.7GB used / 230.5 free ? confirm via powershell

```
Name Root UsedGB
```

G G:\ 517.4

260. assistant (2026-06-29T16:22:13.871Z)

G: isn't mounted into WSL yet (only `/mnt/c` and `/mnt/f`), and Windows shows it as the Drive Desktop mount (517 GB used) ? so `G:\My Drive\KheISutraGuru\Sports Videos Corpus` becomes `/mnt/g/My Drive/KheISutraGuru/Sports Videos Corpus` once you `sudo mount -t drvfs G: /mnt/g` (one-time, like F:). The folder doesn't exist yet ? it gets created on first use.

Before I answer your questions and adopt that root, let me re-read the current CLI so I can wire in a reliable `**`add`**` command (the missing piece that makes "add the next video" actually tool-driven):

261. user (2026-06-29T16:22:15.094Z)

```
236         per_tier_bytes: Dict[str, int] = {t: 0 for t in tiers}
237         for v in vids:
238             for t in tiers:
239                 if v.get("tiers", {}).get(t) == "full" and v.get("full"):
240                     per_tier_bytes[t] += v["full"].get("size", 0)
241         for t, b in per_tier_bytes.items():
242             print(f" {t}: {b/1e9:.1f} GB full-res")
243         viol = n_plus_one_violations(m, tiers)
244         print(f"\nN+1 violations (golden/ratified with <2 durable verified copies):
{len(viol)}")
245         for n in viol[:50]:
246             print(f" ! {n}")
247         return 0
248
249
250     def cmd_verify(args: argparse.Namespace, tiers: Dict[str, Tier]) -> int:
251         tier = tiers[args.tier]
252         m = load_manifest(args.manifest)
253         missing, mismatch, ok = [], [], 0
254         for v in m.get("videos", []):
255             if v.get("tiers", {}).get(tier.name) != "full" or not v.get("full"):
256                 continue
257             if tier.kind == "fs":
258                 p = os.path.join(tier.root, v["full"]["rel_path"])
259                 if not os.path.isfile(p):
260                     missing.append(v["canonical_name"])
261                 elif md5_full(p) != v["video_id"]:
262                     mismatch.append(v["canonical_name"])
263             else:
264                 ok += 1
265         else:
266             print(f" (verify for {tier.kind} uses crc32c via the backend ? TODO wire
{tier.kind})")
267         print(f"# verify {tier.name}: {ok} ok, {len(missing)} missing, {len(mismatch)} hash-
mismatch")
268         for n in missing:
269             print(f" MISSING {n}")
270         for n in mismatch:
271             print(f" CORRUPT {n}")
272         return 1 if (missing or mismatch) else 0
273
274
275     def cmd_evict(args: argparse.Namespace, tiers: Dict[str, Tier]) -> int:
276         m = load_manifest(args.manifest)
277         by_name = {v["canonical_name"]: v for v in m.get("videos", [])}
278         rc = 0
279         for name in args.names:
280             v = by_name.get(name)
281             if not v:
```

```

282         print(f" ? {name}: not in manifest")
283         rc = 1
284         continue
285     ok, reason = can_evict(v, args.from_tier, tiers,
require_ratified=args.require_ratified)
286     if not ok:
287         print(f" REFUSE {name}: {reason}")
288         rc = 1
289         continue
290     tier = tiers[args.from_tier]
291     rel = v["full"]["rel_path"]
292     if tier.kind == "gcs":
293         _run(["gcloud", "storage", "rm", f"{tier.root.rstrip('/')}/{rel}"],
args.dry)
294     elif tier.kind == "fs":
295         _run(["rm", "-f", os.path.join(tier.root, rel)], args.dry)
296     elif tier.kind == "rclone":
297         _run(["rclone", "delete", f"{tier.root.rstrip('/')}/{rel}"], args.dry)
298     if not args.dry:
299         v.setdefault("lifecycle", {})[f"gcs_full_state"] = "evicted" if tier.kind ==
"gcs" else v["lifecycle"].get("gcs_full_state", "never")

```

262. user (2026-06-29T16:22:20.368Z)

```

300         v.setdefault("tiers", {})[args.from_tier] = "proxy" if tier.kind == "gcs"
else "absent"
301         print(f" EVICT {name}: {reason}")
302     if not args.dry:
303         save_manifest(args.manifest, m)
304     return rc
305
306
307 def cmd_stub(verb: str) -> int:
308     print(f"`{verb}` backend is specified in STORAGE_MAP.md §8 and not yet wired in v0.1
"
309         f"(shells to rsync/rclone/gcloud/ffmpeg). Contributions welcome.")
310     return 0
311
312
313 # ----- CLI
314 def main(argv: Optional[List[str]] = None) -> int:
315     ap = argparse.ArgumentParser(prog="storagectl", description=__doc__.splitlines()[0])
316     ap.add_argument("--config", default="vault.local.json", help="tier config (default
vault.local.json)")
317     ap.add_argument("--manifest", default="manifests/corpus_index.json")
318     ap.add_argument("--dry", action="store_true", help="print actions, change nothing")
319     sub = ap.add_subparsers(dest="cmd", required=True)
320
321     p = sub.add_parser("dedup", help="report (and optionally quarantine/delete)
duplicate videos on an fs tier")
322     p.add_argument("tier")
323     p.add_argument("--quarantine", action="store_true")
324     p.add_argument("--delete", action="store_true")
325
326     sub.add_parser("status", help="per-tier bytes + N+1 violations")
327
328     pv = sub.add_parser("verify", help="check a tier matches the manifest by hash")
329     pv.add_argument("tier")
330
331     pe = sub.add_parser("evict", help="drop full-res from a tier (N+1 guarded)")
332     pe.add_argument("names", nargs="+")
333     pe.add_argument("--from", dest="from_tier", required=True)
334     pe.add_argument("--require-ratified", action="store_true", default=True)
335     pe.add_argument("--force-unratified", dest="require_ratified", action="store_false")
336

```

```

337     for verb in ("index", "sync", "stage", "proxy"):
338         sub.add_parser(verb, help=f"{verb} (see STORAGE_MAP.md §8)")
339
340     args = ap.parse_args(argv)
341     tiers = load_config(args.config) if os.path.exists(args.config) else {}
342
343     if args.cmd == "dedup":
344         return cmd_dedup(args, tiers)
345     if args.cmd == "status":
346         return cmd_status(args, tiers)
347     if args.cmd == "verify":
348         return cmd_verify(args, tiers)
349     if args.cmd == "evict":
350         return cmd_evict(args, tiers)
351     return cmd_stub(args.cmd)
352
353
354 if __name__ == "__main__":
355     raise SystemExit(main())
356

```

263. assistant (2026-06-29T16:24:12.205Z)

Now I'll implement a reliable `add` command (copy ? re-hash-verify ? manifest update ? N+1 check) plus a real `index`, so adding a video is tool-driven and safe. Making the `storagectl` edits:

264. user (2026-06-29T16:24:14.947Z)

The file `/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

265. user (2026-06-29T16:24:30.412Z)

The file `/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

266. user (2026-06-29T16:24:39.014Z)

The file `/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

267. assistant (2026-06-29T16:24:41.860Z)

Now the `add`/`index` commands and the manifest-to-tiers writer (so metadata lands on G: too):

268. user (2026-06-29T16:25:01.329Z)

The file `/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

269. assistant (2026-06-29T16:25:07.338Z)

Now wire `add`/`index` into the CLI:

270. user (2026-06-29T16:25:13.140Z)

The file `/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

271. user (2026-06-29T16:25:18.912Z)

The file `/home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py` has

been updated successfully. (file state is current in your context ? no need to Read it back)

272. assistant (2026-06-29T16:25:58.455Z)

Now update the package exports, the example config with the G: root, the spec, and add tests for the new logic:

273. user (2026-06-29T16:26:02.681Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/__init__.py has been updated successfully. (file state is current in your context ? no need to Read it back)

274. assistant (2026-06-29T16:28:13.036Z)

Important point, and it's exactly the right instinct ? I'll bake that in now rather than promise it later. Let me refactor the sink handling into a ****pluggable backend registry**** so adding S3/Azure later = ***one new class + one line***, with every command untouched:

275. user (2026-06-29T16:28:18.367Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

276. user (2026-06-29T16:28:47.505Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

277. assistant (2026-06-29T16:28:51.640Z)

Now route the commands through the backend registry (so ``add`/`verify`/`evict`/`index`` are sink-agnostic):

278. user (2026-06-29T16:28:58.030Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

279. user (2026-06-29T16:29:10.476Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

280. user (2026-06-29T16:29:21.249Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

281. assistant (2026-06-29T16:29:39.137Z)

Now make ``index`` backend-agnostic too:

282. user (2026-06-29T16:29:47.291Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

283. assistant (2026-06-29T16:29:57.201Z)

Now the config (G: root) and an end-to-end test for ``add`` that proves the verified-copy + manifest-to-tiers + N+1 path:

284. assistant (2026-06-29T16:30:01.871Z)

```
{
  "_comment": "Copy to vault.local.json and point each tier at YOUR mount/remote. ONE-TIME:
mount the Windows drives into WSL -> `sudo mount -t drvfs F: /mnt/f` and `sudo mount -t drvfs G:
/mnt/g`. A tier 'root' is the CANONICAL root that directly contains videos/full, videos/proxy,
derived/, manifests/. 'kind' selects the sink backend (fs | gcs | rclone); a future S3/Azure
sink is one new Backend class in storagectl + a tier entry here (or just use kind=rclone with an
s3:/azure: remote today).",
  "tiers": [
    { "name": "F",    "kind": "fs",  "root": "/mnt/f/[Khelsutra] GoPro Hero Black 12/corpus",
"durable": true },
    { "name": "G",    "kind": "fs",  "root": "/mnt/g/My Drive/KhelSutraGuru/Sports Videos
Corpus", "durable": true },
    { "name": "GCS",  "kind": "gcs", "root": "gs://khelsutra-rally-corpus",
"durable": false }
  ]
}
```

285. user (2026-06-29T16:30:02.903Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/vault.example.json has been updated successfully. (file state is current in your context ? no need to Read it back)

286. assistant (2026-06-29T16:30:25.391Z)

""End-to-end tests for `add` / `index` / `upsert\_video` over real (temp) filesystem tiers.

These exercise the reliability contract: copy ? re-hash-verify ? manifest upsert ? N+1 check ? manifest written to every durable tier. Backends are swappable, so the fs path stands in for any sink that implements the same contract.

""

import json

from storagectl import storagectl as S

```
def _cfg(tmp_path, *tiers):
    p = tmp_path / "vault.json"
    p.write_text(json.dumps({"tiers": list(tiers)}))
    return str(p)
```

```
def _fs(name, root, durable=True):
    return {"name": name, "kind": "fs", "root": str(root), "durable": durable}
```

----- upsert_video (pure)

```
def test_upsert_adds_then_merges_tiers():
    m = {"videos": []}
    assert S.upsert_video(m, {"canonical_name": "a", "video_id": "a" * 32,
                              "tiers": {"F": "full"}, "verified": {"F": "d1"}}) == "added"
    # adding the SAME video on a second tier merges, doesn't duplicate or clobber F:
    assert S.upsert_video(m, {"canonical_name": "a", "video_id": "a" * 32,
                              "tiers": {"G": "full"}, "verified": {"G": "d2"}}) == "updated"
    assert len(m["videos"]) == 1
    v = m["videos"][0]
    assert v["tiers"] == {"F": "full", "G": "full"}
    assert v["verified"] == {"F": "d1", "G": "d2"}
```

----- add (integration)

```
def test_add_copies_verifies_reaches_n_plus_one_and_writes_metadata(tmp_path):
    f_root, g_root = tmp_path / "F", tmp_path / "G"
    f_root.mkdir(); g_root.mkdir()
```

```

src = tmp_path / "GX010108.MP4"
src.write_bytes(b"some golden match footage bytes" * 1000)
cfg = _cfg(tmp_path, _fs("F", f_root), _fs("G", g_root),
            {"name": "GCS", "kind": "gcs", "root": "gs://x", "durable": False})
man = tmp_path / "manifest.json"

rc = S.main(["--config", cfg, "--manifest", str(man),
            "add", str(src), "--name", "adarsh_avi_singles", "--golden"])
assert rc == 0 # both durable tiers verified -> N+1 met

m = json.loads(man.read_text())
v = m["videos"][0]
assert v["canonical_name"] == "adarsh_avi_singles" and v["golden"] is True
assert v["tiers"] == {"F": "full", "G": "full"}
assert v["video_id"] == v["full"]["md5"]
# the real bytes landed on both tiers, byte-identical to source
assert (f_root / "videos/full/adarsh_avi_singles.mp4").read_bytes() == src.read_bytes()
assert (g_root / "videos/full/adarsh_avi_singles.mp4").read_bytes() == src.read_bytes()
# metadata travels with the data: manifest written into each durable tier
assert (f_root / "manifests/corpus_index.json").exists()
assert (g_root / "manifests/corpus_index.json").exists()

def test_add_single_tier_warns_n_plus_one(tmp_path):
    f_root = tmp_path / "F"; f_root.mkdir()
    src = tmp_path / "clip.MP4"; src.write_bytes(b"x" * 500)
    cfg = _cfg(tmp_path, _fs("F", f_root))
    man = tmp_path / "m.json"
    rc = S.main(["--config", cfg, "--manifest", str(man), "add", str(src), "--name", "c", "--
to", "F"])
    assert rc == 1 # only one durable copy -> N+1 NOT met (non-zero exit is the signal)

def test_add_is_idempotent(tmp_path):
    f_root, g_root = tmp_path / "F", tmp_path / "G"
    f_root.mkdir(); g_root.mkdir()
    src = tmp_path / "c.MP4"; src.write_bytes(b"y" * 800)
    cfg = _cfg(tmp_path, _fs("F", f_root), _fs("G", g_root))
    man = tmp_path / "m.json"
    a1 = ["--config", cfg, "--manifest", str(man), "add", str(src), "--name", "c"]
    assert S.main(a1) == 0
    assert S.main(a1) == 0 # second run: already-present+verified, still one entry
    assert len(json.loads(man.read_text())["videos"]) == 1

def test_index_reconciles_canonical_tier(tmp_path):
    f_root = tmp_path / "F"
    (f_root / "videos/full").mkdir(parents=True)
    (f_root / "videos/full/clipA.mp4").write_bytes(b"a" * 1234)
    cfg = _cfg(tmp_path, _fs("F", f_root))
    man = tmp_path / "m.json"
    assert S.main(["--config", cfg, "--manifest", str(man), "index", "F"]) == 0
    v = json.loads(man.read_text())["videos"][0]
    assert v["canonical_name"] == "clipA" and v["tiers"] == {"F": "full"}

```

287. user (2026-06-29T16:30:26.421Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/tests/test_add.py (file state is current in your context ? no need to Read it back)

288. assistant (2026-06-29T16:30:50.935Z)

Now verify the whole refactor ? ruff + the full test suite (old safety tests + new add/index/upsert tests):

289. user (2026-06-29T16:30:57.005Z)

ruff

```
72 | def test_add_is_idempotent(tmp_path):
73 |     f_root, g_root = tmp_path / "F", tmp_path / "G"
74 |     f_root.mkdir(); g_root.mkdir()
75 |     ^
76 |     src = tmp_path / "c.MP4"; src.write_bytes(b"y" * 800)
77 |     cfg = _cfg(tmp_path, _fs("F", f_root), _fs("G", g_root))
78 |
```

E702 Multiple statements on one line (semicolon)

```
--> tests/test_add.py:75:29
73 |     f_root, g_root = tmp_path / "F", tmp_path / "G"
74 |     f_root.mkdir(); g_root.mkdir()
75 |     src = tmp_path / "c.MP4"; src.write_bytes(b"y" * 800)
76 |     ^
77 |     cfg = _cfg(tmp_path, _fs("F", f_root), _fs("G", g_root))
78 |     man = tmp_path / "m.json"
```

Found 5 errors.

import + CLI smoke

backends: ['fs', 'gcs', 'rclone']

CLI ok

tests

```
..... [100%]
16 passed in 0.42s
```

290. assistant (2026-06-29T16:31:18.050Z)

Tests pass (16), backends register correctly. Ruff flagged 5 semicolon lines in my test file (E702) ? `storagectl.py` itself is clean. Let me fix the test formatting:

291. assistant (2026-06-29T16:31:34.349Z)

"""End-to-end tests for `add` / `index` / `upsert\_video` over real (temp) filesystem tiers.

These exercise the reliability contract: copy ? re-hash-verify ? manifest upsert ? N+1 check ? manifest written to every durable tier. Backends are swappable, so the fs path stands in for any sink that implements the same contract.

"""

import json

from storagectl import storagectl as S

```
def _cfg(tmp_path, *tiers):
    p = tmp_path / "vault.json"
    p.write_text(json.dumps({"tiers": list(tiers)}))
    return str(p)
```

```
def _fs(name, root, durable=True):
    return {"name": name, "kind": "fs", "root": str(root), "durable": durable}
```

----- upsert_video (pure)

```
def test_upsert_adds_then_merges_tiers():
    m = {"videos": []}
    assert S.upsert_video(m, {"canonical_name": "a", "video_id": "a" * 32,
                              "tiers": {"F": "full", "verified": {"F": "d1"}}}) == "added"
```



```
# adding the SAME video on a second tier merges, doesn't duplicate or clobber F:
assert S.upsert_video(m, {"canonical_name": "a", "video_id": "a" * 32,
                        "tiers": {"G": "full"}, "verified": {"G": "d2"}}) == "updated"
assert len(m["videos"]) == 1
v = m["videos"][0]
assert v["tiers"] == {"F": "full", "G": "full"}
assert v["verified"] == {"F": "d1", "G": "d2"}
```

----- add (integration)

```
def test_add_copies_verifies_reaches_n_plus_one_and_writes_metadata(tmp_path):
    f_root = tmp_path / "F"
    g_root = tmp_path / "G"
    f_root.mkdir()
    g_root.mkdir()
    src = tmp_path / "GX010108.MP4"
    src.write_bytes(b"some golden match footage bytes" * 1000)
    cfg = _cfg(tmp_path, _fs("F", f_root), _fs("G", g_root),
               {"name": "GCS", "kind": "gcs", "root": "gs://x", "durable": False})
    man = tmp_path / "manifest.json"

    rc = S.main(["--config", cfg, "--manifest", str(man),
                "add", str(src), "--name", "adarsh_avi_singles", "--golden"])
    assert rc == 0 # both durable tiers verified -> N+1 met

    m = json.loads(man.read_text())
    v = m["videos"][0]
    assert v["canonical_name"] == "adarsh_avi_singles" and v["golden"] is True
    assert v["tiers"] == {"F": "full", "G": "full"}
    assert v["video_id"] == v["full"]["md5"]
    # the real bytes landed on both tiers, byte-identical to source
    assert (f_root / "videos/full/adarsh_avi_singles.mp4").read_bytes() == src.read_bytes()
    assert (g_root / "videos/full/adarsh_avi_singles.mp4").read_bytes() == src.read_bytes()
    # metadata travels with the data: manifest written into each durable tier
    assert (f_root / "manifests/corpus_index.json").exists()
    assert (g_root / "manifests/corpus_index.json").exists()

def test_add_single_tier_warns_n_plus_one(tmp_path):
    f_root = tmp_path / "F"
    f_root.mkdir()
    src = tmp_path / "clip.MP4"
    src.write_bytes(b"x" * 500)
    cfg = _cfg(tmp_path, _fs("F", f_root))
    man = tmp_path / "m.json"
    rc = S.main(["--config", cfg, "--manifest", str(man), "add", str(src), "--name", "c", "--
to", "F"])
    assert rc == 1 # only one durable copy -> N+1 NOT met (non-zero exit is the signal)

def test_add_is_idempotent(tmp_path):
    f_root = tmp_path / "F"
    g_root = tmp_path / "G"
    f_root.mkdir()
    g_root.mkdir()
    src = tmp_path / "c.MP4"
    src.write_bytes(b"y" * 800)
    cfg = _cfg(tmp_path, _fs("F", f_root), _fs("G", g_root))
    man = tmp_path / "m.json"
    argv = ["--config", cfg, "--manifest", str(man), "add", str(src), "--name", "c"]
    assert S.main(argv) == 0
    assert S.main(argv) == 0 # second run: already-present+verified, still one entry
    assert len(json.loads(man.read_text())["videos"]) == 1
```

```
def test_index_reconciles_canonical_tier(tmp_path):
    f_root = tmp_path / "F"
    (f_root / "videos/full").mkdir(parents=True)
    (f_root / "videos/full/clipA.mp4").write_bytes(b"a" * 1234)
    cfg = _cfg(tmp_path, _fs("F", f_root))
    man = tmp_path / "m.json"
    assert S.main(["--config", cfg, "--manifest", str(man), "index", "F"]) == 0
    v = json.loads(man.read_text())["videos"][0]
    assert v["canonical_name"] == "clipA" and v["tiers"] == {"F": "full"}
```

292. user (2026-06-29T16:31:35.427Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/tests/test_add.py has been updated successfully. (file state is current in your context ? no need to Read it back)

293. assistant (2026-06-29T16:32:06.725Z)

Now updating the spec ? tier roots, the `add` verb, the metadata-travels rule, and the pluggable-sink design:

294. user (2026-06-29T16:32:14.279Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/STORAGE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

295. assistant (2026-06-29T16:33:51.962Z)

Excellent catch ? and you're right to flag both. WSL-over-Google-Drive is genuinely unreliable, so the tool must run ****natively on Windows**** for the G: tier. Let me audit and fix the portability gaps. I found two real ones: `FsBackend.remove`/`dedup` shell out to Unix `rm`/`mv` (break on Windows), and `subprocess` won't resolve `gcloud`?`gcloud.cmd` on Windows. Fixing all three:

296. user (2026-06-29T16:34:00.184Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

297. assistant (2026-06-29T16:34:04.454Z)

Now the two call sites that used Unix `rm`/`mv`:

298. user (2026-06-29T16:34:08.100Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

299. user (2026-06-29T16:34:15.770Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/storagectl/storagectl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

300. assistant (2026-06-29T16:34:37.380Z)

Now finishing the spec (metadata-travels rule, the `add` verb, and the cross-platform/Windows note), plus making the example config Windows-native:

301. user (2026-06-29T16:34:40.446Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/STORAGE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

302. user (2026-06-29T16:34:47.156Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/STORAGE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

303. user (2026-06-29T16:35:00.193Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/STORAGE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

304. assistant (2026-06-29T16:35:15.294Z)

Making the example config Windows-native (since data loads run on Windows) and adding a Platforms note to the README:

305. assistant (2026-06-29T16:35:18.901Z)

```
{
  "_comment": "Per-machine config ? copy to vault.local.json (gitignored). storagectl is pure-
Python and runs natively on Windows, macOS, and Linux/WSL. RECOMMENDED: run it NATIVELY ON
WINDOWS for data loads ? WSL's drvfs over Google Drive (G:) File Stream is unreliable, so use
the native G:\\My Drive\\... path. Windows roots shown below; WSL equivalents are /mnt/f/... and
/mnt/g/My Drive/... (G: often unavailable under WSL). 'kind' selects the sink backend (fs | gcs
| rclone); a future native S3/Azure sink is one new Backend class + a tier entry here (or use
kind=rclone with an s3:/azure: remote today).",
  "tiers": [
    { "name": "F",    "kind": "fs",  "root": "F:\\[Khelsutra] GoPro Hero Black 12\\corpus",
"durable": true },
    { "name": "G",    "kind": "fs",  "root": "G:\\My Drive\\KhelSutraGuru\\Sports Videos Corpus",
"durable": true },
    { "name": "GCS",  "kind": "gcs", "root": "gs://khelsutra-rally-corpus",
"durable": false }
  ]
}
```

306. user (2026-06-29T16:35:19.945Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/vault.example.json has been updated successfully. (file state is current in your context ? no need to Read it back)

307. user (2026-06-29T16:35:33.639Z)

The file /home/avidullu/projects/khelsutra-guru/rally-corpus-vault/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

308. assistant (2026-06-29T16:35:39.555Z)

All edits in. Let me verify the full thing is clean (ruff + tests), then commit and push:

309. user (2026-06-29T16:35:48.571Z)

ruff

All checks passed!

tests

..... [100%]
16 passed in 0.71s

add (dry-run) smoke against temp tiers proves cross-platform fs path + N+1

<frozen runpy>:128: RuntimeWarning: 'storagectl.storagectl' found in sys.modules after import of package 'storagectl', but prior to execution of 'storagectl.storagectl'; this may result in unpredictable behaviour

```
[mov,mp4,m4a,3gp,3g2,mj2 @ 0x55ea9b83a500] moov atom not found
/tmp/tmp.NiV70hGzkr/src.MP4: Invalid data found when processing input
# add testclip (0.00 GB) <- /tmp/tmp.NiV70hGzkr/src.MP4
  hashing source (video_id = full MD5)...
RUN copy /tmp/tmp.NiV70hGzkr/src.MP4 -> /tmp/tmp.NiV70hGzkr/F/videos/full/testclip.mp4
```

```
(+verify)
  F (fs): copied + verified
  RUN copy /tmp/tmp.NiV70hGzkR/src.MP4 -> /tmp/tmp.NiV70hGzkR/G/videos/full/testclip.mp4
(+verify)
  G (fs): copied + verified
  manifest: added; video_id d8ac8184b1fd9fe52a8f328d7cfd5184; durable copies ['F', 'G'] -> N+1
met ?
  wrote manifest: /tmp/tmp.NiV70hGzkR/m.json
  wrote manifest: /tmp/tmp.NiV70hGzkR/F/manifests/corpus_index.json
  wrote manifest: /tmp/tmp.NiV70hGzkR/G/manifests/corpus_index.json
--- verify F ---
<frozen runpy>:128: RuntimeWarning: 'storagectl.storagectl' found in sys.modules after import
of package 'storagectl', but prior to execution of 'storagectl.storagectl'; this may result in
unpredictable behaviour
# verify F: 1 ok, 0 missing, 0 hash-mismatch
```

310. assistant (2026-06-29T16:36:08.302Z)

Everything green ? ruff clean, 16 tests pass, and the live smoke shows `add` copying + `re-hash-verifying` to F: and G:, writing the manifest to every durable tier, and confirming `N+1 met` (the fprobe warning is just because my smoke "video" was random bytes ? it degraded gracefully as designed). Committing and pushing:

311. user (2026-06-29T16:36:17.190Z)

pushed:
63a74bf feat(storagectl): reliable add/index, pluggable sink backends, cross-platform fs ops
c41545a feat: scaffold rally-corpus-vault ? storage map + storagectl + safety core

312. assistant (2026-06-29T16:37:20.124Z)

Pushed (`63a74bf`). All four of your threads are now addressed in the repo ? here's the consolidated answer.

Cross-platform & the Windows/Drive question (your latest)

I audited it and **fixed two real portability bugs**, so it's now genuinely Windows ? macOS ? Linux interoperable:

- Replaced Unix `rm`/`mv` shell-outs with stdlib `os.remove`/`shutil.move`.
- External tools now resolve via PATH (`gcloud` ? `gcloud.cmd` on Windows) instead of failing.
- Pure-Python otherwise; manifest paths are POSIX-style so they're identical on every OS.

Recommendation matches yours: run `storagectl` natively on Windows for data loads (F:, `G:\My Drive\KhelSutraGuru\Sports Videos Corpus`, and `gcloud` all work as normal Windows paths there). WSL stays for the GPU/eval work. `vault.example.json` now ships Windows-native roots. One honesty caveat: I can't run a true Windows test from this WSL session (no Windows Python here) ? the logic is portable and unit-tested (16 green), but your first Windows run is the real validation.

Pluggable sinks (your earlier message) ? done

A tier's `kind` selects a `Backend`; **adding S3/Azure later = one `Backend` subclass + one line in `BACKENDS`**, every command untouched. S3/Azure/Drive are reachable **today** with zero code via `kind=rclone` + a remote. Data also lands on G: configurably (it's just a tier), and the manifest is written into **every durable tier's** `manifests/` so metadata travels with the data.

What the repo supports right now

```
Verb	Status
`add` / `index` / `verify` / `status` / `dedup` / `evict`	? implemented + tested (16 tests, ruff clean)
`sync` / `stage` / `proxy`	? stubbed (print "not wired") ? for bulk migration + GCS lifecycle
```

`add` (copy?verify?manifest?N+1) covers the "next video" flow fully. `sync`/`stage`/`proxy` are for the one-time bulk migration of the existing 175 videos + GCS staging ? I'll build `sync` when you start the bootstrap.

Pending items (apart from video indexing & data moves)

1. **The launch GO/NO-GO call** ? yours to make (my rec: NO-GO on a product launch until ship-gate #172 + WASB weight provenance).
2. **Three filed issues to action when you choose**:
[#397](https://github.com/Khelsutra/badminton-highlight-indexer/issues/397) stray dirs .
[#398](https://github.com/Khelsutra/badminton-highlight-indexer/issues/398) served-gate harness bug .
[#399](https://github.com/Khelsutra/badminton-highlight-indexer/issues/399) launch-PR process hardening.
3. **Minor**: `reportlab` not installed (2 PDF tests skip); R-series doc-cluster archival deferred per DOC_STATUS.

Runbook ? add the next video reliably

One-time (on Windows): install Python 3.10+, `pip install -e .` the cloned repo, ensure `gcloud` on PATH; copy `vault.example.json` ? `vault.local.json` with your roots; make sure `G:\My Drive\KhelSutraGuru\Sports Videos Corpus` exists.

Each new video:

1. Offload the GoPro card ? drop the file anywhere (e.g. `F:\?\corpus\\_ingest\`).
2. Choose the canonical name ? golden clip ? the corpus name (`adarsh\_avi\_singles`); otherwise `- 3. Run one command:
```  
storagectl add "F:\?\\_ingest\GX010210.MP4" --name 2026-07-01\_GX010210 --capture-date 2026-07-01 --device "GoPro Hero Black 12"  
```  
(add `--golden` for a labelled/eval clip). It hashes the source, copies to **F:** and **G:** **videos/full/<name>.mp4**, **re-hashes** each copy to prove it isn't corrupt, upserts the manifest, writes the manifest into F: and G: **manifests/**, and prints **N+1 met** ? or warns + exits non-zero if any copy failed.
- 4. Later, when training needs it: `storagectl stage <name> --to GCS`; after the run is accepted, `storagectl evict <name> --from GCS --require-ratified` (refuses unless F:+G: both hash-verify).
- 5. Anytime: `storagectl status` (N+1 health) . `storagectl verify G` (hash-check Drive).

What makes it reliable: every copy is verified by re-hash, a tier is never recorded unless its hash matched, N+1 is enforced in code, and `evict` can't break it.

Want me to implement `sync` next so we can do the one-time bulk migration (normalize the existing 175 videos into the canonical layout + mirror F:?G:), or hold here?

313. user (2026-06-29T16:41:00.931Z)

and the hash is stored as a json with the data in G?

314. assistant (2026-06-29T16:41:52.751Z)

Yes ? and let me be precise about **where** and **what**.

What's on G: today. When `add` (or `index`) runs, it writes the full manifest to **G:\My Drive\KhelSutraGuru\Sports Videos Corpus\manifests\corpus_index.json** ? co-located with the **videos/** data in the same corpus tree (and the same file also lands in F:'s **manifests/** and the repo). That JSON holds, per video, the **video_id** = full MD5, the per-file **md5**, **size**, **crc32c** (when a backend returns it), **ffprobe** metadata, plus tier-presence and lifecycle. The smoke test earlier confirmed it: `wrote manifest: ?/G/manifests/corpus_index.json`. An entry looks like:

```
```json
{
 "canonical_name": "2026-07-01_GX010210",
 "video_id": "d8ac8184b1fd9fe52a8f328d7cfd5184",
```

```

"full": { "rel_path": "videos/full/2026-07-01_GX010210.mp4",
 "size": 10611615335, "md5": "d8ac?5184",
 "duration_s": 1490, "fps": 59.94, "res": "3840x2160" },
"tiers": { "F": "full", "G": "full" },
"verified": { "F": "2026-07-01", "G": "2026-07-01" },
"lifecycle": { "gcs_full_state": "never", "trained_ratified": false }
}
...

```

So the hashes live as JSON **\*\*in the same Drive corpus folder\*\*** as the videos ? just in a ``manifests/`` subfolder, not literally beside each ``.mp4``.

**\*\*One robustness upgrade worth considering.\*\*** Right now it's a **\*single aggregated\*** ``corpus_index.json``. I'd recommend **\*\*also\*\*** dropping a tiny per-video **\*\*sidecar\*\*** next to each file ? ``videos/full/<name>.json`` (or ``.md5``) ? carrying just that one video's hash + metadata. Benefits:

- Each video is **\*\*self-describing\*\***: you can verify one file from its own sidecar even if the global manifest is lost/corrupt.
- On Drive, an ``add`` then syncs a **\*\*small new sidecar\*\*** instead of rewriting the whole growing manifest each time.
- Avoids any single-file write contention as the corpus grows.

The aggregated manifest stays (great for querying/replication); the sidecars are the durable per-file truth. It's a ~15-line change to ``add`/`verify`/`index``.

Want me to add the per-video sidecar so every video carries its own hash JSON right next to it on G:?

#### 315. user (2026-06-29T16:49:09.412Z)

not right away but pls file a bug on the repo for the same. Right now please ensure all the work is merged and we will just complete the pending items and all it done.  
 You mentioned "The launch GO/NO-GO call ? yours to make (my rec: NO-GO on a product launch until ship-gate #172 + WASB weight provenance)." -- But we merged that PR, so it is now picked up?  
 Right now we are still in pure development phase of the product so its ok but I want to understand.

#### 316. assistant (2026-06-29T16:49:41.890Z)

Three things: file the sidecar bug, confirm everything's merged/pushed, and clear up the launch-vs-merge confusion (good question ? they're very different).

#### 317. user (2026-06-29T16:49:47.559Z)

**rally-corpus-vault: branch / sync / clean?**

**master...origin/master**

63a74bf feat(storagectl): reliable add/index, pluggable sink backends, cross-platform fs ops  
 c41545a feat: scaffold rally-corpus-vault ? storage map + storagectl + safety core

**badminton-highlight-indexer: branch / sync / tracked-changes?**

**master...origin/master**

39dff1d feat(config): promote inactivity\_temporal\_density to default-ON (P2b) (#396)  
 (untracked junk dirs below are issue #397, not pending work):  
 ?? gcs:/  
 ?? gs:/  
 ?? local:/

#### 318. user (2026-06-29T16:49:55.044Z)

<https://github.com/Khelsutra/rally-corpus-vault/issues/1>

#### 319. assistant (2026-06-29T16:50:34.107Z)

All landed and clean ? nothing left uncommitted or unpushed:

- **rally-corpus-vault**: `master == origin/master`, both commits pushed.
- **badminton-highlight-indexer**: `master == origin/master`, PR #396 merged. (The only untracked items are the `gcs:/ gs:/ local:/` junk dirs, which are tracked by issue #397 ? not pending work.)
- **Sidecar bug filed**: [rally-corpus-vault#1](https://github.com/Khelsutra/rally-corpus-vault/issues/1) (deferred, as you asked).

## Merge ? launch ? the part you wanted to understand

These are two different altitudes, and merging #396 did **not** "pick up" the launch decision:

- **PR #396** was one small dev change ? flipping a config default (``inactivity_temporal_density`` ? on), a do-no-harm fix to rally-end trimming. Merging it ships *that one improvement* into the code. Normal development.
- **The launch GO/NO-GO** is a product-level milestone ? "are we ready to release the rally-detection product to real/paying users?" That's gated on things #396 doesn't touch: the **ship-gate quality target** (#172, still unset) and **WASB weight provenance** (licensing).

So the NO-GO was never "blocked from merging #396" ? #396 merged fine. The NO-GO means "don't flip the *product* to GA yet." Think of it as: merging dev PRs = **building the car**; the launch call = the **road-legal certification** before selling it. You keep building freely; certification is a separate, later gate.

**And in pure-development phase**, NO-GO is exactly the expected, correct state ? there's nothing to launch yet, and the gate simply isn't met because the work to meet it (agree the quality bar, clear the weight licensing) hasn't been done. My launch-readiness report wasn't asking you to decide today; it was a snapshot of "here's the gap between today's dev state and a launchable one" so you know what's left. Nothing auto-triggers from a merge ? when you actually approach release, you'd set #172's target, measure the detector against it, resolve WASB provenance, **then** make the GO call.

## So "all done" ? what that leaves

The build work for this session is fully landed. The genuinely-remaining items are **decisions/cleanups**, not pending code:

- The launch call ? a *future* decision (not a task now).
- Tracked tickets to action when you choose: #397, #398, #399, rally-corpus-vault#1.
- The corpus **bootstrap / data moves** (migrate the 175 videos ? canonical, dedup the 251.8 GB, F:G: mirror, golden?GCS) ? gated on your go; needs me to build ``sync`` first.

You're clear to keep developing. Nothing is half-finished or stuck.

### 320. user (2026-06-29T16:56:34.034Z)

the per video sidecar idea is good and please also add a note that the same sidecar can also carry the application information including golden set if that video was used to generate golden set data or the location of the proxy files if there is a low res dup with it. It will need some thinking because this sidecar will then be created in the valut, updated in the data collector and read and updated again by the training and eval systems. -- So pls find a way to park this pointer so that I don't forget it when we pick it up!

So you may wrap up with your "3. Minor: reportlab not installed (2 PDF tests skip); R-series doc-cluster archival deferred per DOC\_STATUS." and then also complete the other "Three filed issues to action when you choose: #397 stray dirs · #398 served-gate harness bug · #399 launch-PR process hardening." feel free to merge them with test coverage and ensuring it doesn't drop. And when doing the doc archival, pls make sure that all the completed docs are also appropriately moved to the archives/ folder.

Thanks!

### 321. assistant (2026-06-29T16:59:25.587Z)

This is a meaty batch ? and the cross-system sidecar contract is a genuinely important design pointer. Let me set up tracking, park the sidecar idea durably (issue + memory), and kick off a

parallel investigation of the four code/doc tasks so I implement them correctly.

### 322. user (2026-06-29T16:59:31.187Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

### 323. user (2026-06-29T16:59:52.900Z)

<https://github.com/Khelsutra/rally-corpus-vault/issues/1#issuecomment-4835053354>

### 324. user (2026-06-29T16:59:56.340Z)

#### memory dir contents

```
ls: cannot access '/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory': No such file or directory
```

#### MEMORY.md (if exists)

```
cat: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory/MEMORY.md: No such file or directory
```

### 325. assistant (2026-06-29T17:00:13.768Z)

Issue #1 updated with the cross-system contract scope. The memory dir doesn't exist yet ? I'll create it via Write. Parking two durable pointers (the sidecar contract + the new repo) and launching the parallel investigation:

### 326. assistant (2026-06-29T17:00:22.127Z)

---

```
name: corpus-sidecar-cross-system-contract
description: Per-video sidecar JSON is evolving into a cross-system metadata contract (vault creates, collector updates, training/eval read+update); design parked in rally-corpus-vault#1.
metadata:
 type: project

```

The recorded-video corpus will get a **per-video sidecar JSON** next to each

``videos/full/<name>.mp4``

(in addition to the aggregated ``manifests/corpus_index.json``). Beyond identity+hashes

(``video_id`/MD5,`

size, crc32c, ffprobe), it should carry **application info**: golden-set provenance (was this video

used to generate golden data ? which golden manifest/labels, labelling status), and **proxy linkage**

(low-res proxy location + its own md5 + ``source_video_id``).

**Why it's parked, not built:** it's a shared contract written/read by multiple systems and needs

field-ownership + merge discipline before coding:

- **rally-corpus-vault** CREATES it (on ``add``: identity, hashes, tiers, proxy).
- **sports-data-collector** UPDATES it (golden provenance, labelling, label-file location).
- **training/eval (badminton-highlight-indexer)** READ (resolve video/proxy) + UPDATE (``trained_ratified``, runs consumed, derived-artifact/trajectory locations).

Canonical ``schema_version``-ed schema lives in `[[rally-corpus-vault]]`; other repos import/validate it.

**How to apply:** when the corpus track resumes, design the schema + merge policy FIRST (no last-writer

clobber across systems), then implement in ``storagectl` add/index/verify`. Tracked at ``rally-corpus-vault#1`` (owner asked to not forget it).

### 327. user (2026-06-29T17:00:23.203Z)



File created successfully at: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory/corpus-sidecar-cross-system-contract.md (file state is current in your context ? no need to Read it back)

### 328. assistant (2026-06-29T17:00:38.193Z)

---

```
name: rally-corpus-vault
description: New isolated repo (github.com/Khelsutra/rally-corpus-vault) ? the corpus
storage/custody + cost-control layer for recorded videos across F:/Google Drive/GCS.
metadata:
 type: project

```

`rally-corpus-vault` (created 2026-06-29, private under Khelsutra) is the storage-management sibling to `sports-data-collector` / `rally-annotator`. It owns the **recorded-video storage map**: one medium-agnostic canonical layout (`videos/full`, `videos/proxy`, `derived/`, `manifests/corpus\_index.json`) replicated across tiers ? **F** (local GoPro source, durable), **G** (durable backup, golden+ratified only), **GCS** (disposable training cache, proxies resident + full-res staged-then-evicted).

Key invariant: **3** copies, **N+1** ? `evict`/delete refuse unless ?2 durable copies hash-verify (`can\_evict`). Identity = `video\_id` = full MD5 (== badminton `compute\_video\_id`).

`storagectl` CLI: `add`/`index`/`verify`/`status`/`dedup`/`evict` implemented + tested; `sync`/`stage`/`proxy` stubbed (build `sync` for the one-time bulk migration of the existing 175 F: videos). **Pluggable sinks** ? tier `kind` (fs/gcs/rc1one) ? a `Backend`; adding S3/Azure = one subclass + one `BACKENDS` line. **Cross-platform** (Win/macOS/Linux); **run natively on Windows for the G: (Drive) tier** (WSL drvfs over Drive is unreliable). G: root = `G:\My Drive\KhelSutraGuru\Sports Videos Corpus`. F: must be mounted in WSL via `sudo mount -t drvfs F: /mnt/f`.

Seeded from the 2026-06-29 F: scan: 175 unique videos, 41 dup groups / **251.8 GB reclaimable**, 15/15 golden full-res on F: but **0/15 full-res in GCS** (migration backlog). Next corpus work (owner-gated): build `sync`, normalize+dedup F:, mirror F:G:, upload golden?GCS. See [[corpus-sidecar-cross-system-contract]].

### 329. user (2026-06-29T17:00:39.264Z)

File created successfully at: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory/rally-corpus-vault.md (file state is current in your context ? no need to Read it back)

### 330. assistant (2026-06-29T17:00:47.046Z)

## Memory index

- [rally-corpus-vault](rally-corpus-vault.md) ? new repo: the corpus storage/custody layer across F:/Drive/GCS (storagectl, N+1, pluggable sinks).
- [corpus-sidecar-cross-system-contract](corpus-sidecar-cross-system-contract.md) ? per-video sidecar JSON as a cross-system contract (vault/collector/training/eval); parked design, rally-corpus-vault#1.

### 331. user (2026-06-29T17:00:48.077Z)

File created successfully at: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory/MEMORY.md (file state is current in your context ? no need to Read it back)

### 332. assistant (2026-06-29T17:00:59.843Z)

Memory parked (sidecar contract + the new repo, indexed in MEMORY.md). Now launching the parallel investigation of the four wrap-up tasks so the PRs are precise:

### 333. user (2026-06-29T17:01:41.333Z)

Workflow launched in background. Task ID: wi7ax1914

Summary: Investigate fixes for badminton-highlight-indexer issues #397/#398/#399 + the doc-archival plan

Transcript dir: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/subagents/workflows/wf\_b0aa0abe-638

Script file: /home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/workflows/scripts/wrapup-investigate-wf\_b0aa0abe-638.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/workflows/scripts/wrapup-investigate-wf\_b0aa0abe-638.js"} to iterate without resending the script.)

Run ID: wf\_b0aa0abe-638

To resume after editing the script: Workflow({scriptPath:

"/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/workflows/scripts/wrapup-investigate-wf\_b0aa0abe-638.js", resumeFromRunId: "wf\_b0aa0abe-638"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

### 334. assistant (2026-06-29T17:02:00.318Z)

Done with the parking and kickoff:

- **\*\*Sidecar contract parked\*\*** ? expanded scope added to [rally-corpus-vault#1](https://github.com/Khelsutra/rally-corpus-vault/issues/1) (the cross-system lifecycle: vault creates ? collector updates golden/labelling ? training/eval read+update, with field-ownership/merge as the hard design problem), plus two durable **\*\*agent memories\*\*** (`rally-corpus-vault` + `corpus-sidecar-cross-system-contract`) so it survives across sessions.  
- **\*\*Investigation running\*\*** ? a background workflow is reading the code to produce exact fix plans for #397/#398/#399 + a conservative doc-archival assessment (which docs are genuinely terminal vs still gated).

Once it returns, I'll execute the five PRs **\*\*one at a time, lowest-risk first\*\*** (likely: reportlab ? #397 ? #399 ? #398 ? doc-archival last, since archival touches the most inbound refs), running `run\_all\_tests.py` + ruff + mypy + coverage after each and merging only when green ? so coverage never drops. I'll report each as it lands.

Standing by for the investigation to finish.

### 335. user (2026-06-29T17:08:27.029Z)

```
<task-notification>
<task-id>wi7ax1914</task-id>
<tool-use-id>toolu_01R64tPhF6By189WwaGPbKs6</tool-use-id>
<output-file>/tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/wi7ax1914.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Investigate fixes for badminton-highlight-indexer issues #397/#398/#399 + the doc-archival plan" completed</summary>
<result>{"plan": "All facts confirmed: stray dirs untracked, all three #399 live-doc link targets verified (the `4-` slug fix, the archived PER_VIDEO target exists, the `../archives/research/` AUDIO target exists), `promotion.py` has both functions, and `tools/leak_scan.py` + a `leak-
```

scan` CI job exist as the precedent for the link-checker. The doc-archival investigation reports zero archive-ready docs (cluster is gated).\n\nI have everything needed. Producing the final merged plan.\n\n# Badminton-Highlight-Indexer Wrap-Up ? Ordered Implementation Plan\n\n5 work items, \*\*one PR each, all branched off `master`, no stacking\*\*. Each PR runs the full gate before merge:\n\n```\ngit checkout master &&& git pull &&& git checkout -b <branch>\n...edits...\npython run\_all\_tests.py &&& ruff check . &&& mypy . &&&\npython run\_all\_tests.py --cov # coverage must not drop\n# then merge\n```\n\nVerified against the tree at `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer` (on `master`). Line anchors below are confirmed accurate as of this checkout.\n\n--\n\n## Recommended MERGE ORDER (lowest-risk first)\n\n| # | PR | Branch | Risk | Why this slot |\n|---|---|---|---|---|\n| 1 | \*\*397 stray-dirs\*\* | `fix/397-stray-scheme-dirs` | \*\*Lowest\*\* | 2 one-line edits + .gitignore; identity for local inputs; new test fails-then-passes. Also cleans the repo so later PRs branch clean. |\n| 2 | \*\*398 served-gate\*\* | `fix/398-served-gate-typed-cfg` | \*\*Low\*\* | Eval-helper only; production byte-identical. |\n| 3 | \*\*reportlab-uncap\*\* | ? | \*\*BLOCKED\*\* | \*\*DO NOT DO ? see below.\*\* No reportlab cap exists in this repo. Skip or convert to a no-op/closed issue. |\n| 4 | \*\*399 process-hardening\*\* | `chore/399-launch-pr-process` | \*\*Low-Med\*\* | Net-new PR template + CI link-job + 3 live doc-link fixes + 1 cross-link. Touches CI; land after the code fixes are green. |\n| 5 | \*\*doc-archival\*\* | ? | \*\*N/A ? DO NOT ARCHIVE\*\* | Cluster is gated; zero archive-ready docs. Only the 3 small independent fixes below. \*\*Last\*\* because its edits touch many inbound refs. |\n\nIf reportlab-uncap is dropped, effective merge order is \*\*397 ? 398 ? 399 ? doc-archival-housekeeping\*\*.\n\n--\n\n## PR 1 ? 397 Stray scheme-prefixed report dirs

Risk: LOWEST\n\n\*\*Root cause (confirmed & reproduced):\*\* `obsreport/io.py:48-49` does `os.path.dirname(os.path.abspath(video\_path))`; `abspath("gcs://b/x.mp4")` collapses `//?` and, lacking a leading `/`, joins under cwd ? &lt;repo\_root&gt;/gcs:/b/x.mp4, then `os.makedirs` creates it. The raw URI leaks in at two callers; the worker already passes the resolved path.\n\n\*\*Files to change\*\* (all absolute):\n- `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py` - Line \*\*953\*\* (API path): `video\_path=req.video\_path,` ? `video\_path=local\_video,` - Line \*\*1725\*\* (CLI path): `video\_path=args.video\_path,` ? `video\_path=local\_video,` - `local\_video` is in scope at both (assigned at 804 before the try@828; and 1658 before the try@1662). Mirrors the already-correct worker (`backend/worker/\_\_main\_\_.py:298`). DB provenance unchanged (`db.add\_video` still stores the source URI).\n- `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.gitignore` ? append (anchored to repo root):\n\n```\n/gcs:\n/gs:\n/local:\n/s3:\n/gdrive:\n```\n\n\*\*Tests:\*\* ADD

`tests/test\_api\_endpoints.py::test\_process\_uri\_input\_writes\_no\_scheme\_dir\_at\_repo\_root` ? drives `/api/process` with `gs://`, `gcs://`, `local://` through the un-mocked `emit\_indexer\_report` and asserts no scheme dir lands at cwd. Fails on current code, passes after the fix. \*\*Guard with `pytest.importorskip("obsreport")`\*\* for CI portability (writer is a no-op without it).\n\n\*\*Cleanup (run from repo root, after merge or as the first commit):\*\* the 3 dirs are untracked (`git status` shows `?? gcs:/ ?? gs:/ ?? local:/`), 48 regenerable report artifacts only:\n\n```\nnrm -rf -- 'gcs:' 'gs:' 'local:'\n```\n\nCoverage: net \*\*up\*\* (new test exercises a previously-untested writer path on the API+CLI branches). \*\*Do NOT\*\* edit the sibling `sports-obsreport` repo ? the 2-line host fix is sufficient and lower-risk (the obsreport `scheme://` guard is a noted-but-out-of-scope option).\n\n--\n\n## PR 2 ? 398 served\_gate\_a passes a dict where production passes a typed config

Risk: LOW\n\n\*\*Root cause (confirmed):\*\* `served\_gate\_a.served\_idx\_cfg` returns a plain dict and feeds it straight into the production seams `seg\_enrich\_candidate\_stats(...)` / `seg\_select\_for\_handoff(...)` , which reach `fusion\_hybrid.py:135 fcfg = idx\_cfg.fusion` ? `AttributeError: 'dict' object has no attribute 'fusion'`. Production always passes a typed `IndexingConfig` (via `\_normalize\_config` ? `AppConfig(\*\*config)`). \*\*Fix = Option A\*\* (return a typed config; exercises the real contract + `extra='forbid'`). Option B (make the seams tolerate dict-or-typed) is \*\*rejected\*\* ? it weakens the very contract the litmus exists to test.\n\n\*\*Files to change:\*\*\n- `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served\_gate\_a.py` - 1. Add import after line 33 (from `backend.eval.metrics import Interval, Score, score`): `from backend.config.models import IndexingConfig`\n- 2. `\_served\_idx\_cfg` (lines 43-57): return signature `-> IndexingConfig` and `return IndexingConfig(fusion={...}, wasb={...}, dead\_time\_merge\_gap=PROD\_MERGE\_GAP, min\_action\_window\_duration=PROD\_MIN\_WINDOW)` (keyword args; nested dicts coerce).\n- 3. In `served\_handoff\_windows` (lines 79-92): delete the `idx\_cfg\_typed: Any = idx\_cfg` # type: ignore` silencer and pass `idx\_cfg` directly to both seams.\n- Leave the `Dict`/`Any` typing imports as-is (still used elsewhere). \*\*No change to `fusion\_hybrid.py`\*\* (line 135 is correct against the typed contract).\n\n\*\*Tests:\*\*\n- Existing 5 tests in `tests/test\_served\_gate\_a\_regression.py` become RUNNABLE+GREEN; \*\*no assertion edits\*\*. Pinned bands stay valid (re-verified on the 15-clip golden set: served ?F1 ?0.0233, f1\_on/f1\_off

```
0.9017, seg_on 0.4696 ? seg_off 0.5825; PROPOSAL ?F1 0.0287). **Do NOT re-pin** ? bands only
change when the corpus changes (test docstring already says so).\n- **Recommended ADD** (CI-
coverable without golden data): a fast synthetic unit test asserting `_served_idx_cfg(2)`
returns an `IndexingConfig` with `.fusion.min_crossings == 2`, and that
`_enrich_candidate_stats` accepts it without `AttributeError` on a tiny synthetic trajectory.
The golden-gated tests can't cover this regression in CI; this one can.\n\n**Coverage:**
unchanged in CI by default (the 5 golden tests still skip without a manifest); the optional
synthetic test adds real CI coverage of the dict-vs-typed regression. To run the golden suite on
the owner box only:\n```\nSERVED_GATE_A_MANIFEST=/home/avidullu/projects/khelsutra-
guru/badminton-highlight-indexer/output/p2b_golden/manifest_local.json \\\n python -m pytest
tests/test_served_gate_a_regression.py -q\n```\n\n**Do NOT** make CI depend on golden data (keeps
the `_golden_specs()` >= 6` gate + `importorskip("cv2")`). One behavior note:
`extra='forbid'` now turns a future typo'd served key into a hard error at construction
(intended stricter contract).\n\n--\n\n## PR 3 ? reportlab-uncap · Risk: BLOCKED ? DO NOT
IMPLEMENT AS SPECIFIED\n\n**Flag:** This work item has **no investigation result in the provided
JSON** and **no verifiable target in the tree**. I searched exhaustively and found **no
reportlab version cap anywhere**:\n- Not in `requirements.txt` (reportlab is not even listed) or
`pyproject.toml` `[project.dependencies]` / `optional-dependencies`\n- Not in
`.github/workflows/*.yaml`, `deploy/**`, any `constraints*.txt` (none exist), or the sibling
`sports-obsreport/pyproject.toml`\n- The only references are `backend/eval/ablation_pdf.py`
(lazy `from reportlab.lib import ...`, unversioned) and two `pytes
... (truncated 57016 chars, full result in /tmp/claude-1000/-home-avidullu-projects-khelsutra-
guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/wi7ax1914.output)</result>
<usage><agent_count>5</agent_count><subagent_tokens>260863</subagent_tokens><tool_uses>97</tool_
uses><duration_ms>405628</duration_ms></usage>
</task-notification>
```

### 336. assistant (2026-06-29T17:09:26.482Z)

Investigation complete with precise, verified fix plans. Key headlines: #397 root cause is `obsreport/io.py` `abspath` on a URI leaking via two `backend/main.py` callers; #398 is the dict-vs-typed-config fix (Option A, production unaffected); \*\*reportlab has no actual cap\*\* (it's just not installed? the tests correctly skip); and critically, \*\*doc-archival found zero archive-ready docs\*\* (the cluster is still gated). Let me read the full plan to get the exact #399 link list and the archival reasoning before I touch anything:

### 337. user (2026-06-29T17:09:28.035Z)

File content (33051 tokens) exceeds maximum allowed tokens (25000). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.

### 338. assistant (2026-06-29T17:09:45.248Z)

The file is too big to read whole. Let me extract just the sections I still need (the #399 link list + the archival reasoning) via a parse:

### 339. user (2026-06-29T17:09:50.879Z)

===== PLAN (PR4 onward) =====

## PR 4 ? #399 Launch-PR process hardening · Risk: LOW-MED

Four coordinated parts. \*\*Scope decision: fix the 3 LIVE broken links only; treat the 49 `docs/archives/\*\*` links as historical-by-design (DOC\_STATUS §2 step 5) and exclude `docs/archives/\*\*` from the committed checker.\*\* Do NOT mass-rewrite the 49 archived `../` links.

**\*\*Files to change:\*\***

- `/home/avidullu/.../docs/I18N\_PLAN.md` ? line 4 anchor: `( #open-decisions-for-the-owner )` ? `( #4-open-decisions-for-the-owner )` (heading `## 4. Open decisions (for the owner)` slugs to `4-open-decisions-for-the-owner` ? \*\*verified\*\*).
- `/home/avidullu/.../docs/QUALITY\_ITERATIONS.md` ? line 621: `( PER\_VIDEO\_OUTPUT\_LAYOUT.md )` ? `( archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md )` (\*\*target exists\*\*).
- `/home/avidullu/.../docs/data\_pipeline/VIDEO\_RECORDING\_GUIDELINES.md` ? line 10:

```
`(archives/research/AUDIO_RALLY_DETECTION_PLAN.md)` ?
`(..../archives/research/AUDIO_RALLY_DETECTION_PLAN.md)` (doc lives one level deep; **target
exists at `docs/archives/research/...`**).
- `/home/avidullu/.../docs/RALLY_DETECTION_FIXES_PLAN.md` ? §8 (line ~110-111): make the
"(promote iff tolF1 does not regress)" rule a link to `../backend/eval/promotion.py` (file
exists; `served_regresses` @38, `promote_if_served_safe` @117 ? **verified**). The phrase wraps
across 110-111; fold the link text across the wrap.
- `/home/avidullu/.../CLAUDE.md` ? add one bullet to "Workflow conventions" (after the
PROJECT_DOC_TEMPLATE bullet, ~line 68): a launch/bit-flip PR MUST run the DOC_STATUS §2 archival
checklist in the same PR, enforced by the new PR template.
- `/home/avidullu/.../.github/PULL_REQUEST_TEMPLATE.md` ? **NEW** (none exists): the "Launch /
bit-flip PRs ? archival gate" checklist block (verify-vs-code ? flip status ? update inbound
refs ? keep lineage ? `git mv` to archives ? record in §3).
- `/home/avidullu/.../tools/check_md_links.py` ? **NEW**, stdlib-only, mirrors the existing
`tools/leak_scan.py` (run as `python -m tools.check_md_links`); **excludes `docs/archives/`**.
- `/home/avidullu/.../.github/workflows/ci.yml` ? add a `link-check` job (sibling of the
existing `leak-scan` job @38, same `python -m tools.X` shape, no pip install).
```

**\*\*Tests:\*\*** ADD (recommended) `tests/test\_check\_md\_links.py` ? tmp\_path fixtures: missing path
flagged, dead anchor flagged, valid link passes, http/mailto/fenced-code skipped,
`docs/archives/` excluded.

**\*\*Coverage:\*\*** net up (new tool + its test). The CI link-check job fails only on **\*\*live\*\***-doc
breakage.

**\*\*Do NOT:\*\*** mass-rewrite the 49 archived links; cover `backend/deploy/scripts` READMEs (out of
the issue's root+`docs/` scope); use a multi-template `.github/PULL\_REQUEST\_TEMPLATE/` directory
(single-file form is correct here).

---

## PR 5 ? Doc-archival · Risk: N/A ? DO NOT ARCHIVE ANY DOC NOW

**\*\*Hard finding:\*\*** `archive\_ready\_docs = []`. Zero docs are archive-ready. The whole
R-series/quality cluster is **\*\*ACTIVE and gated\*\*** (DOC\_STATUS §4 finding #3). P2b (#396) promoted
the operational `inactivity\_temporal\_density` flag (do-no-harm, ?tolF1 +0.010, CI spans 0,
p=0.18) ? it is **\*\*NOT\*\*** the R2 tolF1 metric-gate ablation and **\*\*NOT\*\*** the §7.1 single-number
attribution, so it does **\*\*NOT\*\*** close the gate.

**\*\*Explicitly DO NOT archive / DO NOT `git mv`:\*\***

- `RALLY\_DETECTION\_FIXES\_PLAN.md`, `RALLY\_DETECTION\_CODE\_REVIEW.md` ? DONE in content but their
own headers defer archival to the cluster gate. Keep in place.
- `RALLY\_QUALITY\_RESEARCH.md`, `R2\_EVAL\_METRIC\_DESIGN.md`, `RALLY\_DETECTION\_GAP\_CLOSURE.md`,
`PROMINENT\_COURT\_DETECTION.md`, `REAL\_FOOTAGE\_VALIDATION\_REPORT.md`,
`RALLY\_DETECTION\_QUALITY\_REPORT.md`, `R1\_MILESTONE\_LAUNCH\_REPORT.md` ? DoD not met / part of the
gated cluster.
- `QUALITY\_ITERATIONS.md`, `HOW\_RALLY\_DETECTION\_WORKS.md` ? **\*\*LIVING\*\***, "Keep" per DOC\_STATUS
§3b/§3c ? **\*\*never\*\*** archive as a whole.

When the gate eventually clears, archive the **\*\*whole cluster as one unit\*\*** via DOC\_STATUS §2
steps 1-6 (verify ? flip status ? update inbound refs `README.md:1/2/4/7` + `NEXT\_STEPS.md`
items 8/9/9b + DOC\_STATUS §3 rows ? keep lineage ? `git mv` to `docs/archives/past\_projects/` ?
record). **\*\*Not now.\*\***

**\*\*This PR's actual scope = 3 small independent housekeeping fixes\*\*** (do NOT touch any doc's
location):

1. `docs/DOC\_STATUS.md` §3 ? add tracking rows for `RALLY\_DETECTION\_FIXES\_PLAN.md` (DONE /
gated-to-R-cluster) and `RALLY\_DETECTION\_CODE\_REVIEW.md` (terminal snapshot / gated) ? currently
untracked.
2. `docs/NEXT\_STEPS.md` line 25 (item 9b) ? STALE: still says PR #393 must do the A/B flip "only
if tolF1 doesn't regress," but #396 already did it and promoted default-ON. Update/remove 9b;
re-check item 8 @ line 21.
3. `docs/RALLY\_DETECTION\_FIXES\_PLAN.md` header ? STALE self-claim that it is "indexed in
docs/README.md" (grep of README finds no entry). Remove/correct.

```
Files to change: `docs/DOC_STATUS.md`, `docs/NEXT_STEPS.md`,
`docs/RALLY_DETECTION_FIXES_PLAN.md`. **Tests:** none (docs only); gate is ruff/mypy/tests pass
+ coverage unchanged. **Last in the order** ? these edits touch the doc cross-ref graph, so land
them after everything else is green.
```

---

## Cross-cutting flags

- **\*\*reportlab-uncap** is unverifiable in this repo ? block/clarify before any code (PR 3).**\*\***
- **\*\*No doc is archive-ready** ? PR 5 must NOT `git mv` anything; it is 3 staleness fixes only.**\*\***
- For #399, fix the **\*\*3 live\*\*** links only; the **\*\*49 archived\*\*** links are historical-by-design ?  
exclude `docs/archives/\*\*` from the committed checker rather than rewriting them.
- For #398, do **\*\*not\*\*** re-pin bands and do **\*\*not\*\*** make CI depend on golden data.
- For #397, do **\*\*not\*\*** edit the sibling `sports-obsreport` repo; the 2-line host fix suffices.

Plan source files referenced are all under `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/` (and the sibling `/home/avidullu/projects/khelsutra-guru/sports-obsreport/src/obsreport/io.py:48-49` as the documented root-cause writer for #397, left unmodified).

### 340. assistant (2026-06-29T17:11:10.554Z)

I have the complete, verified plan. Two important findings to flag up front: **\*\***(1) reportlab has no version cap**\*\*** ? it's simply not installed, and the 2 PDF tests skip by design (optional lazy-imported feature). **\*\***(2) Zero docs are archive-ready**\*\*** ? the R-series quality cluster is still gated (P2b promoted the operational flag but is *\*not\** the R2 metric-gate ablation that closes it), and `RALLY\_DETECTION\_FIXES\_PLAN.md`'s own header already defers its archival to that gate. So I will **\*\*not\*\*** `git mv` anything; the "archival" item becomes 3 staleness fixes instead. I'll explain this fully at the end.

Let me refine the plan and start PR 1 (#397). First confirming the exact `backend/main.py` edit sites:

### 341. user (2026-06-29T17:11:17.466Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

### 342. user (2026-06-29T17:11:28.350Z)

#### local\_video assignments + the two emit/report call sites

```
804: local_video = storage.acquire_local_input(
807: video_id = compute_video_id(local_video)
813: val_results, val_context = registry.run_all_with_context(local_video, config)
890: video_id, local_video, req.player_pool, req.max_frames
1658: local_video = storage.acquire_local_input(args.video_path, storage_cfg)
1660: video_id = compute_video_id(local_video)
1666: local_video, global_config
1705: video_id, local_video, max_frames=args.max_frames
```

#### context around 953 (API) and 1725 (CLI) ? confirm video\_path= lines

```
953: video_path=req.video_path,
1725: video_path=args.video_path,
```

### 343. assistant (2026-06-29T17:11:39.830Z)

Both confirmed ? `local\_video` (the resolved local path) is in scope at both call sites. Let me read the surrounding context to confirm these are the report-emit calls (not DB provenance, which must keep the URI):

### 344. user (2026-06-29T17:11:40.593Z)

```
944 (response or {}).get("status") if isinstance(response, dict) else None,
945)
```

```

946 # Always emit the per-video observability report (next to the video, or to
947 # report.output_dir). Never raises. Surface the paths in the response. (The
compute-path
948 # proof lives on the job result + the [COMPUTE] log, not the report ?
obsreport's record_run
949 # doesn't surface arbitrary run signals.)
950 paths = emit_indexer_report(
951 db=db,
952 video_id=video_id,
953 video_path=req.video_path,
954 config=config,
955 val_results=val_results,
956 val_context=val_context,
957 segmenter_requested=req.segmenter,
958 segmenter_actual=segmenter_actual,
959 was_reingest=was_reingest,

```

#### 345. user (2026-06-29T17:11:41.369Z)

```

1716 f"[SUCCESS] Process complete. Indexed {len(segments)} play segments
inside database: {db_path}"
1717)
1718 # Destroy-AFTER-confirm: keep new on success, restore prior on
empty/failure.
1719 _finalize_reingest(db_instance, video_id, segments, play_wm, cand_wm)
1720 finally:
1721 # Always emit the per-video observability report. Never raises.
1722 paths = emit_indexer_report(
1723 db=db_instance,
1724 video_id=video_id,
1725 video_path=args.video_path,
1726 config=global_config,
1727 val_results=val_results,
1728 val_context=val_context,
1729 segmenter_requested=args.segmenter,
1730 segmenter_actual=segmenter_actual,
1731 was_reingest=was_reingest,

```

#### 346. assistant (2026-06-29T17:12:33.904Z)

Both are the `emit\_indexer\_report` calls ? passing the raw URI  
(`req.video\_path`/`args.video\_path`) is what leaks the scheme dir; `local\_video` is the resolved  
path and matches the already-correct worker. Starting PR 1: branching, fixing both call sites,  
cleaning the junk:

#### 347. user (2026-06-30T02:46:32.320Z)

Switched to a new branch 'fix/397-stray-scheme-dirs'  
**delete the untracked junk dirs (regenerable report artifacts)**  
(no untracked left)  
**.gitignore tail**

**Throwaway repro scripts + PR-body / handoff scratch notes (already excluded from  
ruff + pytest; never meant to be committed -- keep them out of `git status` too).**

```

scratch/
.venv/

```

**Private leak-scan name-list (real player/venue names for LOCAL `tools.leak\_scan --na  
scanning -- NEVER committed; CI runs the name-free structural guard).**

```

.leakscan_private

```

#### 348. user (2026-06-30T02:46:33.418Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**349. user (2026-06-30T02:46:49.888Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**350. user (2026-06-30T02:46:58.612Z)**

<tool\_use\_error>File has not been read yet. Read it first before writing to it.</tool\_use\_error>

**351. user (2026-06-30T02:47:07.168Z)**

```
40 *.db
41 *.db-journal
42 *.db-wal
43 output/
44 output_dir/
45 backend/static/highlights_*.mp4
46 backend/static/*_compiled.mp4
47
48 # Local generated artifacts (review reels, eval CSVs, e2e/proof outputs) -- regenerable,
never committed
49 artifacts/
50
51 # OS generated files
52 .DS_Store
53 Thumbs.db
54 desktop.ini
55
56 # Coverage data
57 .coverage
58 .coverage.*
59 htmlcov/
```

**352. assistant (2026-06-30T02:47:16.688Z)**

Now retry the .gitignore append (file is read now):

**353. user (2026-06-30T02:47:21.789Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.gitignore has been updated successfully. (file state is current in your context ? no need to Read it back)

**354. assistant (2026-06-30T02:47:43.849Z)**

Code + gitignore done. Now the regression test ? let me see how the existing tests drive `/api/process` and the report writer, so I write a faithful one:

**355. user (2026-06-30T02:47:51.962Z)**

**test files touching the process endpoint / emit\_indexer\_report / obsreport**

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_profile_parity.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_worker.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_locale_analytics.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_job_worker_loop.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_edge_auth.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_run_signals.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_cloud_dispatch.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_reporting.py
```



```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_reporting_harvest.py
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_async_jobs.py
```

## how emit\_indexer\_report is imported/defined

### existing process-endpoint test shape (first hit)

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:1:"""TestClient coverage for the FastAPI seam (previously zero ? the compiler SyntaxError
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:11:pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:12:from fastapi.testclient import TestClient
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:34: client = TestClient(app)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:179:def test_process_success_with_mocked_segmenter(env):
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:186: "/api/process", json={"video_path": str(vid_file), "segmenter": "x"}
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:195:def test_process_records_locale_for_analytics(env):
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:196: """v7: /api/process persists the SPA UI locale on the job ? count_jobs_by_locale (PMF analytics)."""
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:203: "/api/process",
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:211:def test_process_validation_failure_preserves_prior_segments(env):
```

## 356. assistant (2026-06-30T02:48:06.775Z)

There's an existing `test\_process\_success\_with\_mocked\_segmenter` harness, and the junk path `~/test\_process\_local\_scheme\_uri\_0/?` implies an existing `local://` test that \*produces\* the scheme dir ? my fix likely changes its behavior. Let me find it and read the harness:

## 357. user (2026-06-30T02:48:08.558Z)

### find the local-scheme test (it produced the junk)

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_workspace.py:41:
assert join_uri("local:///srv/data", VID) == f"local:///srv/data/{VID}"
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_input_source.py:30: ("local://abs/x.mp4", "gcs", "gs://r", "local://abs/x.mp4"),
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_input_source.py:65: for p in ("/data/clip.mp4", "rel/clip.mp4", "clip.mp4", "gs://b/k", "local://x"):
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_input_source.py:73: assert input_is_local("local://x.mp4") is True
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_input_source.py:109: # local:// strips to the bare key (so callers stat ref.key, not the raw URI)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_input_source.py:110: assert resolve_input_ref("local:///data/x.mp4").key == "/data/x.mp4"
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_storage.py:38: ("local://abs/x.mp4", "local", "", "/abs/x.mp4"),
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_storage.py:106:
assert all(u.startswith("local://") for u in uris)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py:767:def test_process_local_scheme_uri_accepted(env):
```

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py:768: """An explicit local://<existing file> resolves to
the bare path: the existence gate stats the
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py:775: # Forward-slash local:// URI (parse strips the
scheme to the bare path).
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py:776: uri = "local://" + str(vid).replace("\\", "/")
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py:782:def
test_process_local_scheme_uri_missing_still_404(env):
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py:783: """local://<missing> still 404s ? the existence gate
runs on the resolved local key."""
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py:786: "/api/process", json={"video_path":
"local:///no/such/ls-missing.mp4"}

```

### 358. user (2026-06-30T02:48:09.337Z)

```

24 def submit(self, fn, *args, **kwargs):
25 fn(*args, **kwargs)
26
27
28 @pytest.fixture
29 def env(tmp_path, monkeypatch):
30 db = Database(str(tmp_path / "t.db"))
31 cfg = AppConfig()
32 monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
33 app = m.create_app(cfg, db)
34 client = TestClient(app)
35 return client, db, tmp_path, monkeypatch
36
37
38 def _video(tmp_path, name="m.mp4"):
39 p = tmp_path / name
40 p.write_bytes(b"not-a-real-video-but-hashable")
41 return p
42
43
44 def _pass():
45 return [ValidationResult(validator_name="x", passed=True, message="ok", details={})]
46
47
48 def _fail():
49 return [
50 ValidationResult(
51 validator_name="blur", passed=False, message="blurry", details={}
52)
53]
54
55
56 class _FakeSeg:
57 def __init__(self, db, cfg):
58 self.db = db
59
60 def process_video(self, video_id, video_path, *a, **k):
61 sid = self.db.add_play_segment(video_id, 0.0, 5.0)
62 return [
63 {
64 "id": sid,
65 "start_time": 0.0,
66 "end_time": 5.0,
67 "duration": 5.0,
68 "metadata": {},

```

```

69 }
70]
71
72
73 def test_api_config_ok(env):
74 client = env[0]
75 r = client.get("/api/config")
76 assert r.status_code == 200
77 assert r.json()["sport"] == "badminton"
78
79
80 def test_api_config_redacts_secret_shaped_fields(env):
81 """P0b: the unauthenticated /api/config must never serialize a credential-shaped
value.
82 AppConfig stores no secrets by design (verified by
83 test_redact_does_not_mask_any_real_appconfig_field), so inject a fake object whose

```

### 359. assistant (2026-06-30T02:48:35.475Z)

`test\_process\_local\_scheme\_uri\_accepted` (line 767) is exactly the test that *\*produced\** the junk. The cleanest regression coverage is to extend it (and a sibling) to assert no scheme dir is left at the repo root. Let me read it and the mocked-segmenter harness:

### 360. user (2026-06-30T02:48:36.248Z)

```

179 def test_process_success_with_mocked_segmenter(env):
180 """#16: submit returns 202 + job_id; the (inline) job carries the old response."""
181 client, db, tmp_path, mp = env
182 vid_file = _video(tmp_path)
183 mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
184 mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
185 r = client.post(
186 "/api/process", json={"video_path": str(vid_file), "segmenter": "x"}
187)
188 assert r.status_code == 202
189 job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
190 assert job["status"] == "done"
191 assert job["result"]["status"] == "success"
192 assert len(job["result"]["play_segments"]) == 1
193
194

```

### 361. user (2026-06-30T02:48:40.441Z)

```

766
767 def test_process_local_scheme_uri_accepted(env):
768 """An explicit local://<existing file> resolves to the bare path: the existence gate
stats the
769 RESOLVED key (not the raw URI) so it is accepted, and the segmenter sees the
stripped path."""
770 client, db, tmp_path, mp = env
771 vid = _video(tmp_path, "ls.mp4")
772 mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
773 mp.setattr(m.segmenter_registry, "get", lambda name: _CaptureSeg)
774 _CaptureSeg.last_path = None
775 # Forward-slash local:// URI (parse strips the scheme to the bare path).
776 uri = "local://" + str(vid).replace("\\", "/")
777 r = client.post("/api/process", json={"video_path": uri, "segmenter": "x"})
778 assert r.status_code == 202
779 assert _CaptureSeg.last_path == str(vid).replace("\\", "/")
780
781
782 def test_process_local_scheme_uri_missing_still_404(env):
783 """local://<missing> still 404s ? the existence gate runs on the resolved local

```

```
key."""
784 client = env[0]
785 r = client.post(
786 "/api/process", json={"video_path": "local:///no/such/ls-missing.mp4"}
787)
788 assert r.status_code == 404
789
790
```

### 362. assistant (2026-06-30T02:49:32.949Z)

I have the harness. The existing `local:///` test already drives the exact bug path through `emit\_indexer\_report`. Let me add a dedicated regression test asserting no scheme dir leaks, and confirm `os`/`obsreport` availability:

### 363. user (2026-06-30T02:49:39.299Z)

#### os imported in the test file?

```
4:import os
5:import subprocess
6:import sys
9:import pytest
14:import backend.main as m
```

#### obsreport importable here?

```
obsreport OK 0.1.3
```

### 364. user (2026-06-30T02:49:44.783Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_api\_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

### 365. assistant (2026-06-30T02:50:16.142Z)

`os` is imported and obsreport (0.1.3) is available, so the test actually runs here. Let me verify it passes with the fix **and** genuinely catches the regression (by temporarily reverting just `main.py`):

### 366. user (2026-06-30T02:50:36.835Z)

#### (1) WITH fix ? expect pass, no junk:

```
/home/avidullu/projects/khelsutra-guru/.venv/lib/python3.12/site-
packages/fastapi/testclient.py:1: StarletteDeprecationWarning: Using `httpx` with
`starlette.testclient` is deprecated; install `httpx2` instead.
 from starlette.testclient import TestClient as TestClient # noqa
```

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
3 passed, 41 deselected, 1 warning in 4.09s
junk at repo root?
 none ?
```

#### (2) revert main.py only -> expect the new test to FAIL (proves it catches the bug):

```
/home/avidullu/projects/khelsutra-guru/.venv/lib/python3.12/site-
packages/fastapi/testclient.py:1: StarletteDeprecationWarning: Using `httpx` with
`starlette.testclient` is deprecated; install `httpx2` instead.
 from starlette.testclient import TestClient as TestClient # noqa
```

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== short test summary info =====
FAILED tests/test_api_endpoints.py::test_process_uri_input_writes_no_scheme_dir_at_repo_root
1 failed, 43 deselected, 1 warning in 2.56s
```

#### (3) fix restored -> expect pass again:

```
from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
1 passed, 43 deselected, 1 warning in 2.57s
junk after?
none ?
```

#### 367. assistant (2026-06-30T02:50:46.011Z)

Verified ? the test passes with the fix, **\*\*fails without it\*\*** (catches the regression), and leaves no junk. Now the full gate for PR 1 (suite + ruff + mypy):

#### 368. user (2026-06-30T02:52:25.393Z)

##### full suite

1504 passed, 10 skipped, 1 warning in 89.91s (0:01:29)

```
=====
RESULT: PASS - all tests green
=====
```

##### ruff

All checks passed!

##### mypy

```
backend/pipeline/validators/base.py:44: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend/providers/__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 138 source files
```

##### final junk check

none ?

#### 369. assistant (2026-06-30T02:52:41.690Z)

PR 1 gate green: **\*\*1504 passed\*\*** (+1 from the new test), 10 skipped, ruff + mypy clean, no junk. Committing, opening, and merging:

#### 370. user (2026-06-30T02:53:03.426Z)

```
remote: https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/fix/397-stray-
scheme-dirs
remote:
PR #400 created
```

##### re-sync local master

```
Deleted branch fix/397-stray-scheme-dirs (was 3d0ef0f).
959c630 fix(#397): write per-video report to the resolved local path, not the raw URI (#400)
```

#### 371. assistant (2026-06-30T02:54:10.339Z)

**\*\*PR #400 (#397) merged\*\*** ? master `959c630`. On to PR 2 (#398).

Let me branch and re-read the exact `served\_gate\_a.py` edit sites:

#### 372. user (2026-06-30T02:54:15.166Z)

Switched to a new branch 'fix/398-served-gate-typed-cfg'

#### 373. user (2026-06-30T02:54:15.220Z)

```
27 from dataclasses import dataclass
28 from typing import Any, Dict, List, Optional, Tuple
29
30 from backend.eval import segmentation_metrics as _segm
31 from backend.eval.calibrate_wasb import load_golden_gts
32 from backend.eval.metrics import Interval, Score, score
```

```

33 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
34 from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
35
36 # Production windowing operating point (config.json indexing defaults ? the values
37 # TrajectoryHybridSegmenter.process_video reads for the served fusion run).
38 PROD_VELOCITY_THRESH = 0.02
39 PROD_MERGE_GAP = 2.0
40 PROD_MIN_WINDOW = 2.0
41
42
43 def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:
44 """The ``indexing`` config dict the served FusionSegmenter seams read ? mirrors
45 config.json's ``indexing.fusion`` block, with Gate A at ``min_crossings`` and the
46 person cue OFF (so no GPU / torch is touched)."""
47 return {
48 "fusion": {
49 "substrate": "wasb",
50 "min_crossings": min_crossings,
51 "min_players": 0,
52 "cue_flags": {},
53 },
54 "wasb": {"velocity_threshold": PROD_VELOCITY_THRESH},
55 "dead_time_merge_gap": PROD_MERGE_GAP,
56 "min_action_window_duration": PROD_MIN_WINDOW,
57 }
58
59
60 def served_handoff_windows(
61 trajectory_csv: str, fps: float, frame_width: float, min_crossings: int
62) -> List[Interval]:
63 """Run the PRODUCTION FusionSegmenter decision path on a cached trajectory and
64 return the
65 windows that would reach the AI handoff (i.e. the served rally candidates).
66 Uses the real ``FusionSegmenter`` seam methods ? ``_enrich_candidate_stats``
67 (computes
68 ``net_crossings``) and ``_select_for_handoff`` (Gate A) ? over windows from the
69 production
70 is OFF.
71 """
72 points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
73 windows = TrackNetRunner.trajectory_to_action_windows(
74 points,
75 fps=fps,
76 frame_width=frame_width,
77 velocity_thresh=PROD_VELOCITY_THRESH,
78 merge_gap=PROD_MERGE_GAP,
79 min_window_duration=PROD_MIN_WINDOW,
80)
81 idx_cfg = _served_idx_cfg(min_crossings)
82 # Construct the real served segmenter (db=None, config carries the indexing block).
83 We only
84 # call the pure decision seams ? no process_video, no DB, no provider ? exactly as
85 the
86 # registry constructs it for `name`/`description`.
87 seg = FusionSegmenter(None, {"indexing": idx_cfg, "sport": "badminton"})
88 # The served per-window stats (incl. net_crossings) via the production enrichment
89 hook.
90 idx_cfg_typed: Any = idx_cfg # type: ignore[annotation-type-mismatch]
91 window_stats = [
92 seg._enrich_candidate_stats(
93 cw, points, fps, frame_width, video_path="", idx_cfg=idx_cfg_typed
94)
95 for cw in windows
96]

```

```

91]
92 keep = seg._select_for_handoff(windows, window_stats, idx_cfg_typed)

```

#### 374. assistant (2026-06-30T02:54:35.273Z)

I have the exact edit sites. Before editing, let me confirm how `FusionSegmenter.\_\_init\_\_` handles `config["indexing"]` ? passing a typed object there could break the constructor even if the seams want it typed:

#### 375. user (2026-06-30T02:54:41.398Z)

##### FusionSegmenter / base \_\_init\_\_ + how config['indexing'] is read

```

10: is explicitly enabled (`indexing.fusion.cue_flags.person`), the person-cue features
16:- ``_select_for_handoff`` ? optional served Gate A/B: when ``indexing.fusion.min_crossings``
31:from backend.config.models import FusionConfig, IndexingConfig
47: def __init__(self, db: Any, config: Any):
69: self, idx_cfg: IndexingConfig
72: idx_cfg = IndexingConfig(**idx_cfg)
75: cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
81: wcfg = WasbConfig.from_indexing_cfg(idx_cfg)
88: self, idx_cfg: IndexingConfig
96: "indexing.fusion.substrate='wasb' (no native TrackNet runner yet)."
103: cfg = NativeWasbConfig.from_indexing_cfg(idx_cfg)
118: idx_cfg: IndexingConfig,
177: self._person = PersonDetector(self.config)
179: frame_skip = self.config.indexing.yolo_frame_skip
203: idx_cfg: IndexingConfig,

```

##### base class of FusionSegmenter

```

41:class FusionSegmenter(TrajectoryHybridSegmenter):
48: super().__init__(db, config)

```

##### how the base/trajectory\_hybrid handles config['indexing'] in init

```

/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:11:- ``family`` ? config section
under ``indexing.*``, output subdir, and
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:29:from backend.config.models import
IndexingConfig
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:50: #: config section under
`indexing` (velocity_threshold lives there), the
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:57: self, idx_cfg:
IndexingConfig
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:63: self, idx_cfg:
IndexingConfig
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:74: self, idx_cfg:
IndexingConfig
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:78: ``RALLY_DETECTOR_IMPL`` env
var (takes precedence) or ``indexing.detector_impl``:
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:88: idx_cfg =
IndexingConfig(**idx_cfg)
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:102: return
StubDetectorRunner(StubConfig.from_indexing_cfg(idx_cfg)), {
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:145: idx_cfg: "IndexingConfig",
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:162: idx_cfg: "IndexingConfig",
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:190: idx_cfg =

```

```
self.config.indexing
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:230: #
indexing.detector_impl="stub") so the pipeline runs GPU/WSL-free; otherwise
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/segmenters/trajectory_hybrid.py:243: # Bounded-windowing over-
merge fix (w1): 0 = OFF (unchanged). See IndexingConfig.
```

#### 376. assistant (2026-06-30T02:55:54.869Z)

The seams (`\_enrich\_candidate\_stats`) read `idx\_cfg.fusion` typed with no dict-coercion, while production passes a typed `IndexingConfig` ? so Option A (return typed) is correct, and the seams already accept it. Making the three edits:

#### 377. user (2026-06-30T02:55:57.876Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served\_gate\_a.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 378. user (2026-06-30T02:56:10.130Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served\_gate\_a.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 379. user (2026-06-30T02:56:23.094Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served\_gate\_a.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 380. assistant (2026-06-30T02:56:28.413Z)

Now the synthetic, golden-free regression test (CI-coverable since only golden \*data\* is absent in CI, not cv2):

#### 381. user (2026-06-30T02:56:36.020Z)

<tool\_use\_error>File has not been read yet. Read it first before writing to it.</tool\_use\_error>

#### 382. assistant (2026-06-30T02:56:45.478Z)

Let me re-read that section to satisfy the edit gate:

#### 383. user (2026-06-30T02:56:46.470Z)

```
80
81 _SPECS = _golden_specs()
82 _skip = pytest.mark.skipif(
83 not _SPECS,
84 reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
85)
86
87
88 # ----- served-path
litmus
89
90
91 @_skip
```

#### 384. user (2026-06-30T02:56:59.700Z)



The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_served\_gate\_a\_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

### 385. assistant (2026-06-30T02:57:14.264Z)

Now validate PR 2 ? the golden-free synthetic test (CI path) **\*\*and\*\*** the 5 golden tests on real data (which were crashing), to confirm the fix works and the pinned bands still hold:

### 386. user (2026-06-30T02:57:50.932Z)

#### (1) WITHOUT golden env (mimics CI): synthetic runs, 5 golden skip

```
SKIPPED [1] tests/test_served_gate_a_regression.py:112: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:127: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:144: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:156: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/human_lovo_manifest.json'); runs on the owner box
SKIPPED [1] tests/test_served_gate_a_regression.py:176: golden trajectories absent (manifest
'/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/human_lovo_manifest.json'); runs on the owner box
1 passed, 5 skipped in 2.12s
```

#### (2) WITH golden env: all run ? validates the fix on real data + bands hold (no re-pin)

```
..... [100%]
6 passed in 27.86s
```

### 387. assistant (2026-06-30T02:57:59.850Z)

PR 2 validated beautifully: the synthetic test runs in CI (golden-free), and **\*\*all 6 pass with golden data in 27.9s\*\*** ? the 5 previously-crashing litmus tests now actually exercise the production served path on real footage, **\*\*and the pinned bands hold (no re-pin)\*\***. Full gate:

### 388. user (2026-06-30T02:59:42.066Z)

#### suite

```
=====
RESULT: PASS - all tests green
=====
```

#### ruff

All checks passed!

#### mypy

```
backend/eval/served_gate_a.py:57: error: Argument "wasb" to "IndexingConfig" has incompatible
type "dict[str, float]"; expected "WasbIndexConfig" [arg-type]
Found 2 errors in 1 file (checked 138 source files)
```

### 389. assistant (2026-06-30T03:00:17.096Z)

mypy wants the constructor args typed (Pydantic coerces dicts at runtime, but the static type expects the sub-models). Let me confirm the sub-model class names/fields and use them:

### 390. user (2026-06-30T03:00:23.196Z)

#### IndexingConfig.fusion / .wasb field types + the sub-model classes

```
56: velocity_threshold: float = 0.02
67: class WasbIndexConfig(BaseModel):
68: """Typed config for the live WASB shuttle substrate (indexing.wasb).
```

```

85: velocity_threshold: float = 0.02
112:class FusionConfig(BaseModel):
116: shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']`` is
118: threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court mask
123: # Which trajectory substrate seeds the windows (shuttle stays primary).
124: substrate: Literal["wasb", "tracknet"] = "wasb"
126: # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
127: # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
128: velocity_threshold: float = 0.02
131: cue_flags: dict[str, bool] = Field(default_factory=dict)
134: min_crossings: int = Field(default=0, ge=0)
137: min_players: int = Field(default=0, ge=0)
162: yolo_velocity_threshold: float = 0.15
238: wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
239: fusion: FusionConfig = Field(default_factory=FusionConfig)

```

#### full mypy errors on served\_gate\_a

```

backend/eval/served_gate_a.py:51: error: Argument "fusion" to "IndexingConfig" has incompatible
type "dict[str, object]"; expected "FusionConfig" [arg-type]
backend/eval/served_gate_a.py:57: error: Argument "wasb" to "IndexingConfig" has incompatible
type "dict[str, float]"; expected "WasbIndexConfig" [arg-type]

```

#### 391. assistant (2026-06-30T03:00:35.391Z)

Confirmed: `fusion: FusionConfig`, `wasb: WasbIndexConfig`. Constructing the typed sub-models so it's mypy-clean and fully typed:

#### 392. user (2026-06-30T03:00:36.842Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served\_gate\_a.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 393. user (2026-06-30T03:00:46.748Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served\_gate\_a.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 394. assistant (2026-06-30T03:01:02.612Z)

Now `Any`/`Dict` are still used (in `aggregate`/`evaluate\_video` signatures), so no unused imports. Re-running mypy + ruff + the served tests to confirm clean:

#### 395. user (2026-06-30T03:01:38.255Z)

##### mypy

```

backend/providers/__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 138 source files

```

##### ruff

All checks passed!

##### served tests with golden (fix still works + bands hold)

```

..... [100%]
6 passed in 27.06s

```

#### 396. assistant (2026-06-30T03:01:57.669Z)

mypy + ruff clean, golden tests pass. Running the full suite once more (the change is type-equivalent, but disciplined per the gate), then committing PR 2:

#### 397. user (2026-06-30T03:03:56.789Z)

##### full suite

1505 passed, 10 skipped, 1 warning in 87.39s (0:01:27)  
RESULT: PASS - all tests green  
remote:  
PR #401  
Deleted branch fix/398-served-gate-typed-cfg (was 28b4d81).  
c8a493d fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a dict) (#401)

### 398. assistant (2026-06-30T03:04:26.049Z)

**\*\*PR #401 (#398) merged\*\*** ? `c8a493d`, suite **\*\*1505 passed\*\***. Two issues down. Now PR #399 (the largest). Let me set up: branch and read the scaffolding I'll mirror (the existing `leak\_scan` tool + CI job) and the exact link-fix sites.

### 399. user (2026-06-30T03:04:31.090Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

### 400. user (2026-06-30T03:04:48.933Z)

Switched to a new branch 'chore/399-launch-pr-process'

#### the 3 live broken-link lines

**\*\*Owner decision required\*\*** before any code lands ? see [Open decisions](#open-decisions-for-the-owner).  
([PER\_VIDEO\_OUTPUT\_LAYOUT.md](PER\_VIDEO\_OUTPUT\_LAYOUT.md)) so each video's artifacts are self-contained.

> **\*\*Update (2026-06-07):\*\*** **\*\*AUDIO is now a first-class capture requirement\*\*** (new row below), per ADR-007 ? we will use the shuttle "thwack" as a rally cue (audio+visual fusion lifts recall). This is a hard "record with audio on, mic unobstructed" ask, not a placeholder; efficacy on real GoPro audio is being validated (experiment E1 in [AUDIO\_RALLY\_DETECTION\_PLAN.md](archives/research/AUDIO\_RALLY\_DETECTION\_PLAN.md)).

#### confirm targets exist

/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md  
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/research/AUDIO\_RALLY\_DETECTION\_PLAN.md

#### l18N\_PLAN heading 4 (anchor target)

177:## 4. Open decisions (for the owner)

#### leak\_scan.py size + ci.yml leak-scan job

```
195 /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tools/leak_scan.py
22: runs-on: ubuntu-latest
34: # PII / attribution leak guard (tools/leak_scan.py): fails on a committed 32-hex MD5
38: leak-scan:
40: runs-on: ubuntu-latest
48: run: python -m tools.leak_scan
56: runs-on: ubuntu-latest
74: runs-on: ubuntu-latest
88: runs-on: ubuntu-latest
144: runs-on: ${{ matrix.os }}
171: runs-on: ubuntu-latest
```

### 401. assistant (2026-06-30T03:05:06.449Z)

Link sites and targets confirmed. Let me read the tool/CI patterns to mirror and the remaining edit sites:

### 402. user (2026-06-30T03:05:07.024Z)

```
1 """Leak-scan guard ? refuse to commit obvious PII / attribution leaks into the code
surface.
2
3 A *structural* CI guard (deliberately NOT a name detector): it scans the TRACKED text
files under
```

```

4 the code/data roots for
5
6 (a) a 32-hex MD5 token ? the ``video_id`` shape
7 (``backend/utils/hashing.py::compute_video_id``);
8 16-hex ``fx`` fixture slugs and 12-hex ``gt_hash`` values do NOT match, by
9 design.
10 (b) a personal absolute user path ? ``C:\\Users\\<name>``, ``/home/<name>/``,
11 ``/Users/<name>/`` ?
12 where ``<name>`` is a real username (documented placeholders like
13 ``<name>``/``<user>`` are OK).
14
15 Optionally, given a PRIVATE (gitignored) name-list via ``--names`` (default
16 ``<root>/leakscan_private``
17 if present), it ALSO flags those tokens ? so the owner can scan for real player/venue
18 names locally
19 while CI runs only the name-free structural guard (you can't commit a name-list to scan
20 for).
21
22 A line carrying ``# leakscan-allow`` is skipped (escape hatch for a legitimate token,
23 e.g. an MD5
24 test-vector). Run ``python -m tools.leak_scan`` (exit 1 on any finding). SCOPE = the
25 shipped-code
26 surface (``backend/ training/ scripts/ tools/ eval_baselines/``); ``tests/`` (legit
27 example
28 fixtures), ``docs/``, and gitignored ``scratch/``/``output/`` are NOT scanned (scan
29 those locally
30 with ``--names`` against a private list if desired).
31 """
32
33 from __future__ import annotations
34
35 import argparse
36 import re
37 import subprocess
38 import sys
39 from pathlib import Path
40 from typing import List, Optional, Sequence, Tuple
41
42 #: Shipped-code + committed-data surfaces scanned. NOT ``tests/`` (fixtures legitimately
43 embed
44 #: example paths/usernames/MD5s ? noise for a structural guard; scan those locally with
45 --names),
46 #: and NOT ``docs/``/``scratch/``/``output/`` (intentional operator notes).
47 SCAN_ROOTS: Tuple[str, ...] = (
48 "backend",
49 "training",
50 "scripts",
51 "tools",
52 "eval_baselines",
53)
54
55 #: This tool + its test legitimately contain sample patterns ? never scan them.
56 SELF_EXCLUDE: Tuple[str, ...] = ("tools/leak_scan.py", "tests/test_leak_scan.py")
57
58 #: A run of EXACTLY 32 hex chars (the MD5 video_id shape). The negative look-around
59 means a 16-hex
60 #: ``fx`` slug, a 12-hex ``gt_hash``, or a 40/64-hex SHA do NOT match.
61 _MD5_RE = re.compile(r"(?![0-9a-fA-F])[0-9a-fA-F]{32}(?![0-9a-fA-F])")
62 _WIN_USER_RE = re.compile(r"[A-Za-z]:[\\/]Users[\\/]+([A-Za-z0-9._-]+)")
63 _NIX_USER_RE = re.compile(r"(?:home|Users)/([A-Za-z0-9._-]+)")
64
65 #: Tokens that are a DOCUMENTED placeholder / generic example, not a real username ? so
66 a doc'd
67 #: path like ``C:\\Users\\x`` or ``/home/u`` does not trip the guard.
68 _PLACEHOLDERS = frozenset(

```

```

54 {
55 "<name>",
56 "<user>",
57 "<username>",
58 "<you>",
59 "you",
60 "youruser",
61 "yourname",
62 "user",
63 "username",
64 "x",
65 "u",
66 "a",
67 "b",
68 "c",
69 "n",
70 "foo",
71 "bar",
72 "baz",
73 "test",
74 "testuser",
75 "tmp",
76 "home",
77 "example",
78 }
79)
80
81 _ALLOW = "# leakscan-allow"
82 _TEXT_EXT = {
83 ".py",
84 ".json",
85 ".txt",
86 ".csv",
87 ".yaml",
88 ".yml",
89 ".cfg",
90 ".ini",
91 ".toml",
92 ".md",
93 }
94
95
96 def _tracked_files(root: Path) -> List[str]:
97 """The tracked files under SCAN_ROOTS (via ``git ls-files`` so only committed paths
are scanned)."""
98 out = subprocess.run(
99 ["git", "ls-files", *SCAN_ROOTS], cwd=str(root), capture_output=True, text=True
100)
101 return [f.strip() for f in out.stdout.splitlines() if f.strip()]
102
103
104 def _load_names(path: Optional[Path]) -> List[str]:
105 if path is None or not path.is_file():
106 return []
107 return [
108 ln.strip()
109 for ln in path.read_text(encoding="utf-8", errors="replace").splitlines()
110 if ln.strip() and not ln.lstrip().startswith("#")
111]
112
113
114 def scan_text(text: str, *, names: Sequence[str] = ()) -> List[Tuple[int, str]]:
115 """Return ``(lineno, reason)`` findings for one file's text (1-based line
numbers)."""
116 findings: List[Tuple[int, str]] = []

```

```

117 lnames = [n.lower() for n in names if n]
118 for i, line in enumerate(text.splitlines(), start=1):
119 if _ALLOW in line:
120 continue
121 if _MD5_RE.search(line):
122 findings.append((i, "32-hex token (video_id / MD5 shape)"))
123 for m in _WIN_USER_RE.finditer(line):
124 if m.group(1).lower() not in _PLACEHOLDERS:
125 findings.append(
126 (i, f"personal windows user path (?Users\\{m.group(1)}?)")
127)
128 break
129 for m in _NIX_USER_RE.finditer(line):
130 if m.group(1).lower() not in _PLACEHOLDERS:
131 findings.append((i, f"personal POSIX user path (?/{m.group(1)}/?)"))
132 break
133 low = line.lower()
134 for n in lnames:
135 if n in low:
136 findings.append((i, f"private name-list token {n!r}"))
137 break
138 return findings
139
140
141 def scan_repo(
142 root: Path, *, names_file: Optional[Path] = None
143) -> List[Tuple[str, int, str]]:
144 """Scan the tracked code-surface files under ``root``; return ``(relpath, lineno,
145 reason)``. """
146 names = _load_names(names_file)
147 results: List[Tuple[str, int, str]] = []
148 for rel in _tracked_files(root):
149 if rel in SELF_EXCLUDE or Path(rel).suffix.lower() not in _TEXT_EXT:
150 continue
151 try:
152 text = (root / rel).read_text(encoding="utf-8", errors="replace")
153 except OSError:
154 continue
155 for lineno, reason in scan_text(text, names=names):
156 results.append((rel, lineno, reason))
157 return results
158
159 def main(argv: Optional[Sequence[str]] = None) -> int:
160 ap = argparse.ArgumentParser(
161 description="Leak-scan guard: 32-hex MD5 + personal paths (+ optional private
162 name-list). "
163)
164 ap.add_argument(
165 "--root",
166 type=Path,
167 default=Path(__file__).resolve().parents[1],
168 help="repo root (default: the parent of tools/)",
169)
170 ap.add_argument(
171 "--names",
172 type=Path,
173 default=None,
174 help="optional gitignored private name-list (default: <root>/leakscan_private
175 if present)",
176)
177 args = ap.parse_args(list(argv) if argv is not None else None)
178 root: Path = args.root
179 names_file = args.names if args.names is not None else (root / ".leakscan_private")
180 findings = scan_repo(root, names_file=names_file)

```

```

179 if not findings:
180 print(
181 "[leak-scan] clean ? no MD5 / personal-path / private-name leaks in the code
surface."
182)
183 return 0
184 print(
185 f"[leak-scan] {len(findings)} potential leak(s) "
186 f"(add '# leakscan-allow' on a line to suppress a legitimate one):",
187 file=sys.stderr,
188)
189 for rel, lineno, reason in findings:
190 print(f" {rel}:{lineno}: {reason}", file=sys.stderr)
191 return 1
192
193
194 if __name__ == "__main__":
195 raise SystemExit(main())
196

```

#### 403. assistant (2026-06-30T03:06:26.910Z)

I have the `leak\_scan.py` pattern to mirror. Now writing the link-checker tool (stdlib, excludes `docs/archives/\*\*`, GitHub-style anchor slugs):

#### 404. assistant (2026-06-30T03:06:49.111Z)

"""Markdown internal-link checker ? fail CI on a broken internal link in a LIVE doc.

Scans tracked ``\*.md`` files (via ``git ls-files``) and, for each ``[text](dest)`` / ``![alt](dest)`` link whose destination is INTERNAL (a repo path and/or a ``#anchor``), verifies that

- the target file exists (resolved relative to the linking file, or the repo root for ``/abs``), and
- any ``#anchor`` matches a GitHub-style heading slug in the target markdown file.

SKIPPED: external links (``http(s)``/``mailto``/``tel``/``ftp``/protocol-relative ``//``), links inside fenced code blocks, and any file under ``docs/archives/\*\*`` ? archived docs are frozen snapshots whose relative links are historical-by-design (see ``docs/DOC\_STATUS.md`` §2). Mirrors ``tools/leak\_scan.py``:  
run ``python -m tools.check\_md\_links`` (exit 1 on any broken internal link in a live doc).  
"""

```
from __future__ import annotations
```

```

import argparse
import re
import subprocess
import sys
from pathlib import Path
from typing import Dict, List, Optional, Sequence, Set, Tuple

```

```

#: Source docs under these prefixes are NOT scanned ? archived = frozen snapshots whose relative
#: links are historical-by-design (DOC_STATUS.md §2 step 5).
EXCLUDE_PREFIXES: Tuple[str, ...] = ("docs/archives/",)
#: This tool + its test legitimately embed sample/broken links ? never scan them.
SELF_EXCLUDE: Tuple[str, ...] = ("tools/check_md_links.py", "tests/test_check_md_links.py")

```

```

_LINK_RE = re.compile(r"!?\[[^\]]*\]\(\s*([^\)]+?)\s*\)")
_HEADING_RE = re.compile(r"^(#{1,6})\s+(..*?)\s*#\s*$")
_FENCE_RE = re.compile(r"\s*(?:``|~~~)")
#: a scheme (``http``, ``mailto``, ?) or protocol-relative ``//`` ? external, not our concern.

```

```
_EXTERNAL_RE = re.compile(r"^(?:[a-zA-Z][a-zA-Z0-9+.\-]*:|//)")
```

```
def slugify(heading: str) -> str:
 """GitHub-style heading ? anchor slug: lowercase, drop punctuation (keep word chars/spaces/
 hyphens), collapse whitespace to single hyphens."""
 s = re.sub(r"^\w\s-", "", heading.strip().lower())
 return re.sub(r"\s+", "-", s).strip("-")
```

```
def heading_slugs(text: str) -> Set[str]:
 """The set of GitHub anchor slugs for a markdown file's headings (with -1/-2 dedup
 suffixes)."""
 counts: Dict[str, int] = {}
 out: Set[str] = set()
 in_fence = False
 for line in text.splitlines():
 if _FENCE_RE.match(line):
 in_fence = not in_fence
 continue
 if in_fence:
 continue
 m = _HEADING_RE.match(line)
 if not m:
 continue
 base = slugify(m.group(2))
 n = counts.get(base, 0)
 out.add(base if n == 0 else f"{base}-{n}")
 counts[base] = n + 1
 return out
```

```
def _split_dest(dest: str) -> Tuple[str, str]:
 """``path#anchor`` (with an optional ``title`` suffix) ? ``(path, anchor)``."""
 dest = dest.strip().split()[0] # drop a trailing "title"
 path, _, anchor = dest.partition("#")
 return path, anchor
```

```
def check_file(repo: Path, rel: str, slug_cache: Dict[str, Set[str]]) -> List[Tuple[int, str]]:
 md = repo / rel
 base = md.parent
 findings: List[Tuple[int, str]] = []
 in_fence = False
 for i, line in enumerate(md.read_text(encoding="utf-8", errors="replace").splitlines(), 1):
 if _FENCE_RE.match(line):
 in_fence = not in_fence
 continue
 if in_fence:
 continue
 for m in _LINK_RE.finditer(line):
 dest = m.group(1).strip()
 if not dest or _EXTERNAL_RE.match(dest):
 continue
 path, anchor = _split_dest(dest)
 if path == "":
 target = md # pure same-file anchor
 elif path.startswith("/"):
 target = repo / path.lstrip("/")
 else:
 target = (base / path).resolve()
 if path != "" and not target.exists():
 findings.append((i, f"broken link -> {dest} (no file {path})"))
 continue
 if anchor and target.suffix.lower() == ".md" and target.is_file():
```



```

 key = str(target)
 if key not in slug_cache:
 slug_cache[key] = heading_slugs(
 target.read_text(encoding="utf-8", errors="replace")
)
 if slugify(anchor) not in slug_cache[key]:
 findings.append((i, f"broken anchor -> {dest} (no heading #{anchor})"))
 return findings

def tracked_md(repo: Path) -> List[str]:
 out = subprocess.run(
 ["git", "ls-files", "*.md"], cwd=str(repo), capture_output=True, text=True
)
 files = sorted({f.strip() for f in out.stdout.splitlines() if f.strip()})
 return [
 f for f in files if not f.startswith(EXCLUDE_PREFIXES) and f not in SELF_EXCLUDE
]

def scan_repo(repo: Path) -> List[Tuple[str, int, str]]:
 slug_cache: Dict[str, Set[str]] = {}
 results: List[Tuple[str, int, str]] = []
 for rel in tracked_md(repo):
 try:
 for lineno, reason in check_file(repo, rel, slug_cache):
 results.append((rel, lineno, reason))
 except OSError:
 continue
 return results

def main(argv: Optional[Sequence[str]] = None) -> int:
 ap = argparse.ArgumentParser(
 description="Check internal markdown links in LIVE docs (excludes docs/archives/)."
)
 ap.add_argument(
 "--root",
 type=Path,
 default=Path(__file__).resolve().parents[1],
 help="repo root (default: the parent of tools/)",
)
 args = ap.parse_args(list(argv) if argv is not None else None)
 results = scan_repo(args.root)
 if not results:
 print("[check-md-links] clean ? no broken internal links in live docs.")
 return 0
 print(f"[check-md-links] {len(results)} broken internal link(s):", file=sys.stderr)
 for rel, lineno, reason in results:
 print(f" {rel}:{lineno}: {reason}", file=sys.stderr)
 return 1

if __name__ == "__main__":
 raise SystemExit(main())

```

**405. user (2026-06-30T03:06:50.217Z)**

File created successfully at: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tools/check\_md\_links.py (file state is current in your context ? no need to Read it back)

**406. assistant (2026-06-30T03:06:57.604Z)**

Now run the checker to see all live broken links it flags (before fixing), so I fix the complete

set:

#### 407. user (2026-06-30T03:07:01.713Z)

```
[check-md-links] 9 broken internal link(s):
 README.md:347: broken anchor -> #q5-why-did-the-video-fail-with-no-keyframes--packet-
 timestamps-could-be-extracted (no heading #q5-why-did-the-video-fail-with-no-keyframes--packet-
 timestamps-could-be-extracted)
 deploy/digitalocean/README.md:12: broken link -> ../../docs/DECOUPLED_COMPUTE/README.md (no
 file ../../docs/DECOUPLED_COMPUTE/README.md)
 docs/ANALYZERS_AND_RECONCILERS.md:494: broken anchor -> #13-prior-art-terminology--course-
 corrections-literature-review-2026-06-14 (no heading #13-prior-art-terminology--course-
 corrections-literature-review-2026-06-14)
 docs/ANALYZERS_AND_RECONCILERS.md:538: broken anchor -> #13-prior-art-terminology--course-
 corrections-literature-review-2026-06-14 (no heading #13-prior-art-terminology--course-
 corrections-literature-review-2026-06-14)
 docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md:10: broken anchor -> #alternative--a-persistent-l4-vm
 (no heading #alternative--a-persistent-l4-vm)
 docs/CREATING_REELS.md:11: broken anchor -> #6-run-it-in-the-cloud-advanced--optional (no
 heading #6-run-it-in-the-cloud-advanced--optional)
 docs/I18N_PLAN.md:4: broken anchor -> #open-decisions-for-the-owner (no heading #open-
 decisions-for-the-owner)
 docs/QUALITY_ITERATIONS.md:621: broken link -> PER_VIDEO_OUTPUT_LAYOUT.md (no file
 PER_VIDEO_OUTPUT_LAYOUT.md)
 docs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md:10: broken link ->
 archives/research/AUDIO_RALLY_DETECTION_PLAN.md (no file
 archives/research/AUDIO_RALLY_DETECTION_PLAN.md)
exit: 1
```

#### 408. assistant (2026-06-30T03:08:12.264Z)

9 flagged, but 6 are **false positives** ? my `slugify` collapses consecutive spaces  
(`s+`-), while GitHub converts **each** space to a hyphen, so a heading with ` /` (e.g.  
"keyframes / packet") yields a real `--` anchor. The investigation independently found only 3  
real breaks, confirming this. Fixing the slug algorithm to match GitHub exactly:

#### 409. user (2026-06-30T03:08:17.411Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-  
indexer/tools/check\_md\_links.py has been updated successfully. (file state is current in your  
context ? no need to Read it back)

#### 410. user (2026-06-30T03:08:27.828Z)

##### re-run after slug fix (expect exactly the 3 real broken)

```
[check-md-links] 4 broken internal link(s):
 deploy/digitalocean/README.md:12: broken link -> ../../docs/DECOUPLED_COMPUTE/README.md (no
 file ../../docs/DECOUPLED_COMPUTE/README.md)
 docs/I18N_PLAN.md:4: broken anchor -> #open-decisions-for-the-owner (no heading #open-
 decisions-for-the-owner)
 docs/QUALITY_ITERATIONS.md:621: broken link -> PER_VIDEO_OUTPUT_LAYOUT.md (no file
 PER_VIDEO_OUTPUT_LAYOUT.md)
 docs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md:10: broken link ->
 archives/research/AUDIO_RALLY_DETECTION_PLAN.md (no file
 archives/research/AUDIO_RALLY_DETECTION_PLAN.md)
exit: 1
```

#### 411. assistant (2026-06-30T03:08:49.083Z)

The slug fix cleared the 6 false positives, leaving **4** real broken links ? one more than the  
investigation found: `deploy/digitalocean/README.md:12` points at  
`docs/DECOUPLED\_COMPUTE/README.md`, which was archived. My tool scans the whole repo (better  
than excluding `deploy/`), so I'll fix all 4. Let me confirm the archived target and the exact  
line:

#### 412. user (2026-06-30T03:08:54.301Z)

##### deploy link line 12

```
machine ? the **compute** half of
[`docs/DECOUPLED_COMPUTE/`](../../docs/DECOUPLED_COMPUTE/README.md).
```

##### archived DECOUPLED\_COMPUTE README exists?

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/archives/past_projects/DECOUPLED_COMPUTE/README.md
```

##### any other DECOUPLED\_COMPUTE README locations

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/archives/past_projects/DECOUPLED_COMPUTE/README.md
```

#### 413. assistant (2026-06-30T03:09:13.331Z)

Confirmed target. Now fixing all 4 broken links ? reading each line region first (Edit gate):

#### 414. user (2026-06-30T03:09:14.327Z)

```
11 A storage-free, secret-free way to get the pipeline running on a DigitalOcean (or
Paperspace)
```

```
12 machine ? the **compute** half of
[`docs/DECOUPLED_COMPUTE/`](../../docs/DECOUPLED_COMPUTE/README.md).
```

#### 415. user (2026-06-30T03:09:15.247Z)

```
3 **Status:** Design (P0 deliverable; phases P1+ are owner-gated, not started).
4 **Owner decision required** before any code lands ? see [Open decisions](#open-
decisions-for-the-owner).
```

#### 416. user (2026-06-30T03:09:16.195Z)

```
620 **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
621 (PER_VIDEO_OUTPUT_LAYOUT.md) so each video's artifacts are
self-contained.
```

#### 417. user (2026-06-30T03:09:17.020Z)

```
9 >
10 > **Update (2026-06-07):** **AUDIO is now a first-class capture requirement** (new row
below), per ADR-007 ? we will use the shuttle "thwack" as a rally cue (audio+visual fusion lifts
recall). This is a hard "record with audio on, mic unobstructed" ask, not a placeholder;
efficacy on real GoPro audio is being validated (experiment E1 in
[AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md)).
```

#### 418. assistant (2026-06-30T03:09:25.179Z)

Making all 4 link fixes:

#### 419. user (2026-06-30T03:09:28.238Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/deploy/digitalocean/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 420. user (2026-06-30T03:09:35.744Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/I18N\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 421. user (2026-06-30T03:09:44.486Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/QUALITY\_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

context ? no need to Read it back)

#### 422. user (2026-06-30T03:09:53.110Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/data\_pipeline/VIDEO\_RECORDING\_GUIDELINES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 423. assistant (2026-06-30T03:10:03.381Z)

All 4 fixed. Re-running the checker to confirm clean:

#### 424. user (2026-06-30T03:10:09.016Z)

[check-md-links] clean ? no broken internal links in live docs.  
exit: 0

#### FIXES\_PLAN §8 line for the rule->code link

```
over-seg guard PASS** (net 0.397?0.378). Formal gate: `served_regresses(eps=0.0)=False`,
`promote_if_served_safe(structural=True) ? promote`. Per the rule (promote iff tolF1 does not
regress) ? **PROMOTED** to default-ON. The stricter honest-LIFT bar reports "no significant
lift"
```

#### 425. assistant (2026-06-30T03:10:27.995Z)

Link checker is clean (exit 0). Now the rule?code link, plus reading the CLAUDE.md conventions and the CI leak-scan job to mirror:

#### 426. user (2026-06-30T03:10:29.548Z)

```
108 **2.97 ? 2.72 s**; seg_count_ratio **?0.103** (p=0.006), split_rate **?0.010**
(p=0.016); **R1
109 over-seg guard PASS** (net 0.397?0.378). Formal gate:
`served_regresses(eps=0.0)=False`,
110 `promote_if_served_safe(structural=True) ? promote`. Per the rule (promote iff tolF1
does not
111 regress) ? **PROMOTED** to default-ON. The stricter honest-LIFT bar reports "no
significant lift"
112 (CI spans 0) but **no regression** ? i.e. a do-no-harm correctness fix, not a
measured win.
```

#### 427. user (2026-06-30T03:10:30.426Z)

```
60 on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
61 DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute
only**. The
62 `deploy/digitalocean/*` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
63
64 ## Workflow conventions
65 - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
66 - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
67 - Never push to the default branch directly; descriptive commit messages.
68 - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
69 - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
70 who has the golden corpus + labelled videos; verify pure-logic paths locally.
71
```

#### 428. user (2026-06-30T03:10:30.985Z)

```
34 # PII / attribution leak guard (tools/leak_scan.py): fails on a committed 32-hex MD5
35 # (the video_id shape) or a personal absolute user path in the shipped-code surface
36 # (backend/training/scripts/tools/eval_baselines; tests/ excluded ? legit fixtures).
Real
37 # names are scanned only via a gitignored private list locally; CI runs the structural
guard.
38 leak-scan:
39 name: Leak scan (PII / attribution guard)
40 runs-on: ubuntu-latest
41 timeout-minutes: 10
42 steps:
43 - uses: actions/checkout@v6
44 - uses: actions/setup-python@v6
45 with:
46 python-version: "3.13"
47 - name: Leak scan
48 run: python -m tools.leak_scan
49
50 # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
51 # per-module quarantine list ? ratchet by removing entries as modules clean
52 # up. Typed core deps are installed so mypy sees their real signatures
53 # (without them ignore_missing_imports would Any-stub half the checks).
54 typecheck:
55 name: Mypy (lenient baseline)
```

#### 429. assistant (2026-06-30T03:11:23.732Z)

I have all the patterns. Now the rule?code link, the CLAUDE.md convention, and the CI link-check job:

#### 430. user (2026-06-30T03:11:27.957Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 431. user (2026-06-30T03:11:39.956Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/CLAUDE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 432. user (2026-06-30T03:11:48.966Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.github/workflows/ci.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 433. assistant (2026-06-30T03:11:56.684Z)

Now the PR template and the link-checker test:

#### 434. assistant (2026-06-30T03:12:01.577Z)

```
<!-- Describe the change and link the issue it closes. -->
```

## Summary

## Checks

```
- [] `python run_all_tests.py` green; `ruff check .` + `mypy` clean; coverage not lowered.
```

- [ ] Internal markdown links valid (`python -m tools.check\_md\_links`).

## Launch / "bit-flip" PRs only

A PR that flips a default / "launch bit" so a tracked project reaches terminal state MUST also run the `docs/DOC\_STATUS.md` §2 archival due-diligence on that project's docs, **\*\*in this same PR\*\***:

- [ ] Verified each doc's claims against the code (tagged load-bearing facts `[verified]`).
- [ ] Flipped the doc Status ? DELIVERED/DONE/REJECTED + a completion note (what shipped, PR, paths).
- [ ] Updated all inbound references (`docs/README.md`, `CLAUDE.md`, `docs/NEXT\_STEPS.md`, cross-doc links).
- [ ] Kept the ADR forward/back lineage.
- [ ] `git mv` the terminal docs to `docs/archives/?` and recorded the move in `DOC\_STATUS.md` §3.
- [ ] Fixed any internal-link breakage from the move (`link-check` CI job green).

\_(Not a launch/bit-flip PR? Delete this section.)\_

### 435. user (2026-06-30T03:12:02.665Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.github/PULL\_REQUEST\_TEMPLATE.md (file state is current in your context ? no need to Read it back)

### 436. assistant (2026-06-30T03:12:17.580Z)

"""Tests for tools/check\_md\_links.py ? the internal-markdown-link CI guard."""

from tools import check\_md\_links as C

```
def _check(tmp_path, files):
 """Write {relpath: body} under tmp_path, then run check_file on each .md (no git needed)."""
 for rel, body in files.items():
 p = tmp_path / rel
 p.parent.mkdir(parents=True, exist_ok=True)
 p.write_text(body, encoding="utf-8")
 cache: dict = {}
 out = []
 for rel in files:
 if rel.endswith(".md"):
 out += [(rel, ln, r) for ln, r in C.check_file(tmp_path, rel, cache)]
 return out
```

## ----- slug / heading parsing

```
def test_slugify_matches_github_double_hyphen_on_removed_punct():
 # '/' is dropped but its surrounding spaces each become a hyphen (NO collapsing) -> '--'.
 assert C.slugify("Q5: Why A / B?") == "q5-why-a--b"
 assert C.slugify("4. Open decisions (for the owner)") == "4-open-decisions-for-the-owner"
```

```
def test_heading_slugs_dedup_and_skips_fences():
 s = C.heading_slugs("# Foo\n## Foo\n```\n# Not A Heading\n```\n")
 assert "foo" in s and "foo-1" in s and "not-a-heading" not in s
```

## ----- link checking

```
def test_valid_link_and_anchor_pass(tmp_path):
 files = {
 "a.md": "See [b](b.md#sec-two) and [self](#top).\n# Top\n",
 "b.md": "# Intro\n## Sec Two\n",
 }
```

```

assert _check(tmp_path, files) == []

def test_missing_file_flagged(tmp_path):
 f = _check(tmp_path, {"a.md": "[x](nope.md)\n"})
 assert len(f) == 1 and "broken link" in f[0][2]

def test_dead_anchor_flagged(tmp_path):
 f = _check(tmp_path, {"a.md": "[x](#no-such)\n# Real\n"})
 assert len(f) == 1 and "broken anchor" in f[0][2]

def test_external_and_fenced_are_skipped(tmp_path):
 body = "[h](https://x.com/y) [m](mailto:a@b.c)\n```\n[code](nope.md)\n```\n"
 assert _check(tmp_path, body and {"a.md": body}) == []

def test_link_to_existing_archived_file_is_ok(tmp_path):
 files = {"live.md": "[a](archives/old.md)\n", "archives/old.md": "# Old\n"}
 assert _check(tmp_path, files) == []

def test_title_suffix_is_stripped(tmp_path):
 files = {"a.md": "[x](b.md "a title")\n", "b.md": "# B\n"}
 assert _check(tmp_path, files) == []

```

#### 437. user (2026-06-30T03:12:18.612Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_check\_md\_links.py (file state is current in your context ? no need to Read it back)

#### 438. assistant (2026-06-30T03:12:30.030Z)

Now validate the new test + link checker + ruff on the new files:

#### 439. user (2026-06-30T03:12:35.061Z)

##### new test

```

..... [100%]
8 passed in 0.15s

```

##### link checker (clean?)

[check-md-links] clean ? no broken internal links in live docs.

##### ruff on new + changed files

All checks passed!

#### 440. assistant (2026-06-30T03:12:50.783Z)

8 passed, checker clean, ruff clean. Let me tidy one sloppy expression in the test, then run the full gate:

#### 441. user (2026-06-30T03:12:53.991Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test\_check\_md\_links.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 442. user (2026-06-30T03:14:38.615Z)

##### full suite

1513 passed, 10 skipped, 1 warning in 87.52s (0:01:27)  
 RESULT: PASS - all tests green

**ruff .**

All checks passed!

**mypy**

Success: no issues found in 138 source files

**link-check**

[check-md-links] clean ? no broken internal links in live docs.

#### 443. assistant (2026-06-30T03:14:56.902Z)

PR #399 gate fully green: **\*\*1513 passed\*\*** (+8), ruff + mypy + link-check all clean. Committing and merging:

#### 444. user (2026-06-30T03:15:24.093Z)

**changed files (sanity):**

M .github/workflows/ci.yml  
M CLAUDE.md  
M deploy/digitalocean/README.md  
M docs/I18N\_PLAN.md  
M docs/QUALITY\_ITERATIONS.md  
M docs/RALLY\_DETECTION\_FIXES\_PLAN.md  
M docs/data\_pipeline/VIDEO\_RECORDING\_GUIDELINES.md  
?? .github/PULL\_REQUEST\_TEMPLATE.md  
?? tests/test\_check\_md\_links.py  
?? tools/check\_md\_links.py

remote:

PR #402

Deleted branch chore/399-launch-pr-process (was a6d9fff).

af1ccbe chore(#399): launch-PR process hardening ? link-check CI, PR template, doc cross-links (#402)

#### 445. assistant (2026-06-30T03:16:13.315Z)

**\*\*PR #402 (#399) merged\*\*** ? `af1ccbe`. Three issues fully closed. Now the last code item ? the doc-staleness fixes (the "archival" item reshaped, since nothing is archive-ready). Let me verify each stale claim before editing:

#### 446. user (2026-06-30T03:16:17.958Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

#### 447. user (2026-06-30T03:16:31.785Z)

**FIXES\_PLAN header 'indexed in README' claim**

```
> **Status:** `DONE` . **Owner:** `Agent` . **Created:** `2026-06-19` . **Last updated:**
`2026-06-29`
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (archival deferred ? see the 2026-06-29
P2b note)
> **Tracking anchors:** S7 progress tracker is the source of truth; indexed in `docs/README.md`.
> **Relation to existing docs:** Extends `RALLY_DETECTION_QUALITY_REPORT.md`.
>
```

**is FIXES\_PLAN actually referenced in README.md or docs/README.md?**

NOT referenced -> claim is stale

**NEXT\_STEPS items 8 / 9b region**

5. **\*\*[Claude] Per-video workspace P1?P6\*\*** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` \* [the `output/chunks\_` de-hardcode base = ? done via #282]\* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

6. ? **\*\*Ops-Agent codified (#291)\*\*** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])



7. **\*\*[owner]** Set the ship-gate target (#172)**\*\*** ? needs corpus growth.  
8. **\*\*[owner-side GPU]** R-series: promote-or-reject `R2` toLF1 as the gate**\*\*** (the ablation) ? needs GPU-extracted golden features.

**\*\*P2 ? later / gated / deferred\*\***

9. Rally-quality **\*\*P3/P4\*\*** (person-fusion LOVO, audio cue) ? needs GPU golden features.  
9b. **\*\*[owner · GPU golden]** Promote the R4 density fix (finding B / P2 in `RALLY\_DETECTION\_FIXES\_PLAN.md`). **\*\*PR #393** landed it behind `inactivity\_temporal\_density` (**\*\*default-OFF\*\***, zero-regression no-op in prod). To ship: golden-set A/B with the flag ON vs OFF at the **\*\*SERVED\*\*** windowing (the eval `windowingPreset` carries it); flip the config default to True in a one-line PR only if toLF1 doesn't regress. Fixes a sparse-tracking quirk where one fast sample after a detection gap held the rally tail open (R4 is default-ON in prod).  
10. **\*\*M0.4 scorer swap\*\*** ? TemporalMaxer (per the 2026-06-17 external review).  
11. **\*\*Multisport\*\*** (table-tennis first, then pickleball).

## DOC STATUS §3a register header + a couple rows (format) + does it list FIXES\_PLAN/CODE\_REVIEW?

### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)

Doc	Status	Archive-ready?	Next action
--- --- --- ---			
`COMPUTE_DECOUPLED_SERVING/`   <b>**IN PROGRESS**</b> ? M0 approved + <b>**M1?M4 SHIPPED**</b> (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21)   No   M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)			
`RALLY_QUALITY_RESEARCH.md`   Foundation, <b>**tracked &amp; ACTIVE**</b> (§7)   No   Drives the R-series; complete when §7.1 single quality number is attributed			
`R2_EVAL_METRIC_DESIGN.md`   IMPLEMENTED (augmenting)   No (gates R4 + dual-eval; ablation pending)   Promote-or-reject toLF1 as the gate (ablation) ? then archive with the R-cluster			
`R1_MILESTONE_LAUNCH_REPORT.md`   LAUNCHED (#204) ? terminal record   Soon (with the R-cluster)   Keep with R-series for narrative; archive when the track wraps			
`RALLY_DETECTION_GAP_CLOSURE.md`   DRAFT (owner-gated)   No   G1?G3 levers; part of the R/quality cluster			
`PER_VIDEO_WORKSPACE.md`   DRAFT, IN PROGRESS (P0 = #264)   No   ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once)			
`MULTI_SIGNAL_FUSION_PLAN.md`   P1 DONE; P2+ owner-gated   No   P3/P4 (learned fusion, audio) ? needs GPU golden features			
`EXPERIMENT_HARNESS.md`   P1 IMPLEMENTED   No   A/B + promotion gate; supports the quality track			
`PROMINENT_COURT_DETECTION.md`   DRAFT (owner-gated)   No   C-series; multi-track selection design			
`EXECUTION_POLICY.md`   P1 SHIPPED; P3b next   No   Async job policy; reused by Cloud Run dispatch			
`GOLDEN_REGRESSION_FIXTURES.md`   IN PROGRESS (owner-gated)   No   Golden eval fixtures			
<b>**Data layer**</b> ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`)   LIVING (data, <b>**no-ML**</b> )   n/a   Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora <i>*vendor*</i> docs archived ? `archives/roora-vendor/`.			
`CODE_CLEANUP_AUDIT_2026-06-24.md`   <b>**IN PROGRESS**</b> ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next.   No   Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify			
`DESKTOP_APP_PLAN.md`   DRAFT (owner-gated, nothing built)   No   P0 compute-feasibility spike; relates to truly-local profile			
neither tracked in DOC_STATUS			

### 448. assistant (2026-06-30T03:17:22.395Z)

All 3 staleness claims verified real: the FIXES\_PLAN header claims it's "indexed in docs/README.md" (it isn't), NEXT\_STEPS 9b still says PR #393 must do the A/B flip (#396 already promoted it), and DOC\_STATUS §3a tracks neither RALLY\_DETECTION doc. Branching and reading the exact edit sites:

### 449. user (2026-06-30T03:17:28.177Z)

Switched to a new branch 'chore/doc-staleness-rally-cluster'

**450. user (2026-06-30T03:17:28.225Z)**

```
38 | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile
(finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates
with M8's v6 cost-ledger ? version bump authored once) |
39 | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion,
audio) ? needs GPU golden features |
```

**451. user (2026-06-30T03:17:28.229Z)**

```
24 9. Rally-quality **P3/P4** (person-fusion LOV0, audio cue) ? needs GPU golden features.
25 9b. **[owner · GPU golden] Promote the R4 density fix (finding B / P2 in
`RALLY_DETECTION_FIXES_PLAN.md`).** PR #393 landed it behind `inactivity_temporal_density`
(**default-OFF**, zero-regression no-op in prod). To ship: golden-set A/B with the flag ON vs
OFF at the **SERVED** windowing (the eval `WindowingPreset` carries it); flip the config default
to True in a one-line PR only if tolF1 doesn't regress. Fixes a sparse-tracking quirk where one
fast sample after a detection gap held the rally tail open (R4 is default-ON in prod).
```

**452. assistant (2026-06-30T03:18:03.770Z)**

I have the exact sites. Reading the FIXES\_PLAN header for its edit, then applying all three staleness fixes:

**453. user (2026-06-30T03:18:04.847Z)**

```
9 > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`.
```

**454. user (2026-06-30T03:18:13.581Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY\_DETECTION\_FIXES\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

**455. user (2026-06-30T03:18:25.514Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

**456. user (2026-06-30T03:18:35.905Z)**

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/DOC\_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

**457. assistant (2026-06-30T03:18:48.038Z)**

All 3 staleness fixes applied. Running the gate (link-check is the relevant one for doc edits, plus the suite for discipline):

**458. user (2026-06-30T03:20:19.230Z)**

**link-check (no broken links introduced)**

[check-md-links] clean ? no broken internal links in live docs.

**ruff**

All checks passed!

**suite**

1513 passed, 10 skipped, 1 warning in 83.45s (0:01:23)

RESULT: PASS - all tests green

**459. assistant (2026-06-30T03:20:33.793Z)**

Gate green (link-check + ruff clean, suite 1513). Committing the doc-staleness PR:

#### 460. user (2026-06-30T03:20:57.343Z)

remote:

PR #403

failed to delete remote branch chore/doc-staleness-rally-cluster: HTTP 404: Reference does not exist (https://api.github.com/repos/Khelsutra/badminton-highlight-indexer/git/refs/heads/chore%2Fdoc-staleness-rally-cluster)

Deleted branch chore/doc-staleness-rally-cluster (was 8b777f6).

6e973af docs: fix stale rally-cluster claims (no archival ? cluster still gated) (#403)

#### 461. assistant (2026-06-30T03:21:21.024Z)

**\*\*PR #403 merged\*\*** ? `6e973af` (the remote-branch-delete 404 is benign ? the squash auto-deleted it). All four PRs are in. Let me do a final verification ? confirm the issues auto-closed and master is clean:

#### 462. user (2026-06-30T03:21:29.382Z)

##### master last 5 commits

6e973af docs: fix stale rally-cluster claims (no archival ? cluster still gated) (#403)

af1ccbe chore(#399): launch-PR process hardening ? link-check CI, PR template, doc cross-links (#402)

c8a493d fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a dict) (#401)

959c630 fix(#397): write per-video report to the resolved local path, not the raw URI (#400)

39dfff1d feat(config): promote inactivity\_temporal\_density to default-ON (P2b) (#396)

##### open issues (397/398/399 should be CLOSED; only later ones remain)

388 OPEN refactor: migrate async-job plumbing

(\_get\_executor/\_drain\_due\_jobs/\_arm\_wake\_timer) out of main.py 2026-06-26T13:31:49Z

384 OPEN ci: single-source the shared offline guard lanes + coverage floor via a workflow\_call reusable 2026-06-25T17:23:52Z

349 OPEN WASB native: decode is CPU-bound & single-threaded in streaming mode ? L4 GPU ~80% idle (no NVDEC/parallel decode) 2026-06-23T16:24:58Z

340 OPEN [cost] /api/process re-runs the paid segmenter for an already-processed video ? refresh/"Try again" re-bills Gemini bug 2026-06-22T18:21:52Z

332 OPEN [safety] Content-safety pre-flight litmus: flag non-sports / NSFW video BEFORE sending it to Gemini 2026-06-22T14:18:42Z

331 OPEN [perf] Hosted processing is CPU-bound (ffmpeg/cv2) ? measured per-phase breakdown + levers (GPU/NVENC, parallel extraction, async compile) 2026-06-22T14:14:15Z

330 OPEN [optimization] MD5 segmentation reuse ? skip Gemini re-segmentation when the same video (video\_id) is re-uploaded 2026-06-22T14:11:32Z

328 OPEN Surface the per-video runlog / execution trace to the user (endpoint + UI), not just server logs 2026-06-22T13:57:58Z

327 OPEN Allow gs:// / s3:// / gdrive-api:// ingest on an exposed (jailed) box ? scheme-exempt recognized storage backends from the path-jail 2026-06-22T13:54:42Z

326 OPEN Hosted-app observability: request tracing + replay for debugging production CUJ failures 2026-06-22T13:54:40Z

##### confirm #397/#398/#399 closed:

#397: CLOSED

#398: CLOSED

#399: CLOSED

master clean + synced?

master...origin/master

#### 463. assistant (2026-06-30T03:21:55.563Z)

Everything verified clean: all 4 PRs on master, **\*\*#397/#398/#399 auto-closed\*\***, master synced. Done ? here's the close-out.

464. user (2026-06-30T03:21:58.861Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

465. assistant (2026-06-30T03:22:19.065Z)

## Wrap-up ? all four items closed out

**\*\*4 PRs merged\*\*** (each gated green before merge; the test suite grew **\*\*1503 ? 1513\*\***, coverage up):

```
| PR | Issue | What landed |
|----|----|----|
| [#400](https://github.com/Khelsutra/badminton-highlight-indexer/pull/400) | **#397** | Report writer now uses the *resolved local path*, not the raw URI ? no more `gcs:/ gs:/ local:/` dirs. `.gitignore` guard + junk deleted + a regression test (verified it fails pre-fix). |
| [#401](https://github.com/Khelsutra/badminton-highlight-indexer/pull/401) | **#398** | `served_gate_a` passes a **typed `IndexingConfig`** ? the 5 golden litmus tests that were *crashing* now actually run and **pass on real footage** (bands unchanged); added a golden-free synthetic CI test. |
| [#402](https://github.com/Khelsutra/badminton-highlight-indexer/pull/402) | **#399** | New **`link-check` CI** (`tools/check_md_links.py`, GitHub-exact slugs) ? it found & I fixed **4** live broken links; PR template + CLAUDE.md archival convention; rule?code cross-link; tests. |
| [#403](https://github.com/Khelsutra/badminton-highlight-indexer/pull/403) | ? | Doc-staleness fixes (DOC_STATUS rows, stale NEXT_STEPS 9b, FIXES_PLAN header claim).|
```

**\*\*#397/#398/#399 are closed\*\***, master is clean and synced.

## Two honest calls you should know about

**\*\*1. Archival ? I did *\*not\** move any doc, and shouldn't have.\*\*** The conservative finding (per the repo's own ``DOC_STATUS.md`` §2/§4) is that **\*\*zero docs are archive-ready\*\***: the R-series quality cluster is still ACTIVE/gated ? archival is blocked until R2 tolF1 is promoted-or-rejected via the *\*ablation\** (NEXT\_STEPS #8) and the §7.1 single quality number is attributed. P2b (#396) promoted the operational flag but is **\*\*not\*\*** that gate, and ``RALLY_DETECTION_FIXES_PLAN``'s own header already defers its archival to the cluster. Archiving now would violate the protocol, contradict the doc's own header, and orphan inbound refs. So instead I made the docs *\*honest\** (#403) and added DOC\_STATUS tracking rows. When the gate clears, the whole R-series archives as one unit ? and the new PR template + ``link-check`` CI now *\*enforce\** that the future launch/bit-flip PR does that archival and fixes hrefs. **\*\*If you want to override the gate and archive ``RALLY_DETECTION_FIXES_PLAN`` now anyway ? your call as owner ? just say so and I'll do it properly.\*\***

**\*\*2. reportlab ? it's a non-issue, not a cap.\*\*** There's no version cap anywhere; reportlab simply isn't installed, and the 2 PDF tests ``importorskip`` it **\*\*by design\*\*** (PDF report is an optional, lazy-imported feature). I deliberately did *\*not\** add a heavyweight dependency to CI just to un-skip 2 rendering tests. If you'd like that bit of coverage, ``reportlab`` + ``pypdf`` are both BSD (license-clean) and I can add them to a dev/test extra ? say the word.

## State of play

- **\*\*Sidecar contract\*\*** parked: [rally-corpus-vault#1](https://github.com/Khelsutra/rally-corpus-vault/issues/1) (expanded scope) + agent memory.
- The only thing still on your side is the **\*\*launch GO/NO-GO\*\*** ? unchanged (NO-GO pending ship-gate #172 + WASB provenance), and entirely fine in dev phase.

Nothing is half-finished. Thanks ? that was a good batch.

466. user (2026-06-30T03:53:14.275Z)

thanks and NO-GO for launch

467. assistant (2026-06-30T03:53:33.615Z)

Confirmed: **\*\*NO-GO for launch.\*\*** That's the right call and matches the evidence ? the rally-detection product stays in development; nothing ships to external/paying users yet.

For when you want to revisit it, the path to a future GO is exactly two unblockers (neither urgent in dev phase):

1. **\*\*Set the ship-gate target (#172)\*\*** and measure the detector against it.
2. **\*\*Resolve the WASB weight provenance\*\*** (the commercial-clean licensing question).

Everything from this session is landed and clean ? P2b promoted (#396), the four follow-up PRs merged (#400?#403), the corpus-vault repo + storage map in place, and the sidecar/migration work parked with durable pointers. Nothing is pending on my side.

If you'd like, I can record the NO-GO decision + the two unblockers as a short note in `NEXT\_STEPS.md` (or a memory) so the next session has it front-and-center ? otherwise we're done here. Thanks ? this was a great run. ?

#### 468. user (2026-06-30T03:57:57.216Z)

a note on the decision nwould be good

#### 469. assistant (2026-06-30T03:58:29.971Z)

Let me record it in `NEXT\_STEPS.md` (the canonical project home for ship-gate items) and a memory. First reading the top to place it well, and branching:

#### 470. user (2026-06-30T03:58:31.949Z)

```
1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized
pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
owned-model & rally-quality track plan (2026-06-13) is **preserved below, still valid** ? it
is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc
lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only**
(ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ?
one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video
workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273).
The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /
owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 ## ? Prioritized pending ? all tracks
9 **P0 ? highest leverage / unblocks the most**
10 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): ** (i) ** deploy the engine as a reachable
service (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), ** (ii) **
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), ** (iii) ** flip **M9 edge-auth ON** (merged #290,
default-OFF) to gate alpha tenants, ** (iv) ** a **compute PICKER in the UI** ("your machine vs
our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full
record: [[ui-driven-cloud-serving-gap]].
11 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable
batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip
install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-
home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades
(P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ** (1) ** P3 **GPU
validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ?
native-WASB 0-rally tuned-config blocker), ** (2) ** P2b-install (uv/script + truly-local default
config), ** (3) ** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then
P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` /
```

```
`[[next-session-packaging-app]]`.
12 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014
ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged
(default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All $8
gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel
ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12
real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a
$100/?10k cap (D17).
13 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3
matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus
training feed on it.
14 3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).
```

#### 471. user (2026-06-30T03:58:34.544Z)

Switched to a new branch 'docs/launch-decision-no-go'

#### 472. assistant (2026-06-30T03:58:49.420Z)

Good spot for it ? right after "Where we are." Adding a prominent dated launch-decision callout:

#### 473. user (2026-06-30T03:58:55.050Z)

The file /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT\_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 474. assistant (2026-06-30T03:59:06.274Z)

Gate for the doc note (link-check + ruff ? a one-paragraph doc note can't affect the test suite/coverage):

#### 475. user (2026-06-30T03:59:25.262Z)

##### link-check

[check-md-links] clean ? no broken internal links in live docs.

##### ruff

All checks passed!

remote:

PR #404

Deleted branch docs/launch-decision-no-go (was 01fc11e).

4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)

#### 476. assistant (2026-06-30T03:59:41.642Z)

\*\*PR #404 merged\*\* ? master `4a7bfbe`. The NO-GO decision is now a dated callout at the top of `NEXT\_STEPS.md`. Adding a matching agent memory so it surfaces in future sessions:

#### 477. assistant (2026-06-30T03:59:49.450Z)

---

name: launch-no-go-decision

description: Owner launch decision (2026-06-30) ? NO-GO for the rally-detection product launch; stay in development.

metadata:

type: project

---

\*\*2026-06-30, owner: NO-GO for an external/commercial rally-detection product launch.\*\* Stay in development ? the expected default for this phase. Recorded as a dated callout at the top of `docs/NEXT\_STEPS.md` (PR #404).

Two unblockers gate a future GO (re-decide only when BOTH clear):

1. **\*\*Ship-gate target (#172)\*\*** ? currently unset; needs corpus growth + the R2 tolF1 ablation (NEXT\_STEPS item 8) to promote-or-reject the metric gate.
2. **\*\*WASB shuttle-detector weight provenance\*\*** ? the commercial-clean licensing question on `wasb\_badminton\_best.pth.tar`.

**\*\*Why / how to apply:\*\*** merging dev PRs ? launching. P2b (#396, promoted `inactivity\_temporal\_density` default-ON) was a do-no-harm quality increment, NOT a launch gate. Don't treat a quality PR merge as moving the launch decision. The detector's current served quality (~0.47 F1@0.5) is the free heuristic floor; Gemini serves the quality tier; the owned model (~0.74 LOVO n=6) would clear a ~0.70 bar but is owner-gated and unmeasured at scale. See `[[rally-corpus-vault]]` for the corpus track that feeds #172.

#### 478. user (2026-06-30T03:59:50.529Z)

File created successfully at: `/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory/launch-no-go-decision.md` (file state is current in your context ? no need to Read it back)

#### 479. assistant (2026-06-30T03:59:59.015Z)

## Memory index

- `[launch-no-go-decision](launch-no-go-decision.md)` ? owner decided NO-GO for the rally-detection product launch (2026-06-30); unblockers = ship-gate #172 + WASB weight provenance.
- `[rally-corpus-vault](rally-corpus-vault.md)` ? new repo: the corpus storage/custody layer across `F:/Drive/GCS (storagectl, N+1, pluggable sinks)`.
- `[corpus-sidecar-cross-system-contract](corpus-sidecar-cross-system-contract.md)` ? per-video sidecar JSON as a cross-system contract (`vault/collector/training/eval`); parked design, rally-corpus-vault#1.

#### 480. user (2026-06-30T04:00:00.063Z)

The file `/home/avidullu/.claude/projects/-home-avidullu-projects-khelsutra-guru/memory/MEMORY.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 481. assistant (2026-06-30T04:00:08.348Z)

The decision is recorded in two places:

- **\*\*In the repo\*\*** ? a dated `? LAUNCH DECISION ? NO-GO (owner, 2026-06-30)` callout at the top of `[`docs/NEXT_STEPS.md`](badminton-highlight-indexer/docs/NEXT_STEPS.md)`, merged as PR #404. It states the decision, the two unblockers (ship-gate #172 + WASB weight provenance), and the "merging dev PRs ? launching" point.
- **\*\*In agent memory\*\*** ? ``launch-no-go-decision`` (indexed in `MEMORY.md`), so any future session surfaces it immediately.

That's the full close-out: P2b promoted, the four follow-up PRs (#400?#403) merged, the corpus-vault repo + storage map shipped, the sidecar contract parked, and now the launch NO-GO on record. Master is clean and synced; nothing pending on my side. Thanks ? good run. ?